**Open Geospatial Consortium**

Submission Date: <yyyy-mm-dd>

Approval Date: <yyyy-mm-dd>

Publication Date: <2021-05-15>

External identifier of this OGC® document: http://www.opengis.net/doc/EG/CityGML/3.0

Internal reference number of this OGC® document: 20-066

Version: 0.1

Category: OGC® Engineering Guidance

Editor: Charles Heazel

# OGC City Geography Markup Language (CityGML) 3.0 Conceptual Model Draft Users Guide

**Copyright notice**

Copyright © 2021 Open Geospatial Consortium

To obtain additional rights of use, visit http://www.opengeospatial.org/legal/

**Warning**

This document is not an OGC Standard. This document provides Engineering Guidance on the use of the OGC CityGML: 3.0 Conceptual Model Standard. This document is a non-normative resource and not an official position of the OGC membership. It is subject to change without notice and may not be referred to as an OGC Standard. Further, Engineering Guidance should not be referenced as required or mandatory technology in procurements.

Document type: OGC®Engineering Guidance

Document subtype:

Document stage: Draft

Document language: English

## License Agreement

Permission is hereby granted by the Open Geospatial Consortium, ("Licensor"), free of charge and subject to the terms set forth below, to any person obtaining a copy of this Intellectual Property and any associated documentation, to deal in the Intellectual Property without restriction (except as set forth below), including without limitation the rights to implement, use, copy, modify, merge, publish, distribute, and/or sublicense copies of the Intellectual Property, and to permit persons to whom the Intellectual Property is furnished to do so, provided that all copyright notices on the intellectual property are retained intact and that each person to whom the Intellectual Property is furnished agrees to the terms of this Agreement.

If you modify the Intellectual Property, all copies of the modified Intellectual Property must include, in addition to the above copyright notice, a notice that the Intellectual Property includes modifications that have not been approved or adopted by LICENSOR.

THIS LICENSE IS A COPYRIGHT LICENSE ONLY, AND DOES NOT CONVEY ANY RIGHTS UNDER ANY PATENTS THAT MAY BE IN FORCE ANYWHERE IN THE WORLD.

THE INTELLECTUAL PROPERTY IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, AND NONINFRINGEMENT OF THIRD PARTY RIGHTS. THE COPYRIGHT HOLDER OR HOLDERS INCLUDED IN THIS NOTICE DO NOT WARRANT THAT THE FUNCTIONS CONTAINED IN THE INTELLECTUAL PROPERTY WILL MEET YOUR REQUIREMENTS OR THAT THE OPERATION OF THE INTELLECTUAL PROPERTY WILL BE UNINTERRUPTED OR ERROR FREE. ANY USE OF THE INTELLECTUAL PROPERTY SHALL BE MADE ENTIRELY AT THE USER'S OWN RISK. IN NO EVENT SHALL THE COPYRIGHT HOLDER OR ANY CONTRIBUTOR OF INTELLECTUAL PROPERTY RIGHTS TO THE INTELLECTUAL PROPERTY BE LIABLE FOR ANY CLAIM, OR ANY DIRECT, SPECIAL, INDIRECT OR CONSEQUENTIAL DAMAGES, OR ANY DAMAGES WHATSOEVER RESULTING FROM ANY ALLEGED INFRINGEMENT OR ANY LOSS OF USE, DATA OR PROFITS, WHETHER IN AN ACTION OF CONTRACT, NEGLIGENCE OR UNDER ANY OTHER LEGAL THEORY, ARISING OUT OF OR IN CONNECTION WITH THE IMPLEMENTATION, USE, COMMERCIALIZATION OR PERFORMANCE OF THIS INTELLECTUAL PROPERTY.

This license is effective until terminated. You may terminate it at any time by destroying the Intellectual Property together with all copies in any form. The license will also terminate if you fail to comply with any term or condition of this Agreement. Except as provided in the following sentence, no such termination of this license shall require the termination of any third party end-user sublicense to the Intellectual Property which is in force as of the date of notice of such termination. In addition, should the Intellectual Property, or the operation of the Intellectual Property, infringe, or in LICENSOR's sole opinion be likely to infringe, any patent, copyright, trademark or other right of a third party, you agree that LICENSOR, in its sole discretion, may terminate this license without any compensation or liability to you, your licensees or any other party. You agree upon termination of any kind to destroy or cause to be destroyed the Intellectual Property together with all copies in any form, whether held by you or by any third party.

Except as contained in this notice, the name of LICENSOR or of any other holder of a copyright in all or part of the Intellectual Property shall not be used in advertising or otherwise to promote the sale, use or other dealings in this Intellectual Property without prior written authorization of LICENSOR or such copyright holder. LICENSOR is and shall at all times be the sole entity that may authorize you or any third party to use certification marks, trademarks or other special designations to indicate compliance with any LICENSOR standards or specifications. This Agreement is governed by the laws of the Commonwealth of Massachusetts. The application to this Agreement of the United Nations Convention on Contracts for the International Sale of Goods is hereby expressly excluded. In the event any provision of this Agreement shall be deemed unenforceable, void or invalid, such provision shall be modified so as to make it valid and enforceable, and as so modified the entire Agreement shall remain in full force and effect. No decision, action or inaction by LICENSOR shall be construed to be a waiver of any rights or remedies available to it.

# Table of Contents


1. Introduction . . . 12
2. How To Use This Resource . . . 13
3. Scope . . . 14
4. References . . . 15
5. Terms and Definitions . . . 16
6. Conventions . . . 19
   6.1. Identifiers . . . 19
   6.2. UML Notation . . . 19
7. CityGML Foundations . . . 23
   7.1. Modularization . . . 23
   7.2. General Modelling Principles . . . 25
      7.2.1. Semantic Modelling of Real-World Objects . . . 25
      7.2.2. Class Hierarchy and Inheritance of Properties and Relations . . . 26
      7.2.3. Relationships between CityGML objects . . . 26
      7.2.4. Definition of the Semantics for all Classes, Properties, and Relations . . . 27
   7.3. Representation of Spatial Properties . . . 27
      7.3.1. Geometry and Topology . . . 27
      7.3.2. Prototypic Objects / Scene Graph Concepts . . . 29
      7.3.3. Point Cloud Representation . . . 29
      7.3.4. Coordinate Reference Systems (CRS) . . . 30
   7.4. Appearances . . . 30
   7.5. Modelling Dynamic Data . . . 30
      7.5.1. Versioning and Histories . . . 31
      7.5.2. Dynamizers: Using Time-Series Data for Object Attributes . . . 31
   7.6. Extending CityGML . . . 33
8. CityGML Model . . . 34
   8.1. Core . . . 34
      8.1.1. Key Concepts . . . 34
      8.1.2. ISO Dependencies . . . 36
      8.1.3. City Models and City Objects . . . 43
      8.1.4. Space Concept . . . 45
      8.1.5. Geometry and LOD . . . 48
      8.1.6. CityGML Core UML Model . . . 50
      8.1.7. Data types, Enumerations, and Code lists . . . 50
   8.2. Appearance . . . 52
      8.2.1. Synopsis . . . 52
      8.2.2. Key Concepts . . . 52
      8.2.3. Discussion . . . 53

8.2.4. Material . . . 53
8.2.5. Texture and texture mapping . . . 54
8.2.6. Related concepts . . . 62
8.2.7. UML Model . . . 62
8.2.8. Examples . . . 64
8.3. Bridge Model . . . 64
8.3.1. Synopsis . . . 64
8.3.2. Key Concepts . . . 64
8.3.3. Discussion . . . 64
8.3.4. Bridge and bridge part . . . 69
8.3.5. Bridge construction elements and bridge installations . . . 69
8.3.6. Boundary surfaces . . . 70
8.3.7. Openings . . . 72
8.3.8. Bridge Interior . . . 73
8.3.9. UML Model . . . 74
8.3.10. Examples . . . 75
8.4. Building Model . . . 78
8.4.1. Synopsis . . . 78
8.4.2. Key Concepts . . . 78
8.4.3. Discussion . . . 79
8.4.4. Building Part . . . 83
8.4.5. Outer building installations . . . 87
8.4.6. Boundary surfaces . . . 87
8.4.7. Openings . . . 90
8.4.8. Building Interior . . . 91
8.4.9. Modelling building storeys using CityObjectGroups . . . 93
8.4.10. UML Model . . . 93
8.4.11. Examples . . . 95
8.5. City Furniture . . . 100
8.5.1. Synopsis . . . 100
8.5.2. Key Concepts . . . 100
8.5.3. Discussion . . . 100
8.5.4. UML Model . . . 104
8.5.5. Examples . . . 105
8.6. City Object Group . . . 106
8.6.1. Synopsis . . . 106
8.6.2. Key Concepts . . . 106
8.6.3. Discussion . . . 107
8.6.4. UML Model . . . 107
8.6.5. Examples . . . 108
8.7. Construction . . . 109

8.7.1. Synopsis . . . 109
8.7.2. Key Concepts . . . 109
8.7.3. Discussion . . . 109
8.7.4. Construction Spaces . . . 110
8.7.5. Construction Surfaces . . . 110
8.7.6. Levels of Detail . . . 110
8.7.7. Spaces and Surfaces . . . 114
8.7.8. UML Model . . . 117
8.7.9. Examples . . . 119
8.8. Dynamizer . . . 119
8.8.1. Synopsis . . . 120
8.8.2. Key Concepts . . . 120
8.8.3. Discussion . . . 120
8.8.4. UML Model . . . 121
8.8.5. Examples . . . 122
8.9. Generics . . . 122
8.9.1. Synopsis . . . 122
8.9.2. Key Concepts . . . 122
8.9.3. Discussion . . . 123
8.9.4. UML Model . . . 125
8.9.5. Examples . . . 127
8.10. Land Use . . . 127
8.10.1. Synopsis . . . 127
8.10.2. Key Concepts . . . 128
8.10.3. Discussion . . . 128
8.10.4. UML Model . . . 129
8.10.5. Examples . . . 130
8.11. Point Cloud . . . 130
8.11.1. Synopsis . . . 130
8.11.2. Key Concepts . . . 130
8.11.3. Discussion . . . 131
8.11.4. UML Model . . . 131
8.11.5. Examples . . . 132
8.12. Relief . . . 132
8.12.1. Synopsis . . . 132
8.12.2. Key Concepts . . . 132
8.12.3. Discussion . . . 132
8.12.4. UML Model . . . 134
8.12.5. Examples . . . 135
8.13. Transportation . . . 135
8.13.1. Synopsis . . . 136

8.13.2. Key Concepts . . . 136
8.13.3. Discussion . . . 136
8.13.4. Level of Detail . . . 137
8.13.5. UML Model . . . 140
8.13.6. Examples . . . 142
8.14. Tunnel Model . . . 144
8.14.1. Synopsis . . . 144
8.14.2. Key Concepts . . . 144
8.14.3. Discussion . . . 145
8.14.4. Level of Detail . . . 149
8.14.5. UML Model . . . 150
8.14.6. Examples . . . 152
8.15. Vegetation . . . 155
8.15.1. Synopsis . . . 155
8.15.2. Key Concepts . . . 156
8.15.3. Discussion . . . 156
8.15.4. Level of Detail . . . 157
8.15.5. UML Model . . . 160
8.15.6. Examples . . . 162
8.16. Versioning . . . 162
8.16.1. Synopsis . . . 162
8.16.2. Key Concepts . . . 162
8.16.3. Discussion . . . 163
8.16.4. UML Model . . . 163
8.16.5. Examples . . . 164
8.17. Waterbodies . . . 164
8.17.1. Synopsis . . . 164
8.17.2. Key Concepts . . . 164
8.17.3. Discussion . . . 164
8.17.4. Level of Detail . . . 165
8.17.5. UML Model . . . 166
8.17.6. Examples . . . 167
8.18. Extensions . . . 168
8.18.1. Technical principle of ADEs . . . 169
8.18.2. Example ADE . . . 170
9. CityGML Data Dictionary . . . 172
9.1. ISO Classes . . . 172
9.1.1. Class AnyFeature (ISO 19109:2015) . . . 172
9.1.2. Class CV_DiscreteGridPointCoverage (ISO 19123:2005) . . . 173
9.1.3. Class DirectPosition (ISO 19107: 2003) . . . 173
9.1.4. Class GM_Object (ISO 19107: 2003) . . . 174

9.1.5. Class GM_MultiCurve (ISO 19107: 2003) . . . 175
9.1.6. Class GM_MultiPoint (ISO 19107:2003) . . . 176
9.1.7. Class GM_MultiSurface (ISO 19107:2003) . . . 176
9.1.8. Class GM_Point (ISO 19107:2003) . . . 177
9.1.9. Class GM_Solid (ISO 19107:2003) . . . 177
9.1.10. Class GM_Surface (ISO 19107:2003) . . . 178
9.1.11. Class GM_Tin (ISO 19107:2003) . . . 179
9.1.12. Class GM_TriangulatedSurface (ISO 19107:2003) . . . 179
9.1.13. Class SC_CRS (ISO 19111:2019) . . . 179
9.1.14. Class TM_Position (ISO 19108:2006) . . . 180
9.2. Core . . . 181
9.2.1. Classes . . . 181
9.2.2. Data Types . . . 193
9.2.3. Basic Types . . . 199
9.2.4. Unions . . . 202
9.2.5. Code Lists . . . 203
9.2.6. Enumerations . . . 205
9.3. **Appearance** . . . 206
9.3.1. **Classes** . . . 206
9.3.2. Data Types . . . 211
9.3.3. Basic Types . . . 213
9.3.4. Unions . . . 213
9.3.5. Code Lists . . . 213
9.3.6. Enumerations . . . 213
9.4. CityFurniture . . . 214
9.4.1. Classes . . . 215
9.4.2. Data Types . . . 215
9.4.3. Basic Types . . . 215
9.4.4. Unions . . . 215
9.4.5. Code Lists . . . 216
9.4.6. Enumerations . . . 216
9.5. CityObjectGroup . . . 216
9.5.1. Classes . . . 216
9.5.2. Data Types . . . 218
9.5.3. Basic Types . . . 218
9.5.4. Unions . . . 218
9.5.5. Code Lists . . . 218
9.5.6. Enumerations . . . 219
9.6. Dynamizer . . . 219
9.6.1. Classes . . . 219
9.6.2. Data Types . . . 224

9.6.3. Basic Types . . . 229
9.6.4. Unions . . . 229
9.6.5. Code Lists . . . 229
9.6.6. Enumerations . . . 230
9.7. Generics . . . 231
9.7.1. Classes . . . 231
9.7.2. Data Types . . . 233
9.7.3. Basic Types . . . 237
9.7.4. Unions . . . 237
9.7.5. Code Lists . . . 237
9.7.6. Enumerations . . . 240
9.8. LandUse . . . 240
9.8.1. Classes . . . 240
9.8.2. Data Types . . . 240
9.8.3. Basic Types . . . 241
9.8.4. Unions . . . 241
9.8.5. Code Lists . . . 241
9.8.6. Enumerations . . . 241
9.9. PointCloud . . . 242
9.9.1. Classes . . . 242
9.9.2. Data Types . . . 242
9.9.3. Basic Types . . . 243
9.9.4. Unions . . . 243
9.9.5. Code Lists . . . 243
9.9.6. Enumerations . . . 243
9.10. Relief . . . 243
9.10.1. Classes . . . 243
9.10.2. Data Types . . . 247
9.10.3. Basic Types . . . 248
9.10.4. Unions . . . 248
9.10.5. Code Lists . . . 248
9.10.6. Enumerations . . . 248
9.11. Transportation . . . 248
9.11.1. Classes . . . 248
9.11.2. Data Types . . . 259
9.11.3. Basic Types . . . 262
9.11.4. Unions . . . 262
9.11.5. Code Lists . . . 262
9.11.6. Enumerations . . . 268
9.12. Vegetation . . . 268
9.12.1. Classes . . . 269

9.12.2. Data Types . . . 271
9.12.3. Basic Types . . . 271
9.12.4. Unions . . . 271
9.12.5. Code Lists . . . 271
9.12.6. Enumerations . . . 272
9.13. Versioning . . . 273
9.13.1. Classes . . . 273
9.13.2. Data Types . . . 274
9.13.3. Basic Types . . . 275
9.13.4. Unions . . . 275
9.13.5. Code Lists . . . 276
9.13.6. Enumerations . . . 276
9.14. WaterBody . . . 277
9.14.1. Classes . . . 277
9.14.2. Data Types . . . 279
9.14.3. Basic Types . . . 280
9.14.4. Unions . . . 280
9.14.5. Code Lists . . . 280
9.14.6. Enumerations . . . 281
9.15. Construction . . . 281
9.15.1. Classes . . . 281
9.15.2. Data Types . . . 291
9.15.3. Basic Types . . . 296
9.15.4. Unions . . . 296
9.15.5. Code Lists . . . 296
9.15.6. Enumerations . . . 298
9.16. Bridge . . . 300
9.16.1. Classes . . . 300
9.16.2. Data Types . . . 304
9.16.3. Basic Types . . . 306
9.16.4. Unions . . . 306
9.16.5. Code Lists . . . 306
9.16.6. Enumerations . . . 308
9.17. Building . . . 309
9.17.1. Classes . . . 309
9.17.2. Data Types . . . 316
9.17.3. Basic Types . . . 319
9.17.4. Unions . . . 319
9.17.5. Code Lists . . . 319
9.17.6. Enumerations . . . 322
9.18. Tunnel . . . 322

9.18.1. Classes . . . 322
9.18.2. Data Types . . . 327
9.18.3. Basic Types . . . 328
9.18.4. Unions . . . 328
9.18.5. Code Lists . . . 328
9.18.6. Enumerations . . . 331
Annex A: Revision History . . . 332
Annex B: Glossary . . . 333
B.1. ISO Concepts . . . 334
B.2. Abbreviated Terms . . . 338
Annex C: Bibliography . . . 340

## i. Abstract

CityGML is an open conceptual data model for the storage and exchange of virtual 3D city models. It is defined through a Unified Modeling Language (UML) object model. This UML model extends the ISO Technical Committee 211 (TC211) conceptual model standards for spatial and temporal data. Building on the ISO fundation assures that the man-made features described in the City Models share the same spatial-temporal universe as the surrounding countryside within which they reside.

The aim of the development of CityGML is to reach a common definition of the basic entities, attributes, and relations of a 3D city model. This is especially important with respect to the cost-effective sustainable maintenance of 3D city models, allowing the reuse of the same data in different application fields.

## ii. Keywords

The following are keywords to be used by search engines and document catalogues.

ogcdoc, OGC document, CityGML, 3D city models

## iii. Preface

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. The Open Geospatial Consortium shall not be held responsible for identifying any or all such patent rights.

Recipients of this document are requested to submit, with their comments, notification of any relevant patent claims or other intellectual property rights of which they may be aware that might be infringed by any implementation of the standard set forth in this document, and to provide supporting documentation.

## iv. Submitting organizations

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

Organization name(s)

## v. Submitters

All questions regarding this submission should be directed to the editor or the submitters:

Name Affiliation

# Chapter 1. Introduction

An increasing number of cities and companies are building virtual 3D city models for different application areas like urban planning, mobile telecommunication, disaster management, 3D cadastre, tourism, vehicle and pedestrian navigation, facility management and environmental simulations. Furthermore, in the implementation of the European Environmental Noise Directive (END, 2002/49/EC) 3D geoinformation and 3D city models play an important role.

In recent years, most virtual 3D city models have been defined as purely graphical or geometrical models, neglecting the semantic and topological aspects. Thus, these models could almost only be used for visualisation purposes but not for thematic queries, analysis tasks, or spatial data mining. Since the limited reusability of models inhibits the broader use of 3D city models, a more general modelling approach had to be taken in order to satisfy the information needs of the various application fields.

CityGML is a common semantic information model for the representation of 3D urban objects that can be shared over different applications. The latter capability is especially important with respect to the cost-effective sustain-able maintenance of 3D city models, allowing the possibility of selling the same data to customers from different application fields. The targeted application areas explicitly include city planning, architectural design, tourist and leisure activities, environmental simulation, mobile telecommunication, disaster management, homeland securi-ty, real estate management, vehicle and pedestrian navigation, and training simulators.

CityGML is an open conceptual data model for the storage and exchange of virtual 3D city models. It is defined through a Unified Modeling Language (UML) object model. This UML model extends the ISO Technical Committee 211 (TC211) conceptual model standards for spatial and temporal data. Building on the ISO fundation assures that the man-made features described in the City Models share the same spatial-temporal universe as the surrounding countryside within which they reside.

CityGML defines the classes and relations for the most relevant topographic objects in cities and regional models with respect to their geometrical, topological, semantical, and appearance properties. “City” is broadly defined to comprise not just built structures, but also elevation, vegetation, water bodies, “city furniture”, and more. Included are generalisation hierarchies between thematic classes, aggregations, relations between objects, and spatial properties. CityGML is applicable for large areas and small regions and can represent the terrain and 3D objects in different levels of detail simultaneously. Since either simple, single scale models without topology and few semantics or very complex multi-scale models with full topology and fine-grained semantical differenti-ations can be represented, CityGML enables lossless information exchange between different GI systems and users.

# Chapter 2. How To Use This Resource

The Users Guide to the CityGML 3.0 Conceptual Model Standard is not intended to be read from start to finnish. Rather, it is a resource structured to provide quick answers to questions which an implementer may have about the CityGML 3.0 Standard.

The CityGML 3.0 Standard includes hyperlinks which can be used to navigate directly to relevant sections of the Users Guide.

# Chapter 3. Scope

This document provides Engineering Guidance on the use of the CityGML 3.0 Conceptual Model Standard.

The OGC Conceptual Model Standard specifies the representation of virtual 3D city and landscape models. The CityGML 3.0 Conceptual Model is expected to be the basis for a number of future Encoding Standards in which subsets of the Conceptual Model can be implemented. These Encoding Standards will enable both storage and exchange of data.

The CityGML 3.0 Conceptual Model Standard was designed to be concise and easy to use. As a result, most non-normative content has been removed. The purpose of this Users Guide is to capture that non-normative content and make it easy to access if and when needed.

# Chapter 4. References

The following normative documents contain provisions that, through reference in this text, constitute provisions of this Users Guide. For dated references, subsequent amendments to, or revisions of, any of these publications do not apply. For undated references, the latest edition of the document referred to applies.

- IETF: RFC 2045 & 2046, Multipurpose Internet Mail Extensions (MIME). (November 1996),
- IETF: RFC 3986, Uniform Resource Identifier (URI): Generic Syntax. (January 2005)
- INSPIRE: D2.8.III.2 Data Specification on Buildings – Technical Guidelines. European Commission Joint Research Centre.
- ISO: ISO 19101-1:2014, Geographic information - Reference model - Part 1: Fundamentals
- ISO: ISO 19103:2015, Geographic Information – Conceptual Schema Language
- ISO: ISO 19105:2000, Geographic information – Conformance and testing
- ISO: ISO 19107:2003, Geographic Information – Spatial Schema
- ISO: ISO 19108:2002/Cor 1:2006, Geographic information – Temporal schema — Technical Corrigendum 1
- ISO: ISO 19109:2015, Geographic Information – Rules for Application Schemas
- ISO: ISO 19111:2019, Geographic information – Referencing by coordinates
- ISO: ISO 19123:2005, Geographic information — Schema for coverage geometry and functions
- ISO: ISO 19156:2011, Geographic information – Observations and measurements
- ISO: ISO/IEC 19505-2:2012, Information technology — Object Management Group Unified Modeling Language (OMG UML) — Part 2: Superstructure
- ISO/IEC 19507:2012, Information technology — Object Management Group Object Constraint Language (OCL)
- ISO: ISO/IEC 19775-1:2013 Information technology — Computer graphics, image processing and environmental data representation — Extensible 3D (X3D) — Part 1: Architecture and base components
- Khronos Group Inc.: COLLADA – Digital Asset Schema Release 1.5.0
- OASIS: Customer Information Quality Specifications - extensible Address Language (xAL), Version v3.0
- OGC: The OpenGIS® Abstract Specification Topic 5: Features, OGC document 08-126
- OGC: The OpenGIS™ Abstract Specification Topic 8: Relationships Between Features, OGC document 99-108r2
- OGC: The OpenGIS™ Abstract Specification Topic 10: Feature Collections, OGC document 99-110

# Chapter 5. Terms and Definitions

For the purposes of this document, the following additional terms and definitions apply.

**2D data**
geometry of features is represented in a two-dimensional space
NOTE In other words, the geometry of 2D data is given using (X,Y) coordinates.
[INSPIRE D2.8.III.2, definition 1]

**2.5D data**
geometry of features is represented in a three-dimensional space with the constraint that, for each (X,Y) position, there is only one Z
[INSPIRE D2.8.III.2, definition 2]

**3D data**
Geometry of features is represented in a three-dimensional space.
NOTE In other words, the geometry of 2D data is given using (X,Y,Z) coordinates without any constraints.
[INSPIRE D2.8.III.2, definition 3]

**application schema**
A set of conceptual schema for data required by one or more applications. An application schema contains selected parts of the base schemas presented in the ORM Information Viewpoint. Designers of application schemas may extend or restrict the types defined in the base schemas to define appropriate types for an application domain. Application schemas are information models for a specific information community.
OGC Definitions Register at http://www.opengis.net/def/glossary/term/ApplicationSchema

**codelist**
A value domain including a code for each permissible value.

**conceptual model**
model that defines concepts of a universe of discourse
[ISO 19101-1:2014, 4.1.5]

**conceptual schema**

1. formal description of a conceptual model
   [ISO 19101-1:2014, 4.1.6]
2. base schema. Formal description of the model of any geospatial information. Application schemas are built from conceptual schemas.
   OGC Definitions Register at http://www.opengis.net/def/glossary/term/ConceptualSchema

**Implementation Specification**
Specified on the OGC Document Types Register at http://www.opengis.net/def/doc-type/is

**levels of detail**
quantity of information that portrays the real world
NOTE The concept comprises data capturing rules of spatial object types, the accuracy and the types

of geometries, and other aspects of a data specification. In particular, it is related to the notions of scale and resolution.
[INSPIRE Glossary]

**life-cycle information**
set of properties of a spatial object that describe the temporal characteristics of a version of a spatial object or the changes between versions
[INSPIRE Glossary]

**Platform (Model Driven Architecture)**
the set of resources on which a system is realized.
[Object Management Group, Model Driven Architecture Guide rev. 2.0]

**Platform Independent Model**
a model that is independent of a spacific platform
[Object Management Group, Model Driven Architecture Guide rev. 2.0]

**Platform Specific Model**
a model of a system that is defined in terms of a specific platform
[Object Management Group, Model Driven Architecture Guide rev. 2.0]

**Universally Unique Identifier**
A 128-bit value generated in accordance with this Recommendation | International Standard, or in accordance with some historical specifications, and providing unique values between systems and over time. [ISO/IEC 9834-8:2014, Rec. ITU-T X.667 (10/2012)]

**universe of discourse**
view of the real or hypothetical world that includes everything of interest
[ISO 19101-1:2014, definition 4.1.38]

**Abbreviated Terms**

The following abbreviated terms are used in this document:

The list of acronyms needs to be reviewed once all sections have been updated.

- 2D Two Dimensional
- 3D Three Dimensional
- AEC Architecture, Engineering, Construction
- ALKIS German National Standard for Cadastral Information
- ATKIS German National Standard for Topographic and Cartographic Information
- B-Rep Boundary Representation
- bSI buildingSMART International
- CAD Computer Aided Design
- COLLADA Collaborative Design Activity
- CSG Constructive Solid Geometry

- DTM Digital Terrain Model
- DXF Drawing Exchange Format
- EuroSDR European Spatial Data Research Organisation
- ESRI Environmental Systems Research Institute
- FM Facility Management
- GDF Geographic Data Files
- GDI-DE Spatial Data Infrastructure Germany (Geodateninfrastruktur Deutschland)
- GDI NRW Geodata Infrastructure North-Rhine Westphalia
- GML Geography Markup Language
- IAI International Alliance for Interoperability (now buildingSMART International (bSI))
- IETF Internet Engineering Task Force
- IFC Industry Foundation Classes
- ISO International Organization for Standardisation
- LOD Level of Detail
- NBIMS National Building Information Model Standard
- OASIS Organisation for the Advancement of Structured Information Standards
- OGC Open Geospatial Consortium
- OSCRE Open Standards Consortium for Real Estate
- SIG 3D Special Interest Group 3D of the GDI-DE
- TC211 ISO Technical Committee 211
- TIC Terrain Intersection Curve
- TIN Triangulated Irregular Network
- UML Unified Modeling Language
- URI Uniform Resource Identifier
- UUID Universally Unique Identifier
- VRML Virtual Reality Modeling Language
- W3C World Wide Web Consortium
- W3DS OGC Web 3D Service
- WFS OGC Web Feature Service
- X3D Open Standards XML-enabled 3D file format of the Web 3D Consortium
- XML Extensible Markup Language
- xAL OASIS extensible Address Language

# Chapter 6. Conventions

## 6.1. Identifiers

The normative provisions in this document are denoted by the URI

http://www.opengis.net/eg/CityGML-1/3.0

All requirements and conformance tests that appear in this document are denoted by partial URIs relative to this base.

## 6.2. UML Notation

The CityGML Conceptual Model (CM) Standard is presented in this document through diagrams using the Unified Modeling Language (UML) static structure diagram (see Booch et al. 1997). The UML notations used in this standard are described in the diagram in Figure 1.

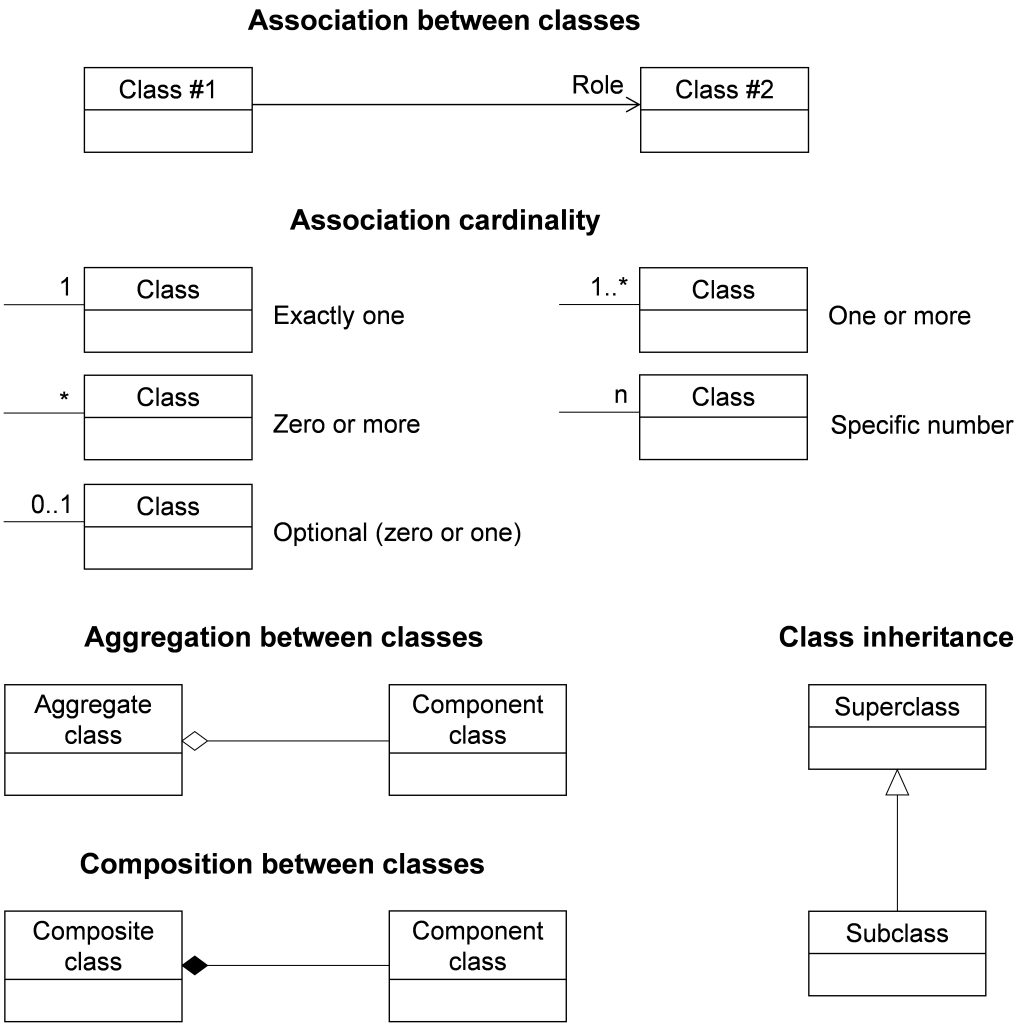



*Figure 1. UML notation (see ISO TS 19103, Geographic information - Conceptual schema language).*

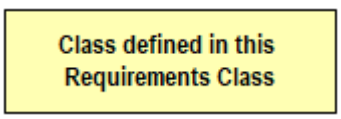
All associations between model elements in the CityGML Conceptual Model are uni-directional. Thus, associations in the model are navigable in only one direction. The direction of navigation is depicted by an arrowhead. In general, the context an element takes within the association is

indicated by its role. The role is displayed near the target of the association. If the graphical representation is ambiguous though, the position of the role has to be drawn to the element the association points to.

The following stereotypes are used in this model:

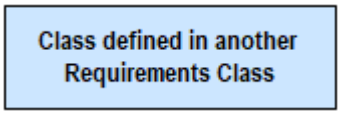
- «ApplicationSchema» denotes a conceptual schema for data required by one or more applications. In the CityGML Conceptual Model, every module is defined as a separate application schema to allow for modularization.
- «FeatureType» represents features that are similar and exhibit common characteristics. Features are abstractions of real-world phenomena and have an identity.
- «TopLevelFeatureType» denotes features that represent the main components of the conceptual model. Top-level features may be further semantically and spatially decomposed and substructured into parts.
- «Type» denotes classes that are not directly instantiable, but are used as an abstract collection of operation, attribute and relation signatures. The stereotype is used in the CityGML Conceptual Model only for classes that are imported from the ISO standards 19107, 19109, 19111, and 19123.
- «ObjectType» represents objects that have an identity, but are not features.
- «DataType» defines a set of properties that lack identity. A data type is a classifier with no operations, whose primary purpose is to hold information.
- «Enumeration» enumerates the valid attribute values in a fixed list of named literal values. Enumerations are specified in the CityGML Conceptual Model.
- «BasicType» defines a basic data type.
- «CodeList» enumerates the valid attribute values. In contrast to Enumeration, the list of values is open and, thus, not given inline in the CityGML UML Model. The allowed values can be provided within an external code list.
- «Union» is a list of attributes. The semantics are that only one of the attributes can be present at any time.
- «Property» denotes attributes and association roles. This stereotype does not add further semantics to the conceptual model, but is required to be able to add tagged values to the attributes and association roles that are relevant for the encoding.
- «Version» denotes that the value of an association role that ends at a feature type is a specific version of the feature, not the feature in general.

In order to enhance the readability of the CityGML UML diagrams, classes are depicted in different colors. The following coloring scheme is applied:

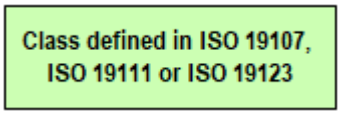



Classes painted in yellow belong to the Requirements Class which is subject of discussion in that clause of the standard in which the UML diagram is given. For example, in the context of [rc_core_section], which introduces the *CityGML Core* module, the yellow color is used to denote

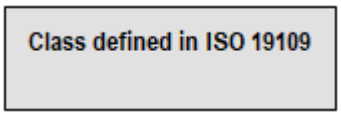
classes that are defined in the *CityGML Core* Requirements Class. Likewise, the yellow classes shown in the UML diagram in [rc_building-model_section] are associated with the *Building* Requirements Class that is subject of discussion in that chapter.



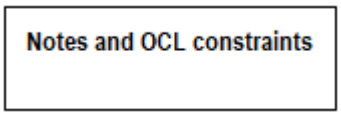

Classes painted in blue belong to a Requirements Class different to that associated with the yellow color. In order to explicitly denote to which Requirements Class these classes belong, their class names are preceded by the UML package name of that Requirements Class. For example, in the context of the *Building* Requirements Class, classes from the *CityGML Core* and the *Construction* Requirements Classes are painted in blue and their class names are preceded by *Core* and *Construction*, respectively.




Classes painted in green are defined in the ISO standards 19107, 19111, or 19123. Their class names are preceded by the UML package name, in which the classes are defined.




Classes painted in grey are defined in the ISO standard 19109. In the context of this standard, this only applies to the class *AnyFeature*. *AnyFeature* is an instance of the metaclass *FeatureType* and acts as super class of all classes in the CityGML UML model with the stereotype «FeatureType». A metaclass is a class whose instances are classes.




The color white is used for notes and Object Constraint Language (OCL) constraints that are provided in the UML diagrams.

The example UML diagram in Figure 2 demonstrates the UML notation and coloring scheme used throughout this standard. In this example, the yellow classes are associated with the *CityGML Building* module, the blue classes are from the *CityGML Core* and *Construction* modules, and the green class depicts a geometry element defined by ISO 19107.

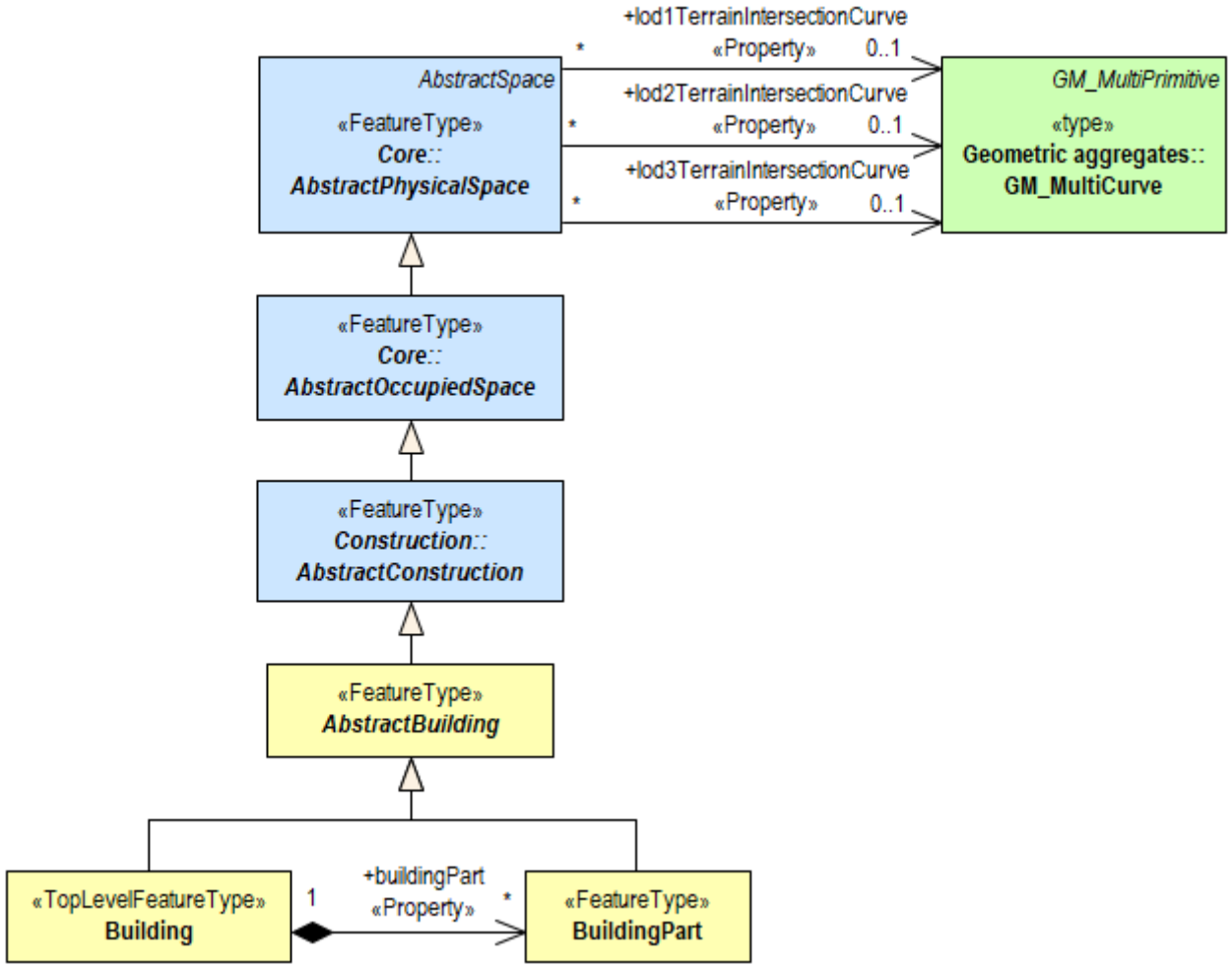



*Figure 2. Example UML diagram demonstrating the UML notation and coloring scheme used throughout the CityGML Standard.*

# Chapter 7. CityGML Foundations

This standard defines an open CityGML Conceptual Model (CM) for the storage and exchange of virtual 3D city and landscape models. These models include the most relevant entities of the urban space like buildings, roads, railways, tunnels, bridges, city furniture, water bodies, vegetation, and the terrain. The conceptual schema specifies how and into which parts and pieces physical objects of the real world should be decomposed and classified. All objects can be represented with respect to their semantics, 3D geometry, 3D topology, appearances, and their changes over time. Different spatial representations can be provided for each object (outdoor and indoor) in four predefined Levels of Detail (LOD 0-3). The CityGML 3.0 Conceptual Model (Chapter 8) is formally specified using UML class diagrams, complemented by a data dictionary (Chapter 9) providing the definitions and explanations of the object classes and attributes. This Conceptual Model is the basis for multiple encoding standards, which map the concepts (or subsets thereof) onto exchange formats or database structures for data exchange and storage.

While the CityGML Conceptual Model can be used for 3D visualization purposes, its special merits lie in applications that go beyond visualization such as decision support, urban and landscape planning, urban facility management, Smart Cities, navigation (both indoor and outdoor), Building Information Modelling (especially for as-built documentation), integration of city and BIM models, assisted and autonomous driving, and simulations in general (cf. Kolbe 2009). A comprehensive overview on the many different applications of virtual 3D city models is given in [Biljecki et al. 2015]. Many of the applications already use and some even require using CityGML.

In the CityGML CM, all 3D city objects can easily be enriched with thematic data. For example, street objects can be enriched with information about traffic density, speed limit, number of lanes etc., or buildings can be enriched by information on the heating and electrical energy demand, numbers of households and inhabitants, the appraised building value etc. Even building parts such as individual roof or wall surfaces can be enriched with information e.g. about solar irradiation and thermal insulation parameters. For many application domains specific extensions of the CityGML CM have already been created (cf. Biljecki et al. 2018).

## 7.1. Modularization

The CityGML Conceptual Model provides models for the most important types of objects within virtual 3D city and landscape models. These feature types have been identified to be either required or important in many different application areas. However, implementations are not required to support the complete CityGML model in order to be conformant to the standard. Implementations may employ a subset of constructs according to their specific information needs. For this purpose, modularisation is applied to the CityGML CM.

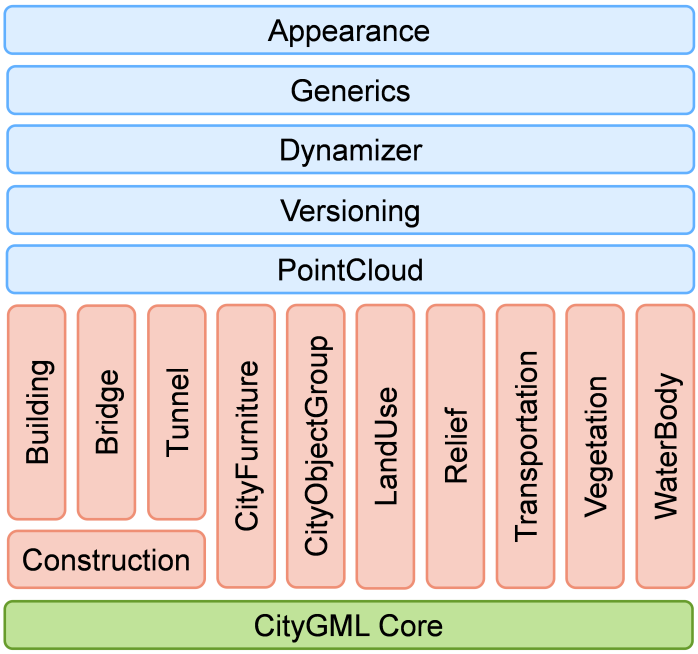



*Figure 3. CityGML 3.0 module overview. The vertical boxes show the different thematic modules. Horizontal modules specify concepts that are applicable to all thematic modules.*

The CityGML conceptual model is thematically decomposed into a *Core module* and different kinds of *extension modules* as shown in Figure 3. The Core module (shown in green) comprises the basic concepts and components of the CityGML CM and, thus, must be implemented by any conformant system. Each red colored module covers a specific thematic field of virtual 3D city models.

The CityGML CM introduces the following eleven thematic extension modules: *Building*, *Bridge*, *Tunnel*, *Construction*, *CityFurniture*, *CityObjectGroup*, *LandUse*, *Relief*, *Transportation*, *Vegetation*, and *WaterBody*. All three modules *Building*, *Bridge*, and *Tunnel* model civil structures and share common concepts that are grouped within the *Construction* module. The five blue colored extension modules add specific modelling aspects that can be used in conjunction with all thematic modules:

- The *Appearance* module contains the concepts to represent appearances (like textures and colours) of city objects.
- The *PointCloud* module provides concepts to represent the geometry of city objects by 3D point clouds.
- The *Generics* module defines the concepts for generic objects, attributes, and relationships.
- *Versioning* adds concepts for the representation of concurrent versions, real world object histories and feature histories.
- The *Dynamizer* module contains the concepts to represent city object properties by time series data and to link them with sensors, sensor data services or external files.

Each CityGML encoding can specify support for a subset of the CityGML modules only. If a module is supported by an encoding, then all concepts should be mapped. However, the encoding specification can define so-called *null mappings* to restrict the use of specific elements of the conceptual model in an encoding. Null mappings can be expressed in an encoding specification for individual feature types, properties, and associations defined within a CityGML module. This means that the corresponding element will not be included in the respective encoding.

Note that also CityGML applications do not have to support all modules. Applications can also decide to only support a specific subset of CityGML modules. For example, when an application only has to work with building data, only the modules *Core*, *Construction*, and *Building* would have to be supported.

## 7.2. General Modelling Principles

### 7.2.1. Semantic Modelling of Real-World Objects

Real-world objects are represented by geographic features according to the definition in ISO 19109. Geographic features of the same type (e.g. buildings, roads) are modelled by corresponding feature types that are represented as classes in the Conceptual Model (CM). The objects within a 3D city model are instances of the different feature types.

In order to distinguish and reference individual objects, each object has unique identifiers. In the CityGML 3.0 CM, each geographic feature has the mandatory *featureID* and an optional *identifier* property. The *featureID* is used to distinguish all objects and possible multiple versions of the same real-world object. The *identifier* is identical for all versions of the same real-world object and can be used to reference specific objects independent from their actual object version. The *featureID* is unique within the same CityGML dataset, but it is generally recommended to use globally unique identifiers like UUID values or identifiers maintained by an organization such as a mapping agency. Providing globally unique and stable identifiers for the *identifier* attribute is recommended. This means these identifiers should remain stable over the lifetime of the real-world object.

CityGML feature types typically have a number of spatial and non-spatial properties (also called attributes) as well as relationships with other feature or object types. Note that a single CityGML object can have different spatial representations at the same time. For example, different geometry objects representing the feature's geometry in different levels of detail or as different spatial abstractions.

Many attributes have simple, scalar values like a number or a character string. However, some attributes are complex. They do not just have a single property value. In CityGML the following types of complex attributes occur:

- *Qualified attribute values*: For example, a measure consists of the value and a reference to the unit of measure, or e.g. for relative and absolute height levels the reference level has to also be named.
- *Code list values*: A code list is a form of enumeration where the valid values are defined in a separate register. The code list values consist of a link or identifier for the register as well as the value from that register which is being used.
- Attributes consisting of a *tuple of different fields and values*: For example, addresses, space occupancy, and others
- Attribute value consisting of a *list of numbers*: For example, representing coordinate lists or matrices

In order to support feature history, CityGML 3.0 introduces bitemporal timestamps for all objects. In CityGML 2.0, the attributes *creationDate* and *terminationDate* are supported. These refer to the time

period in which a specific version of an object is an integral part of the 3D city model. In 3.0, all features can now additionally have the attributes *validFrom* and *validTo*. These represent the lifespan a specific version of an object has in the real-world. Using these two time intervals a CityGML dataset could be queried both for how did the *city* look alike at a specific point in time as well as how did the *city model* look at that time.

The combination of the two types of feature identifiers and bitemporal timestamps enables encoding not only the current version of a 3D city model, but also the model's entire history can be represented in CityGML and possibly exchanged within a single file.

### 7.2.2. Class Hierarchy and Inheritance of Properties and Relations

In CityGML, the specific feature types like *Building*, *Tunnel*, or *WaterBody* are defined as subclasses of more general higher-level classes. Hence, feature types build a hierarchy along specialization / generalization relationships where more specialized feature types inherit the properties and relationships of all their superclasses along the entire generalization path to the topmost feature type *AnyFeature*.

Note: A superclass is the class from which subclasses can be created.

### 7.2.3. Relationships between CityGML objects

In CityGML, objects can be related to each other and different types of relations are distinguished. First of all, complex objects like buildings or transportation objects typically consist of parts. These parts are individual features of their own, and can even be further decomposed. Therefore, CityGML objects can form aggregation hierarchies. Some feature types are marked in the conceptual model with the stereotype *«TopLevelFeatureType»*. These constitute the main objects of a city model and are typically the root of an aggregation hierarchy. Only top-level features are allowed as direct members of a city model. The information about which feature types belong to the top level is required for software packages that want to filter imports, exports, and visualizations according to the general type of a city object (e.g. only show buildings, solitary vegetation objects, and roads). CityGML Application Domain Extensions should also make use of this concept, such that software tools can learn from inspecting their conceptual schema what are the main, i.e. the top-level, feature types of the extension.

Some relations in CityGML are qualified by additional parameters, typically to further specify the type of relationship. For example, a relationship can be qualified with a URI pointing to a definition of the respective relation type in an Ontology. Qualified relationships are used in CityGML, among others, for:

- General relationships between features – association *relatedTo* between city objects,
- User-defined aggregations using *CityObjectGroup*. This relation allows also for recursive aggregations,
- External references – linking of city objects with corresponding entities from external resources like objects in a cadastre or within a BIM dataset.

The CityGML CM contains many relationships that are specifically defined between certain feature types. For example, there is the *boundary* relationship from 3D volumetric objects to its

thematically differentiated 3D boundary surfaces. Another example is the *generalizesTo* relation between feature instances that represent objects on different generalisation levels.

In the CityGML 3.0 CM there are new associations to express topologic, geometric, and semantic relations between all kinds of city objects. For example, information that two rooms are adjacent or that one interior building installation (like a curtain rail) is overlapping with the spaces of two connected rooms can be expressed. The CM also enables documenting that two wall surfaces are parallel and two others are orthogonal. Also distances between objects can be represented explicitly using geometric relations. In addition to spatial relations logical relations can be expressed.

### 7.2.4. Definition of the Semantics for all Classes, Properties, and Relations

The meanings of all elements defined in the CityGML conceptual model are normatively specified in the data dictionary in Chapter 9.

## 7.3. Representation of Spatial Properties

### 7.3.1. Geometry and Topology

Spatial properties of all CityGML feature types are represented using the geometry classes defined in ISO 19107. Spatial representations can have 0-, 1-, 2-, or 3-dimensional extents depending on the respective feature type and Levels of Detail (LOD). The LOD concept is discussed in [levels-of-detail-section] and [geometry-lod-section]. With only a few exceptions, all geometries must use 3D coordinate values. Besides primitive geometries like single points, curves, surfaces, and solids, CityGML makes use of different kinds of aggregations of geometries like spatial aggregates (*MultiPoint*, *MultiCurve*, *MultiSurface*, *MultiSolid*) and composites (*CompositeCurve*, *CompositeSurface*, *CompositeSolid*). Volumetric shapes are represented in ISO 19107 according to the so-called *Boundary Representation* (B-Rep). For further explanation see Foley et al. 2002.

The CityGML Conceptual Model does not put any restriction on the usage of specific geometry types as defined in ISO 19107. For example, 3D surfaces could be represented in a dataset using 3D polygons or 3D meshes such as triangulated irregular networks (TINS) or by non-uniform rational B-spline surfaces (NURBS). However, an encoding may restrict the usage of geometry types. For example, curved lines like B-splines or clothoids, or curved surfaces like NURBS could be disallowed by explicitly defining *null encodings* for these concepts in the encoding specification (c.f. [modularization-section] above).

Note that the conceptual schema of ISO 19107 allows composite geometries to be defined by a recursive aggregation for every primitive type of the corresponding dimension. This aggregation schema allows the definition of nested aggregations (hierarchy of components). For example, a building geometry (*CompositeSolid*) can be composed of the house geometry (*CompositeSolid*) and the garage geometry (*Solid*), while the house's geometry is further decomposed into the roof geometry (*Solid*) and the geometry of the house body (*Solid*). This is illustrated in Figure 4.

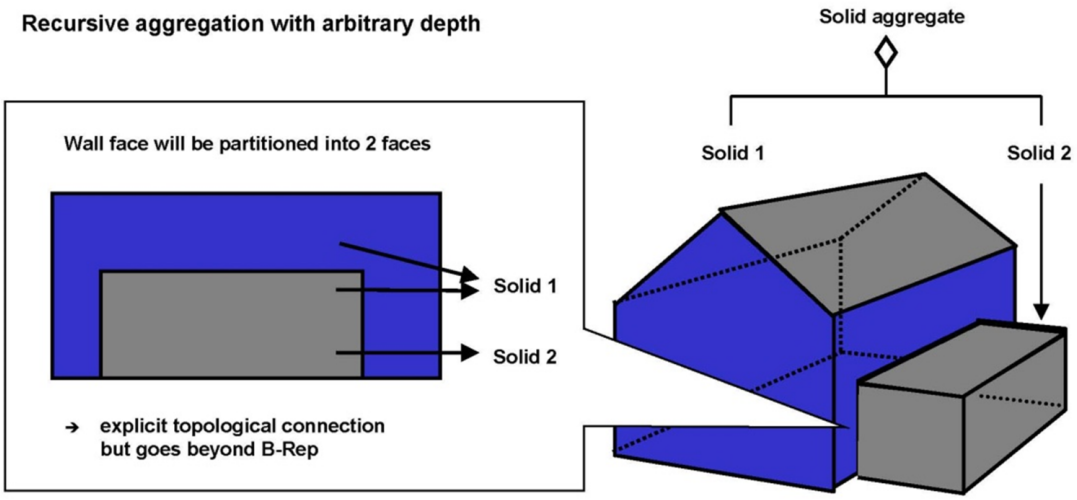



*Figure 4. Recursive aggregation of objects and geometries in CityGML (graphic: IGG Uni Bonn).*

While the CityGML Conceptual Model does not employ the topology classes from ISO 19107, topological relations between geometries can be established by sharing geometries (typically parts of the boundary) between different geometric objects. One part of real-world space can be represented only once by a geometry object and is referenced by all features or more complex geometries which are defined or bounded by this geometry object. Thus redundancy can be avoided and explicit topological relations between parts are maintained.

Basically, there are three cases for sharing geometries:

- First, two different semantic objects may be spatially represented by the same geometry object. For example, if a foot path is both a transportation feature and a vegetation feature, the surface geometry defining the path is referenced by both the transportation object and by the vegetation object.
- Second, a geometry object may be shared between a feature and another geometry. For example, a geometry defining a wall of a building may be referenced twice: By the solid geometry defining the geometry of the building, and by the wall feature.
- Third, two geometries may reference the same geometry, which is in the boundary of both. For example, a building and an adjacent garage may be represented by two solids. The surface describing the area where both solids touch may be represented only once and it is referenced by both solids. As it can be seen from Figure 4, this requires partitioning of the respective surfaces.

In general, B-Rep only considers visible surfaces. However, to make topological adjacency explicit and to allow the possibility of deletion of one part of a composed object without leaving holes in the remaining aggregate, touching elements are included. Whereas touching is allowed, permeation of objects is not in order to avoid the multiple representation of the same space.

Another example of sharing geometry objects that are members of the boundaries in different higher-dimensional geometry objects is the sharing of point geometries or curve geometries, which make up the outer and inner boundaries of a polygon. This means that each point is only represented once, and different polygons could reference this point geometry. The same applies to the representation of curves for transportation objects like roads, whose end points could be shared such as between different road segments to topologically connect them.

Note that the use of topology in CityGML datasets by sharing geometries is optional. Furthermore, an encoding of the CityGML conceptual model might restrict the usage of shared geometries. For example, it might only be allowed to share identical (support) points from different 3D polygons or only entire polygons can be shared between touching solids (like shown in Figure 4).

### 7.3.2. Prototypic Objects / Scene Graph Concepts

In CityGML, objects of equal shape like trees and other vegetation objects, traffic lights and traffic signs can be represented as prototypes which are instantiated multiple times at different locations (see Figure 5). The geometry of prototypes is defined in local coordinate systems. Every instance is represented by a reference to the prototype, a base point in the world coordinate reference system (CRS) and a transformation matrix that facilitates scaling, rotation, and translation of the prototype. The principle is adopted from the concept of scene graphs used in computer graphics standards. Since the ISO 19107 geometry model does not provide support for scene graph concepts, the CityGML class ImplicitGeometry has been introduced (for further description see [geometry-lod-section]). The prototype geometry can be represented using ISO 19107 geometry objects or by referencing an external file containing the geometry in another data format.

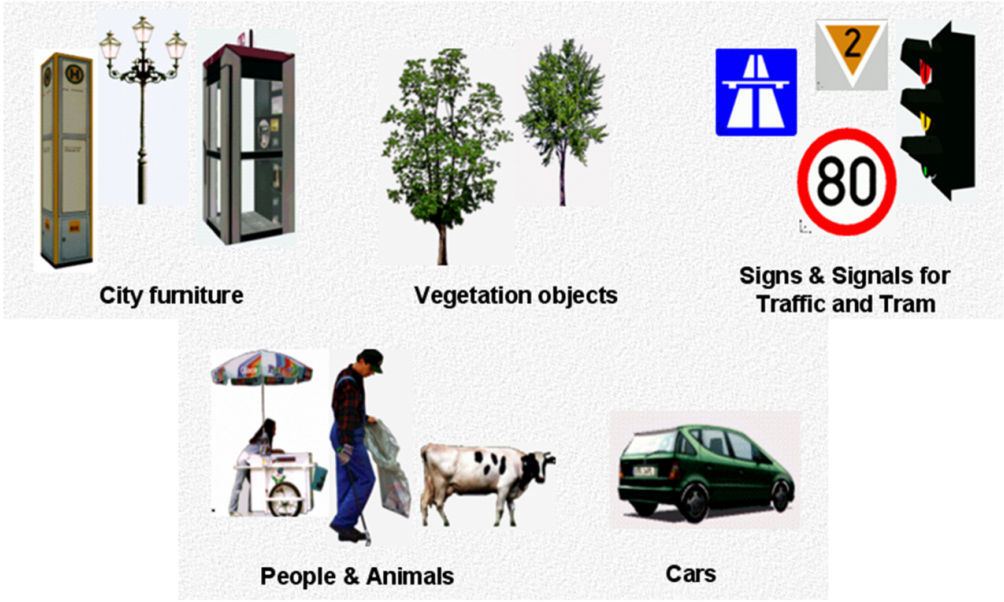



*Figure 5. Examples of prototypic shapes (source: Rheinmetall Defence Electronics).*

### 7.3.3. Point Cloud Representation

In addition to the spatial representations defined in the *Core* module, the geometry of physical spaces and of thematic surfaces can now also be provided by 3D point clouds using MultiPoint geometry. This allows, for example, spatially representing the building hull, a room within a building or a single wall surface just by a point cloud. All thematic feature types including transportation objects, vegetation, city furniture, etc. can also be spatially represented by point clouds. In this way, the ClearanceSpace of a road or railway could, for instance, be modelled directly from the result of a mobile laser scanning campaign. Point clouds can either be included in a CityGML dataset or just reference an external file of some common types such as LAS or LAZ.

### 7.3.4. Coordinate Reference Systems (CRS)

CityGML is about 3D city and landscape models. This means that nearly all geometries use 3D coordinates, where each single point and also the points defining the boundaries of surfaces and solids have three coordinate values (x,y,z) each. Coordinates always have to be given with respect to a coordinate reference system (CRS) that relates them unambiguously with a specific position on the Earth. In contrast to CAD or BIM, each 3D point is absolutely georeferenced, which makes CityGML especially suitable to represent geographically large extended structures like airports, railways, bridges, dams, where the Earth curvature has a significant effect on the object's geometry (for further explanations see Kaden & Clemen 2017).

In most CRS, the (x,y) coordinates refer to the horizontal position of a point on the Earth's surface. The z coordinate typically refers to the vertical height over (or under) the reference surface. Note that depending on the chosen CRS, x and y may be given as angular values like latitude and longitude or as distance values in meters or feet. According to ISO 19111, numerous 3D CRS can be used. This includes global as well as national reference systems using geocentric, geodetic, or projected coordinate systems.

## 7.4. Appearances

In addition to semantics and geometry, information about the appearance of surfaces, i.e. observable properties of the surface, is considered an integral part of virtual 3D city and landscape models. Appearance relates to any surface-based theme such as infrared radiation or noise pollution, not just visual properties like RGB texture images. Consequently, data provided by appearances can be used as input for both, presentation of and analysis in virtual 3D city models.

The CityGML Conceptual Model supports feature appearances for an arbitrary number of themes per city model. Each LOD of a feature can have an individual appearance. Appearances can represent – among others – textures and georeferenced textures. CityGML's appearance model is packaged within the Appearance module (cf. [rc_appearance_section]).

## 7.5. Modelling Dynamic Data

In general, city objects can have properties related to their geometry, topology, semantics, and appearance. All of these properties may change over time. For example, a construction event leads to the change in geometry of a building (i.e. addition of a new building floor or demolition of an existing door). The geometry of an object can be further classified according to its shape, location, and extent, which can also change over time. A moving car object involves changing only the location of the car object. However, a flood incident involves variations in the location and shape of water. There might be other properties, which change with respect to thematic data of city objects such as hourly variations in energy or gas consumption of a building or changing the building usage from residential to commercial. Some properties involve changes in appearances over a time period, such as building textures changing over years or traffic cameras recording videos of moving traffic over definite intervals. 3D city models also represent interrelationships between objects and relations may change over time as well. Hence, it is important to consider that the representation of time-varying data is required to be associated with these different properties. A detailed discussion on the requirements of city model applications regarding the support of dynamic data is given in [Chaturvedi & Kolbe 2019].

The CityGML 3.0 Conceptual Model introduces two concepts to manage dynamic, time-dependent, properties of city models. The *Versioning* module manages changes that are slower in nature. Examples are (1) the history or evolution of cities such as construction or demolition of buildings, and (2) managing multiple versions of the city models.

The *Dynamizer* module manages higher-frequency or dynamic variations of object properties, including variations of (1) thematic attributes such as changes of physical quantities (energy demands, temperature, solar irradiation levels), (2) spatial properties such as change of a feature's geometry, with respect to shape and location (moving objects), and (3) real-time sensor observations. The Dynamizer module allows establishing explicit links from city objects to sensors and sensor data services.

### 7.5.1. Versioning and Histories

As described in [semantic-modelling-section], the bitemporal timestamps of all CityGML feature types allow representing the evolution of the real city and its model over time. The new *Versioning* module extends this concept by the possibility of representing multiple, concurrent versions of the city model. For that purpose, the module defines two new feature types: 1) *Version*, which can be used to explicitly define named states of the 3D city model and denote all the specific versions of objects belonging to such states. 2) *VersionTransition*, which allows to explicitly link different versions of the 3D city model by describing the reason of change and the modifications applied. Details on the versioning concept are given in [Chaturvedi et al. 2015].

This approach not only facilitates the explicit representation of different city model versions, but also allows distinguishing and referring to different versions of city objects in an interoperable exchange format. All object versions could be stored and exchanged within a single dataset. Software systems could use such a dataset to visualize and work with the different versions simultaneously. The conceptual model also takes into account the management of multiple histories or multiple interpretations of the past of a city, which is required when looking at historical city developments and for archaeological applications. In addition, the Versioning module supports collaborative work. All functionality to represent a tree of workspaces as version control systems like *git* or *SVN* is provided. The Versioning module handles versions and version transitions as feature types, which allows the version management to be completely handled using the standard OGC Web Feature Service [Vrenatos 2010]. No extension of the OGC Web Feature Service standard is required to manage the versioning of city models.

### 7.5.2. Dynamizers: Using Time-Series Data for Object Attributes

The new Dynamizer module improves the usability of CityGML for different kinds of simulations as well as to facilitate the integration of devices from the Internet-of-Things (IoT) like sensors with 3D city models. Both, simulations and sensors provide dynamic variations of some measured or simulated properties such as the electricity consumption of a building or the traffic density within a road segment. The variations of the value are typically represented using time-series data. The data sources of the time-series data could be either sensor observations (e.g. from a smart meter), pre-recorded load profiles (e.g. from an energy company), or the results of some simulation run.

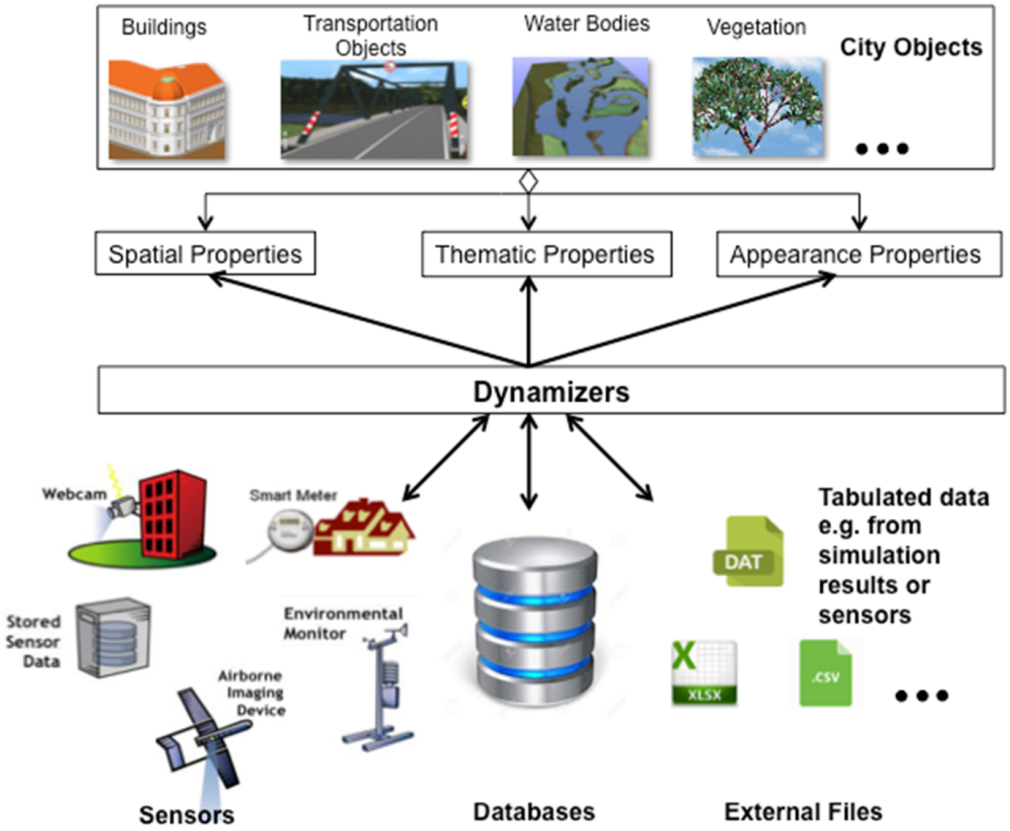



*Figure 6. Dynamizers link timeseries data coming from different sources to specific properties of individual city objects.*

As shown in Figure 6, Dynamizers serve three main purposes:

1. Dynamizer is a data structure to represent dynamic values in different and generic ways. Such dynamic values may be given by (1) tabulation of time/value pairs using its *AtomicTimeseries* class, (2) patterns of time/value pairs based on statistical rules using its *CompositeTimeseries* class, and (3) retrieving observations directly from external sensor/IoT services using its *SensorConnection* class. The values can be obtained from sensor services such as the OGC Sensor Observation Service or OGC SensorThings API, simulation specific databases, and also external files such as CSV or Excel sheets.
2. Dynamizer delivers a method to enhance static city models by adding dynamic property values. A Dynamizer references a specific property (e.g. spatial, thematic or appearance properties) of a specific object within a 3D city model providing dynamic values overriding the static value of the referenced object attribute.
3. Dynamizer objects establish explicit links between sensor/observation data and the respective properties of city model objects that are measured by them. By making such explicit links with city object properties, the semantics of sensor data become implicitly defined by the city model.

Dynamizers are used to inject dynamic variations of city object properties into an otherwise static representation. The advantage in following such an approach is that it allows only selected properties of city models to be made dynamic. If an application does not support dynamic data, the application simply does not allow/include these special types of features.

Dynamizers have already been implemented as an Application Domain Extension (ADE) for CityGML 2.0 and were employed in the OGC Future City Pilot Phase 1. More details about

Dynamizers are given in [Chaturvedi & Kolbe 2017].

## 7.6. Extending CityGML

CityGML is designed as a universal information model that defines object types and attributes which are useful for a broad range of applications. In practical applications, the objects within specific 3D city models will most likely contain attributes which are not explicitly modelled in CityGML. Moreover, there might be 3D objects which are not covered by the CityGML CM thematic classes. The CityGML CM provides three different concepts to support the exchange of such data:

1. Generic objects and attributes,
2. Application Domain Extensions, and
3. Code lists.

The concept of generic objects and attributes allows application-specific concepts to be represented in CityGML at runtime. This means that any city object may be augmented by additional attributes and relations, whose names, data types, and values can be provided by a running application without requiring extensions to the CityGML conceptual schema and the respective encodings. Similarly, features not represented by the predefined thematic classes of the CityGML conceptual model may be modelled and exchanged using generic objects. The generic extensions of CityGML are provided by the *Generics* module (cf. [ug-generics-section]).

Application Domain Extensions (ADE) specify additions to the CityGML conceptual model. Such additions comprise the introduction of new properties to existing CityGML feature types such as the energy demand of a building or the definition of additional feature types. The difference between ADEs and generic objects and attributes is, that an ADE has to be defined in an extra conceptual schema (provided in UML) with its own namespace. Encodings have to be extended accordingly. The advantage of this approach is that the extension is formally specified. Extended CityGML datasets can be validated against the CityGML CM and the respective ADE schema. ADEs can be defined (and even standardized) by information communities which are interested in specific application fields. More than one ADE can be used simultaneously in the same dataset. Examples for popular ADEs are the Utility Network ADE [Becker et al. 2011; Kutzner et al. 2018] and the Energy ADE [Nouvel et al. 2015; Agugiaro et al. 2018]. A comprehensive overview of CityGML ADEs is given in [Biljecki et al. 2018]. Further details on ADEs are given in [ug_ade_section].

CityGML can also be extended with regard to the allowed values specified in code lists. Many attributes of CityGML types use a code list as a data type such as, for instance, the attributes *class*, *usage*, and *function* of city objects. A code list defines a value domain including a code for each permissible value. In contrast to fixed enumerations, modifications and extensions to the value domain become possible with code lists. The values for all code lists in CityGML have to be defined externally. This could, for example, be by adopting classifications from global, national, or industrial standards.

# Chapter 8. CityGML Model

## 8.1. Core

| Contributors |
| --- |
| C. Heazel - first draft |

The CityGML Core module defines the basic concepts and components of city models. This rather large body of work is divided into seven sections. These sections build on each other from the fundamental principles specified by the relevant ISO standards up to the full CityGML model. These sections are summarized in Table 1.

*Table 1. CityGML Core Sections*

| | |
| --- | --- |
| Key Concepts | Summarizes the key concepts described in the Core module. |
| The Use of ISO Standards | Describes the use of the ISO 19100 series of International Standards to provide a foundation to the CityGML model. |
| City Models and City Objects | Defines the basic building blocks of the CityGML model. |
| Space Concept | Defines the concepts of space as used in the CityGML model. |
| Geometry and LOD | Defines the geometry and Levels Of Detail concepts. |
| CityGML Core Model | Presents the complete Core model. |
| Types, Enumerations, and Codelist | Defines the little things which make this model work. |

### 8.1.1. Key Concepts

The following is a summary of the key concepts described by the Core Module. This is not an exaustive listing of all of the Core concepts. Rather, it is an introduction those concepts which are essentiall for understanding the role of the Core Module in the CityGML Conceptual Model.

| | |
| --- | --- |
| **NOTE** | Each entry sould include a link to the section which provides the next level of detail. Clicking on the concept name will take you to the data dictionary. A (see also) link will take you to more discussion. |

| | |
| --- | --- |
| **NOTE** | Alternatly, clicking on the topic should send the user to a discussion of this topic. Consider adding a link to the data dictionary as a separate link. |

**CityModel**: (Discussion) The CityModel class is the root class of every CityGML conceptual model. It's primary purpose is to aggregate CityModelMembers.

**CityModelMember**: (Discussion) CityModelMember is a Union of all possible classes which can be included in a city model. Those classes are summarized in Table 2.

*Table 2. City Model Members*

| | |
|---|---|
| AbstractAppearance | Rules for rending the associated objects |
| AbstractCityObject | Physical objects with a location and geometry |
| AbstractFeature | Others |
| AbstractVersion | Version information |
| AbstractVersionTransition | Version information |

**AbstractFeature**: (Discussion) AbstractFeature is a subclass of the AnyFeature class from the ISO General Feature Model. Every «FeatureType» class in a CityGML model is descended from this class. `AbstractFeature` also defines identifying attributes which are common across the model. Finally, `AbstractFeature` allows an ADE to be associated with any «FeatureType» class.

**AbstractFeatureWithLifespan**: (Discussion)This class extends AbstractFeature with temporal properties for the classes. Most Feature classes in the CityGML model are descended from this class. As such they contain the following attributes:

- featureId
- identifier
- name
- description
- creationDate
- terminationDate
- validFrom
- validTo
- adeOfAbstractFeatureWithLifespan

**AbstractAppearance**: (Discussion) The root class for all classes related to the rendering of the model including textures and materials.

**AbstractVersion**: (Discussion) a defined state of a city model consisting of the dedicated versions of all city object instances that belong to the respective city model version.

**AbstractVersionTransition**: (Discussion) describes the change of the state of a city model from one version to another.

**AbstractCityObject**: (Discussion) The superclass of all thematic classes within the CityGML conceptual model. Significant subclasses of `AbstractCityObject` are AbstractSpace and AbstractSpaceBoundary. These subclasses provide CityGML objects with location, geometry and boundary properties.

**AbstractSpace**: (Discussion) Classes which have a volumetric extent in the real world.

**AbstractOccupiedSpace**: (Discussion) Classes which describe spaces that are partially or entirely filled with matter.

**AbstractUnccupiedSpace**: (Discussion) Classes which describe spaces that are entirely or mostly free of matter.

**AbstractLogicalSpace**: (Discussion) Classes which describe spaces that are not bounded by physical surfaces but are defined according to thematic considerations.

**AbstractSpaceBoundary**: (Discussion) Classes which bound a space.

**AbstractThematicSurface**: (Discussion) The superclass for all thematic surfaces. A type of Space Boundary.

**ImplicitGeometry**: (Discussion) A geometry representation where the shape is stored only once as a prototypical geometry, for example a tree or other vegetation object, a traffic light or a traffic sign. This prototypic geometry object can be re-used or referenced many times, wherever the corresponding feature occurs in the 3D city model. A type of Abstract Feature.

### 8.1.2. ISO Dependencies

CityGML builds on the ISO 19100 family of standards. The applicable standards are identified in Figure 7. A Data dictionary is also included for all of the ISO-defined classes explicitly referenced in the CityGML UML model. These data dictionaries are provided for the convenience of the user. The ISO standards are the normative source.

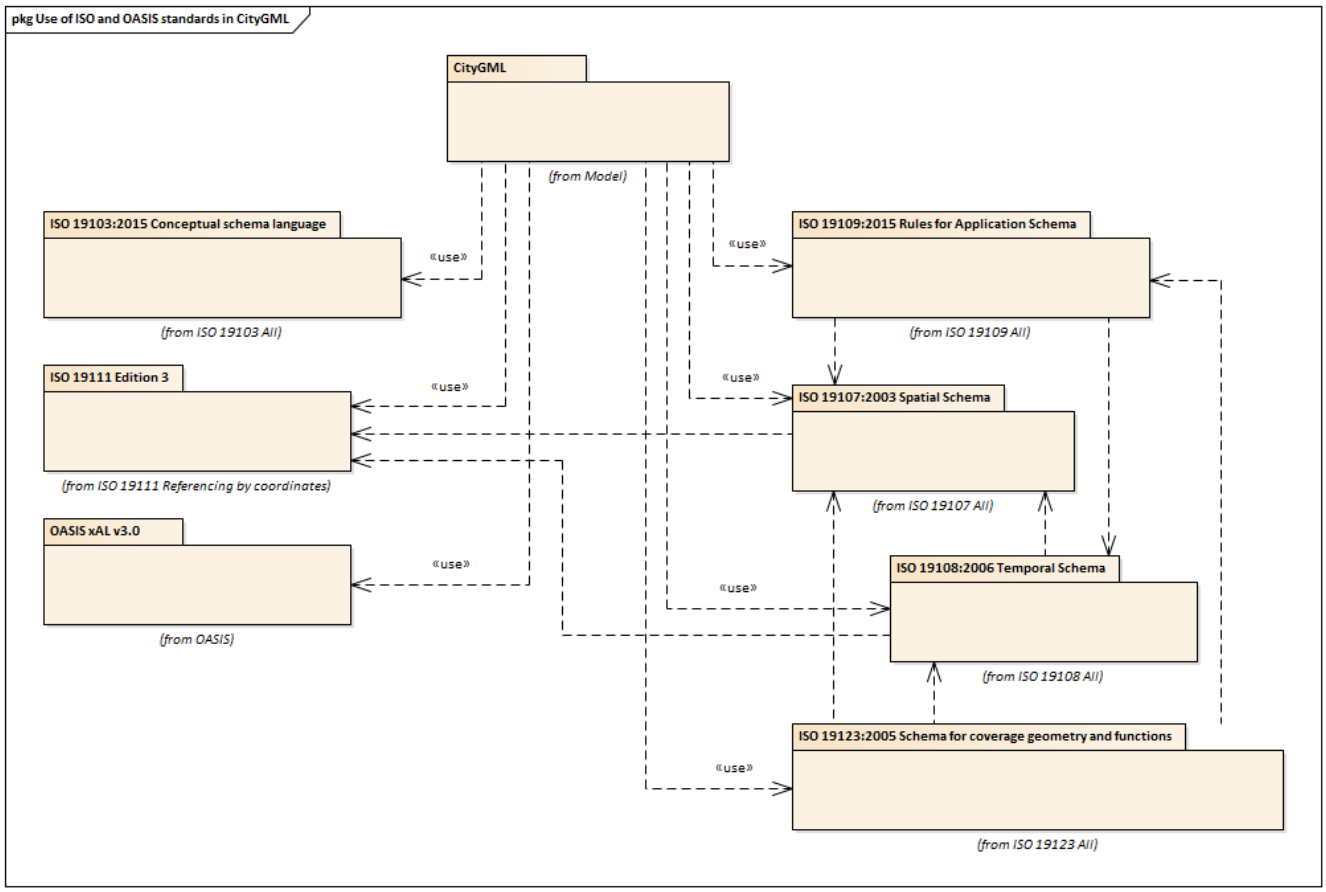



*Figure 7. Use of ISO Standards in CityGML*

#### Classes

The ISO classes explicitly used in the CityGML UML model are introduced in Table 3. Detailed

descriptions are provided in the Data Dictionary.

*Table 3. ISO Classes used in CityGML*

| Class Name | Description |
|---|---|
| AnyFeature | A generalization of all feature types |
| CV_DiscreteGridPointCoverage | A coverage that returns the same feature attribute values for every direct position within any object in its domain. |
| DirectPosition | The coordinates for a position within some coordinate reference system. |
| GM_Object | The root class of the geometric object taxonomy. |
| GM_MultiCurve | An aggregate class containing only instances of GM_OrientableCurve. |
| GM_MultiPoint | An aggregate class containing only points. |
| GM_MultiSurface | An aggregate class containing only instances of GM_OrientableSurface. |
| GM_Point | The basic data type for a geometric object consisting of one and only one point. |
| GM_Solid | The basis for 3-dimensional geometry. The extent of a solid is defined by the boundary surfaces. |
| GM_Surface | The basis for 2-dimensional geometry. |
| GM_Tin | A GM_TriangulatedSurface which uses the Delaunay or similar algorithm. |
| GM_TriangulatedSurface | A GM_PolyhedralSurface that is composed only of triangles |
| SC_CRS | Coordinate reference system which is usually single but may be compound. |
| TM_Position | A union class that consists of one of the data types listed as its attributes. |

#### Geometry

The most common geometry concept found in the CityGML 3.0 Standard is the concept of multi-primitives. These are homogeneous collections of GM_Primitives which are aggregated to form a more compex geometry.

GM_Composites are another form of GM_Primitive collection. These differ from GM_MultiPrimitive in that the collection can be heterogeneous. It should be noted that none of the classes in the CityGML 3.0 Standard are decended from GM_Composites. However, the terms "CompositeCurve", "CompositeSurface", and "CompositeSolid" do appear in the text. The `composit` concept can also be seen in the assocation between spaces and surfaces. Therefore, an explanation of `composits` has been included for completeness.

#### GM_Primitive

GM_Primitive is the abstract root class of the geometric primitives. Its main purpose is to define the basic "boundary" operation that ties the primitives in each dimension together. A geometric primitive (GM_Primitive) is a geometric object that is not decomposed further into other primitives in the system. This includes curves and surfaces, even though they are composed of curve segments

and surface patches, respectively. This composition is a strong aggregation: curve segments and surface patches cannot exist outside the context of a primitive.

NOTE Most geometric primitives are decomposable infinitely many times. Adding a centre point to a line may split that line into two separate lines. A new curve drawn across a surface may divide that surface into two parts, each of which is a surface. This is the reason that the normal definition of primitive as "non-decomposable" is not plausible in a geometry model - the only non-decomposable object in geometry is a point.

### GM_MultiPrimitive

Any geometric object that is used to describe a feature is a collection of geometric primitives. A homogeneous collection of geometric primitives may be a multi-primitive (GM_MultiPrimitive). Geometric complexes have additional properties specific to the type of geometric primitive they aggregate.

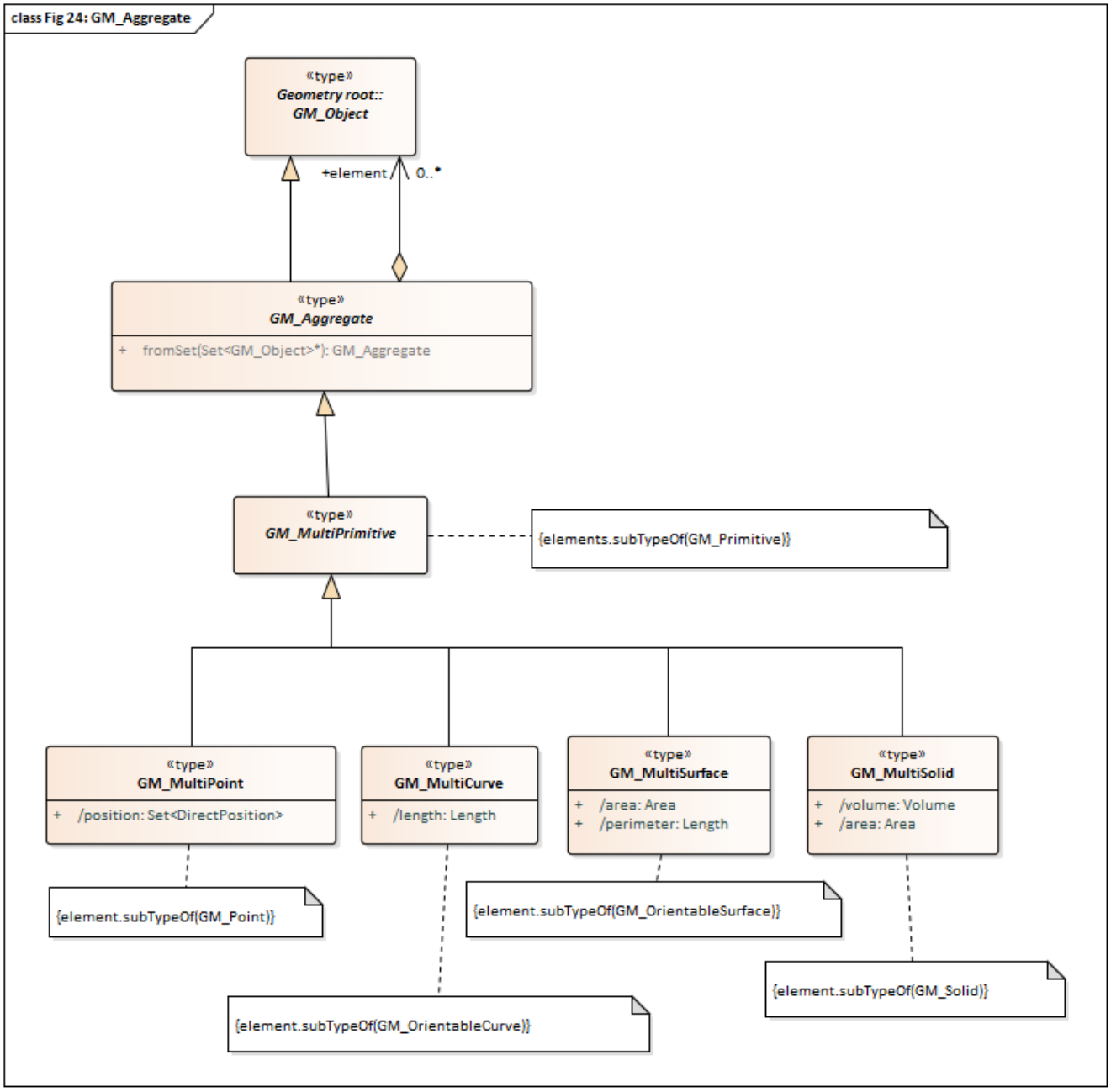



*Figure 8. GM_MultiPrimitive Context Diagram*

### GM_Complex

A GM_Complex is a set of disjoint geometric primitives (GM_Primitive) such that the boundary of each primitive can be represented as the union of other geometric primitives within the complex.

Any geometric object that is used to describe a feature is a collection of geometric primitives. A collection of geometric primitives may be a geometric complex (GM_Complex). Geometric complexes have additional properties such as closure by boundary operations and mutually exclusive component parts.

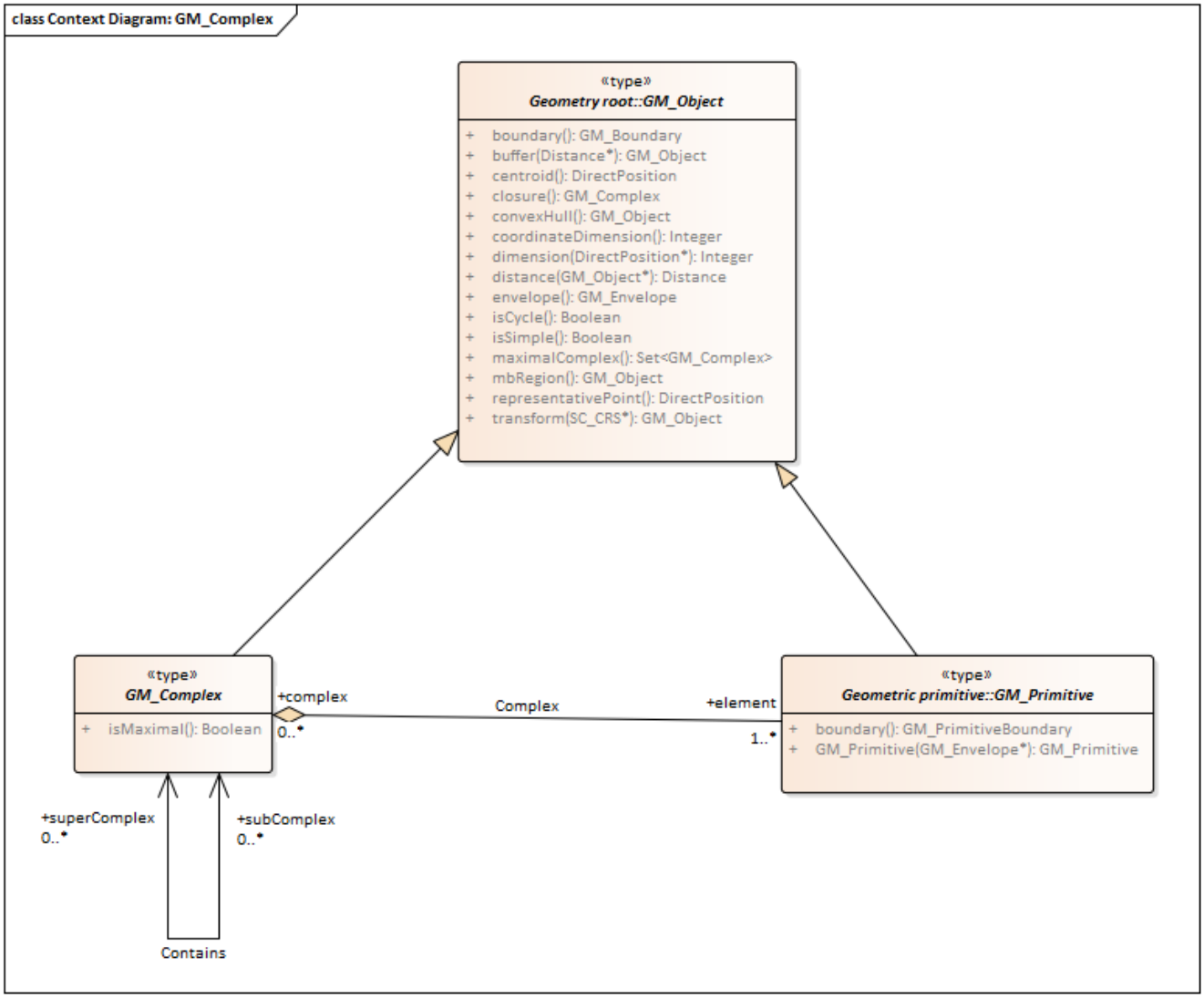



*Figure 9. GM_Complex Context Diagram*

GM_Primitive and GM_Complex share most semantics, in the meaning of operations, attributes and associations. There is an exception in that a GM_Primitive shall not contain its boundary (except in the trivial case of GM_Point where the boundary is empty), while a GM_Complex shall contain its boundary in all cases. This means that if an instantiated object implements GM_Object operations both as GM_Primitive and as a GM_Complex, the semantics of each set theoretic operation is determined by the its name resolution. Specifically, for a particular object such as GM_CompositeCurve, GM_Primitive::contains (returns FALSE for end points) is different from GM_Complex::contains (returns TRUE for end points). Further, if that object is cast as a GM_Primitive value and as a GM_Complex value, then the two values need not be equal as GM_Objects.

GM_Complex aggregates GM_Primitives through the `element` property. Since this is an aggregation, the target GM_Primitive may be associated with more than one GM_Complex.

A GM_Complex object can also have a whole/part relationship with other GM_Complex objects. The `contains` association is used to associate the `superComplex` instance with the `subComplex` instance.

Note that the geometric primitives in the set are mutually exclusive in the sense that no point is interior to more than one primitive. The set is closed under boundary operations, meaning that for each element in the complex, there is a collection (also a complex) of geometric primitives that represents the boundary of that element.

### GM_Composite

GM_Composite is a subclass of GM_Complex. Like GM_Complex, it has an association with GM_Primitives. In this case this is an `composition` association with a `composite` role (GM_Composite) and a `generator` role (GM_Primitive). As with the GM_Complex, the GM_Primitve may be associated with more than one GM_Composite.

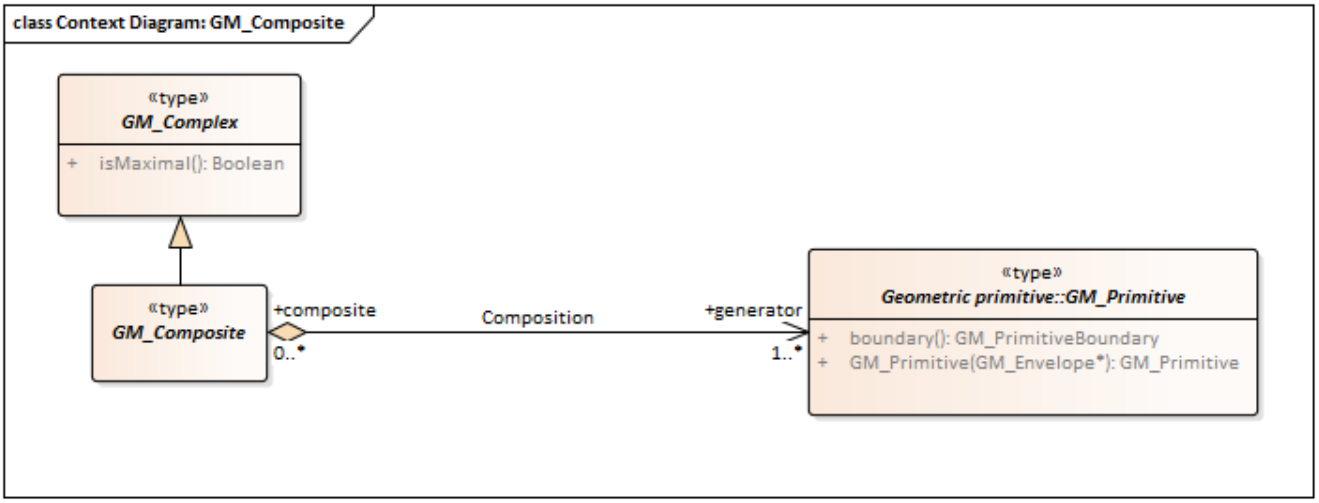



*Figure 10. GM_Composite Context Diagram*

### GM_CompositeSurface

A GM_Composite where the GM_Primitives is a GM_OrientatableSurface.

A GM_CompositeSurface is also a subclass of GM_Primitives is a GM_OrientatableSurface. One of the few examples of multiple inheritance.

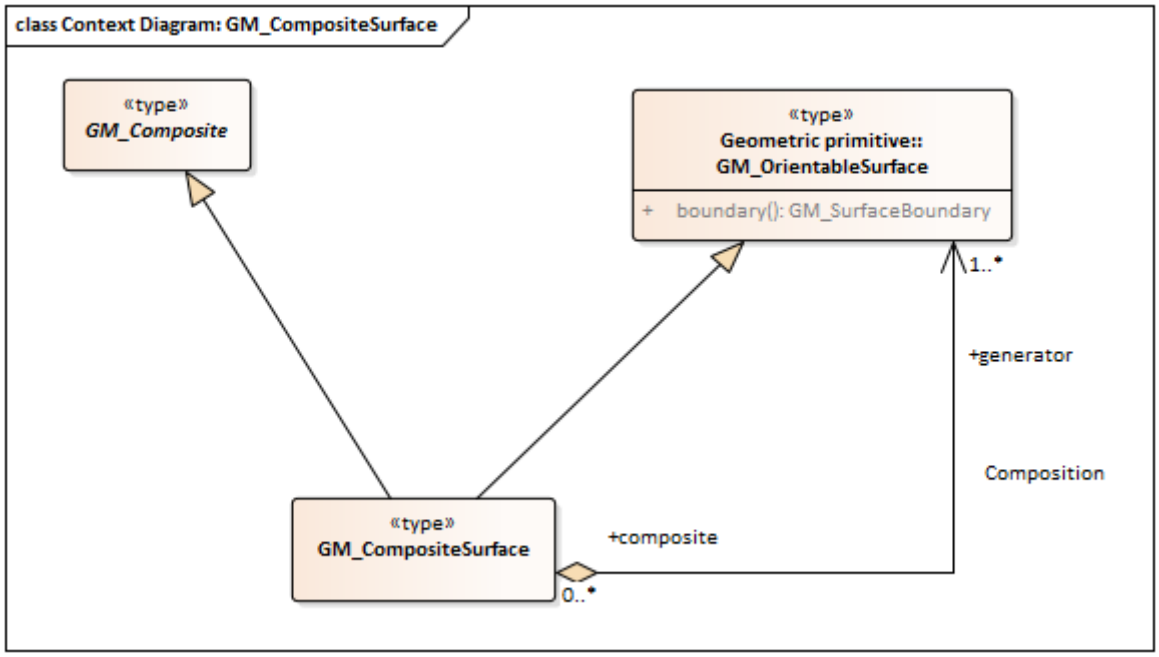



*Figure 11. GM_CompositeSurface Context Diagram*

### GM_OrientatableSurface

GM_OrientableSurface consists of a surface and an orientation inherited from GM_OrientablePrimitive. If the orientation is "+", then the GM_OrientableSurface is a GM_Surface. If the orientation is "-", then the GM_OrientableSurface is a reference to a GM_Surface with an upNormal that reverses the direction for this GM_OrientableSurface, the sense of "the top of the surface" (see 6.4.33.2).

```
GM_OrientableSurface:
{Orientation = "+" implies primitive = self};
{(Orientation = "-" and TransfiniteSet::contains(p : DirectPosition)) implies
(primitive.upNormal(p) = - self.upNormal(p))};
```

### GM_CompositeCurve

A GM_CompositeCurve is a list of geometric curves such that the each geometric curve in the set terminates at the start point of the subsequent curve in the list

The `generator` is a GM_OrientableCurve.

A GM_CompositCurve is also a subclass of GM_OrientatableCurve. One of the few examples of multiple inheritance.

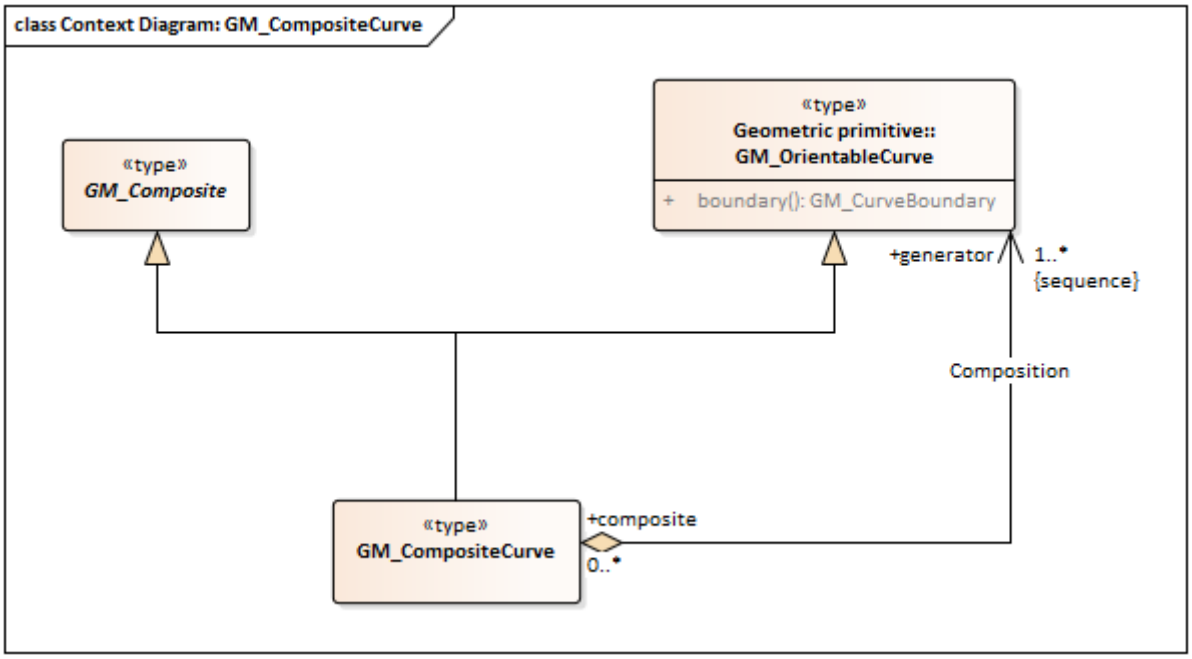



*Figure 12. GM_CompositeCurve Context Diagram*

### GM_OrientatableCurve

GM_OrientableCurve consists of a curve and an orientation inherited from GM_OrientablePrimitive. If the orientation is "+", then the GM_OrientableCurve is a GM_Curve. If the orientation is "-", then the GM_OrientableCurve is related to another GM_Curve with a parameterization that reverses the sense of the curve traversal.

```
GM_OrientableCurve:
{Orientation = "+" implies primitive = self}; +
{Orientation = "-" implies primitive.parameterization(length()-s) =
parameterization(s)};
```

### GM_CompositeSolid

A GM_CompositeSolid is a set of geometric solids adjoining one another along common boundary geometric surfaces.

The `generator` is a GM_Solid.

A GM_CompositSolid is also a subclass of GM_Solid. One of the few examples of multiple inheritance.

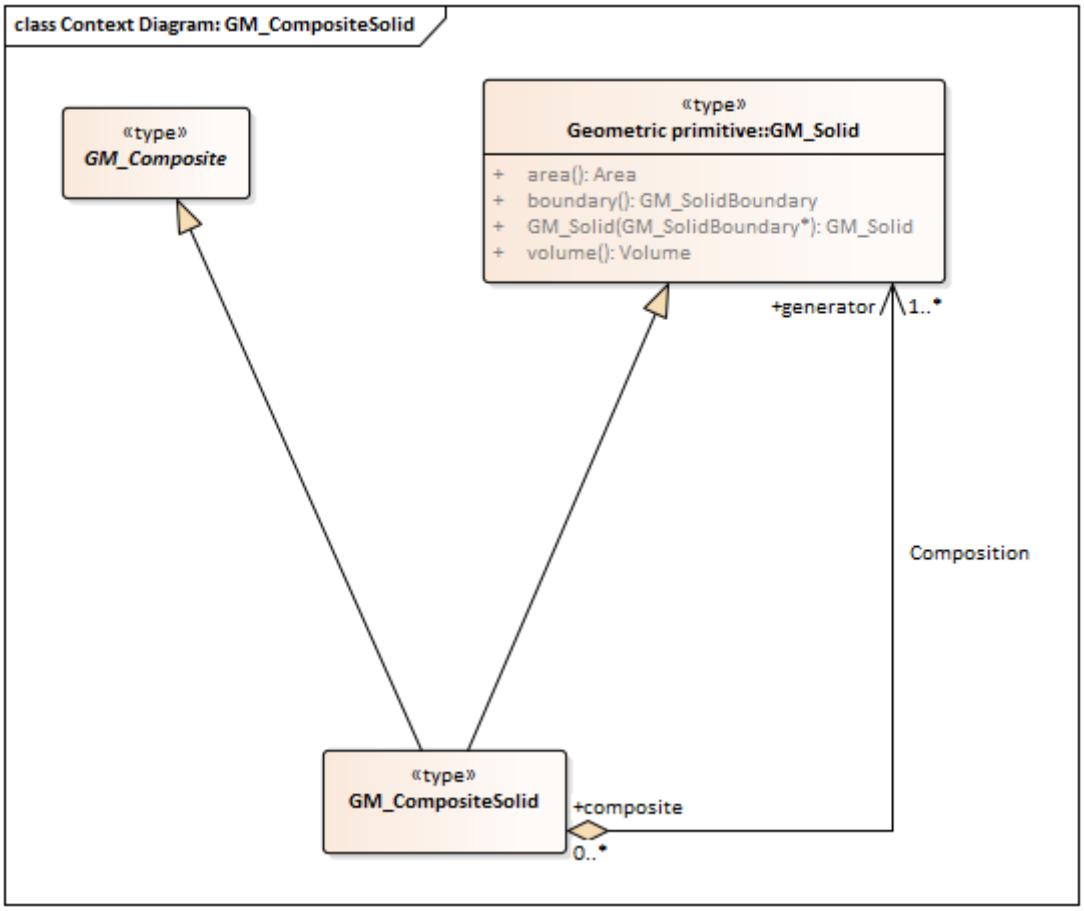



*Figure 13. GM_CompositeSolid Context Diagram*

### 8.1.3. City Models and City Objects

City models are virtual representations of real-world cities and landscapes. A city model aggregates different types of objects, which can be city objects, appearances, different versions of the city model, transitions between different versions of the city model, and feature objects. All objects defined in the CityGML CM are features with lifespan. This allows the optional specification of the real-world and database times for the existence of each feature, as is required by the Versioning module (cf. Section 8.16). Features that define thematic concepts related to cities and landscapes, such as building, bridge, water body, or land use, are referred to as city objects. All city objects define properties that describe the objects in more detail. These static properties can be overridden with time-varying data through Dynamizers (cf. Section 8.8).

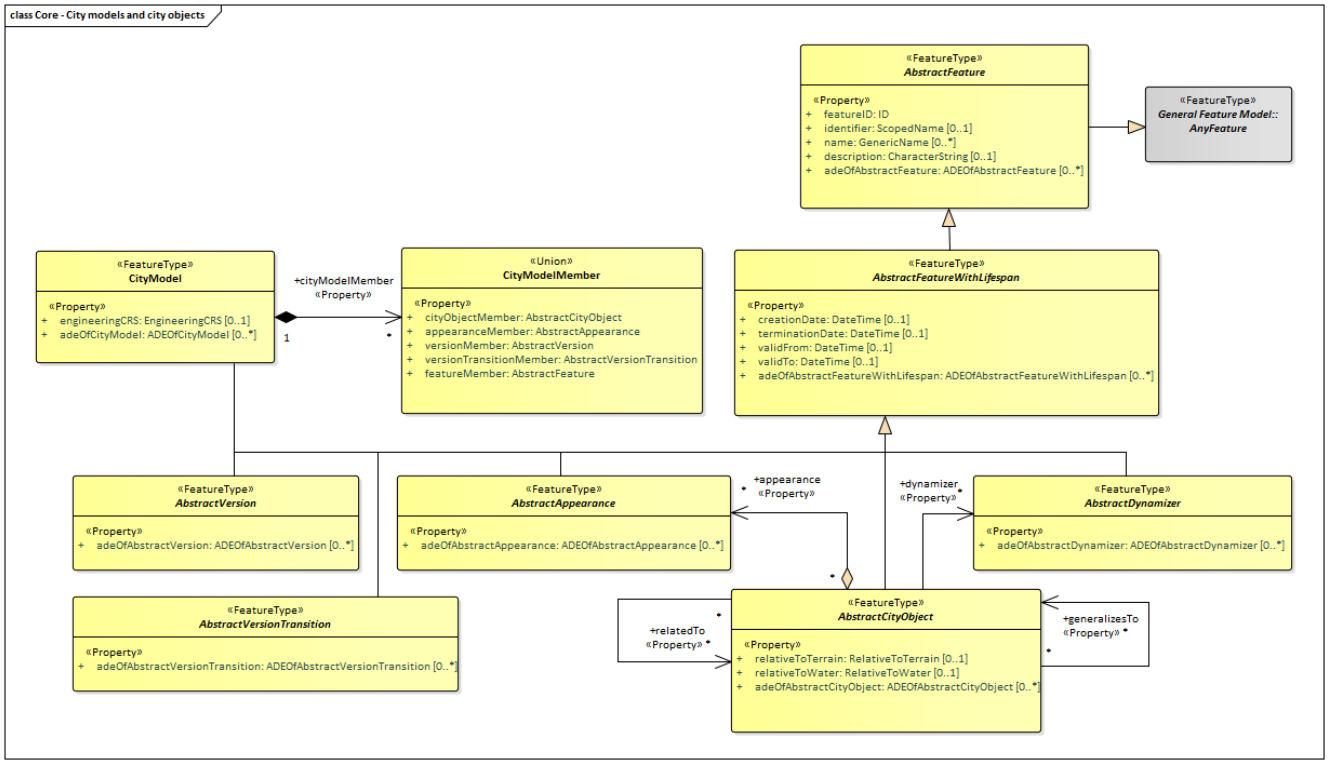



*Figure 14. UML City Models and City Objects*

The City Model and City Object classes in the in the Data Dictionary.

The most important concepts for the City Model user to understand are:

- AbstractFeature (provides identifiing properties)
- AbstractFeatureWithLifespan (adds temporal properties)
- AbstractThematicSurface (adds geometry and city context)
- AbstractOccupiedSpace
- AbstractUnoccupiedSpace
- AbstractLogicalSpace
- AbstractSpaceBoundary (relief)

**CityModel**

**CityModelMember**

**AbstractFeature**

**AbstractFeatureWithLifespan**

**AbstractAppearance**

**AbstractVersion**

**AbstractVersionTransition**

### 8.1.4. Space Concept

All city objects are differentiated into spaces and space boundaries.

Spaces are entities of volumetric extent in the real world. Buildings, water bodies, trees, rooms, and traffic spaces, for instance, have a volumetric extent. Spaces can be classified into physical spaces and logical spaces. Physical spaces, in turn, can be further classified into occupied spaces and unoccupied spaces.

Space boundaries, in contrast, are entities with areal extent in the real world. Space boundaries can be differentiated into different types of thematic surfaces, such as wall surfaces and roof surfaces.

| NOTE | add definition of `boundary surface` and introduce B_Rep. (Foley et al. 2002). In short, the solids in CityGML 3.0 can be represented by surfaces which define the boundaries of the associated shape. These `boundary surfaces` are associated with their AbstractSpace class through the `boundary` association (property). |
|---|---|

AbstractSpaceBoundary is specialized into thematic surfaces. It is the subclassing of the thematic surfaces which gives semantics to the CityGML surfaces. Note that a Closure Surface is also a thematic surface. so a solid can be fully B_Rep enclosed even if part of the surface does not physically exist.

Consider that the AbstractSpaceBoundary is a Multi-Surface. So it stands to reason that a `space boundary` is an AbstractSpaceBoundary which is composed of one or more `Boundary Surfaces`. Per the B_Rep concept, those `Boundary Surfaces` must completely enclose the Abstract Space. This may include the addition of surfaces both internal (but visible) as well as external. Hence, the distinction between internal and external space is moot. Internal surfaces are just a continuation of the space boundary around the solid.

Table 4 lists the surfaces that are allowed as thematic surface boundaries of the space classes defined in the Core module:

*Table 4. Core space classes and their allowed thematic surface boundaries*

| Space class | Allowed space boundaries |
|---|---|
| AbstractLogicalSpace | • Core::AbstractSpaceBoundary and the subclasses: Core::AbstractThematicSurface, Core::ClosureSurface<br>• Generics::GenericThematicSurface<br>• possible classes from ADEs |

| | |
|---|---|
| AbstractOccupiedSpace | • Core::AbstractSpaceBoundary and the subclasses: Core::AbstractThematicSurface, Core::ClosureSurface<br>• Generics::GenericThematicSurface<br>• possible classes from ADEs |
| AbstractPhysicalSpace | • Core::AbstractSpaceBoundary and the subclasses: Core::AbstractThematicSurface, Core::ClosureSurface<br>• Generics::GenericThematicSurface<br>• possible classes from ADEs |
| AbstractSpace | • Core::AbstractSpaceBoundary and the subclasses: Core::AbstractThematicSurface, Core::ClosureSurface<br>• Generics::GenericThematicSurface<br>• possible classes from ADEs |
| AbstractUnoccupiedSpace | • Core::AbstractSpaceBoundary and the subclasses: Core::AbstractThematicSurface, Core::ClosureSurface<br>• Generics::GenericThematicSurface<br>• possible classes from ADEs |

A detailed introduction to the Space concept can be found in [ug-space-and-space-boundaries-section].

In particular, the classification into OccupiedSpace and UnoccupiedSpace might not always be apparent at first sight. Carports, for instance, represent an OccupiedSpace, although they are not closed and most of the space is free of matter, see Figure 15. Since a carport is a roofed, immovable structure with the purpose of providing shelter to objects (i.e. cars), carports are frequently represented as buildings in cadastres. Thus, also in CityGML, a carport should be modelled as an instance of the class Building. Since Building is transitively a subclass of OccupiedSpace, a carport is an OccupiedSpace as well. However, only in LOD1, the entire volumetric region covered by the carport would be considered as physically occupied. In LOD1, the occupied space is defined by the entire carport solid (unless a room would be defined in LOD1 that would model the unoccupied part below the roof); whereas in LOD2 and LOD3, the solids represent more realistically the really physically occupied space of the carport. In addition, for all OccupiedSpaces, the normal vectors of the thematic surfaces like the RoofSurface need to point away from the solids, i.e. consistent with the solid geometry.

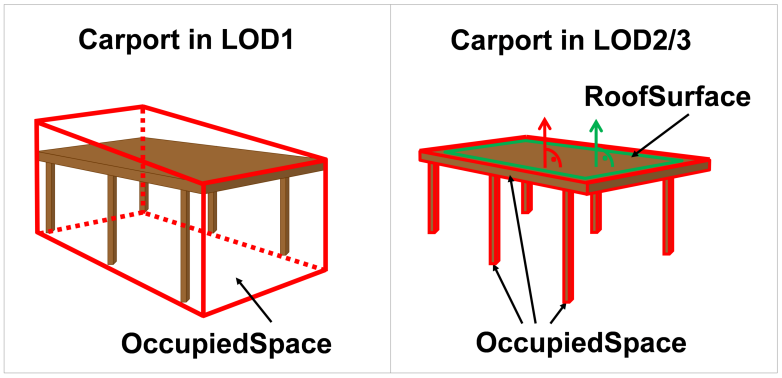



*Figure 15. Representation of a carport as OccupiedSpace in different LODs. The red boxes represent solids, the green area represents a surface. In addition, the normal vectors of the roof solid (in red) and the roof surface (in green) are shown.*

In contrast, a room is a physically unoccupied space. In CityGML, a room is represented by the class BuildingRoom that is a subclass of UnoccupiedSpace. In LOD1, the entire room solid would be considered as unoccupied space, which can contain furniture and installations, though, as is shown in Figure 16. In LOD2 and 3, the solid represents more realistically the really physically unoccupied space of the room (possibly somewhat generalized as indicated in the figure). For all UnoccupiedSpaces, the normal vectors of the bounding thematic surfaces like the InteriorWallSurface need to point inside the object, i.e. opposite to the solid geometry.

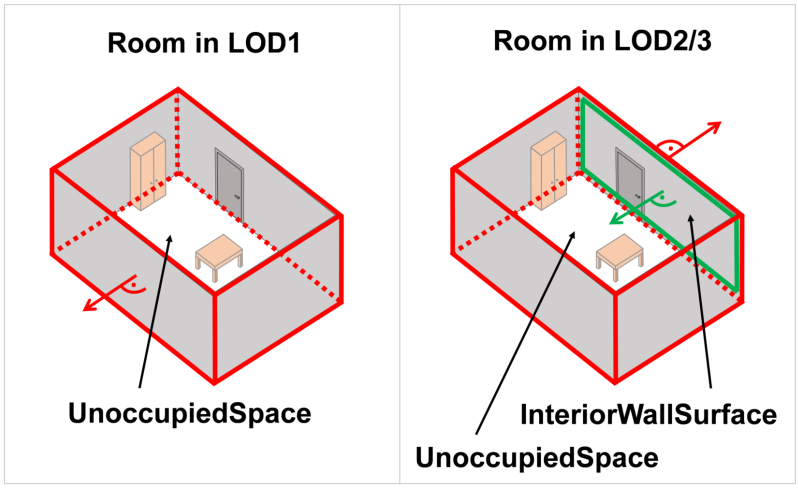



*Figure 16. Representation of a room as UnoccupiedSpace in different LODs. The red boxes represent solids, the green area represents a surface. In addition, the normal vectors of the room solid (in red) and the wall surface (in green) are shown.*

The UML diagram of the Space concept classes is depicted in Figure 17].

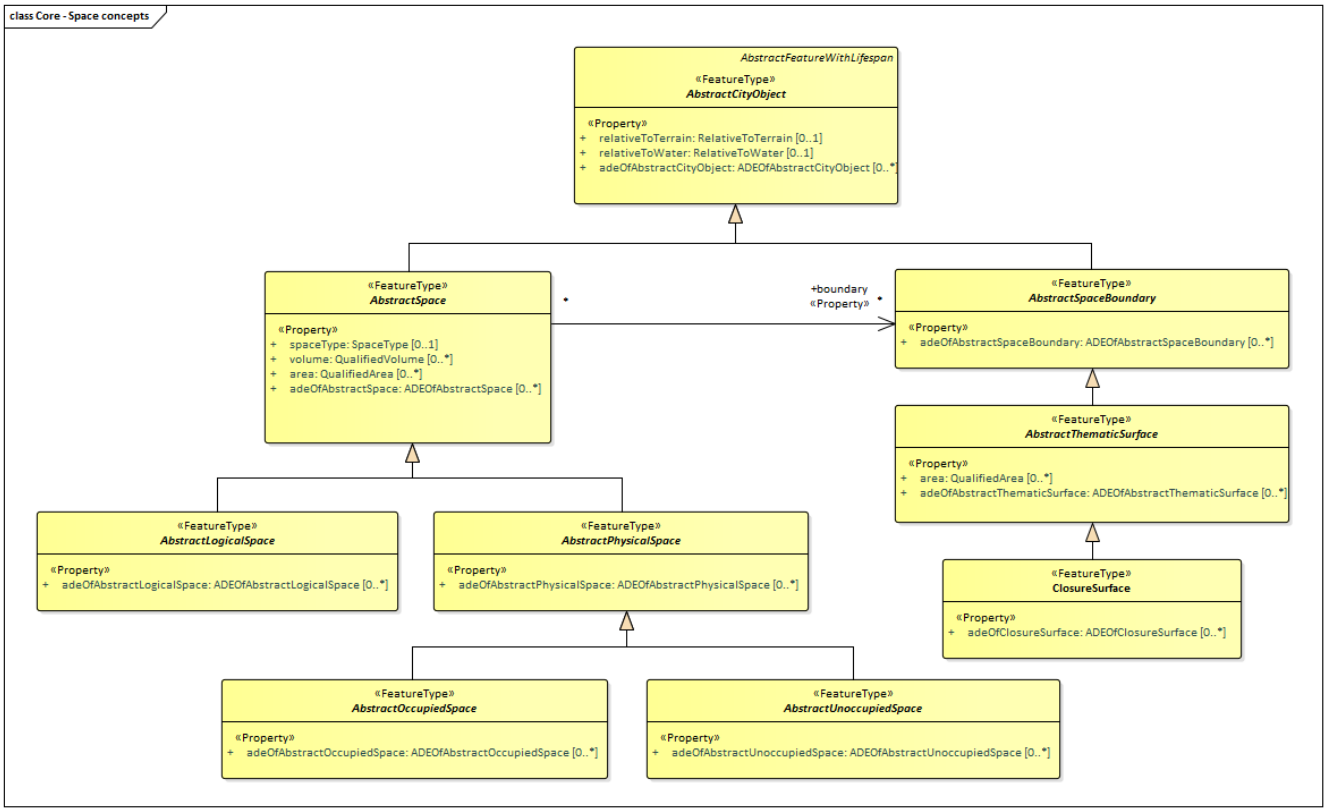



*Figure 17. UML Space Concepts*

The Space Concept classes defined in the CityGML UML model are documented in the Data Dictionary.

### 8.1.5. Geometry and LOD

Spaces and space boundaries can have various geometry representations depending on the Levels of Detail (LOD). Spaces can be spatially represented as single points in LOD0, multi-surfaces in LOD0/2/3, solids in LOD1/2/3, and multi-curves in LOD2/3. Space boundaries can be represented as multi-surfaces in LOD0/2/3 and as multi-curves in LOD2/3. All Levels of Detail allow for the representation of the interior of city objects.

The different Levels of Detail are defined in the following way:

- LOD 0: Volumetric real-world objects (Spaces) can be spatially represented by a single point, by a set of curves, or by a set of surfaces. Areal real-world objects (Space Boundaries) can be spatially represented in LOD0 by a set of curves or a set of surfaces. LOD0 surface representations are typically the result of a projection of the shape of a volumetric object onto a plane parallel to the ground, hence, representing a footprint (e.g. a building footprint or a floor plan of the rooms inside a building). LOD0 curve representations are either the result of a projection of the shape of a vertical surface (e.g. a wall surface) onto a grounding plane or the skeleton of a volumetric shape of longitudinal extent such as a road or river segment.
- LOD 1: Volumetric real-world objects (Spaces) are spatially represented by a vertical extrusion solid, i.e. a solid created from a horizontal footprint by vertical extrusion. Areal real-world objects (Space Boundaries) can be spatially represented in LOD1 by a set of horizontal or vertical surfaces.
- LOD 2: Volumetric real-world objects (Spaces) can be spatially represented by a set of curves, a

set of surfaces, or a single solid geometry. Areal real-world objects (Space Boundaries) can be spatially represented in LOD2 by a set of surfaces. The shape of the real-world object is generalized in LOD2 and smaller details (e.g. bulges, dents, sills, but also structures like e.g. balconies or dormers of buildings) are typically neglected. LOD2 curve representations are skeletons of volumetric shapes of longitudinal extent like an antenna or a chimney.

- LOD 3: Volumetric real-world objects (Spaces) can be spatially represented by a set of curves, a set of surfaces, or a single solid geometry. Areal real-world objects (Space Boundaries) can be spatially represented in LOD3 by a set of surfaces. LOD3 is the highest level of detail and respective geometries include all available shape details.

In addition, the geometry can also be represented implicitly. The shape is stored only once as a prototypical geometry, which then is re-used or referenced, wherever the corresponding feature occurs in the 3D city model.

The thematic classes, such as building, tunnel, road, land use, water body, or city furniture are defined as subclasses of the space and space boundary classes within the thematic modules. Since all city objects in the thematic modules represent subclasses of the space and space boundary classes, they automatically inherit the geometries defined in the Core module.

The UML diagram of the Geometry and LoD concept classes is depicted in Figure 18.

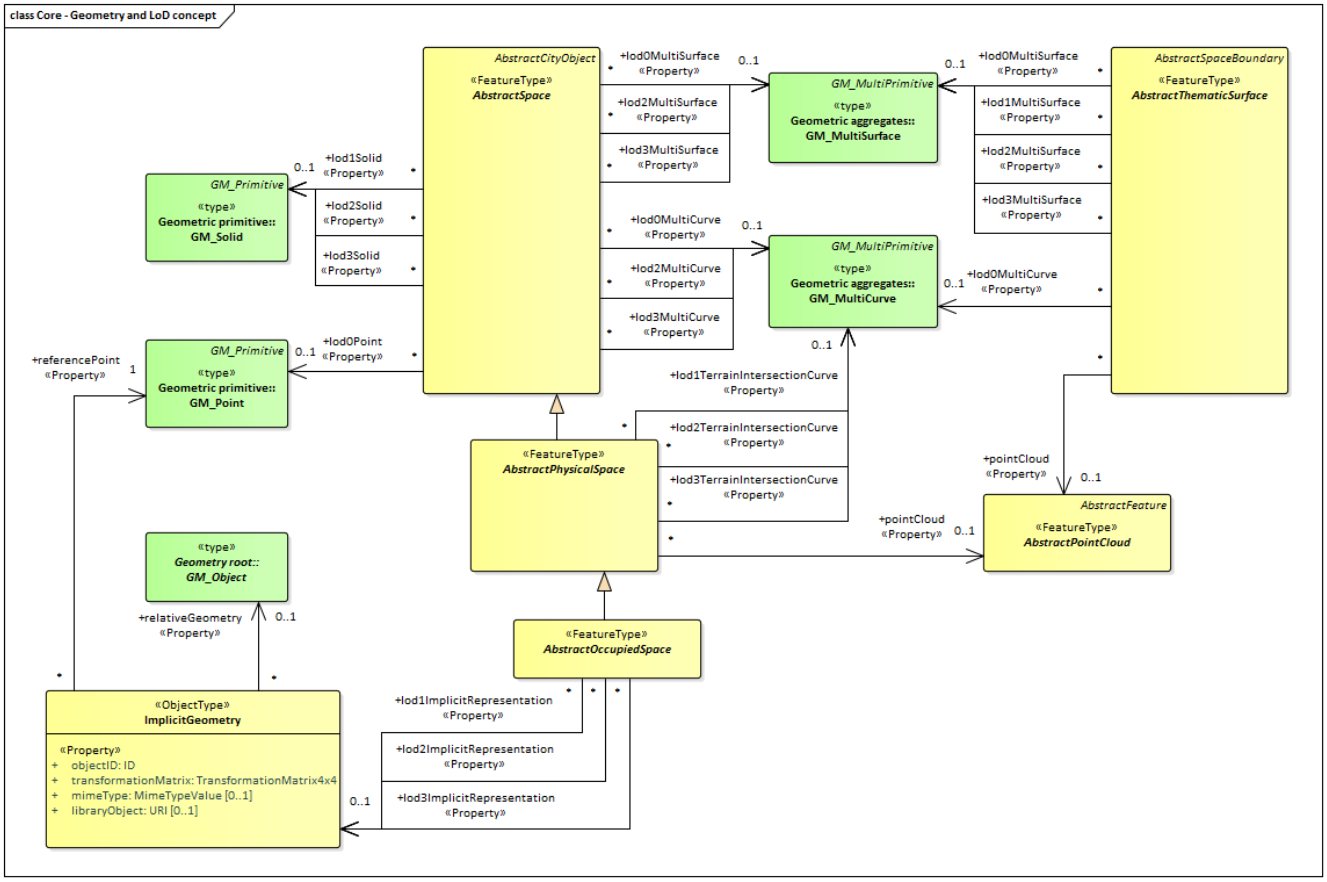



*Figure 18. UML Geometry and LOD Concepts*

The Geometry and LOD Concept classes defined in the CityGML UML model and documented in the Data Dictionary.

Of particular note is the Implicit Geometry concept. Many of the objects encountered in a city

landscape have the same geometry. How many types of street lamps can there be? An Implicit Geometry captures that geometry once, and re-uses that one geometry for all similar street lamp objects.

### 8.1.6. CityGML Core UML Model

The City Model and City Object classes, the Space Concept classes, and the Geometry and LOD classes define the majority of the CityGML Core module. In addition to these concepts, the Core module also specifies that city objects can have relations to other city objects and that they can have address information. All other modules defined in the CityGML model refer to the Core module.

The UML diagram of the complete Core module is depicted in Figure 19.

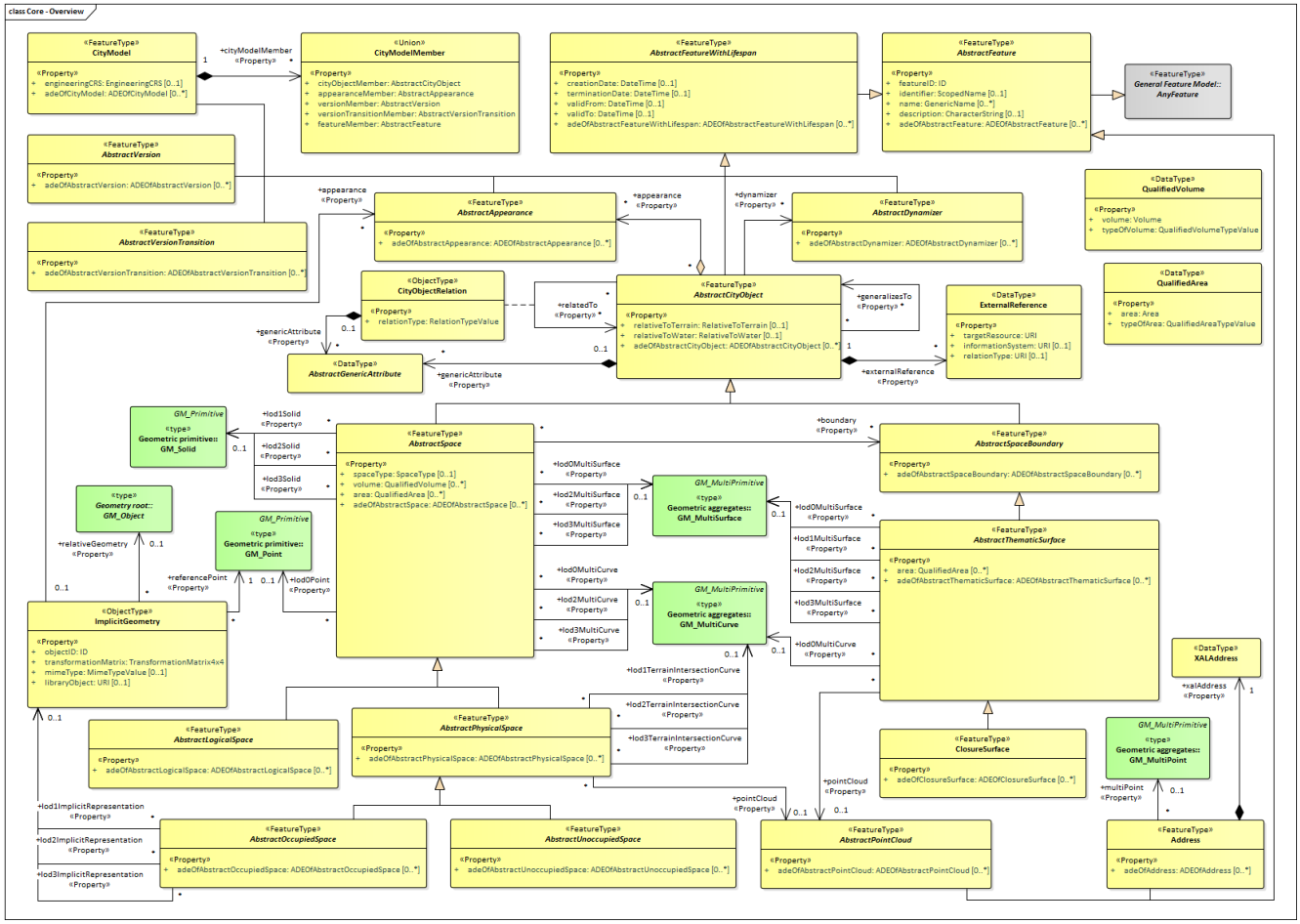



*Figure 19. UML diagram of CityGML's core module.*

The most important of the Core classes have been introduce already in the Key Concepts Section. More details about these classes can be found in the Data Dictionary.

### 8.1.7. Data types, Enumerations, and Code lists

The ADE data types provided for in the Core module are illustrated in Figure 20.

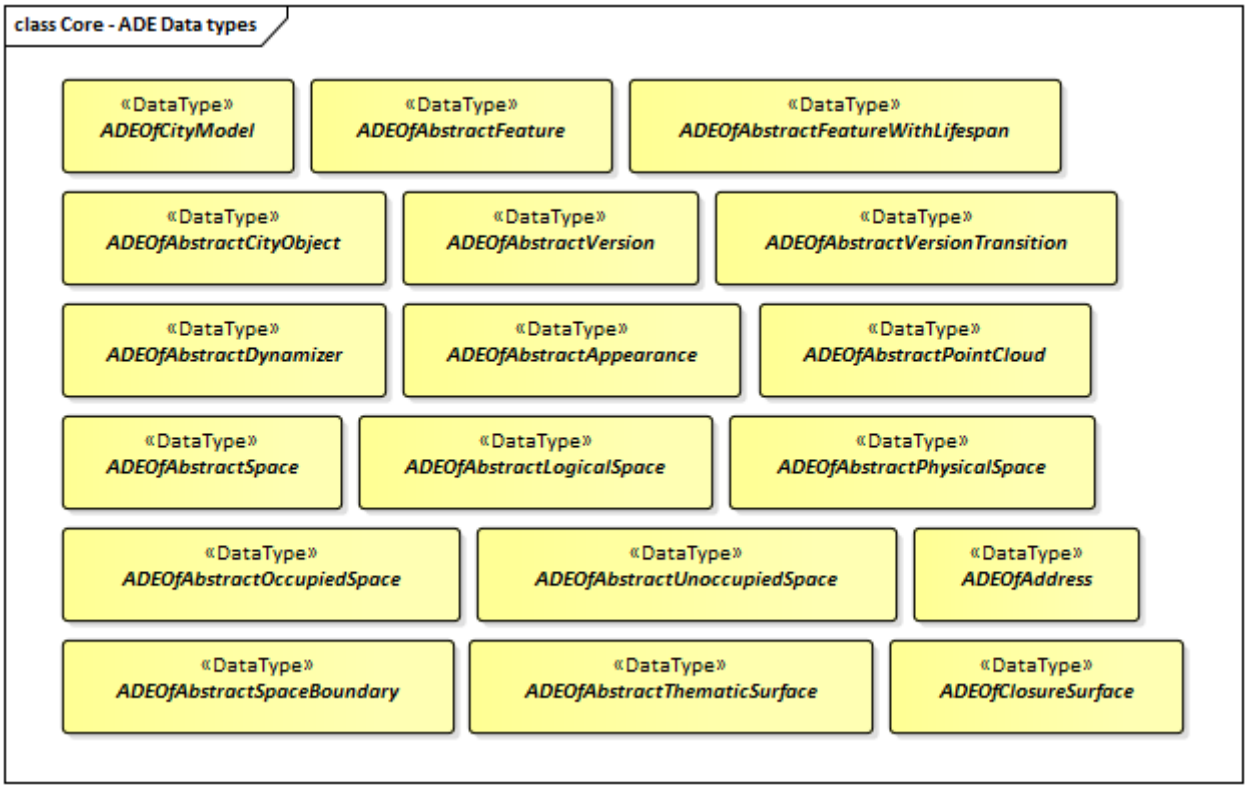



*Figure 20. ADE classes of the CityGML Core module.*

The Data Types, Basic Types, Enumerations, Unions, and Code Lists provided for the Core module are illustrated in Figure 21.

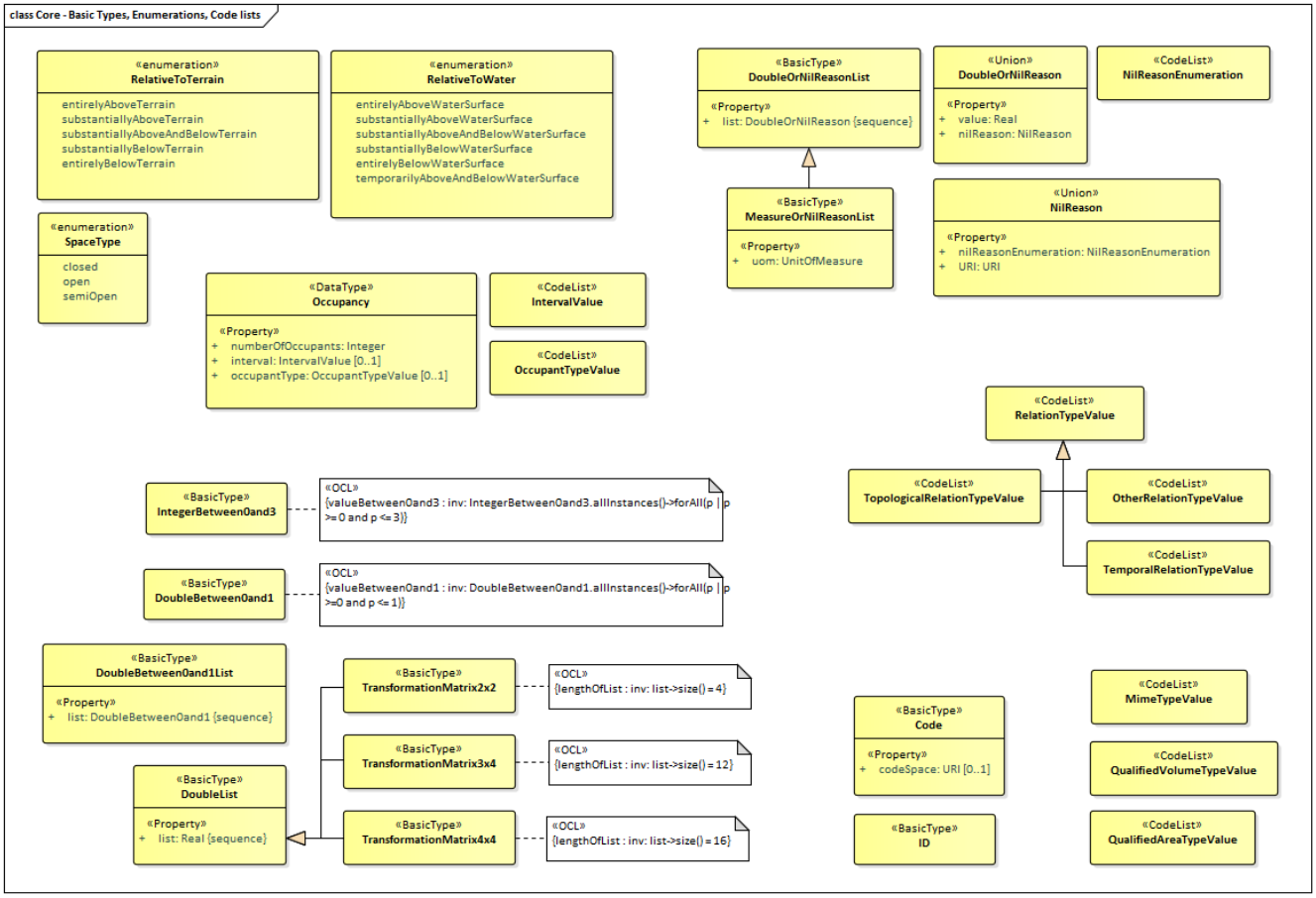



*Figure 21. Basic Types, Enumerations, and Codelists from the CityGML Core module.*

# 8.2. Appearance

## 8.2.1. Synopsis

The Appearance module provides the representation of surface data such as observable properties for surface geometry objects in the form of textures and material.

Appearances are not limited to visual data but represent arbitrary categories called themes such as infrared radiation, noise pollution, or earthquake-induced structural stress. A single surface geometry object may have surface data for multiple themes. Similarly, surface data can be shared by multiple surface geometry objects (e.g. road paving).

Surface data that is constant across a surface is modelled as material based on the material definitions from the X3D and COLLADA standards. Surface data that depends on the exact location within the surface is modelled as a texture. This can either be a parameterized texture (a texture that uses texture coordinates) or a transformation matrix for parameterization, or a georeferenced texture (a texture that uses a planimetric projection).

Each surface geometry object can have both, a material and a texture per theme and side. This allows for providing a constant approximation and a complex measurement of a surface's property simultaneously.

## 8.2.2. Key Concepts

**Appearance**: An Appearance is a collection of surface data, i.e. observable properties for surface geometry objects in the form of textures and material.
A type of AbstractAppearance.

**AbstractSurfaceData**: AbstractSurfaceData is the abstract superclass for different kinds of textures and material. Textures and materials are the primitives used to render 3D objects.
A type of AbstractFeature.

**X3DMaterial**: X3DMaterial defines properties for surface geometry objects based on the material definitions from the standards X3D and COLLADA.
A type of AbstractSurfaceData.

**AbstractTexture**: AbstractTexture is the abstract superclass to represent the common attributes of CityGML textures.
A type of AbstractSurfaceData.

**ParameterizedTexture**: A ParameterizedTexture is a texture that uses texture coordinates or a transformation matrix for parameterization.
A type of AbstractTexture.

**GeoreferencedTexture**: A GeoreferencedTexture is a texture that uses a planimetric projection. It contains an implicit parameterization that is either stored within the image file, an accompanying world file or specified using the orientation and referencePoint elements.
A type of AbstractTexture.

### 8.2.3. Discussion

In addition to spatial properties, CityGML features have appearances – observable properties of the feature's surface. Appearances are not limited to visual data but represent arbitrary categories called themes such as infrared radiation, noise pollution, or earthquake-induced structural stress. Each LOD can have an individual appearance for a specific theme. An appearance is composed of Surface Data (ArstractSurfaceData) objects. A single Surface Data object may have surface data for multiple themes. Similarly, surface data can be shared by multiple City Objects. Finally, surface data values can either be constant across a surface or depend on the exact location within the surface.

A constant surface property is modelled as material. A surface property, which depends on the location within the surface, is modelled as texture. Each City Object can have both a material and a texture per theme and side. This allows for providing both a constant approximation and a complex measurement of a surface's property simultaneously. An application is responsible for choosing the appropriate property representation for its task (e.g. analysis or rendering). A specific mixing is not defined since this is beyond the scope of CityGML. If a City Object is to receive multiple textures or materials, each texture or material requires a separate theme. The mixing of themes or their usage is not defined within CityGML and left to the application.

In CityGML's appearance model, themes are represented by an identifier only. The appearance of a city model for a given theme is defined by a set of Appearance objects referencing this theme. Thus, the Appearance objects belonging to the same theme compose a virtual group. They may be included in different places within a CityGML dataset. Furthermore a single CityGML dataset may contain several themes. An Appearance object collects surface data relevant for a specific theme either for individual features or the whole city model in any LOD. Surface data is represented by objects of class AbstractSurfaceData and its descendents. The relation between a City Object and Surface Data objects is expressed by the `appearance` relation from a City Object to an object of type Surface Data.

### 8.2.4. Material

Materials define light reflection properties being constant for a whole surface geometry object. The definition of the class X3DMaterial is adopted from the X3D and COLLADA specification.

The colors of a material can have three different forms:

- diffuseColor: the color of diffusely reflected light.
- specularColor: the color of a directed reflection.
- emissiveColor: the color of light generated by the surface.

All colors use RGB values with red, green, and blue between 0 and 1.

Transparency is defined separately using the `transparency` property where 0 stands for fully opaque and 1 for fully transparent.

The `ambientIntensity` property defines the minimum percentage of `diffuseColor` that is visible regardless of light sources.

The `shininess` property controls the sharpness of the specular highlight. 0 produces a soft glow

while 1 results in a sharp highlight.

The `isSmooth` property gives a hint for normal interpolation. If this boolean flag is set to true, vertex normals should be used for shading (Gouraud shading). Otherwise, normals should be constant for a surface patch (flat shading).

### 8.2.5. Texture and texture mapping

The abstract base class for textures is AbstractTexture. Textures in CityGML are always raster-based 2D textures. The raster image is specified by the `imageURI` property. This property is an URI which references an arbitrary image data resource, or even a preformatted request for a web service. The image data format can be defined using standard MIME types in the `mimeType` element.

Textures can be qualified by the attribute `textureType`. The `textureType` differentiates between textures, which are specific for a certain object (specific) and prototypic textures being typical for that object surface (typical). Textures may also be classified as unknown.

The specification of texture wrapping is adopted from the COLLADA standard. Texture wrapping is required when accessing a texture outside the underlying image raster. The `wrapMode` property can have one of five values (Figure 22 illustrates the effect of these wrap modes):

1. none – the resulting color is fully transparent
2. wrap – the texture is repeated
3. mirror – the texture is repeated and mirrored
4. clamp – the texture is clamped to its edges
5. border – the resulting color is specified by the borderColor element (RGBA)

In wrap mode `mirror` the texture image is repeated both in horizontal and in vertical direction to fill the texture space similar to wrap mode wrap. Unlike `wrap`, each repetition results from flipping the previous texture part along the repetition direction. This behaviour removes the edge correspondence constraint for wrapped textures and always results in a seamless texture.

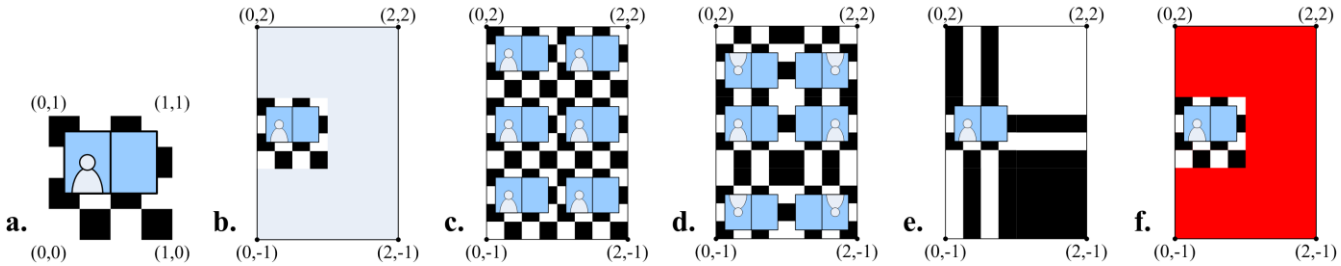



*Figure 22. A texture (a) applied to a facade using different wrap modes: (b) none, (c) wrap, (d) mirror, (e) clamp and (f) border. The border color is red. The numbers denote texture coordinates (image: Hasso-Plattner-Institute).*

#### Georeferenced Texture

AbstractTexture is further specialised according to the texture parameterisation, i.e. the mapping function from a location on the surface to a location in the texture image. CityGML uses the notion of texture space, where the texture image always occupies the region $[0,1]^2$ regardless of the actual image size or aspect ratio. The lower left image corner is located at the origin (some graphics APIs

may use other conventions and require texture coordi-nate conversion). The mapping function must be known for each surface geometry object to receive texture.

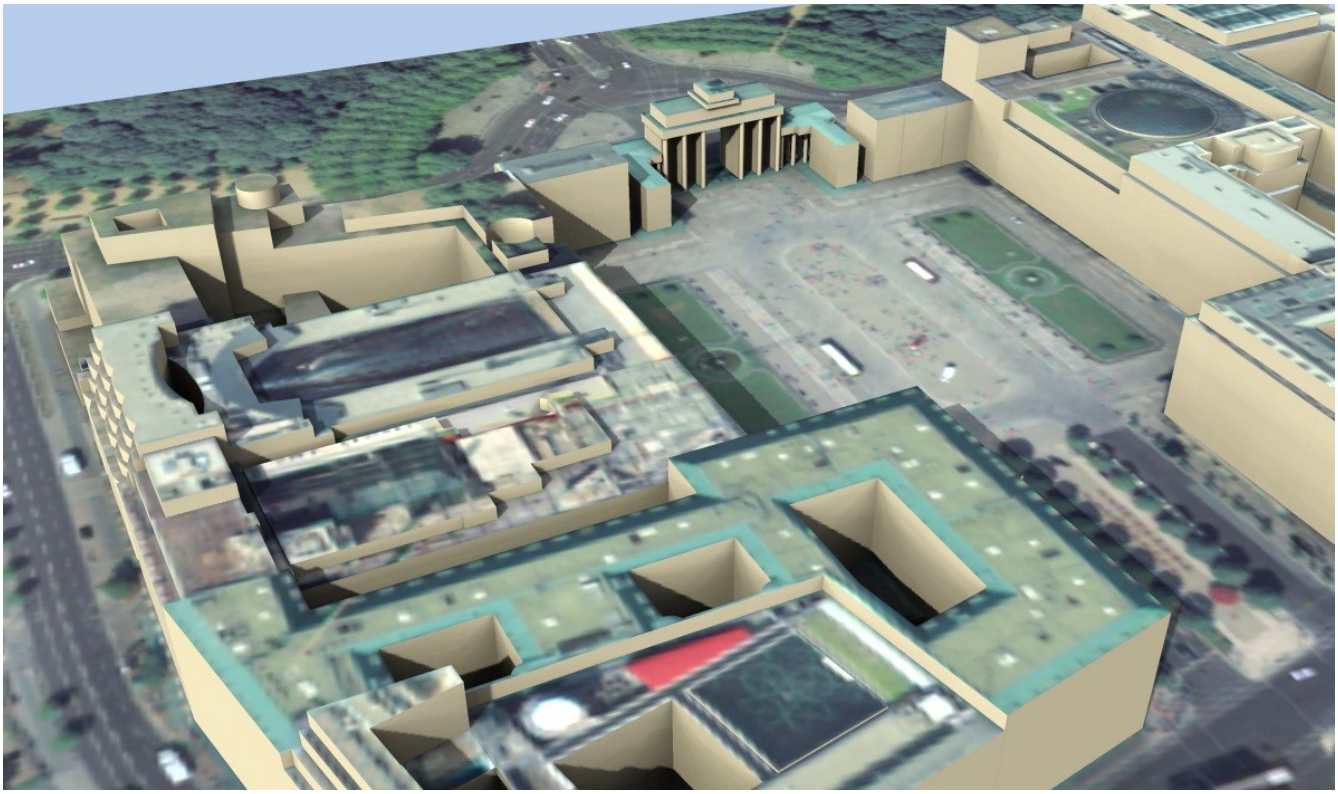


*Figure 23. A georeferenced texture applied to ground and roof surfaces (source: Senate of Berlin, Hasso-Plattner-Institute).*

The class GeoreferencedTexture describes a texture that uses a planimetric projection. Consequently, it does not make sense to texture vertical surfaces using a GeoreferencedTexture. Such a texture has a unique mapping function which is usually provided with the image file (e.g. georeferenced TIFF) of an ESRI World file. The search order for an external georeference is determined by the boolean flag `preferWorldFile`. If this flag is set to true (its default value), a world file is looked for first and only if it is not found the georeference from the image data is used. If `preferWorldFile` is false, the world file is used only if no georeference from the image data is available.

Alternatively, CityGML allows for inline specification of a georeference similar to a World file. This internal georeference specification always takes precedence over any external georeference. The property `referencePoint` identifies the GM_Point which represenbts the location of the center of the upper left image pixel in world space.

If neither an internal nor an external georeference is given the GeoreferencedTexture is invalid.

#### Parameterized Texture

The class ParameterizedTexture describes a texture with target-dependent mapping function.

The target of each mapping function is a subclass of class AbstractTextureParameterization. The `textureParmaterization` property associates an AbstractTextureParameterization object with the the ParameterizedTexture object within which it is contained.

The mapping function is performed through a TextureAssociation object. This object is a UML

Association Class object which is associated with each `textureParmaterization` property. The TextureAssociation class contains one property which is a URI identifying a surface geometry. The AbstractTextureParameterization object is selected based on the URI value in the corresponding TextureAssociation object.

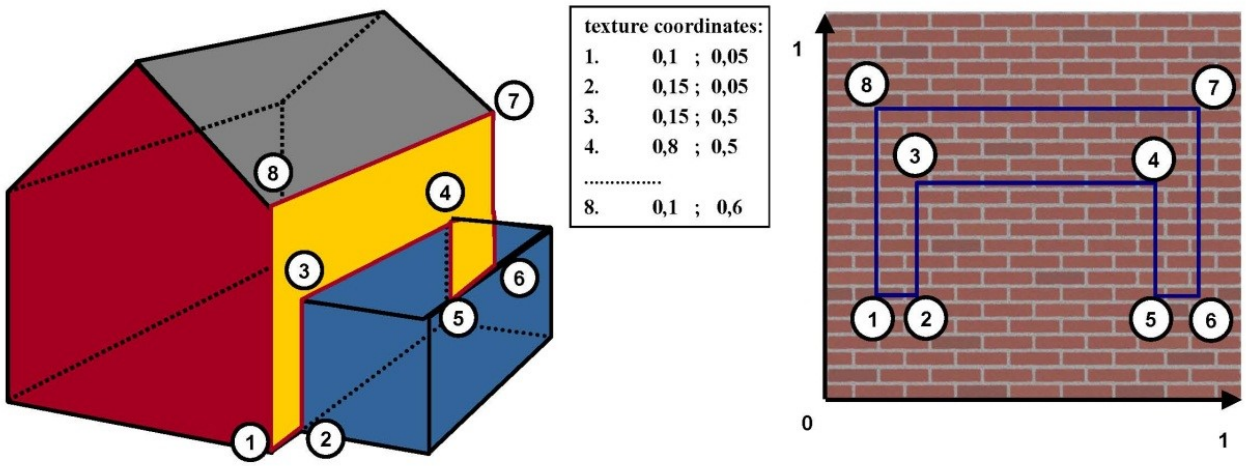



*Figure 24. Positioning of textures using texture coordinates (image: IGG Uni Bonn).*

Texture coordinates are applicable only to polygonal surfaces, whose boundaries are described by GM_MultiSurface or GM_MultiCurve. They define an explicit mapping of a surface's vertices to points in texture space, i.e. each vertex including interior ring vertices must receive a corresponding coordinate pair in texture space (for the notion of coordinates, refer to ISO 19111). These coordinates are not restricted to the [0,1] interval. Texture coordinates for interior surface points are planarly interpolated from the vertices' texture coordinates.

Texture coordinates for a target surface geometry object are specified using class TexCoordList as a texture parameterization object in the texture's target property. Each exterior and interior GM_MultiSurface or GM_MultiCurve composing the boundary of the target surface geometry object requires its own set of texture coordinates. A set of texture coordinates is specified using the `textureCoordinates` element of class TexCoordList. Thus, a TexCoordList contains as many `textureCoordinate` elements as the target surface geometry object contains gml:LinearRings.

| **NOTE** | Linear Rings are a GML construct. What do we replace it with? |
|---|---|

textureCoordinate's mandatory attribute ring provides the gml:id of the respective ring. The content is an ordered list of double values where each two values define a  T s,t texture coordinate pair with s denoting the horizontal and t the vertical texture axis. The list contains one

pair per ring point with the pairs' order corresponding to the ring points' order in the CityGML document (regardless of a possibly flipped surface orientation). If any ring point of a target surface geometry object has no texture coordinates assigned, the mapping is incomplete and the respective surface cannot be textured. In case of aggregated target geometry objects, mapping completeness is determined only for leaf geometry objects.

| **NOTE** | Update the text between this and the previous note. |
|---|---|

*Figure 25. Projecting a photograph (a) onto multiple facades (b) using the worldToTexture transformation. The photograph does not cover the left facade completely. Thus, the texture appears to be clipped. Texture wrapping is set to "none" (source: Senate of Berlin, Hasso-Plattner-Institute).*

| **NOTE** | the rest of this section still needs updating from v 2.0 |
|---|---|

Alternatively, the mapping function can comprise a 3x4 transformation matrix specified by class TexCoordGen. The transformation matrix, specified by the `worldToTexture` element, defines a linear transformation from a spatial location in homogeneous coordinates to texture space. The use of homogeneous coordinates facilitates perspective projections as transformation, e.g. for projecting a photograph into a city model (cf. Figure 25). Texture coordinates □ □T s,t are calculated from a space location □ □T x, y, z as □ □ □ □T T s,t □ s□ q□,t□ q□ with □ □ □ □T T s□,t□,q□ □M□ x, y, z,1 . M denotes the 3x4 transformation matrix. Compared to a general 4x4 transformation, the resulting z component is ignored. Thus, the respective matrix row is omitted. Additionally, the TexCoordGen object uses the `crs` association to identify a SC_CRS object which defines its CRS. A location in world space has to be first transformed into this CRS before the transformation matrix can be applied.

The following construction results in a worldToTexture transformation that mimics the process of taking a photograph by projecting a location in world space (in the city model) to a location in texture space:

| **NOTE** | insert transformation matrix |
|---|---|

In this formula, f denotes the focal length; w and h represent the image sensor's physical dimensions; r □ , u □ , and d □ define the camera's frame of reference as right, up and directional unit vectors expressed in world coordinates; and P stands for the camera's location in world space. Fig. 19 sketches this setting.

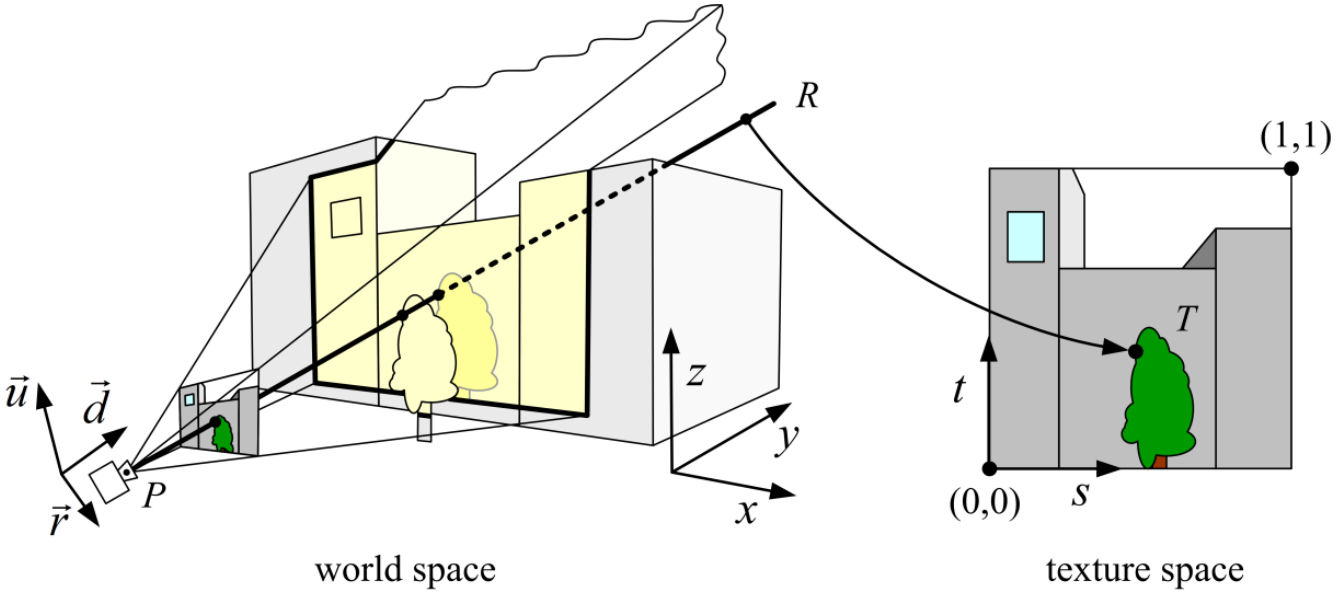



*Figure 26. Projective texture mapping. All points on a ray R starting from the projection center P are mapped to the same point T in texture space (image: Hasso-Plattner-Institute, IGG TU Berlin).*

Alternatively, if the 3x4 camera matrix MP is known (e.g. through a calibration and registration process), it can easily be adopted for use in worldToTexture. MP is derived from intrinsic and extrinsic camera parameters (interior and exterior orientation) and transforms a location in world space to a pixel location in the image. Assuming the upper left image corner has pixel coordinates (0,0), the complete transformation to texture space coordinates can be written as (widthimage and heightimage denote the image size in pixels):

**NOTE** insert formula

Please note, that worldToTexture cannot compensate for radial or other non-linear distortions introduced by a real camera lens.

Another use of worldToTexture is texturing a facade with complex geometry without specifying texture coordinates for each gml:LinearRing. Instead, only the facade's aggregated surface becomes the texture target using a TexCoordGen as parameterization. Then, worldToTexture effectively encodes an orthographic projection of world space into texture space. For the special case of a vertical facade this transformation is given by:

**NOTE** insert formula

```
<math display="block">
  <mrow class="MJX-TeXAtom-ORD">
    <mover>
      <mi mathvariant="normal">&#x2207;</mi>
      <mo stretchy="false">&#x2192;</mo>
    </mover>
  </mrow>
  <mo>&#xD7;</mo>
  <mrow class="MJX-TeXAtom-ORD">
    <mover>
      <mi>F</mi>
```

```
      <mo stretchy="false">&#x2192;</mo>
    </mover>
  </mrow>
  <mo>=</mo>
  <mrow>
    <mo>(</mo>
    <mfrac>
      <mrow>
        <mi mathvariant="normal">&#x2202;</mi>
        <msub>
          <mi>F</mi>
          <mi>z</mi>
        </msub>
      </mrow>
      <mrow>
        <mi mathvariant="normal">&#x2202;</mi>
        <mi>y</mi>
      </mrow>
    </mfrac>
    <mo>&#x2212;</mo>
    <mfrac>
      <mrow>
        <mi mathvariant="normal">&#x2202;</mi>
        <msub>
          <mi>F</mi>
          <mi>y</mi>
        </msub>
      </mrow>
      <mrow>
        <mi mathvariant="normal">&#x2202;</mi>
        <mi>z</mi>
      </mrow>
    </mfrac>
    <mo>)</mo>
  </mrow>
  <mrow class="MJX-TeXAtom-ORD">
    <mi mathvariant="bold">i</mi>
  </mrow>
  <mo>+</mo>
  <mrow>
    <mo>(</mo>
    <mfrac>
      <mrow>
        <mi mathvariant="normal">&#x2202;</mi>
        <msub>
          <mi>F</mi>
          <mi>x</mi>
        </msub>
      </mrow>
      <mrow>
        <mi mathvariant="normal">&#x2202;</mi>
```

```
        <mi>z</mi>
      </mrow>
    </mfrac>
    <mo>&#x2212;</mo>
    <mfrac>
      <mrow>
        <mi mathvariant="normal">&#x2202;</mi>
        <msub>
          <mi>F</mi>
          <mi>z</mi>
        </msub>
      </mrow>
      <mrow>
        <mi mathvariant="normal">&#x2202;</mi>
        <mi>x</mi>
      </mrow>
    </mfrac>
    <mo>)</mo>
  </mrow>
  <mrow class="MJX-TeXAtom-ORD">
    <mi mathvariant="bold">j</mi>
  </mrow>
  <mo>+</mo>
  <mrow>
    <mo>(</mo>
    <mfrac>
      <mrow>
        <mi mathvariant="normal">&#x2202;</mi>
        <msub>
          <mi>F</mi>
          <mi>y</mi>
        </msub>
      </mrow>
      <mrow>
        <mi mathvariant="normal">&#x2202;</mi>
        <mi>x</mi>
      </mrow>
    </mfrac>
    <mo>&#x2212;</mo>
    <mfrac>
      <mrow>
        <mi mathvariant="normal">&#x2202;</mi>
        <msub>
          <mi>F</mi>
          <mi>x</mi>
        </msub>
      </mrow>
      <mrow>
        <mi mathvariant="normal">&#x2202;</mi>
        <mi>y</mi>
      </mrow>
```

```
    </mfrac>
    <mo>)</mo>
  </mrow>
  <mrow class="MJX-TeXAtom-ORD">
    <mi mathvariant="bold">k</mi>
  </mrow>
</math>
```

This equation assumes n  denoting the facade's overall normal vector (normalized, pointing outward, and being parallel to the ground), F denoting the facade's lower left point, and widthf and heightf specifying the facade's dimensions in world units. For the general case of an arbitrary normal vector the facade orientation matrix assumes a form similar to the camera orientation matrix:

**NOTE** insert formula

### 8.2.6. Related concepts

The notion of appearance clearly relates to the generic coverage approach (cf. ISO 19123 and OGC Abstract specification, Topic 6). Surface data can be described as discrete or continuous coverage over a surface as two-dimensional domain with a specific mapping function. Such an implementation requires the extension of GML coverages (as of version 3.1) by suitable mapping functions and specialisation for valid domain and range sets. For reasons of simplicity and comprehensibility both in implementation and usage, CityGML does not follow this approach, but relies on textures and materials as well-known surface property descriptions from the field of computer graphics (cf. X3D, COLLADA specification, Foley et al.). Textures and materials store data as color using an appropriate mapping. If such a mapping is impractical, data storage can be customised using ADEs. A review of coverages for appearance modelling is considered for CityGML beyond version 2.0.0.

Appearance is also related to portrayal. Portrayal describes the composition and symbolisation of a digital model's image, i.e. presentation, while appearance encodes observations of the real object's surface, i.e. data. Even though being based on graphical terms such as textures and materials, surface data is not limited to being input for portrayal, but similarly serves as input or output for analyses on a feature's surface. Consequently, CityGML does not define mixing or composition of themes for portrayal purposes. Portrayal is left to viewer applications or styling specification languages such as OGC Styled Layer Descriptors (SLD) or OGC Symbolo-gy Encoding (SE).

### 8.2.7. UML Model

The UML diagram of the Appearance module is illustrated in Figure 27.

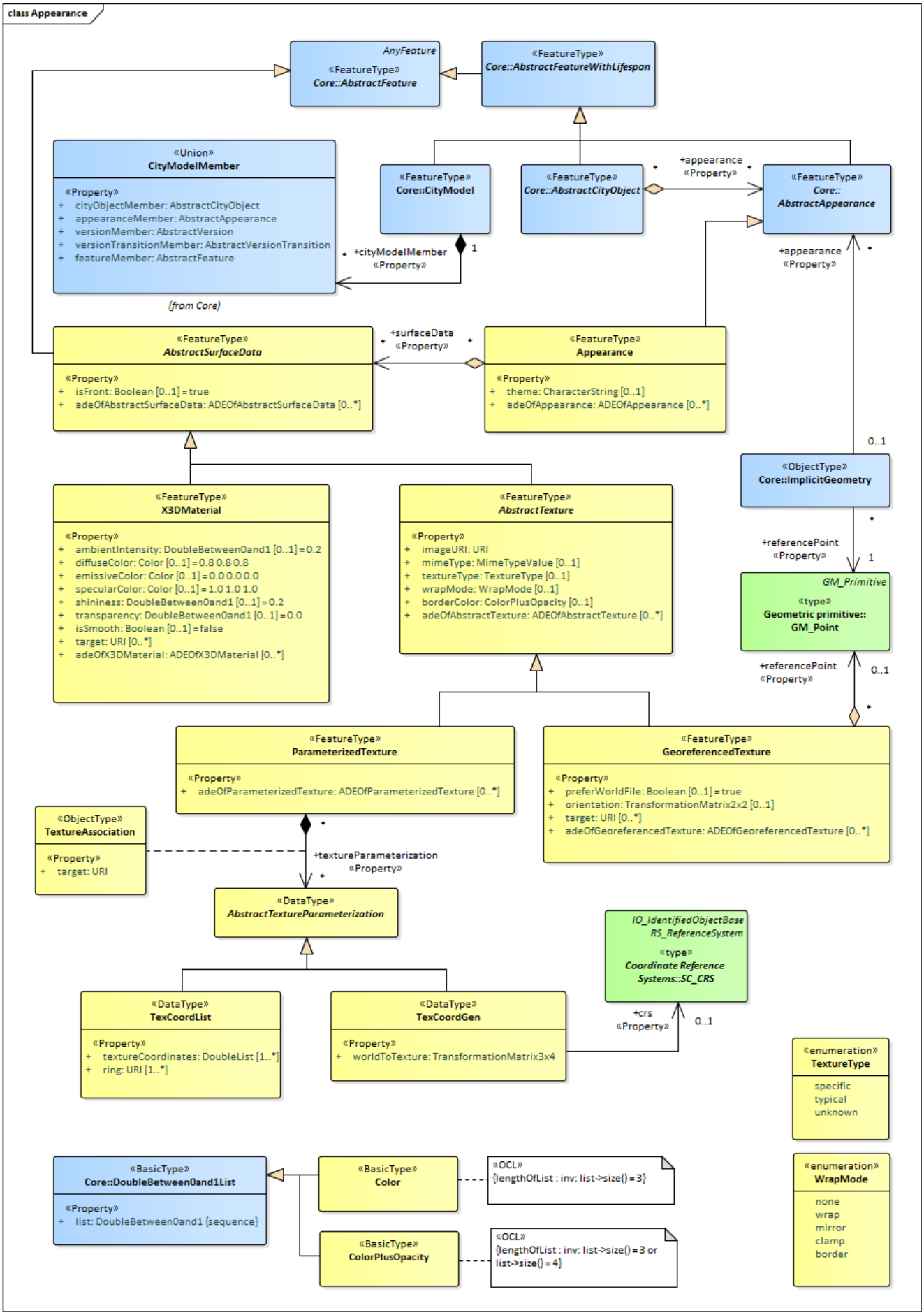



*Figure 27. UML diagram of CityGML's Appearance model.*

The ADE data types provided for the Appearance module are illustrated in Figure 28.

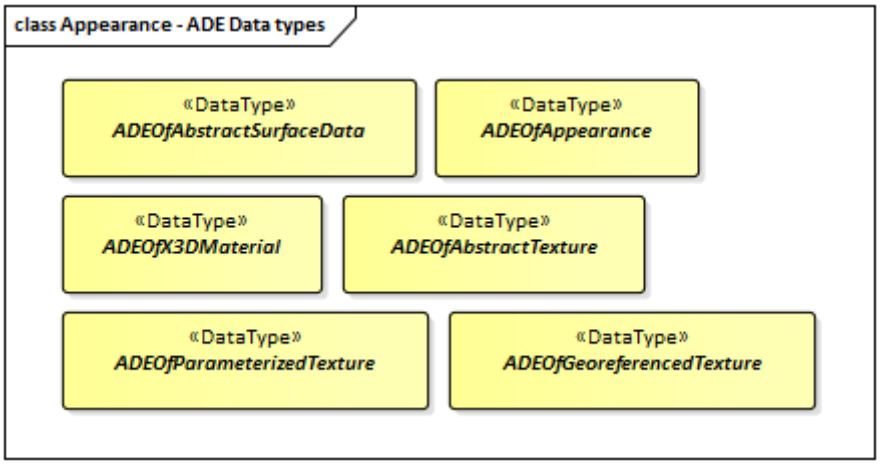



*Figure 28. ADE classes of the CityGML Appearance Module.*

### 8.2.8. Examples

# 8.3. Bridge Model

| Contributors |
| --- |
| Chuck Heazel - first draft |

### 8.3.1. Synopsis

The bridge model allows for the representation of the thematic, spatial and visual aspects of bridges and bridge parts in four levels of detail, LOD 1 – 3.

### 8.3.2. Key Concepts

### 8.3.3. Discussion

| NOTE | From CityGML 3.0 |
| --- | --- |

The Bridge module provides the representation of thematic and spatial aspects of bridges. Bridges are movable or unmovable structures that span intervening natural or built elements. In this way, bridges allow the passage of pedestrians, animals, vehicles, and service(s) above obstacles or between two points at a height above ground. Bridges are represented in the UML model by the top-level feature type *Bridge*, which is the main class of the Bridge module. Bridges can physically or functionally be subdivided into bridge parts. In addition, bridges can be decomposed into structural elements, such as pylons, anchorages, cables, slabs, and beams.

The free space inside bridges is represented by rooms, which allows a virtual accessibility of bridges. Bridges can contain installations and furniture. Installations are permanent parts of a bridge that strongly affect the outer or inner appearance of the bridge and that cannot be moved. Examples are stairways, signals, railings, and lamps. Furniture, in contrast, represent moveable objects of a bridge, like signs, art works, and benches. Bridges can be bounded by different types of surfaces. In this way, the outer structure of bridges can be differentiated semantically into wall surfaces, roof surfaces, ground surfaces, outer floor surfaces, and outer ceiling surfaces, whereas

the visible surface of rooms can be structured into interior wall surfaces, floor surfaces, and ceiling surfaces. Furthermore, the openings of bridges, i.e. windows and doors, can be represented including their corresponding surfaces.

| | |
|---|---|
| **NOTE** | The following text is from CityGML 2.0. It needs to be reviewed and updated. |
| **NOTE** | Many of these concepts have been moved into the Construction module. They should be captured once there. The following text should reference those sections then provide any details specific to bridges. |

The bridge model of CityGML is defined by the thematic extension module Bridge (cf. chapter 7). Figure 27 illustrates examples of bridge models in all LODs.

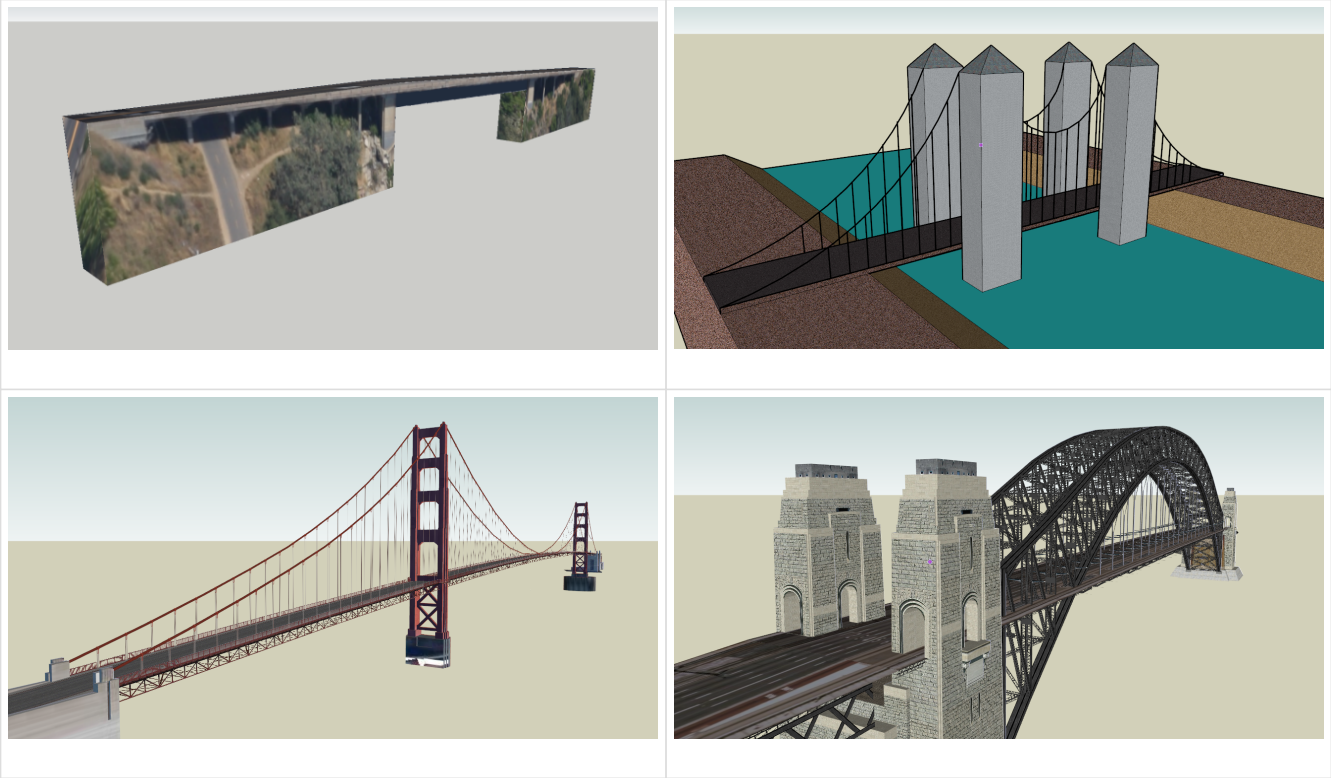


*Figure 29. Examples for bridge models in LOD1 (upper left), LOD2 (upper right), LOD3 (lower left) and LOD4 (lower right) (source: Google 3D warehouse).*

The bridge model was developed in analogy to the building model (cf. chapter 10.3) with regard to structure and attributes. The UML diagram of the bridge model is depicted in Figure 30.

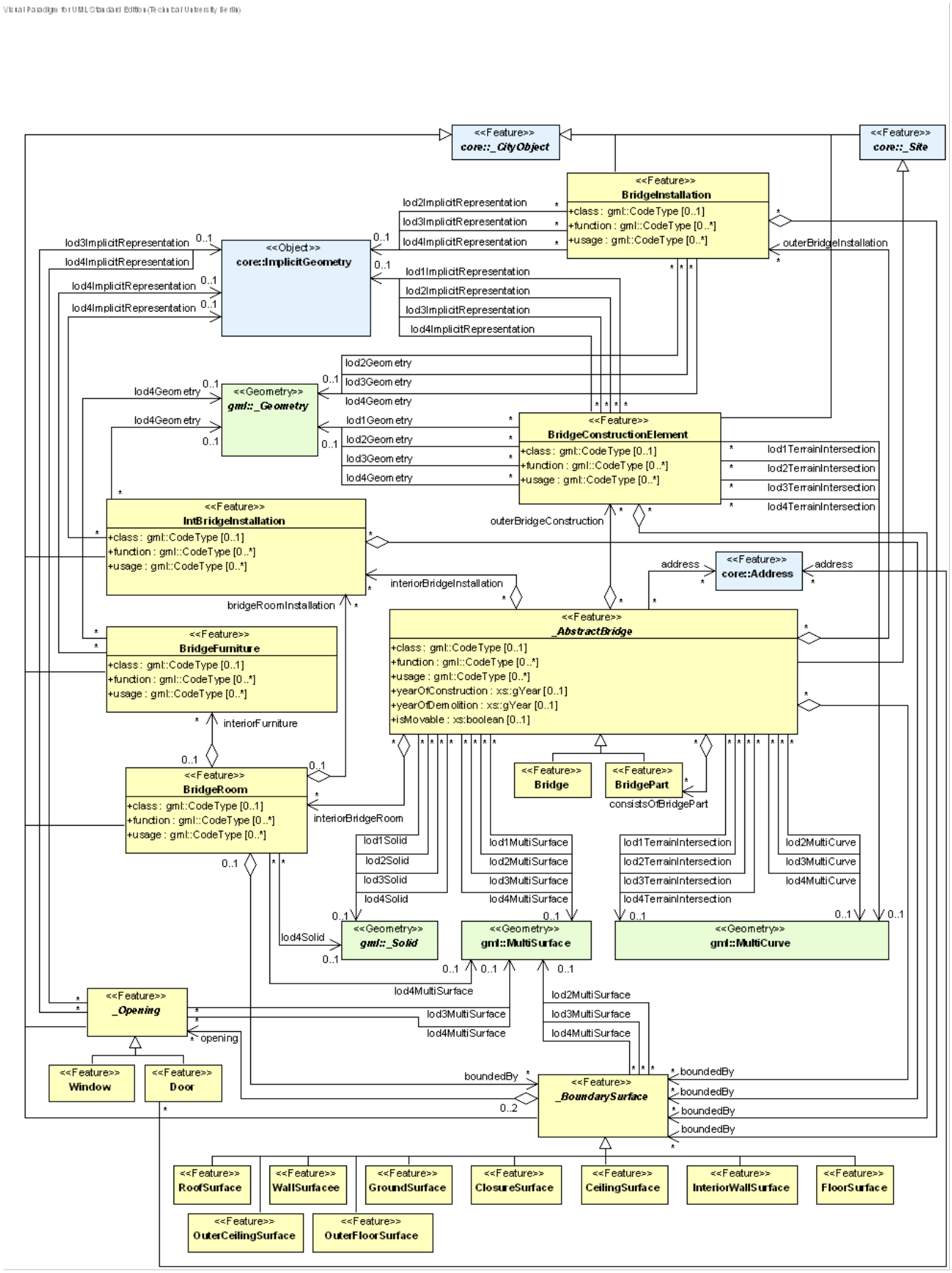



*Figure 30. UML diagram of the bridge model, part one*

A (movable or unmovable) bridge is represented by an object of the class Bridge. This class inherits its attributes and relations from the abstract base class _AbstractBridge. The spatial properties are defined by a solid for each of the four LODs (relations lod1Solid to lod4Solid). In analogy to the

building model, the semantical as well as the geometrical richness increases from LOD1 (blocks model) to LOD3 (architectural model). Simple examples of bridges in each of those LODs are depicted in Fig. 46. Interior structures like rooms are dedicated to LOD4. To cover the case of bridge models where the topology does not satisfy the properties of a solid (essentially water tightness), a multi surface representation is allowed (lod1MultiSurface to lod4MultiSurface). The line where the bridge touches the terrain surface is represented by a terrain intersection curve, which is provided for each LOD (relations lod1TerrainIntersection to lod4TerrainIntersection). In addition to the solid representation of a bridge, linear characteristics like ropes or antennas can be specified geometrically by the lod1MultiCurve to lod4MultiCurve relations. If those characteristics shall be represented semantically, the features BridgeInstalla-tion or BridgeConstructionElement can be used (see section 10.5.2). All relations to semantic objects and geo-metric properties are listed in Tab. 7.

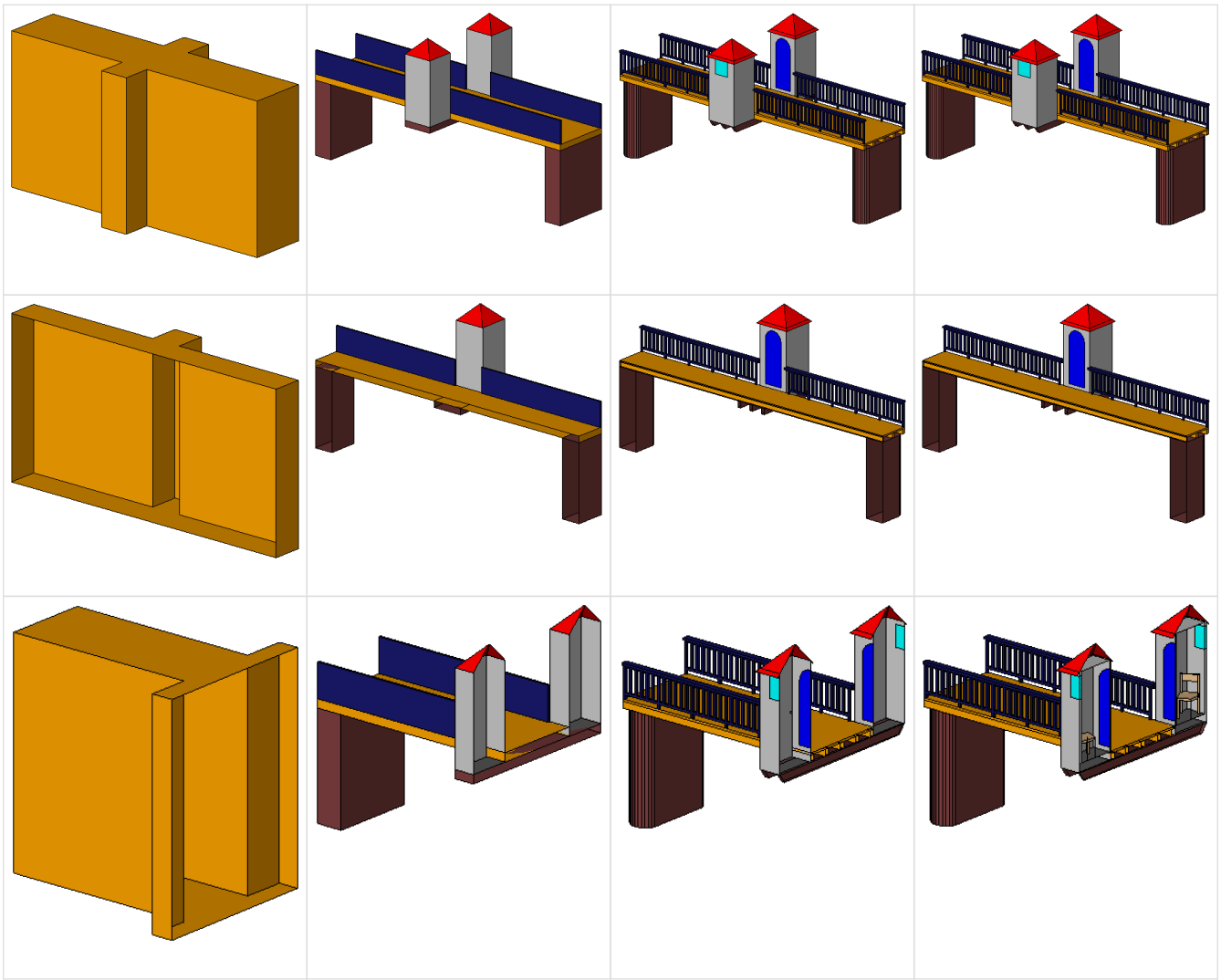


*Figure 31. Bridge model in LOD1 – LOD4. (source: Karlsruhe Institute of Technology (KIT)).*

The semantic attributes of an _AbstractBridge are class, function, usage and is_movable. The attribute class is used to classify bridges, e.g. to distinguish different construction types (cf. Fig. 48). The attribute function allows representing the utilization of the bridge independently of the construction. Possible values may be railway bridge, roadway bridge, pedestrian bridge, aqueduct, etc. The option to denote a usage which is divergent to one of the primary functions of the bridge (function) is given by the attribute usage. The type of these attributes is gml:CodeType, the values of

which can be defined in code lists. The name of the bridge can be represented by the gml:name attribute, which is inherited from the base class gml:_GML via the classes gml:_Feature, _CityObject, and _Site. Each Bridge or BridgePart feature may be assigned zero or more addresses using the address property. The corresponding AddressPropertyType is defined within the CityGML core module (cf. chapter 10.1.4).

*Table 5. Semantic themes of the class _AbstractBridge*

| Geometric / semantic theme | Property type | LOD1 | LOD2 | LOD3 | LOD4 |
|---|---|---|---|---|---|
| Volume part of the bridge shell | gml:SolidType | • | • | • | • |
| Surface part of the bridge shell | gml:MultiSurfaceType | • | • | • | • |
| Terrain intersection curve | gml:MultiCurveType | • | • | • | • |
| Curve part of the bridge shell | gml:MultiCurveType | | • | • | • |
| Bridge parts (chapter 10.5.1) | BridgePartType | • | • | • | • |
| Boundary surfaces (chapter 10.5.3) | AbstractBoundarySurfaceType | | • | • | • |
| Outer bridge installations (chapter 10.5.2) | BridgeInstallationType | | • | • | • |
| Bridge construction elements (chapter 10.5.2) | BridgeConstruction-ElementType | • | • | • | • |
| Openings (chapter 10.5.4) | AbstractOpeningType | | | • | • |
| Bridge rooms (chapter 10.5.5) | BridgeRoomType | | | | • |
| Interior bridge installations | IntBridgeInstallationType | | | | • |

The boolean attribute is_movable is defined to specify whether a bridge is movable or not. The modeling of the geometric aspects of the movement is delayed to later versions of this standard. Some types of movable bridges are depicted in Fig. 47.

**NOTE** the following are annimated GIFs. Currently these do not render for PDF.

[Figure 47 1] | *figures/inwork/Figure_47_1.gif*

[[align="center"] | *figures/inwork/Figure_47_2.gif*

[[align="center"] | *figures/inwork/Figure_47_3.gif*

[[align="center"] | *figures/inwork/Figure_47_4.gif*

[[align="center"] | *figures/inwork/Figure_47_5.gif*

*Figure 32. Examples for movable bridges (source: ISO 6707).*

### 8.3.4. Bridge and bridge part

**Bridge**

**BridgePart**

If some parts of a bridge differ from the remaining bridge with regard to attribute values or if parts like ramps can be identified as objects of their own, those parts can be represented as BridgePart. A bridge can consist of multiple BridgeParts. Like Bridge, BridgePart is a subclass of _AbstractBridge and hence, has the same attrib-utes and relations. The relation consistOfBridgePart represents the aggregation hierarchy between a Bridge (or a BridgePart) and it's BridgeParts. By this means, an aggregation hierarchy of arbitrary depth can be modeled. Each BridgePart belongs to exactly one Bridge (or BridgePart). Similar to the building model, the aggregation structure of a bridge forms a tree. A simple example for a bridge with parts is a twin bridge. Another example is presented in chapter 10.5.6.

**AbstractBridge**

The abstract class _AbstractBridge is the base class of Bridges and BridgeParts. It contains properties for bridge attributes, purely geometric representations, and geometric/semantic representations of the bridge or bridge part in different levels of detail. The attributes describe:

1. The classification of the bridge or bridge part (class), the different intended usages (function), and the different actual usages (usage). The permitted values for these property types can be specified in code lists.
2. The year of construction (yearOfConstruction) and the year of demolition (yearOfDemolition) of the bridge or bridge part. These attributes can be used to describe the chronology of the bridge development within a city model. The points of time refer to real world time.
3. Whether the bridge is movable is specified by the Boolean attribute isMovable.

### 8.3.5. Bridge construction elements and bridge installations

**BridgeConstructionElement**

**BridgeInstallation**

Bridge elements which do not have the size, significance or meaning of a BridgePart can be modelled either as BridgeConstructionElement or as BridgeInstallation. Elements which are essential from a structural point of view are modelled as BridgeConstructionElement, for example structural elements like pylons, anchorages etc. (cf. Fig. 49). A general classification as well as the intended and actual function of the construction element are represented by the attributes class,

function, and usage. The geometry of a BridgeConstructionElement, which may be present in LOD1 to LOD4, is gml:_Geometry. Alternatively, the geometry may be given as ImplicitGe-ometry object. Following the concept of ImplicitGeometry the geometry of a prototype bridge construction element is stored only once in a local coordinate system and referenced by other bridge construction element features (cf. chapter 8.2). The visible surfaces of a bridge construction element can be semantically classified using the concept of boundary surfaces (cf. chapter 10.5.3).

Whereas a BridgeConstructionElement has structural relevance, a BridgeInstallation represents an element of the bridge which can be eliminated without collapsing of the bridge (e.g. stairway, antenna, railing). BridgeInstalla-tions occur in LOD 2 to 4 only and are geometrically representated as gml:_Geometry. Again, the concept of ImplicitGeometry can be applied to BridgeInstallations alternatively, and their visible surfaces can be semanti-cally classified using the concept of boundary surfaces (cf. chapter 10.5.3). The class BridgeInstallation contains the semantic attributes class, function and usage. The attribute class gives a classification of installations of a bridge. With the attributes function and usage, nominal and real functions of the bridge installation can be described. The type of all attributes is gml:CodeType and their values can be defined in code lists.

| | |
|---|---|
| **NOTE** | *BridgeConstructionElements of a suspension bridge.*<br>insert Fig 49 - currently does not render |

## 8.3.6. Boundary surfaces

**BoundarySurface (AbstractConstructionSurface?)**

The thematic boundary surfaces of a bridge are defined in analogy to the building module. _BoundarySurface is the abstract base class for several thematic classes, structuring the exterior shell of a bridge as well as the visible surfaces of rooms, bridge construction elements and both outer and interior bridge installations. It is a subclass of _CityObject and thus inherits all properties like the GML3 standard feature properties (gml:name etc.) and the CityGML specific properties like ExternalReferences. From _BoundarySurface, the thematic classes RoofSur-face, WallSurface, GroundSurface, OuterCeilingSurface, OuterFloorSurface, ClosureSurface, FloorSurface, InteriorWallSurface, and CeilingSurface are derived.

For each LOD between 2 and 4, the geometry of a _BoundarySurface may be defined by a different gml:MultiSurface geometry.

In LOD3 and LOD4, a _BoundarySurface may contain _Openings (cf. chapter 10.5.4) like doors and windows. If the geometric location of _Openings topologically lies within a surface component (e.g. gml:Polygon) of the gml:MultiSurface geometry, these _Openings must be represented as holes within that surface. A hole is repre-sented by an interior ring within the corresponding surface geometry object. According to GML3, the points have to be specified in reverse order (exterior boundaries counter-clockwise and interior boundaries clockwise when looking in opposite direction of the surface's normal vector). If such an opening is sealed by a Door, a Window, or a ClosureSurface, their outer boundary may consist of the same points as the inner ring (denoting the hole) of the surrounding surface. The embrasure surfaces of an Opening belong to the relevant adjacent _BoundarySurface. If, for example a door seals the Opening, the embrasure surface on the one side of the door belongs to the InteriorWallSurface and on the other side to the WallSurface.

Fig. 50 depicts a bridge with RoofSurfaces, WallSurfaces, OuterFloorSurfaces and OuterCeilingSurfaces. Besides Bridges and BridgeParts, BridgeConstructionElements, BridgeInstallations as well as IntBridgeInstalla-tions can be related to _BoundarySurface. _BoundarySurfaces occur in LOD2 to LOD4. In LOD3 and LOD4, such a surface may contain _Openings (see chapter 10.3.4) like doors and windows.

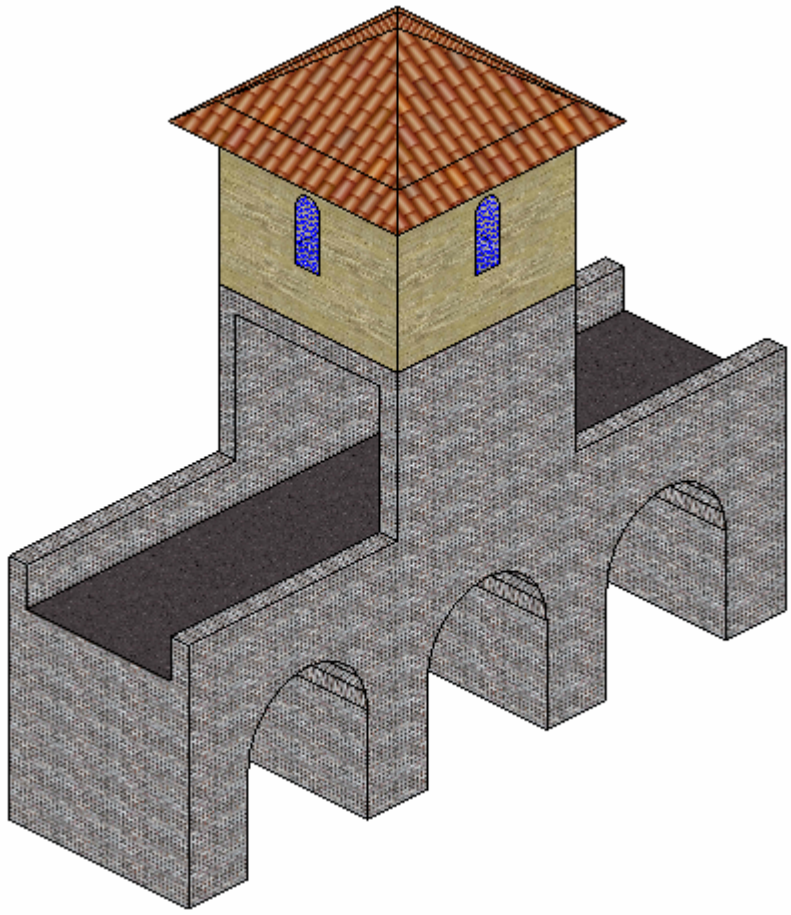


*Figure 33. Different BoundarySurfaces of a bridge.*

| | |
|---|---|
| **NOTE** | need to add annotations to Figure 50 |

| | |
|---|---|
| **NOTE** | Much of the following has been moved to Construction. These sections should only discuss bridge-specific properties of the Construction concepts. |

**GroundSurface**

The ground plate of a bridge or bridge part is modelled by the class GroundSurface. The polygon defining the ground plate is congruent with the bridge's footprint. However, the surface normal of the ground plate is point-ing downwards.

**OuterCeilingSurface**

A mostly horizontal surface belonging to the outer bridge shell and having the orientation pointing downwards can be modeled as an OuterCeilingSurface.

**WallSurface**

All parts of the bridge facade belonging to the outer bridge shell can be modelled by the class WallSurface

**OuterFloorSurface**

A mostly horizontal surface belonging to the outer bridge shell and with the orientation pointing upwards can be modeled as an OuterFloorSurface

**RoofSurface**

The major roof parts of a bridge or bridge part are expressed by the class RoofSurface.

**ClosureSurface**

An opening in a bridge not filled by a door or window can be sealed by a virtual surface called ClosureSurface (cf. chapter 6.4). Hence, bridge with open sides can be virtually closed in order to be able to compute their volume. ClosureSurfaces are also used in the interior bridge model. If two rooms with a different are directly connected without a separating door, a ClosureSurface should be used to separate or connect the volumes of both rooms.

**FloorSurface**

The class FloorSurface must only be used in the LOD4 interior bridge model for modelling the floor of a bridge room.

**InteriorWallSurface**

The class InteriorWallSurface must only be used in the LOD4 interior bridge model for modelling the visible surfaces of the bridge room walls.

**CeilingSurface**

The class CeilingSurface must only be used in the LOD4 interior bridge model for modelling the ceiling of a bridge room.

### 8.3.7. Openings

**Opening**

The class _Opening is the abstract base class for semantically describing openings like doors or windows in outer or inner boundary surfaces like walls and roofs. Openings only exist in models of LOD3 or LOD4. Each _Opening is associated with a gml:MultiSurface geometry. Alternatively, the geometry may be given as ImplicitGeometry object. Following the concept of ImplicitGeometry the geometry of a prototype opening is stored only once in a local coordinate system and referenced by other opening features (see chapter 8.2).

**Window**

The class Window is used for modelling windows in the exterior shell of a bridge, or hatches between adjacent rooms. The formal difference between the classes Window and Door is that – in normal cases – Windows are not specifically intended for the transit of people or vehicles.

**Door**

The class Door is used for modelling doors in the exterior shell of a bridge, or between adjacent rooms. Doors can be used by people to enter or leave a bridge or room. In contrast to a ClosureSurface a door may be closed, blocking the transit of people. A Door may be assigned zero or more addresses. The corresponding Address-PropertyType is defined within the CityGML core module (cf. chapter 10.1.4).

### 8.3.8. Bridge Interior

The classes BridgeRoom, IntBridgeInstallation and BridgeFurniture allow for the representation of the bridge interior. They are designed in analogy to the classes Room, IntBuildingInstallation and BuildingFurniture of the building module and share the same meaning. The bridge interior can only be modeled in LOD4.

**BridgeRoom**

A BridgeRoom is a semantic object for modelling the free space inside a bridge and should be uniquely related to exactly one bridge or bridge part object. It should be closed (if necessary by using ClosureSurfaces) and the geometry normally will be described by a solid (lod4Solid). However, if the topological correctness of the boundary cannot be guaranteed, the geometry can alternatively be given as a MultiSurface (lod4MultiSurface). The surface normals of the outer shell of a GML solid must point outwards. This is important to consider when BridgeRoom surfaces should be assigned Appearances. In this case, textures and colors must be placed on the backside of the corresponding surfaces in order to be visible from the inside of the room.

In addition to the geometrical representation, different parts of the visible surface of a room can be modelled by specialised BoundarySurfaces (FloorSurface, CeilingSurface, InteriorWallSurface, and ClosureSurface; cf. chapter 10.5.3).

**BridgeFurniture**

BridgeRooms may have BridgeFurnitures and IntBridgeInstallations. A BridgeFurniture is a movable part of a room, such as a chair or furniture. A BridgeFurniture object should be uniquely related to exactly one room object. Its geometry may be represented by an explicit geometry or an ImplicitGeometry object. Following the concept of ImplicitGeometry the geometry of a prototype bridge furniture is stored only once in a local coordi-nate system and referenced by other bridge furniture features (see chapter 8.2).

**IntBridgeInstallation**

An IntBridgeInstallation is an object inside a bridge with a specialised function or semantic meaning. In contrast to BridgeFurniture, IntBridgeInstallations are permanently attached to the bridge structure and cannot be moved. Examples for IntBridgeInstallations are stairways, railings

and heaters. Objects of the class IntBridgeInstallation can either be associated with a room (class BridgeRoom), or with the complete bridge / bridge part (class _AbstractBridge, cf. chapter 10.5.1). However, they should be uniquely related to exactly one room or one bridge / bridge part object. An IntBridgeInstallation optionally has attributes class, function and usage. The attribute class, which can only occur once, represents a general classification of the internal bridge component. With the attributes function and usage, nominal and real functions of a bridge installation can be described. For all three attributes the list of feasible values can be specified in a code list. For the geometrical representation of an IntBridgeInstallation, an arbitrary geometry object from the GML subset shown in Fig. 9 can be used. Alter-natively, the geometry may be given as ImplicitGeometry object. Following the concept of ImplicitGeometry the geometry of a prototype interior bridge installation is stored only once in a local coordinate system and refer-enced by other interior bridge installation features (see chapter 8.2). The visible surfaces of an interior bridge installation can be semantically classified using the concept of boundary surfaces (cf. 10.5.3).

## 8.3.9. UML Model

The UML diagram of the Bridge module is depicted in Figure 34. The Bridge module inherits concepts from the Construction module (cf. [ug_construction_section]). The Construction module defines objects that are common to all types of construction, such as the different surface types and the openings.

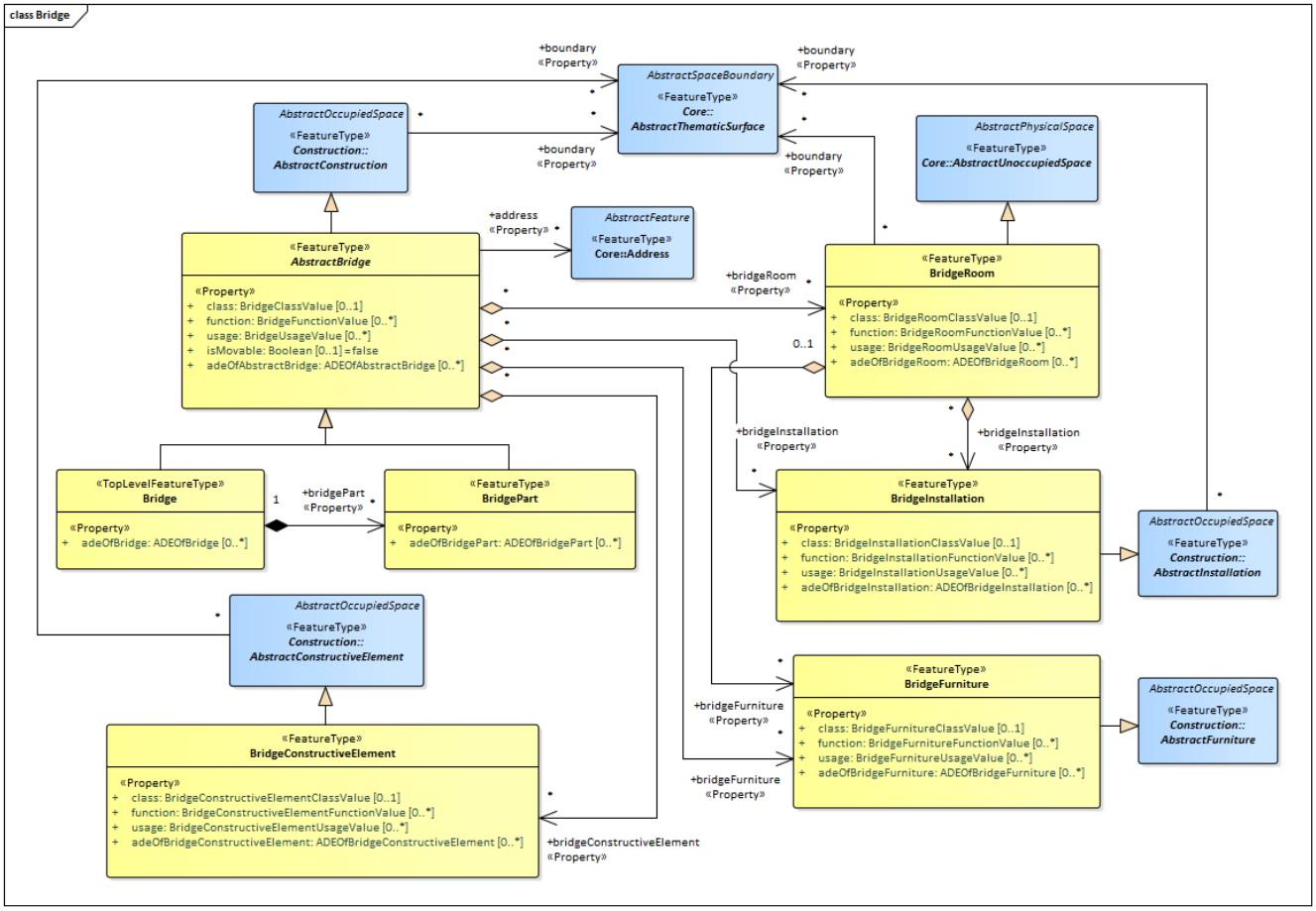



*Figure 34. UML diagram of the Bridge Model.*

The ADE data types provided for the Bridge module are illustrated in Figure 35.

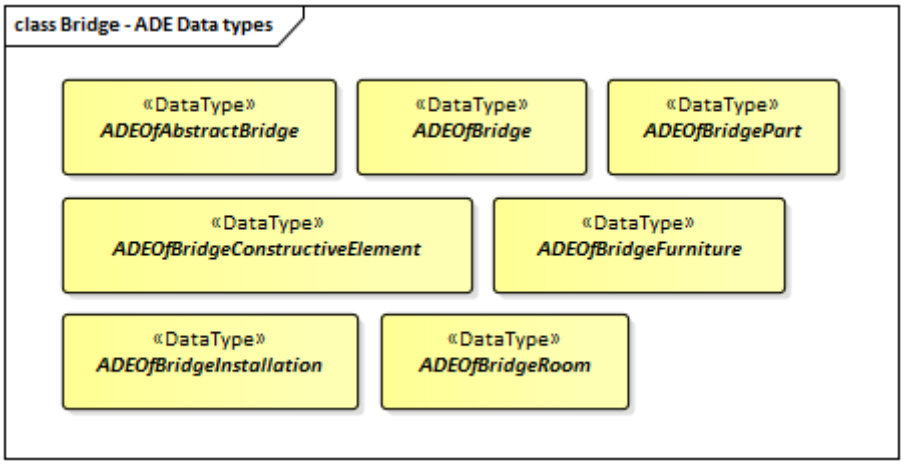



*Figure 35. ADE classes of the CityGML Bridge module.*

The Code Lists provided for the Bridge module are illustrated in Figure 36.

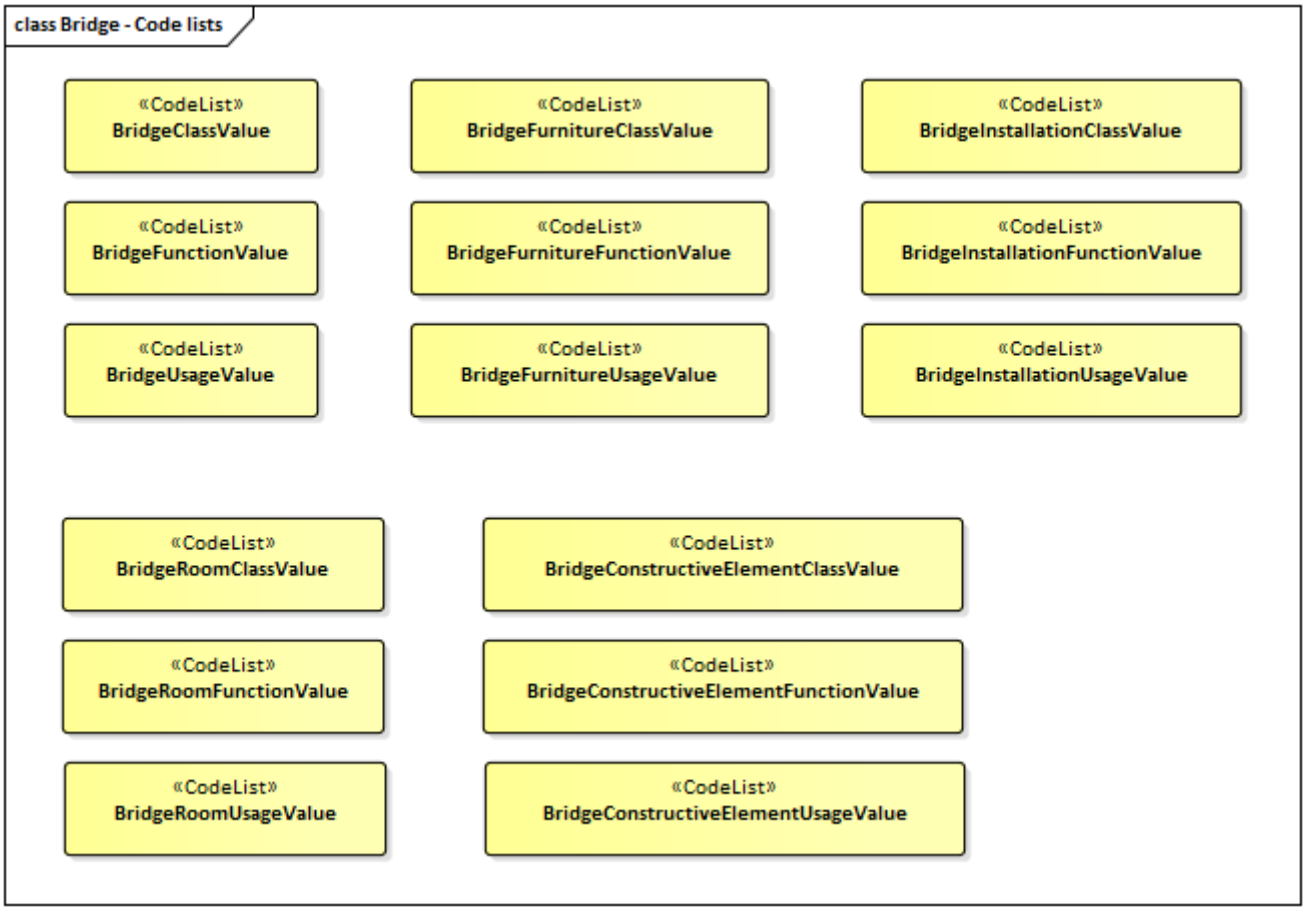



*Figure 36. Codelists from the CityGML Bridge module.*

### 8.3.10. Examples

The bridge of Rees crossing the Rhine in Germany has three bridge parts which are separated by pylons. Fig. 51 (left) depicts the Rees bridge model containing one Bridge feature which consists of three BridgePart features. The pylons, which are structurally essential, are represented by BridgeConstructionElements. On the top of the pylons, four lamps are located which are modeled as BridgeInstallation features (cf. right part of Fig. 51).

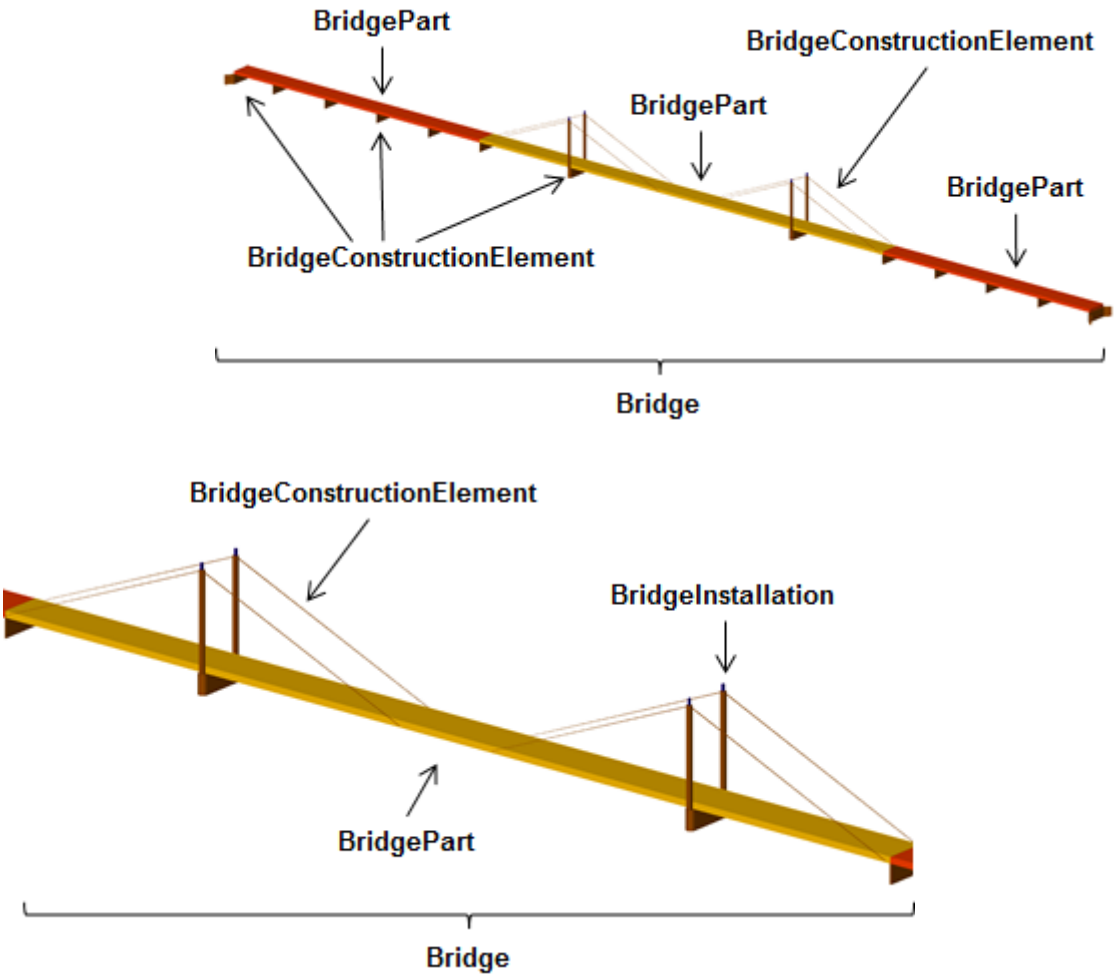



*Figure 37. The bridge of Rees, consisting of a Bridge feature and three BridgePart features (left). The bridge contains BridgeConstructionElement and BridgeInstallation features (right).*

In the following Fig. 52, the main part of the bridge of Rees is shown as photograph on the left side (source: Harald Halfpapp), and the corresponding part of the LOD2 bridge model is depicted on the the right side (source: District of Recklinghausen / KIT).

**NOTE** Figures 52, 53 and 54 are from the images folder.

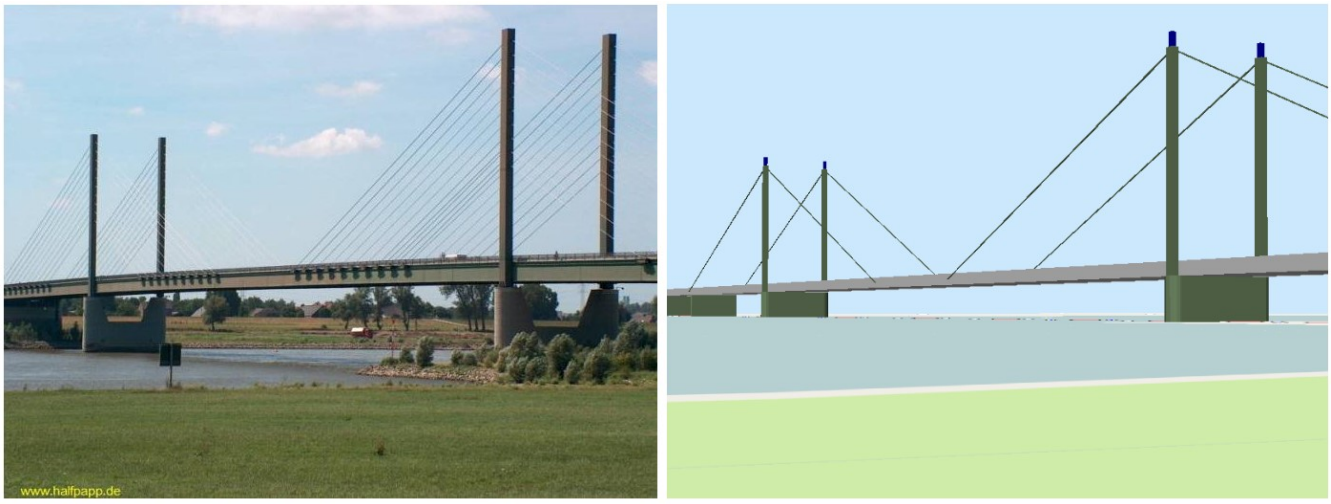



*Figure 38. The bridge of Rees (left photo (source: Harald Halfpapp); right LOD2 model (source: District of Recklinghausen / KIT)).*

There are two bridges crossing the river Rhine at Karlsruhe, Germany. The first one is a two track railway bridge constructed as a truss bridge (cf. Fig. 53 front). The second one is a four lane highway bridge constructed as a cable-stayed bridge (cf. Fig. 53 background).

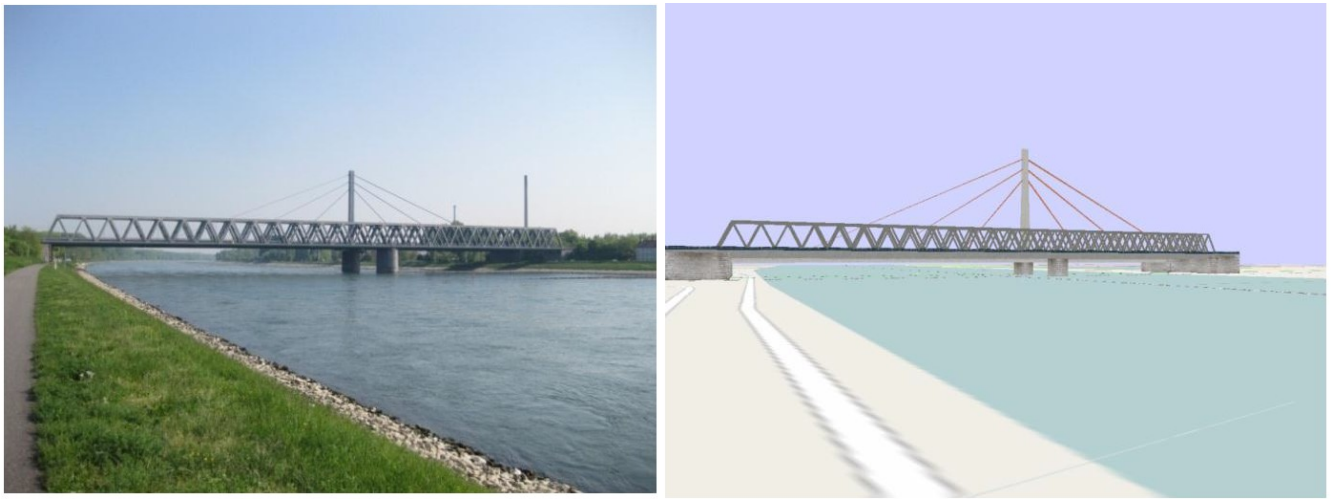


*Figure 39. Bridge over the river Rhine at Karlsruhe (left a photo, right the 3D CityGML model) (source: Karlsruhe Institute of Technology (KIT), courtesy of City of Karlsruhe).*

In CityGML both bridges are modeled as single Bridge object with BridgeConstructionElements and BridgeIn-stallations. The construction elements of the cable stayed bridge are the footings on both river sides and in the middle of the river, as well as the cables and the pylon. The construction elements of the truss bridge are the footings and the truss itself. Both bridges have several railings which are modeled as BridgeInstallation. The bridge “Oberbaumbrücke” shown in Fig. 54 is located in the centre of Berlin crossing the river Spree and serves as example for bridges having interior rooms. The real-world bridge is depicted in the left part of Fig. 54, whereas the corresponding CityGML model is shown on the right. The outer geometry of the bridge is modeled as gml:MultiSurface element (lod4MultiSurface property) and is assigned photorealistic textures. Additionally, the interior rooms located in both bridge towers are represented as BridgeRoom objects with solid geometries (gml:Solid assigned through the lod4Solid property). Due to its geometric accuracy and the representation of the interior structures of both bridge towers, the model is classified as LOD4.

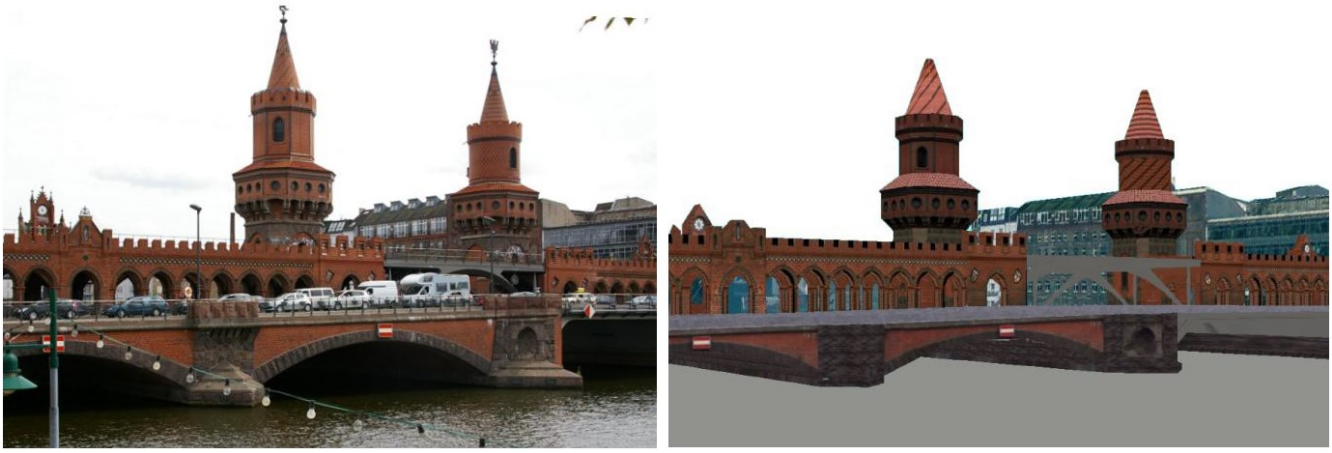


*Figure 40. The bridge“Oberbaumbrücke” in Berlin represented as bridge model in LOD4 (left a photo, right the 3D CityGML model) (source: Berlin Senate of Business, Technology and Women; Business Location Center, Berlin; Technische Universität Berlin; Karlsruhe Institute of Technology (KIT)).*

# 8.4. Building Model

| Contributors |
| --- |
| C. Heazel - first draft - working on better integration with Construction. |

## 8.4.1. Synopsis

The building model is one of the most detailed thematic concepts of CityGML. It allows for the representation of thematic and spatial aspects of buildings and building parts in four levels of detail, LOD0 to LOD3. Figure 29 provides examples of 3D city and building models in LOD1 – 3.

**NOTE** Figure needs to be updated

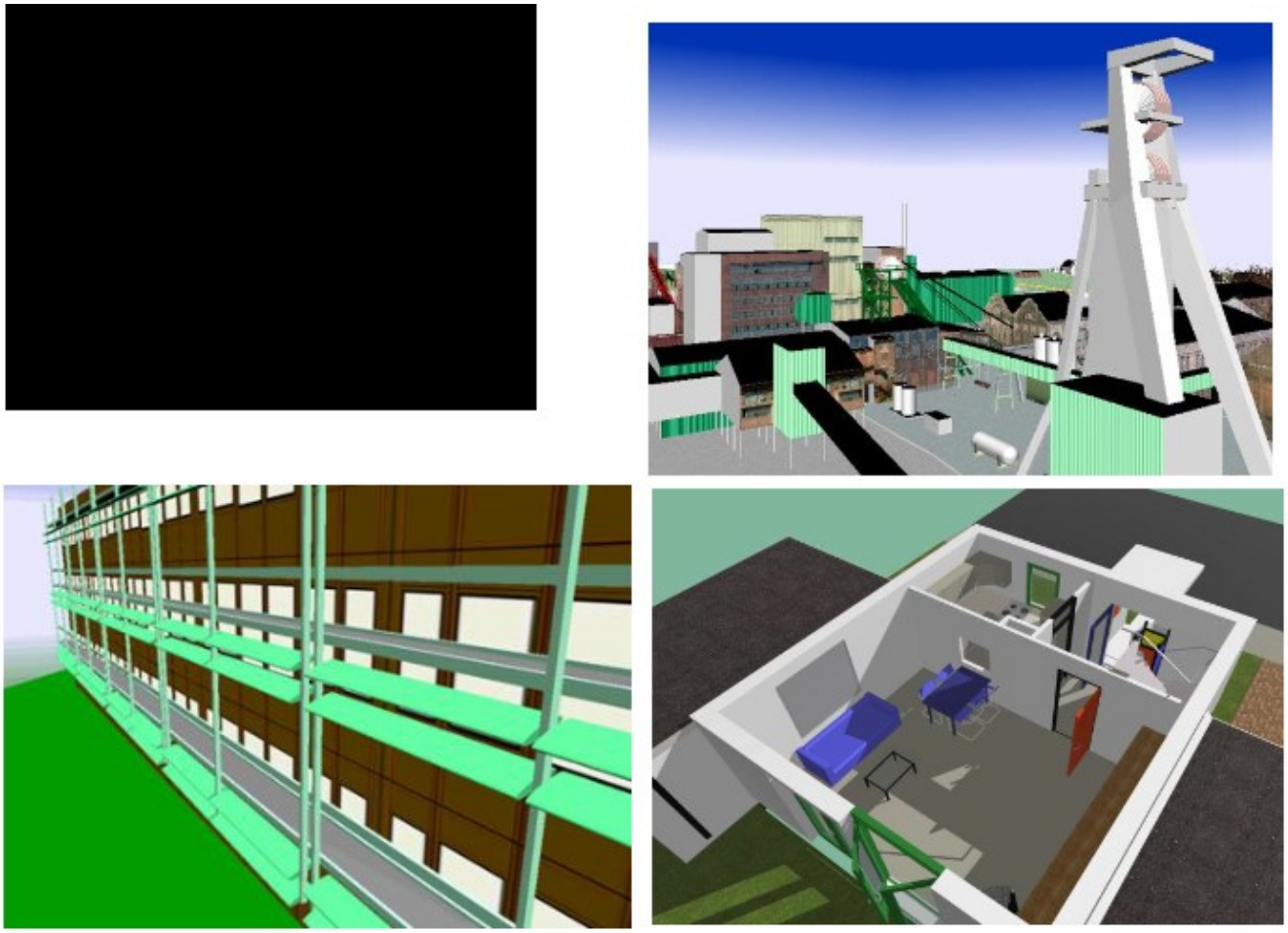


*Figure 41. Examples for city or building models in LOD1 (upper left), LOD2 (upper right), and LOD3 (lower left). (source: District of Recklinghausen, m-g-h ingenieure+architekten GmbH).*

## 8.4.2. Key Concepts

**AbstractBuilding**: An abstract superclass representing the common attributes and associations of the classes Building and BuildingPart.
A type of AbstractConstruction.

**Building**: (Discussion) A free-standing, self-supporting construction that is roofed, usually walled, and can be entered by humans and is normally designed to stand permanently in one place. It is

intended for human occupancy (e.g. a place of work or recreation), habitation and/or shelter of humans, animals or things.
A type of AbstractBuilding.

**BuildingPart**: (Discussion) A physical or functional subdivision of a Building. It would be considered a Building, if it were not part of a collection of other BuildingParts.
A type of AbstractBuilding.

**AbstractBuildingSubdivision**: (Discussion) The abstract superclass for different kinds of logical building subdivisions.
A type of AbstractLogicalSpace.

**BuildingUnit**: (Discussion) A logical subdivision of a Building. BuildingUnits are formed according to some homogeneous property like function, ownership, management, or accessability. They may be separately sold, rented out, inherited, managed, etc.
A type of AbstractBuildingSubdivision.

**Storey**: (Discussion) A horizontal section of a Building. Storeys are not always defined according to the building structure, but can also be defined according to logical considerations.
A type of AbstractBuildingSubdivision.

**BuildingRoom**: (Discussion) A space within a Building or BuildingPart intended for human occupancy (e.g. a place of work or recreation) and/or containment of animals or things. A BuildingRoom is bounded physically and/or virtually (e.g. by ClosureSurfaces or GenericSurfaces).
A type of AbstractUnoccupiedSpace.

**BuildingInstallation**: (Discussion) A permanent part of a Building (inside and/or outside) which has not the significance of a BuildingPart. Examples are stairs, antennas, balconies or small roofs.
A type of AbstractInstallation.

**BuildingFurniture**: (Discussion) An equipment for occupant use, usually not fixed to the building.
A type of AbstractFurniture.

**BuildingConstructiveElement**: (Discussion) An element of a Building which is essential from a structural point of view. Examples are walls, slabs, staircases, beams.
A type of AbstractConstructiveElement.

### 8.4.3. Discussion

| NOTE | From CityGML 3.0 |
|---|---|

The Building module provides the representation of thematic and spatial aspects of buildings. Buildings are free-standing, self-supporting constructions that are roofed and usually walled, and that can be entered by humans and are normally designed to stand permanently in one place. Buildings are intended for human occupancy (e.g. a place of work or recreation), habitation and/or shelter of humans, animals or things. Buildings are represented in the UML model by the top-level feature type *Building*, which is the main class of the Building module. Buildings can physically or functionally be subdivided into building parts, and logically into storeys and building units (e.g. apartments). In addition, buildings can be decomposed into structural elements, such as walls,

slabs, staircases, and beams.

The interior of buildings is represented by rooms. This allows a virtual accessibility of buildings such as for visitor information in a museum (“Location Based Services“), the examination of accommodation standards or the presentation of daylight illumination of a building.

Buildings can contain installations and furniture. Installations are permanent parts of a building that strongly affect the outer or inner appearance of the building and that cannot be moved. Examples are balconies, chimneys, dormers or stairs. In contrast, furniture, represents moveable objects inside a building, like tables and chairs.

Buildings can be bounded by different types of surfaces. In this way, the outer façade of buildings can be differentiated semantically into wall surfaces, roof surfaces, ground surfaces, outer floor surfaces, and outer ceiling surfaces, whereas the visible surface of rooms can be structured into interior wall surfaces, floor surfaces, and ceiling surfaces. Furthermore, the openings of buildings, i.e. windows and doors, can be represented including their corresponding surfaces.

| | |
|---|---|
| **NOTE** | The following content is from CityGML 2.0 with some updates to CityGML 3.0. Some of this content probably belongs in the Construction section now (windows, doord, etc.). There are also some changes from 2.0 to 3.0 (ex. Boundary Surfaces) which I am not clear on. |

The pivotal class of the model is AbstractBuilding, which is a subclass of AbstractConstruction and therefore a descendent of AbstractCityObject. AbstractBuilding is specialised either to a Building or to a BuildingPart. Since an AbstractBuilding consists of BuildingParts, which again are AbstractBuildings, an aggregation hierarchy of arbitrary depth may be realised.

As a descendent of the AbstractCityObject class, an AbstractBuilding inherits all City Objectproperties including:

- The standard Feature properties (`name` etc.),
- The City Object specific properties like `externalReference`,
- Spatial properties:
  - lod[1,2,3]Solid,
  - lod0Point,
  - lod[0,2,3]MultiSurface,
  - lod[0,2,3]MultCurve,
  - lod[1,2,3]TerrainIntersectionCurve,
  - lod[1,2,3]ImplicitRepresenation

Further properties not explicitly covered by _AbstractBuilding may be modelled as generic attributes provided by the CityGML Generics module or using the CityGML Application Domain Extension mechanism.

Building complexes, which consist of a number of distinct buildings like a factory site or hospital complex, should be aggregated using the concept of CityObjectGroups. The main building of the

complex can be denoted by providing “main building” as the role name of the corresponding group member.

Both classes Building and BuildingPart inherit the attributes of AbstractBuilding:

- the `class` of the building,
- the `function` (e.g. residential, public, or industry),
- the `usage`,
- the `year` of construction,
- the `year` of demolition,
- the `roof` type,
- the `measured height`, and
- the number and individual heights of the `storeys` above and below ground.

This set of parameters is suited for roughly reconstructing the three-dimensional shape of a building and can be provided by cadastral systems. Furthermore, Address features can be assigned to Buildings or BuildingParts.

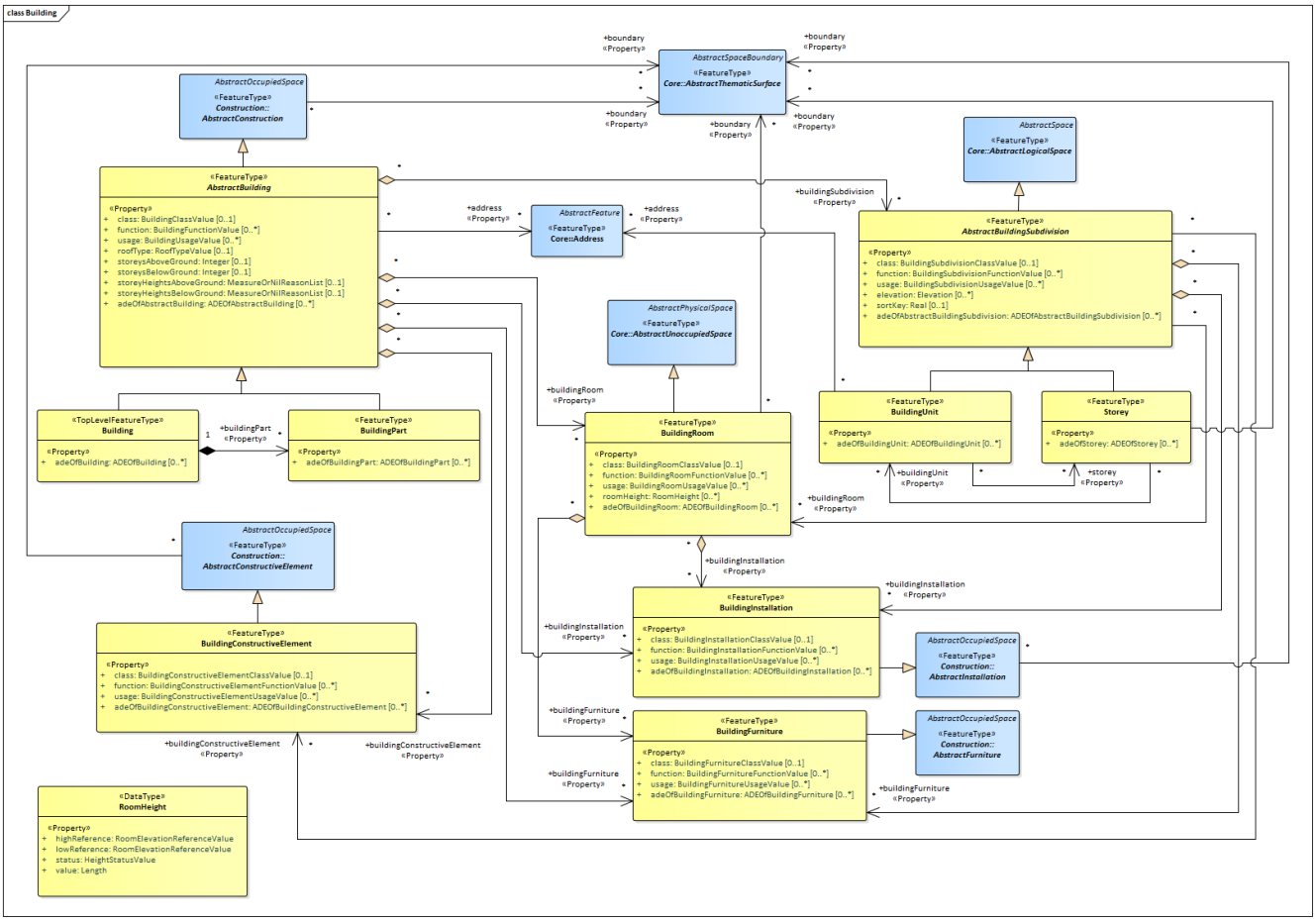



*Figure 42. UML diagram of CityGML’s building model.*

The geometric representation and semantic structure of an AbstractBuilding is shown in [Figure-27]. The model is successively refined from LOD0 to LOD3. Therefore, not all components of a building model are represented equally in each LOD and not all aggregation levels are allowed in each LOD. In CityGML, all object classes are associated to the LODs with respect to the proposed

minimum acquisition criteria for each LOD. An object can be represented simultaneously in different LODs by providing distinct geometries for the corresponding LODs.

In LOD0, the building can be represented by horizontal, 3-dimensional surfaces. These can represent the foot-print of the building and, separately, the roof edge. This allows the easy integration of 2D data into the model. In many countries these 2D geometries readily exist, for example in cadastral or topographic data holdings. Cadastre data typically depicts the shape of the building on the ground (footprints) and topographic data is often a mixture between footprints and geometries at roof level (roof edges), which are often photogrametrically extracted from area/satellite images or derived from airborne laser data. The building model allows the inclusion of both. In this case large overhanging roofs can be modelled as a preliminary stage to more detailed LOD2 and LOD3 depictions. The surface geometries require 3D coordinates, though it is mandated that the height values of all vertices belonging to the same surface are identical. If 2D geometries are imported into any of these two LOD0 geometries, an appropriate height value for all vertices needs to be chosen. The footprint is typically located at the lowest elevation of the ground surface of the building whereas the roof edge representation should be placed at roof level (e.g., eaves height).

In LOD1, a building model consists of a generalized geometric representation of the outer shell. Optionally, a GM_MultiCurve representing the TerrainIntersectionCurve can be specified. This geometric representation is refined in LOD2 by additional GM_MultiSurface and GM_MultiCurve geometries, used for modelling architectural details like roof overhangs, columns, or antennas. In LOD2 and higher LODs the outer facade of a building can also be differentiated semantically by the classes BuildingConstructiveElement and BuildingInstallation. A BuildingConstructiveElement is a part of the building's exterior shell with a special function like wall (WallSur-face), roof (RoofSurface), ground plate (GroundSurface), outer floor (OuterFloorSurface), outer ceiling (Outer-CeilingSurface) or ClosureSurface. The BuildingInstallation class is used for building elements like balconies, chimneys, dormers or outer stairs, strongly affecting the outer appearance of a building. A BuildingInstallation may have the attributes `class`, `function`, and `usage` (cf. Figure 42).

In LOD3, the openings in BuildingConstructiveElement objects (doors and windows) can be represented as thematic objects. In LOD4, the highest level of resolution, also the interior of a building, composed of several rooms, is represented in the building model by the class Room. This enlargement allows a virtual accessibility of buildings, e.g. for visitor information in a museum ("Location Based Services"), the examination of accommodation standards or the presentation of daylight illumination of a building. The aggregation of rooms according to arbitrary, user defined criteria (e.g. for defining the rooms corresponding to a certain storey) is achieved by employing the general grouping concept provided by CityGML (cf. chapter 10.3.6). Interior installations of a building, i.e. objects within a building which (in contrast to furniture) cannot be moved, are represented by the class IntBuildingInstallation. If an installation is attached to a specific room (e.g. radiators or lamps), they are associated with the Room class, otherwise (e.g. in case of rafters or pipes) with _AbstractBuilding. A Room may have the attributes class, function and usage whose value can be defined in code lists (chapter 10.3.8 and annex C.1). The class attribute allows a classification of rooms with respect to the stated function, e.g. commercial or private rooms, and occurs only once. The function attribute is intended to express the main purpose of the room, e.g. living room, kitchen. The attribute usage can be used if the way the object is actually used differs from the function. Both attributes can occur multiple times.

The visible surface of a room is represented geometrically as a Solid or MultiSurface. Semantically,

the surface can be structured into specialised _BoundarySurfaces, representing floor (FloorSurface), ceiling (CeilingSur-face), and interior walls (InteriorWallSurface). Room furniture, like tables and chairs, can be represented in the CityGML building model with the class BuildingFurniture. A BuildingFurniture may have the attributes class, function and usage. Annexes G.1 to G.6 provide example CityGML documents containing a single building model which is subsequently refined from a coarse LOD0 representation up to a semantically rich and geomet-ric-topologically sound LOD4 model including the building interior.

## 8.4.4. Building Part

### Building

The Building class is one of the two subclasses of AbstractBuilding. If a building only consists of one (homo-geneous) part, this class shall be used. A building composed of structural segments differing in, for example the number of storeys or the roof type has to be separated into one Building having one or more additional BuildingPart (see Figure 43). The geometry and non-spatial properties of the central part of the building should be represented in the aggregating Building feature.

### Building Part

The class BuildingPart is derived from AbstractBuilding. It is used to model a structural part of a building (see Figure 43). A BuildingPart object should be uniquely related to exactly one building or building part object.

*Figure 43. Examples of buildings consisting of one and two building parts (source: City of Coburg)*

### AbstractBuilding

The abstract class AbstractBuilding contains properties for building attributes, purely geometric representations, and geometric/semantic representations of the building or building part in different levels of detail. The attributes describe:

1. classification of the building or building part (class), the different intended usages (function), and the different actual usages (usage). The permitted values for these attributes can be specified in code lists.

2. The year of construction (yearOfConstruction) and the year of demolition (yearOfDemolition) of the build-ing or building part. These attributes can be used to describe the chronology of the building development within a city model. The points of time refer to real world time.
3. The roof type of the building or building part (roofType). The permitted values for this attribute can be specified in a code list.
4. The measured relative height (measuredHeight) of the building or building part.
5. The number of storeys above (storeyAboveGround) and below (storeyBelowGround) ground level.
6. The list of storey heights above (storeyHeightsAboveGround) and below (storeyHeightsBelowGround) ground level. The first value in a list denotes the height of the nearest storey wrt. to the ground level and last value the height of the farthest.

Spanning the different levels of detail, the building model differs in the complexity and granularity of the geometric representation and the thematic structuring of the model into components with a special semantic meaning. This is illustrated in [figure-29] and Figure 45, showing the same building in five different LODs. The class AbstractBuilding has a number of properties which are associated with certain LODs.

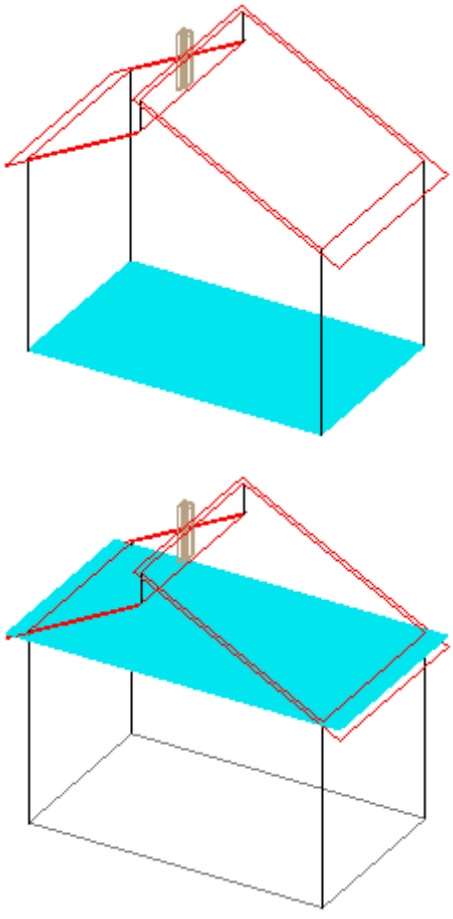


*Figure 44. The two possibilities of modeling a building in LOD0 using horizontal 3D surfaces. On the left, the building footprint (lod0FootPrint) is shown (cyan) which denotes the shape of the building on the ground. The corresponding surface representation is located at ground level. On the right, the lod0RoofEgde representation is illustrated (cyan) which results from a horizontal projection of the building's roof and which is located at the eaves height (source: Karlsruhe Institute of Technology (KIT), courtesy of Franz-Josef Kaiser).*

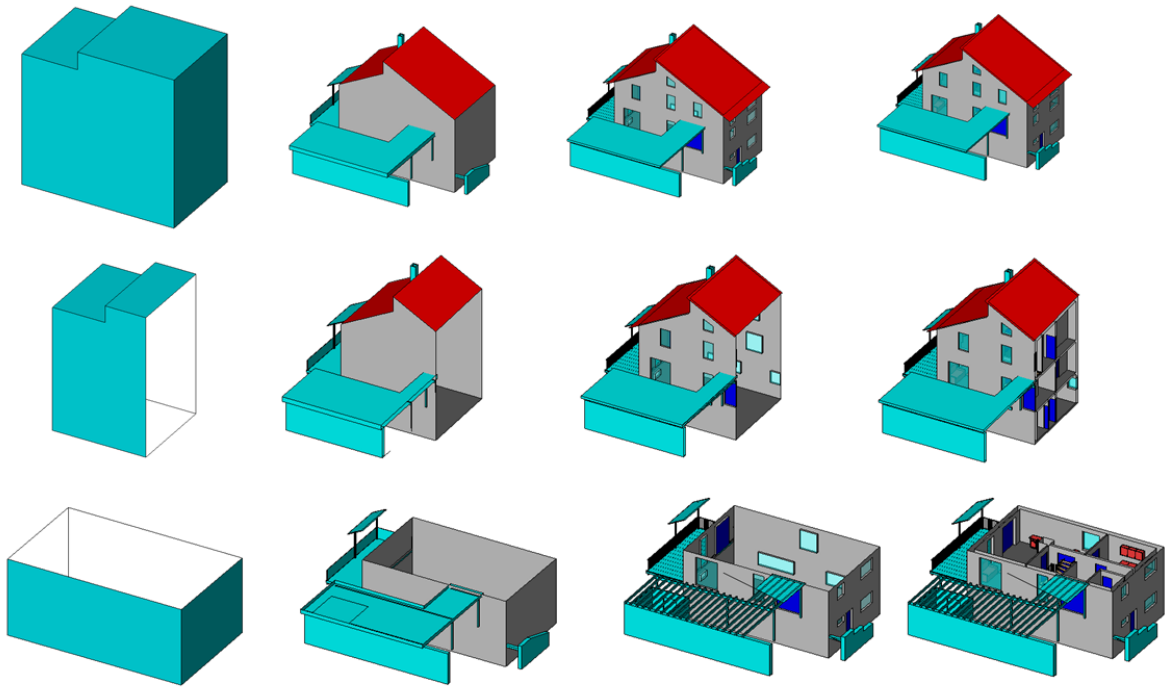


*Figure 45. Building model in LOD1 – LOD4 (source: Karlsruhe Institute of Technology (KIT), courtesy of Franz-Josef Kaiser).<o:p></o:p>*

Tab. 5 shows the correspondence of the different geometric and semantic themes of the building model to LODs. In LOD1 – 3, the volume of a building can be expressed by a GM_Solid geometry and/or a GM_MultiSurface geometry. The definition of a 3D Terrain Intersection Curve (TIC), used to integrate buildings from different sources with the Digital Terrain Model, is also possible in LOD1 – 3. The TIC can – but does not have to – build closed rings around the building or building parts.

In LOD0 (cf. [figure-29]) the building is represented by horizontal surfaces describing the footprint and the roof edge.

In LOD1 (cf. Figure 45), the different structural entities of a building are aggregated to a simple block and not differentiated in detail. The volumetric and surface parts of the exterior building shell are identical and only one of the corresponding properties (lod1Solid or lod1MultiSurface) must be used.

In LOD2 and higher levels of detail, the exterior shell of a building is not only represented geometrically as GM_Solid geometry and/or a GM_MultiSurface geometry, but it can also be composed of semantic objects. The base class for all objects semantically structuring the building shell is **_BoundarySurface** (cf. chapter 10.3.2), which is associated with a GM_MultiSurface geometry. If in a building model there is both a geometric representation of the exterior shell as volume or surface model and a semantic representation by means of thematic **_BoundarySurfaces**, the geometric representation must not explicitly define the geometry, but has to reference the corresponding geometry components of the GM_MultiSurface of the **_BoundarySurface** elements.

*Table 6. Semantic themes of the class _AbstractBuilding*

| Geometric / semantic theme | Property type | LOD0 | LOD1 | LOD2 | LOD3 | LOD4 |
|---|---|---|---|---|---|---|
| Building footprint and roof edge | gml:MultiSurfaceType | • | | | | |
| Volume part of the building shell | gml:SolidType | | • | • | • | • |
| Surface part of the building shell | gml:MultiSurfaceType | | • | • | • | • |
| Terrain intersection curve | gml:MultiCurveType | | • | • | • | • |
| Curve part of the building shell | gml:MultiCurveType | | | • | • | • |
| Building parts | BuildingPartType | | • | • | • | • |
| Boundary surfaces (chapter 10.3.3) | AbstractBoundarySurfaceType | | | • | • | • |
| Outer building installations (chapter 10.3.2) | BuildingInstallationType | | | • | • | • |
| Openings (chapter 10.3.4) | AbstractOpeningType | | | | • | • |
| Rooms (chapter 10.3.5) | RoomType | | | | | • |
| Interior building installations (chapter 10.3.5) | IntBuildingInstallationType | | | | | • |

Apart from BuildingParts, smaller features of the building (“outer building installations”) can also strongly affect the building characteristic. These features are modelled by the class BuildingInstallation (cf. chapter 10.3.2). Typical candidates for this class are chimneys (see. Fig. 30), dormers (see Fig. 28), balconies, outer stairs, or antennas. BuildingInstallations may only be included in LOD2 models, if their extents exceed the proposed minimum dimensions as specified in chapter 6.2. For the geometrical representation of the class Build-ingInstallation, an arbitrary geometry object from the GML subset shown in Fig. 9 can be used.

The class _AbstractBuilding has no additional properties for LOD3. Besides the higher requirements on geomet-ric precision and smaller minimum dimensions, the main difference of LOD2 and LOD3 buildings concerns the class _BoundarySurface (cf. chapter 10.3.3). In LOD3, openings in a building corresponding with windows or doors (see Fig. 30) are modelled by the abstract class _Opening and the derived subclasses Window and Door (cf. chapter 10.3.4).

With respect to the exterior building shell, the LOD4 data model is identical to that of LOD3. But LOD4 pro-vides the possibility to model the interior structure of a building with the classes IntBuildingInstallation and Room (cf. chapter 10.3.5).

Each Building or BuildingPart feature may be assigned zero or more addresses using the address property. The corresponding AddressPropertyType is defined within the CityGML core module (cf. chapter 10.1.4).

## 8.4.5. Outer building installations

**BuildingInstallation**

A BuildingInstallation is an outer component of a building which has not the significance of a BuildingPart, but which strongly affects the outer characteristic of the building. Examples are chimneys, stairs, antennas, balconies or attached roofs above stairs and paths. A BuildingInstallation optionally has attributes class, function and usage. The attribute class - which can only occur once - represents a general classification of the installation. With the attributes function and usage, nominal and real functions of a building installation can be described. For all three attributes the list of feasible values can be specified in a code list. For the geometrical representation of a BuildingInstallation, an arbitrary geometry object from the GML subset shown in Fig. 9 can be used. Alterna-tively, the geometry may be given as ImplicitGeometry object. Following the concept of ImplicitGeometry the geometry of a prototype building installation is stored only once in a local coordinate system and referenced by other building installation features (see chapter 8.2). The visible surfaces of a building installation can be seman-tically classified using the concept of boundary surfaces (cf. 10.3.3). A BuildingInstallation object should be uniquely related to exactly one building or building part object.

## 8.4.6. Boundary surfaces

**BoundarySurface**

| | |
|---|---|
| **NOTE** | Is this now AbstractConstructionSurface? |

_BoundarySurface is the abstract base class for several thematic classes, structuring the exterior shell of a build-ing as well as the visible surfaces of rooms and both outer and interior building installations. It is a subclass of _CityObject and thus inherits all properties like the GML3 standard feature properties (gml:name etc.) and the CityGML specific properties like ExternalReferences. From _BoundarySurface, the thematic classes RoofSur-face, WallSurface, GroundSurface, OuterCeilingSurface, OuterFloorSurface, ClosureSurface, FloorSurface, InteriorWallSurface, and CeilingSurface are derived. The thematic classification of building surfaces is illustrat-ed in Fig. 31 (outer building shell) and Fig. 32 (additional interior surfaces) and subsequently specified.

For each LOD between 2 and 4, the geometry of a _BoundarySurface may be defined by a different GM_MultiSurface geometry.

In LOD3 and LOD4, a _BoundarySurface may contain _Openings (cf. chapter 10.3.4) like doors and windows. If the geometric location of _Openings topologically lies within a surface component (e.g. gml:Polygon) of the gml:MultiSurface geometry, these _Openings must be represented as holes within that surface. A hole is repre-sented by an interior ring within the corresponding surface geometry object. According to GML3, the points have to be specified in reverse order (exterior boundaries counter-clockwise and interior boundaries clockwise when looking in opposite direction of the surface’s normal vector). If such an opening is sealed by a Door, a Window, or a

ClosureSurface, their outer boundary may consist of the same points as the inner ring (denoting the hole) of the surrounding surface. The embrasure surfaces of an Opening belong to the relevant adjacent _BoundarySurface. If, for example a door seals the Opening, the embrasure surface on the one side of the door belongs to the InteriorWallSurface and on the other side to the WallSurface (Fig. 32 on the right).

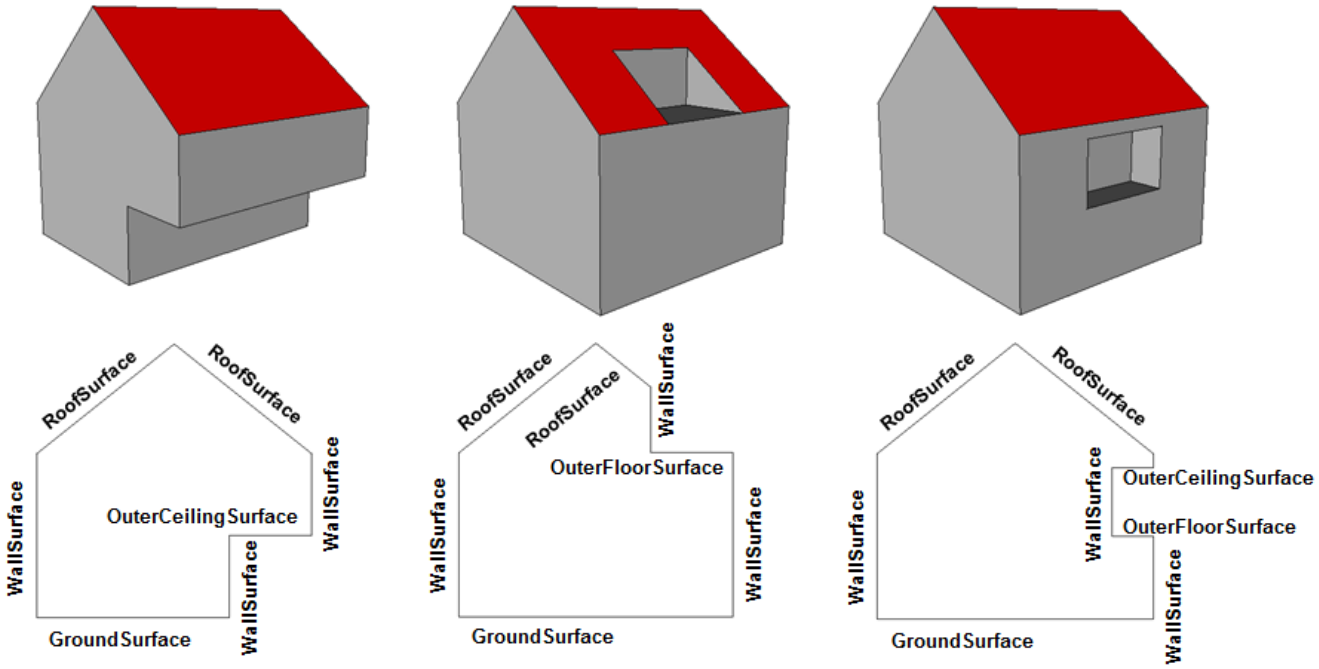
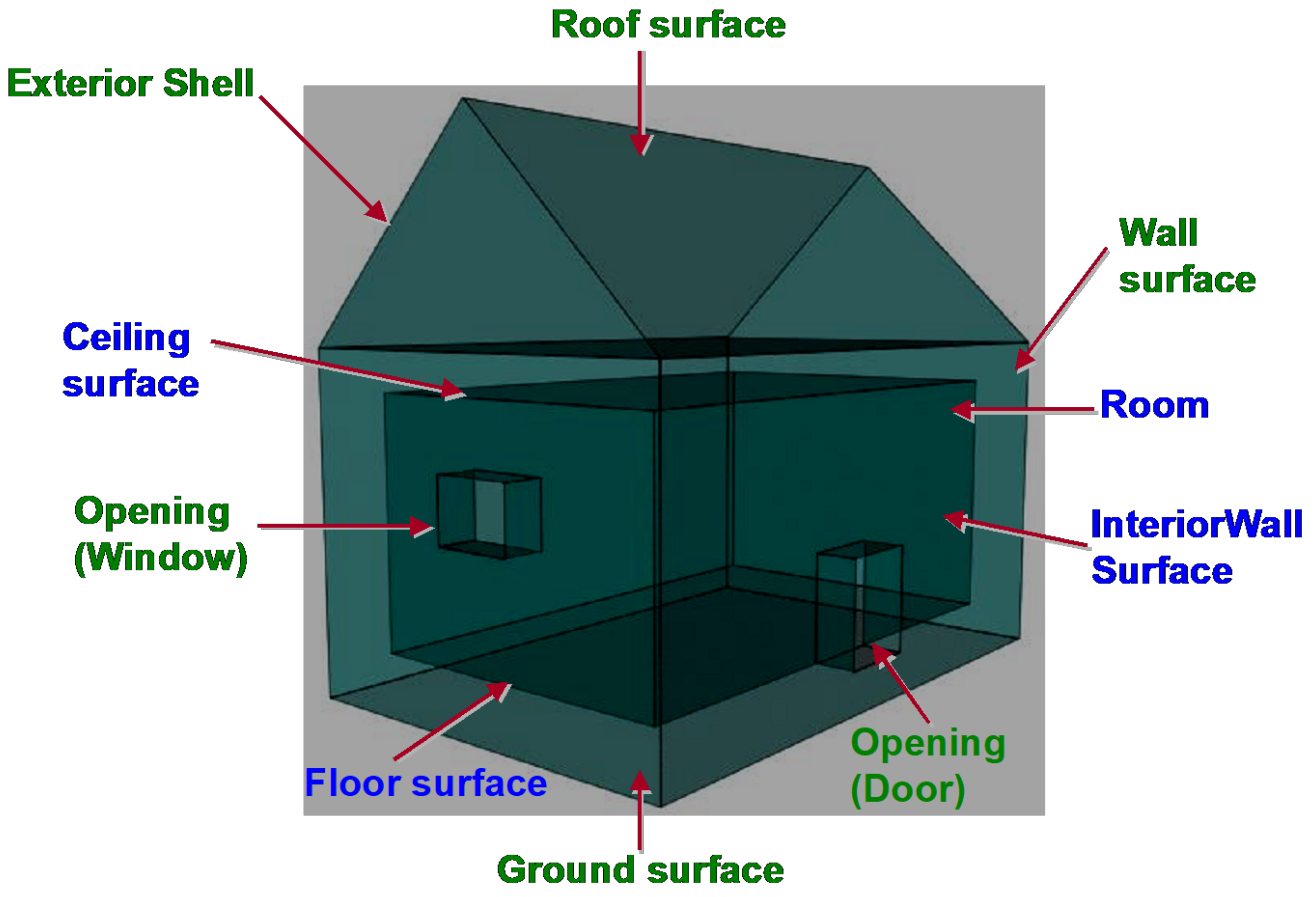



*Figure 46. Examples of the classification of _BoundarySurfaces of the outer building shell (source: Karlsruhe Institute of Technology (KIT))*



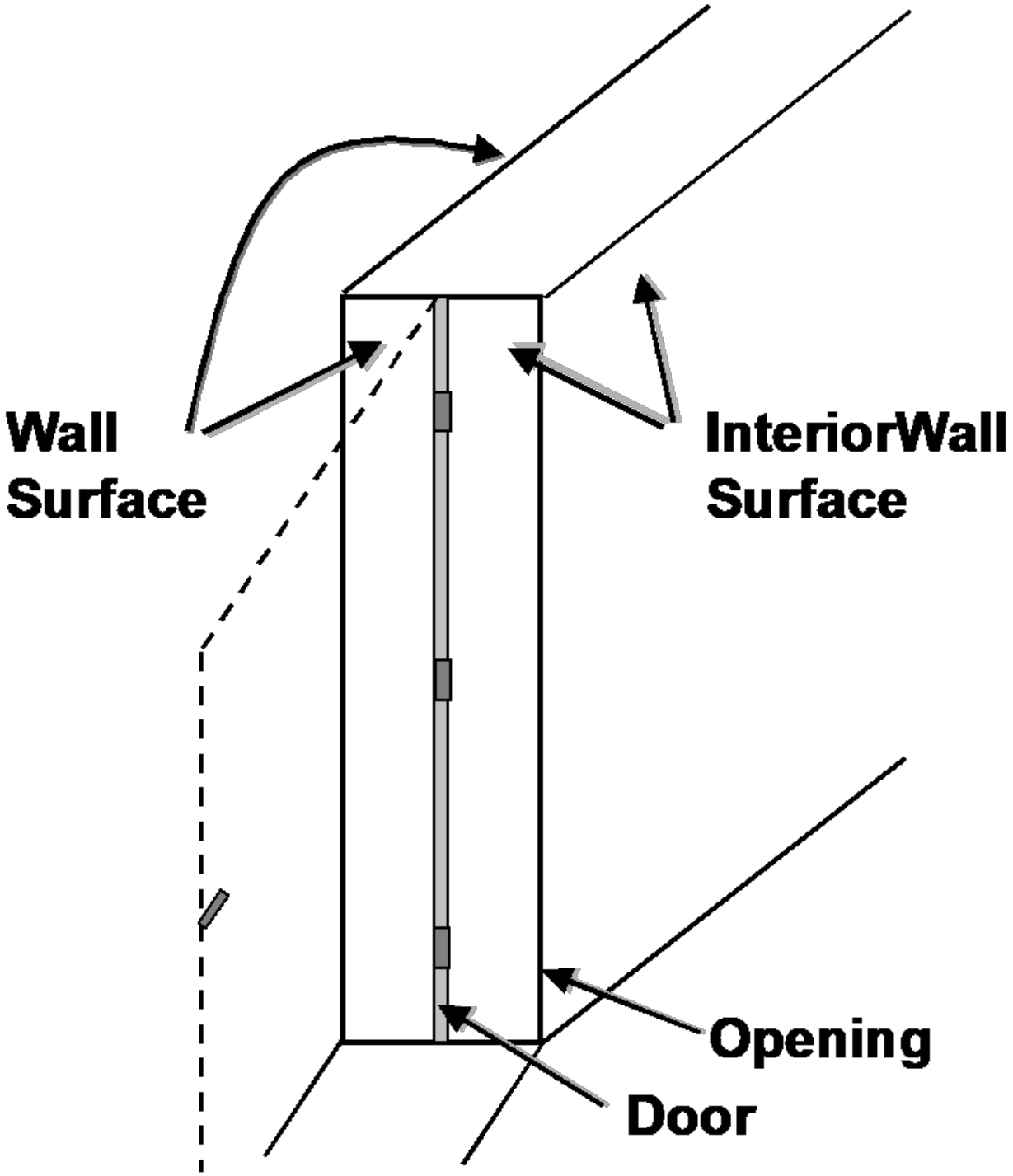



*Figure 47. Classification of BoundarySurfaces (left), in particular for Openings (right) (graphic: IGG Uni Bonn).*

**GroundSurface**

The ground plate of a building or building part is modelled by the class GroundSurface. The polygon defining the ground plate is congruent with the building's footprint. However, the surface normal of the ground plate is pointing downwards.

**OuterCeilingSurface**

A mostly horizontal surface belonging to the outer building shell and having the orientation pointing downwards can be modeled as an OuterCeilingSurface. Examples are the visible part of the ceiling of a loggia or the ceiling of a passage.

**WallSurface**

All parts of the building facade belonging to the outer building shell can be modelled by the class WallSurface.

**OuterFloorSurface**

A mostly horizontal surface belonging to the outer building shell and with the orientation pointing upwards can be modeled as an OuterFloorSurface. An example is the floor of a loggia.

**RoofSurface**

The major roof parts of a building or building part are expressed by the class RoofSurface. Secondary parts of a roof with a specific semantic meaning like dormers or chimneys should be modelled as BuildingInstallation.

**ClosureSurface**

An opening in a building not filled by a door or window can be sealed by a virtual surface called ClosureSurface (cf. chapter 6.4). Hence, buildings with open sides like a barn or a hangar, can be virtually closed in order to be able to compute their volume. ClosureSurfaces are also used in the interior building model. If two rooms with a different function (e.g. kitchen and living room) are directly connected without a separating door, a ClosureSur-face should be used to separate or connect the volumes of both rooms.

**FloorSurface**

The class FloorSurface must only be used in the LOD4 interior building model for modelling the floor of a room.

**InteriorWallSurface**

The class InteriorWallSurface must only be used in the LOD4 interior building model for modelling the visible surfaces of the room walls.

**CeilingSurface**

The class CeilingSurface must only be used in the LOD4 interior building model for modelling the ceiling of a room.

## 8.4.7. Openings

**Opening**

The class _Opening is the abstract base class for semantically describing openings like doors or windows in outer or inner boundary surfaces like walls and roofs. Openings only exist in models of LOD3 or LOD4. Each _Opening is associated with a gml:MultiSurface geometry. Alternatively, the geometry may be given as Implic-itGeometry object. Following the concept of ImplicitGeometry the geometry of a prototype opening is stored only once in a local coordinate system and referenced by other opening features (see chapter 8.2).

**Window**

The class Window is used for modelling windows in the exterior shell of a building, or hatches between adjacent rooms. The formal difference between the classes Window and Door is that – in normal cases – Windows are not specifically intended for the transit of people or vehicles.

**Door**

The class Door is used for modelling doors in the exterior shell of a building, or between adjacent rooms. Doors can be used by people to enter or leave a building or room. In contrast to a ClosureSurface a door may be closed, blocking the transit of people. A Door may be assigned zero or more addresses. The corresponding Address-PropertyType is defined within the CityGML core module (cf. chapter 10.1.4) .

## 8.4.8. Building Interior

**Room**

A Room is a semantic object for modelling the free space inside a building and should be uniquely related to exactly one building or building part object. It should be closed (if necessary by using ClosureSurfaces) and the geometry normally will be described by a solid (lod4Solid). However, if the topological correctness of the boundary cannot be guaranteed, the geometry can alternatively be given as a MultiSurface (lod4MultiSurface). The surface normals of the outer shell of a GML solid must point outwards. This is important to consider when Room surfaces should be assigned Appearances. In this case, textures and colors must be placed on the backside of the corresponding surfaces in order to be visible from the inside of the room.

In addition to the geometrical representation, different parts of the visible surface of a room can be modelled by specialised BoundarySurfaces (FloorSurface, CeilingSurface, InteriorWallSurface, and ClosureSurface cf. chapter 10.3.3).

A special task is the modelling of passages between adjacent rooms. The room solids are topologically connected by the surfaces representing hatches, doors or closure surfaces that seal open doorways. Rooms are defined as being adjacent, if they have common _Openings or ClosureSurfaces. The surface that represents the opening geometrically is part of the boundaries of the solids of both rooms, or the opening is referenced by both rooms on the semantic level. This adjacency implies an accessibility graph, which can be employed to determine the spread of e.g. smoke or gas, but which can also be used to compute escape routes using classical shortest path algorithms (see Fig. 33).

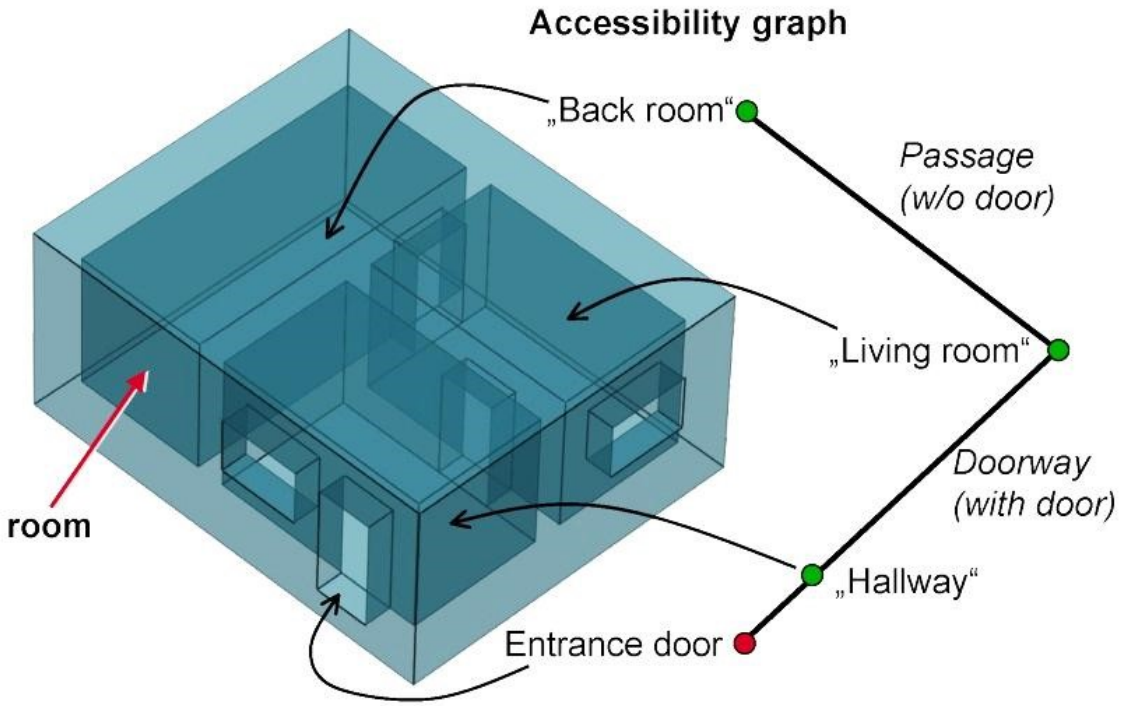



*Figure 48. Accessibility graph derived from topological adjacencies of room surfaces (graphic: IGG Uni Bonn).*

**BuildingFurniture**

Rooms may have BuildingFurnitures and IntBuildingInstallations. A BuildingFurniture is a movable part of a room, such as a chair or furniture. A BuildingFurniture object should be uniquely related to exactly one room object. Its geometry may be represented by an explicit geometry or an ImplicitGeometry object. Following the concept of ImplicitGeometry the geometry of a prototype building furniture is stored only once in a local coordi-nate system and referenced by other building furniture features (see chapter 8.2).

**IntBuildingInstallation**

An IntBuildingInstallation is an object inside a building with a specialised function or semantic meaning. In contrast to BuildingFurniture, IntBuildingInstallations are permanently attached to the building structure and cannot be moved. Typical examples are interior stairs, railings, radiators or pipes. Objects of the class IntBuild-ingInstallation can either be associated with a room (class Room), or with the complete building / building part (class _AbstractBuilding, cf. chapter 10.3.1). However, they should be uniquely related to exactly one room or one building / building part object. An IntBuildingInstallation optionally has attributes class, function and usage. The attribute class, which can only occur once, represents a general classification of the internal building component. With the attributes function and usage, nominal and real functions of a building installation can be described. For all three attributes the list of feasible values can be specified in a code list. For the geometrical representation of an IntBuildingInstallation, an arbitrary geometry object from the GML subset shown in Fig. 9 can be used. Alternatively, the geometry may be given

as ImplicitGeometry object. Following the concept of ImplicitGeometry the geometry of a prototype interior building installation is stored only once in a local coordi-nate system and referenced by other interior building installation features (see chapter 8.2). The visible surfaces of an interior building installation can be semantically classified using the concept of boundary surfaces (cf. 10.3.3).

### 8.4.9. Modelling building storeys using CityObjectGroups

CityGML does currently not provide a specific concept for the representation of storeys as it is available in the AEC/FM standard IFC (IAI 2006). However, a storey can be represented as an explicit aggregation of all build-ing features on a certain height level using CityGML's notion of CityObjectGroups (cf. chapter 10.11). This would include Rooms, Doors, Windows, IntBuildingInstallations and BuildingFurniture. If thematic surfaces like walls and interior walls should also be associated to a specific storey, this might require the vertical fragmenta-tion of these surfaces (one per storey), as in virtual 3D city models they typically span the whole façade.

In order to model building storeys with CityGML's generic grouping concept, a nested hierarchy of CityObject-Group objects has to be used. In a first step, all semantic objects belonging to a specific storey are grouped. The attributes of the corresponding CityObjectGroup object are set as follows:

- The class attribute shall be assigned the value "building separation".
- The function attribute shall be assigned the value "lodXStorey" with X between 1 and 4 in order to de-note that this group represents a storey wrt. a specific LOD.
- The storey name or number can be stored in the gml:name property. The storey number attribute shall be assigned the value "storeyNo_X" with decimal number X in order to denote that this group repre-sents a storey wrt. a specific number.

In a second step, the CityObjectGroup objects representing different storeys are grouped themselves. By using the generic aggregation concept of CityObjectGroup, the "storeys group" is associated with the corresponding Building or BuildingPart object. The class attribute of the storeys group shall be assigned the value "building storeys".

### 8.4.10. UML Model

The UML diagram of the building module is depicted in Figure 49. The Building module inherits concepts from the Construction module. The Construction module defines objects that are common to all types of construction, such as the different surface types and the openings.

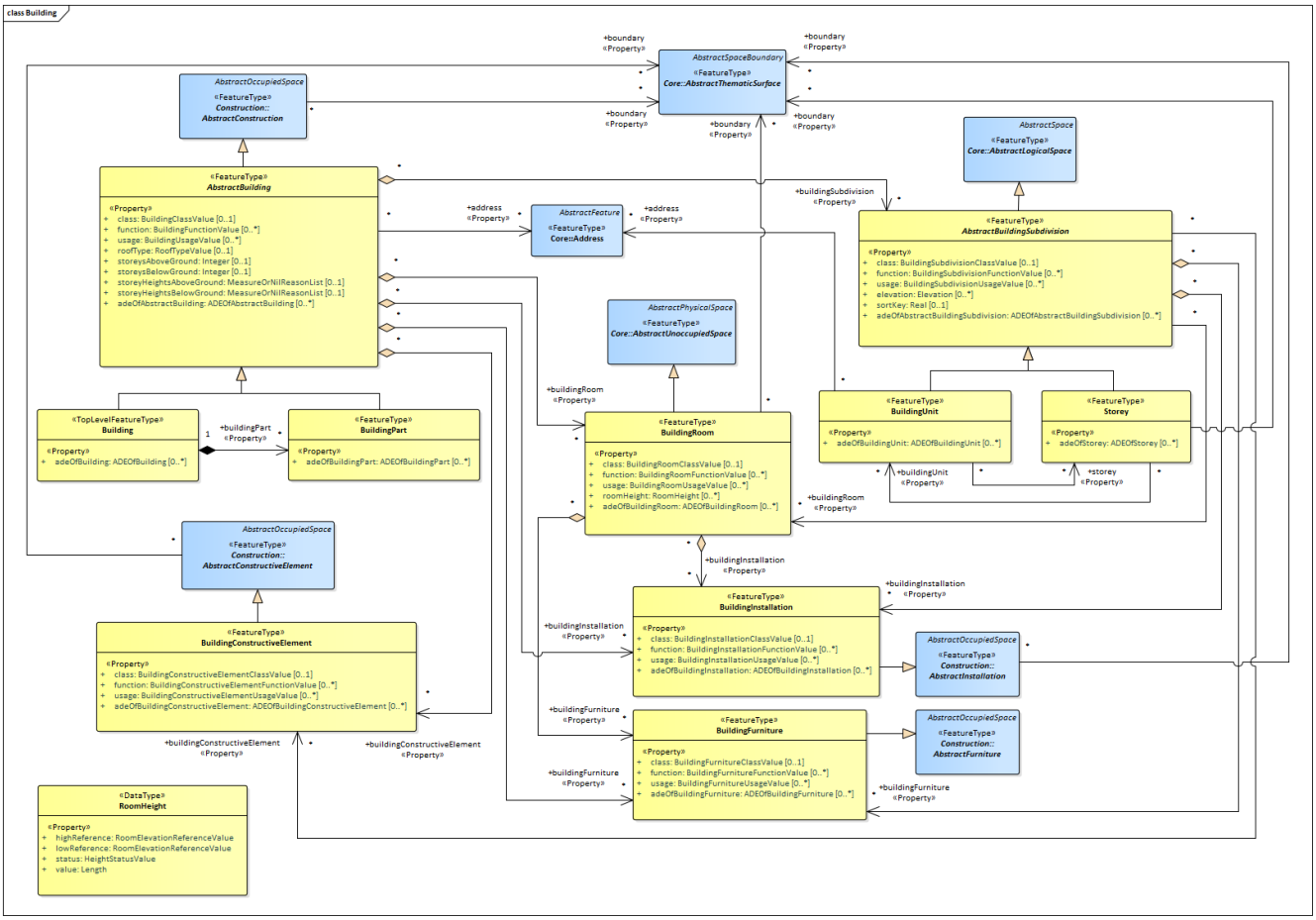



*Figure 49. UML diagram of CityGML's building model.*

The ADE data types provided for the Building module are illustrated in Figure 50.

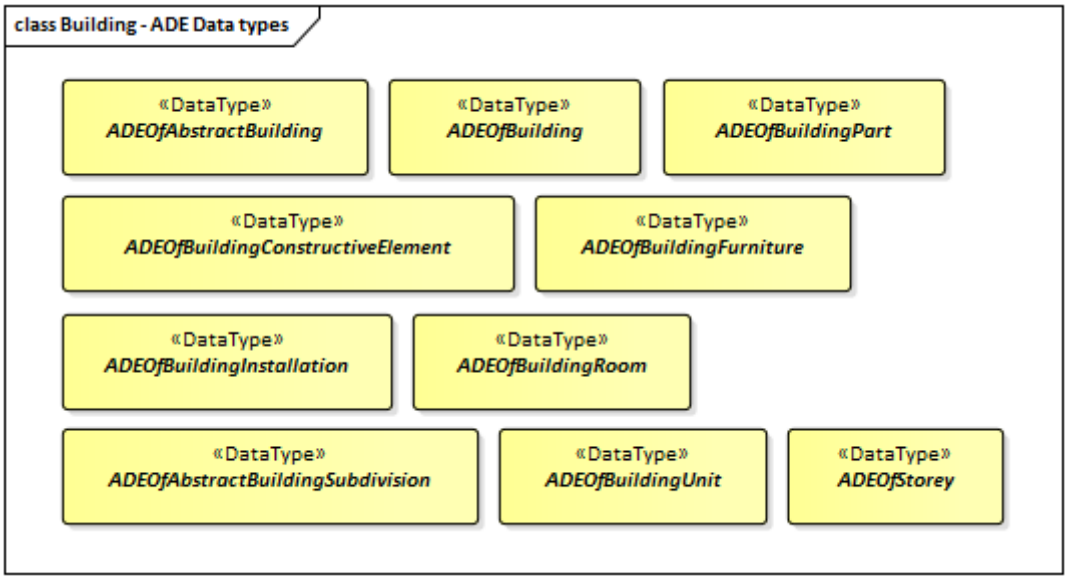



*Figure 50. ADE classes of the CityGML Building module.*

The Code Lists provided for the Building module are illustrated in Figure 51.

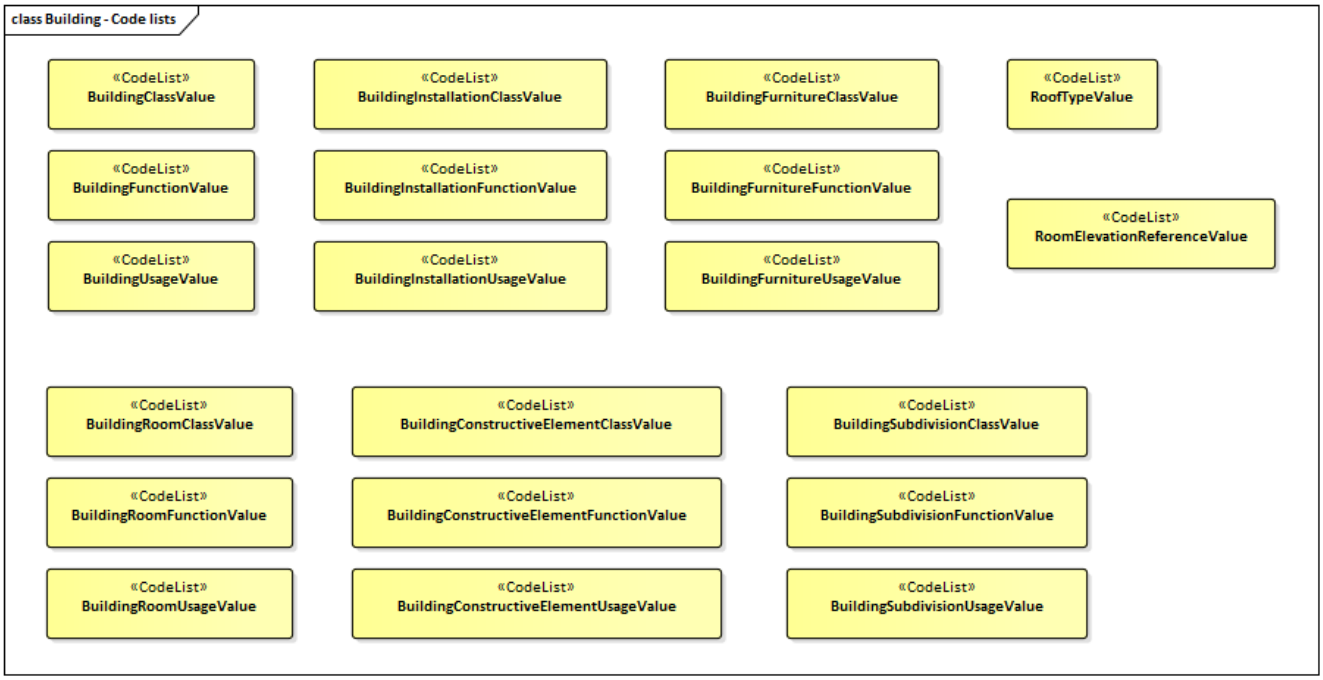



*Figure 51. Codelists from the CityGML Building module.*

## 8.4.11. Examples

The LOD1 model of the Campus North of the Karlruhe Institute of Technology (KIT) shown in Fig. 34 consists of 596 buildings and 187 building parts. The footprint geometries of the buildings are taken from a cadastral information system and extruded by a given height. Buildings with a unique identifier and a single height value are modeled as one building (bldg:Building). Buildings having a unique identifier but different height values are modeled as one building (bldg:Building) with one or more building parts (bldg:BuildingPart). Both buildings and building parts have solid geometries and their height values are additionally represented as thematic attribute (bldg:measuredHeight). Fig. 34 shows an aerial photograph of the KIT Campus North (left) and the CityGML LOD1 model (right).

*Figure 52. LOD1 model of the KIT Campus North (source: Karlsruhe Institute of Technology (KIT)).*

An example for a fully textured LOD2 building model is given in Fig. 35 which shows the Bernhardus church located in the city of Karlsruhe, Germany. On the left side of Fig. 35, a photograph of the church in real world is shown whereas the CityGML building model of the church with photorealistic textures is illustrated on the right. The model is bounded by a ground surface, several wall and roof surfaces. The railing above the church clock is modeled as a building installation (BuildingInstallation).

*Figure 53. Textured LOD2 model of the Bernhardus church in Karlsruhe (source: Karlsruhe Institute of Technology (KIT), courtesy of City of Karlsruhe).*

The model shown in Fig. 36 was derived from a 3D CAD model generated during the planning phase of the building. On the left side of Fig. 36, the building is shown whereas on the right side the LOD3 model is present-ed. The building itself is bounded by wall surfaces, roof surfaces and a ground surface. Doors and windows are modeled including reveals. According to the cadaster data, the car port next to the building is not part of the building. Therefore the car port, the balcony and the chimney are modeled as building installations (BuildingIn-stallation). The model also contains the terrain intersection curve (lod3TerrainIntersection) as planned by the architect.

In order to determine the volume of the building, the geometries of all boundary surfaces, including doors and windows, are referenced by the building solid (lod3Solid) using the XLink mechanism. Consequently, the roof surfaces are split into surfaces representing the roof itself and surfaces representing the roof overhangs.

*Figure 54. Example of a buildingmodeled in the Level of Detail 3. The chimney, the balcony and the car port are modeled as building installations (source: Karlsruhe Institute of Technology (KIT), courtesy of Franz-Josef Kaiser).*

# 8.5. City Furniture

| Contributors |
| --- |
| Chuck Heazel: first draft |

## 8.5.1. Synopsis

The CityFurniture module provides the representation of objects or pieces of equipment that are installed in the outdoor environment for various purposes, such as decoration, explanation or control. City furniture objects are relatively small, immovable objects and usually are of stereotypical form. Examples include road signs, traffic signals, bicycle racks, street lamps, fountains, flower buckets, advertising columns, and benches.

## 8.5.2. Key Concepts

**CityFurniture**: CityFurniture is an object or piece of equipment installed in the outdoor environment for various purposes. Examples include street signs, traffic signals, street lamps, benches, fountains.
A type of AbstractOccupiedSpace.

## 8.5.3. Discussion

| NOTE | The following text needs to be reviewed and updated. |
| --- | --- |

City furniture objects are immovable objects like lanterns, traffic lights, traffic signs, flower buckets, advertising columns, benches, delimitation stakes, or bus stops (Figure 30, Figure 31). City furniture objects can be found in traffic areas, residential areas, on squares or in built-up areas. The modelling of city furniture objects is used for visualisation of, for example city traffic, but also for analysing local structural conditions. The recognition of special locations in a city model is improved by the use of these detailed city furniture objects, and the city model itself becomes more alive and animated.

City furniture objects can have an important influence on simulations of, for example city traffic situations. Navigation systems can be realised, for example for visually handicapped people using a traffic light as routing target. Likewise, city furniture objects are important to plan a heavy vehicle transportation, where the exact position and further conditions of obstacles must be known.

*Figure 55. Real situation showing a bus stop (left). The advertising billboard and the refuge are modelled as CityFurniture objects in the right image (source: 3D city model of Barkenberg).*

*Figure 56. Real situation showing lanterns and delimitation stakes (left). In the right image they are modelled as CityFurniture ob-jects with ImplicitGeometry representations (source: 3D city model of Barkenberg).*

The class CityFurniture may have the attributes `class`, `function` and `usage`. Their possible values can be specified in corresponding code lists. The class `attribute` allows an object classification like traffic light, traffic sign, delimitation stake, or garbage can, and can occur only once. The `function` attribute describes to which thematic area the city furniture object belongs to (e.g. transportation, traffic regulation, architecture), and can occur multiple times. The attribute `usage` denotes the real purpose of the object.

Since CityFurniture is a subclass of AbstractOccupiedSpace and hence is a feature, it inherits the attribute `name`. As with any CityObject, CityFurniture objects may be assigned ExternalReferences and may be augmented by generic attributes using CityGML's Generics module. For ExternalReferences city furniture objects can have links to external thematic databases. Thereby, semantic information of the objects, which cannot be modelled in CityGML, can be transmitted and used in the 3D city model for further processing, for example information from systems of powerlines or pipelines, traffic sign cadaster, or water resources for disaster management.

City furniture objects can be represented in city models with their specific geometry, but in most cases the same kind of object has an identical geometry. The geometry of CityFurniture objects in LOD 1-3 may be represented by an explicit geometry (lodXGeometry where X is between 0 and 3) or an ImplicitGeometry object. Following the concept of ImplicitGeometry, the geometry of a prototype city furniture is stored only once in a local coordinate system and referenced by other city furniture features. Spatial information of city furniture objects can be taken, for example, from city maps or from public and private external information systems.

In order to specify the exact intersection of the DTM with the 3D geometry of a city furniture object, the latter can have a TerrainIntersectionCurve (TIC) for each LOD. This allows for ensuring a smooth transition between the DTM and the city furniture object.

### 8.5.4. UML Model

The UML diagram of the CityFurniture module is depicted in Figure 57.

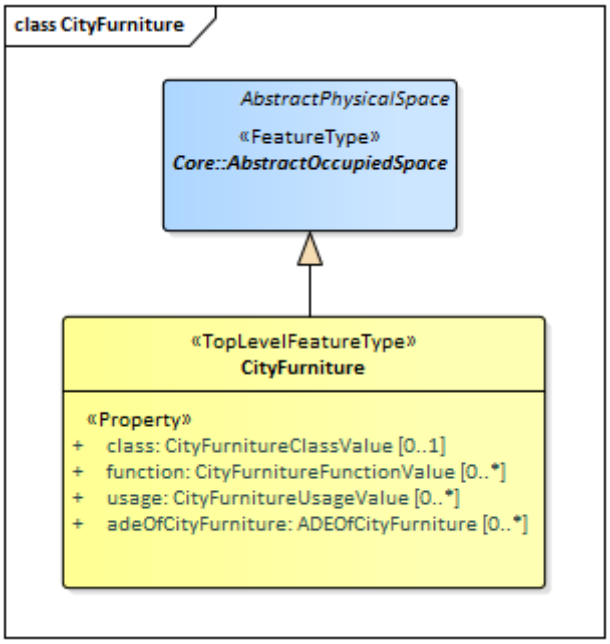



*Figure 57. UML diagram of CityGML's City Furniture model.*

The ADE data types and Code Lists provided for the CityFurniture module are illustrated in Figure 58.

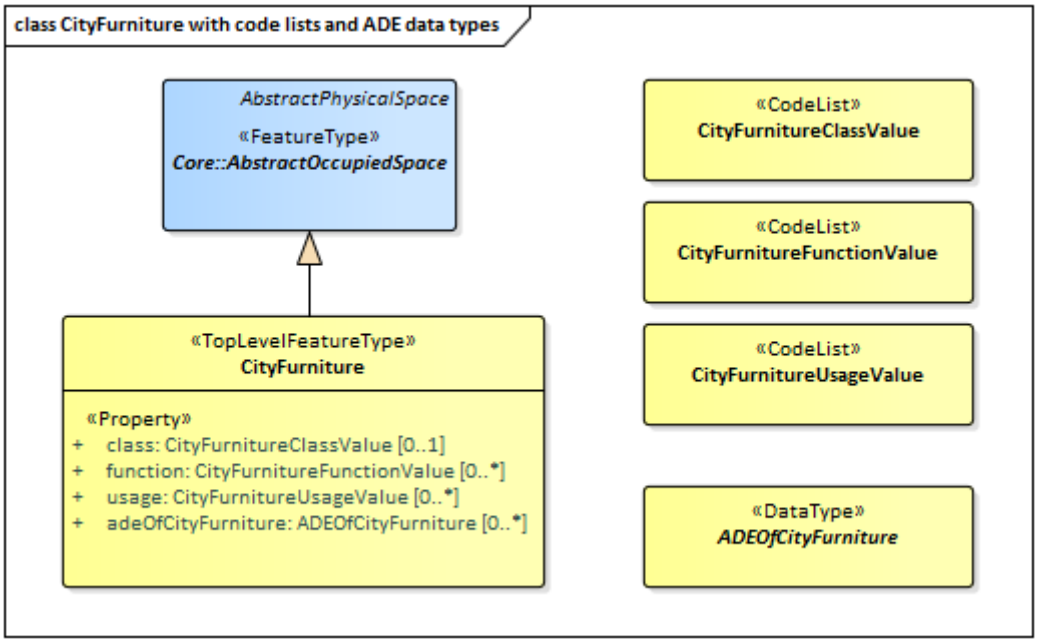



*Figure 58. ADE classes and Code Lists of the CityGML CityFurniture module.*

### 8.5.5. Examples

| NOTE | the following example is from version 2.0 |
|---|---|

The following example of a CityGML dataset is an extract of the model of a delimitation stake in LOD3 and contains the attributes class = 1000 and function = 1520 (delimitation stake) whose coded attribute values are taken from a code list proposed by the SIG 3D (cf. annex C.4). The delimitation stake with the object ID stake0815 has an ExternalReference pointing to a cadastre object within the German ALKIS database (www.adv-online.de) which is identified by the URI urn:adv:oid:DEHE123400007001.

The example dataset shows the geometry of the top surface (gml:id “cover”) and of the left surface (gml:id “surfLeft”) of the stake which are both depicted in Fig. 70. The top surface is assigned a constant material (white color) and the left surface is textured using the texture image stake.gif by denoting the relevant texture coordi-nates. Both surface appearances are modelled using the CityGML Appearance module (cf. chapter 9).

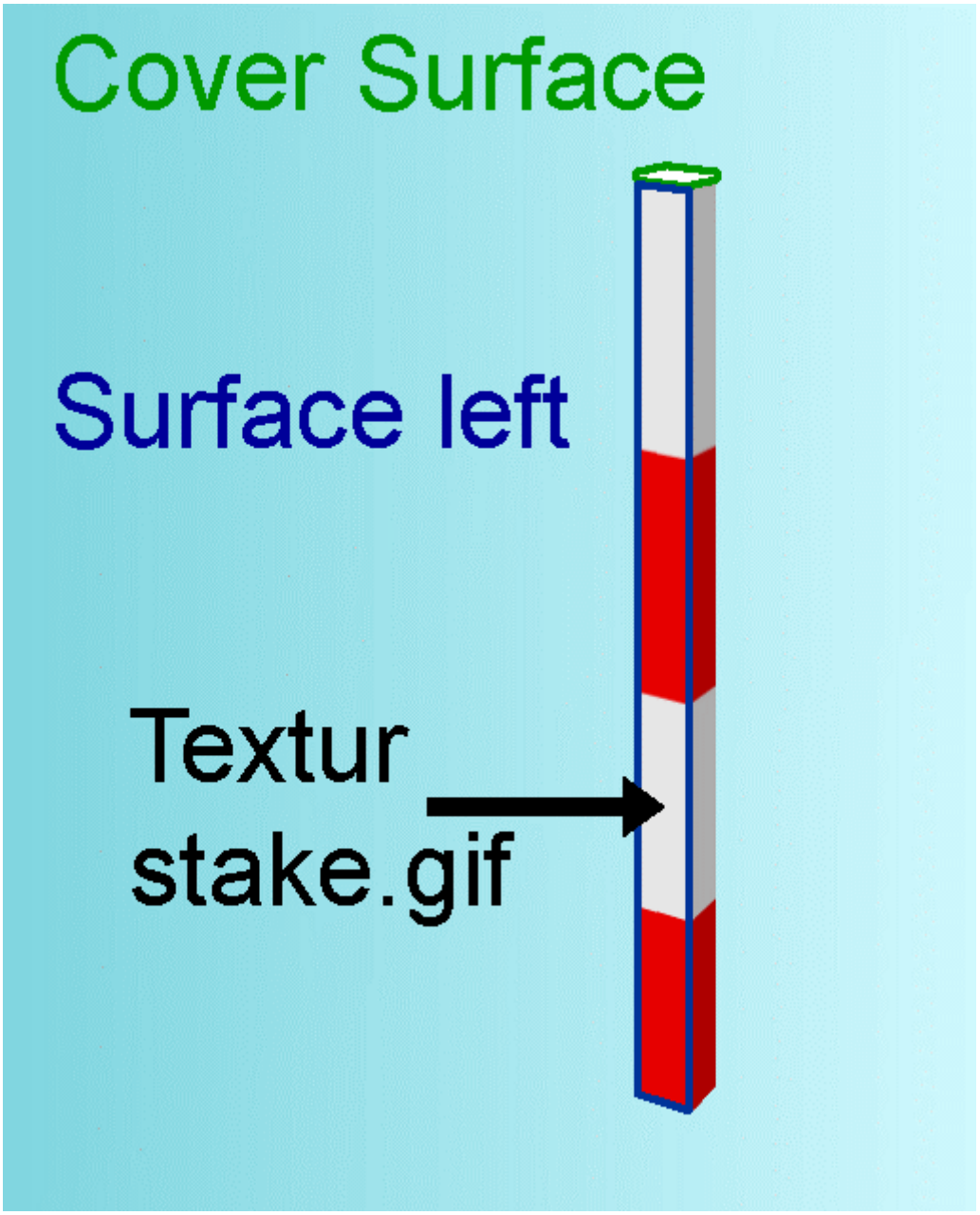



*Figure 59. Example of a simple city furniture object (source: District of Recklinghausen).*

| NOTE | insert example - GML? |
|---|---|

## 8.6. City Object Group

| Contributors |
|---|
| Chuck Heazel: first draft |

### 8.6.1. Synopsis

Aggregates City Objects into a collection of City Objects. That collection is itself a City Object.

### 8.6.2. Key Concepts

**CityObjectGroup**: A CityObjectGroup represents an application-specific aggregation of city objects

according to some user-defined criteria.
A type of AbstractLogicalSpace.

**Role**: Role is a UML Association Class tied to the `groupMember` association between **CityObjectGroup** and AbstractCityObject. It serves to qualify the function of a city object within the CityObjectGroup.

### 8.6.3. Discussion

CityObjectGroups are modelled using the Composite Design Pattern from software engineering (cf. Gamma et al. 1995): CityObjectGroups aggregate City Objects and furthermore are defined as special City Objects. This implies that a group may become a member of another group realizing a recursive aggregation schema. However, in a CityGML instance document it has to be ensured (by the generating application) that no cyclic groupings are included.

The class CityObjectGroup has the optional attributes `class`, `function` and `usage`. The `class` attribute allows a group classification with respect to the stated function and may occur only once. The `function` attribute is intended to express the main purpose of a group, possibly to which thematic area it belongs (e.g. site, building, transportation, architecture, unknown etc.). The attribute `usage` can be used, if the way the object is actually used differs from the function. The `function` and `usage` attributes can occur multiple times.

Each member of a group may be qualified by a Role, reflecting the role each City Object plays in the context of the group. Furthermore, a CityObjectGroup can optionally be assigned an arbitrary ISO geometry object. This may be used to represent a generalised geometry generated from the members geometries.

The `parent` association linking a CityObjectGroup to a CityObject allows for the modelling of a generic hierarchical grouping concept. Named aggregations of components (CityObjects) can be added to specific CityObjects considered as the parent object. The parent association links to the aggregate, while the parts are given by the group members. This concept is used, for example, to represent storeys in buildings.

### 8.6.4. UML Model

The UML diagram of the CityObjectGroup module is depicted in Figure 60.

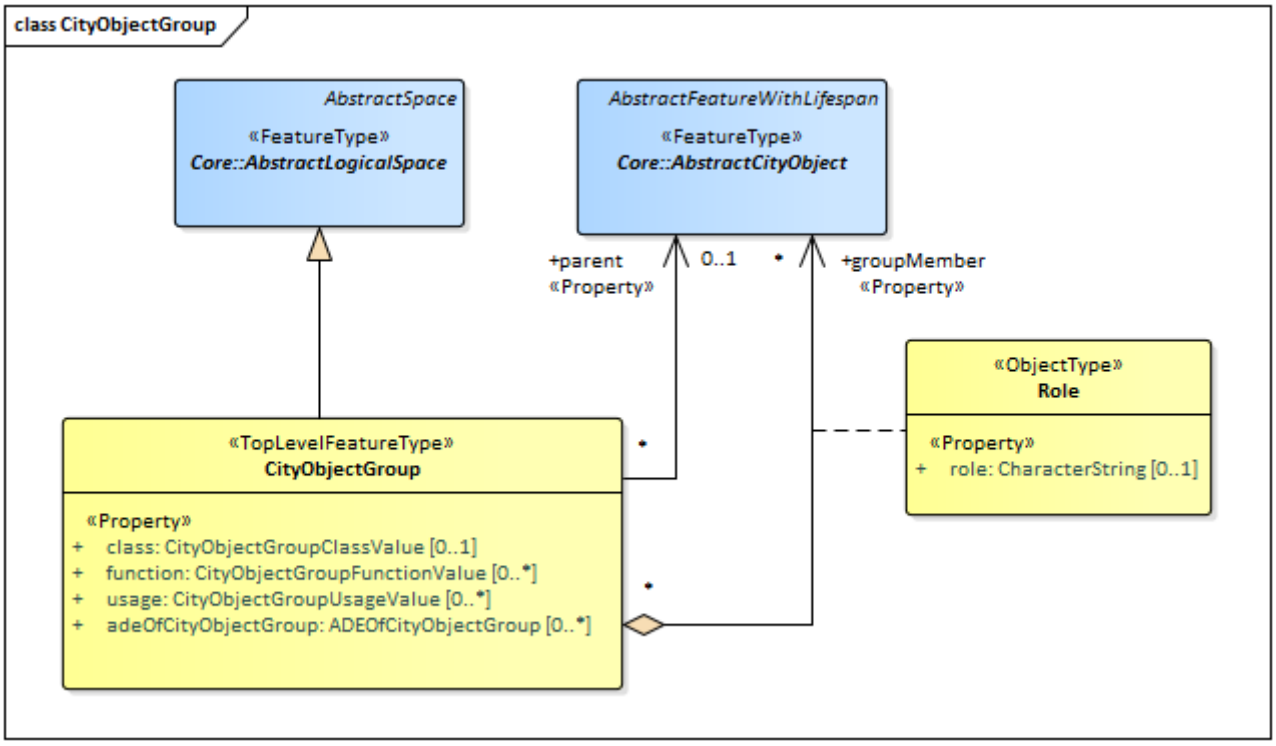



*Figure 60. UML diagram of the City Object Group Model.*

The ADE data types provided for the CityObjectGroup module are illustrated in Figure 61.

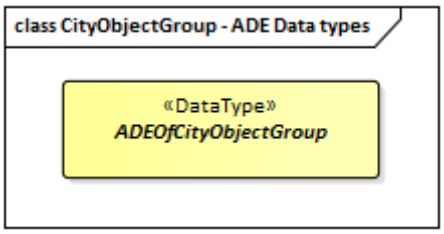



*Figure 61. ADE classes of the CityGML CityObjectGroup module.*

The Code Lists provided for the CityObjectGroup module are illustrated in Figure 62.

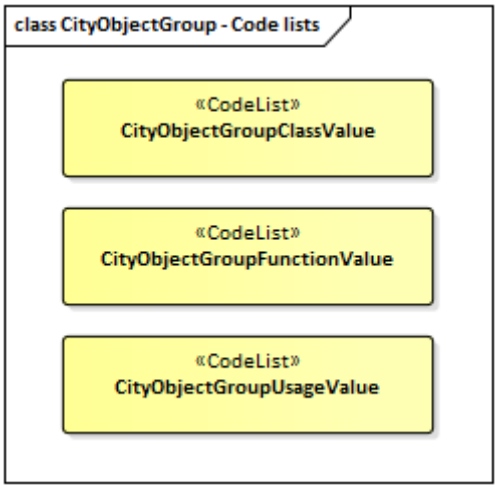



*Figure 62. Codelists from the CityGML CityObjectGroup module.*

## 8.6.5. Examples

# 8.7. Construction

| Contributors |
| --- |
| Chuck Heazel - first draft |

## 8.7.1. Synopsis

The Construction module defines concepts that are common to all forms of constructions.

## 8.7.2. Key Concepts

**AbstractConstruction**: (Discussion) Classes of objects that are manufactured by humans from construction materials, are connected to earth, and are intended to be permanent. A type of AbstractOccupiedSpace

**AbstractConstructionElement**: (Discussion) Classes that represent volumetric elements of a construction. Examples are walls, beams, slabs. A type of AbstractOccupiedSpace.

**AbstractFillingElement**: (Discussion) Classes that represent different kinds of elements that fill the openings of a construction such as windows and doors. A type of AbstractOccupiedSpace.

**AbstractFurniture**: (Discussion) Classes that represent furniture objects of a construction. A type of AbstractOccupiedSpace.

**AbstractInstallation**: (Discussion) Classes that represent installation objects of a construction. A type of AbstractOccupiedSpace.

**AbstractConstructionSurface**: (Discussion) Classes that represent different kinds of surfaces that bound a construction such as roof and wall surfaces. A type of AbstractThematicSurface.

**AbstractFillingSurface**: (Discussion) Classes that represent different kinds of surfaces that seal openings. The openings are filled by filling elements such as window and door surfaces. A type of AbstractThematicSurface.

## 8.7.3. Discussion

The Construction module defines concepts that are common to all forms of constructions. Constructions are objects that are manufactured by humans from construction materials, are connected to earth, and are intended to be permanent. The Construction module focuses on as-built representations of constructions and integrates all concepts that are similar over different types of constructions, in particular buildings, bridges, and tunnels. In addition, for representing man-made structures that are neither buildings, nor bridges, nor tunnels so-called other constructions (e.g. large chimneys or city walls) can be defined.

Furniture, installations, and constructive elements are further concepts that are defined in the Construction module. Installations are permanent parts of a construction that strongly affect the outer or inner appearance of the construction and that cannot be moved (e.g. balconies, chimneys, or stairs), whereas furniture represent moveable objects of a construction (e.g. tables and chairs). Constructive elements allow for decomposing a construction into volumetric components, such as

walls, beams, and slabs.

Constructions and constructive elements can be bounded by different types of surfaces. In this way, the outer structure of constructions and constructive elements can be differentiated semantically into wall surfaces, roof surfaces, ground surfaces, outer floor surfaces, and outer ceiling surfaces, whereas the visible surface of interior spaces can be structured into interior wall surfaces, floor surfaces, and ceiling surfaces.

Furthermore, the openings of constructions, i.e. windows and doors, can be represented as so-called filling elements including their corresponding filling surfaces.

The Construction module defines concepts that are inherited and, where necessary, are specialized by the modules Building, Bridge, and Tunnel (cf. [ug_model_building_section], [ug_model_bridge_section], and [ug_model_tunnel_section]).

### 8.7.4. Construction Spaces

### 8.7.5. Construction Surfaces

### 8.7.6. Levels of Detail

The different Levels of Detail are defined in the following way:

- LOD 0: Volumetric real-world objects (Spaces) can be spatially represented by a single point, by a set of curves, or by a set of surfaces.
  Areal real-world objects (Space Boundaries) can be spatially represented in LOD0 by a set of curves or a set of surfaces. LOD0 surface representations are typically the result of a projection of the shape of a volumetric object onto a plane parallel to the ground, hence, representing a footprint (e.g. a building footprint or a floor plan of the rooms inside a building). LOD0 curve representations are either the result of a projection of the shape of a vertical surface (e.g. a wall surface) onto a grounding plane or the skeleton of a volumetric shape of longitudinal extent such as a road or river segment.
- LOD 1: Volumetric real-world objects (Spaces) are spatially represented by a vertical extrusion solid, i.e. a solid created from a horizontal footprint by vertical extrusion.
  Areal real-world objects (Space Boundaries) can be spatially represented in LOD1 by a set of horizontal or vertical surfaces.
- LOD 2: Volumetric real-world objects (Spaces) can be spatially represented by a set of curves, a set of surfaces, or a single solid geometry.
  Areal real-world objects (Space Boundaries) can be spatially represented in LOD2 by a set of surfaces. The shape of the real-world object is generalized in LOD2 and smaller details (e.g. bulges, dents, sills, but also structures like e.g. balconies or dormers of buildings) are typically neglected.
  LOD2 curve representations are skeletons of volumetric shapes of longitudinal extent like an antenna or a chimney.
- LOD 3: Volumetric real-world objects (Spaces) can be spatially represented by a set of curves, a set of surfaces, or a single solid geometry.
  Areal real-world objects (Space Boundaries) can be spatially represented in LOD3 by a set of

surfaces. LOD3 is the highest level of detail and respective geometries include all available shape details.

A `Construction Model` is successively refined from LOD0 to LOD3. Therefore, not all components of a construction model are represented equally in each LOD and not all aggregation levels are allowed in each LOD. In CityGML, all object classes are associated to the LODs with respect to the proposed minimum acquisition criteria for each LOD. An object can be represented simultaneously in different LODs by providing distinct geometries for the corresponding LODs.

**LOD0**

In LOD0, a `Construction Object` can be represented by two horizontal, 3-dimensional surfaces (AbstractThematicSurface). These two surfaces represent the foot-print of a `Construction Object` and a top surface such as a roof edge (see Figure 32). This allows the easy integration of 2D data into the model.

In many countries these 2D geometries readily exist, for example in cadastral or topographic data holdings. Cadastre data typically depicts the shape of a `Construction Object` on the ground (footprints). Higher level surfaces are often photogrametrically extracted from area/satellite images or derived from airborne laser data. Topographic data is often provides a mixture between footprints and geometries at higher levels.

The surface geometries at LOD0 can be represented using GM_MultiCurve or GM_MultiSurface. 3D coordinates are required for both representations. At LOD0 it is mandated that the height values of all vertices belonging to the same surface are identical. If 2D geometries are imported into any of these two LOD0 geometries, an appropriate height value for all vertices needs to be chosen.

Figure 32 illustrates an LOD0 representation of a Building Constuction. The two horizontal 3D surfaces are colored in cyan. On the left, the building footprint is shown (cyan) which denotes the shape of the building on the ground. The corresponding surface representation is located at ground level. On the right, the Roof Edge representation is illustrated which results from a horizontal projection of the building's roof and which is located at the eaves height

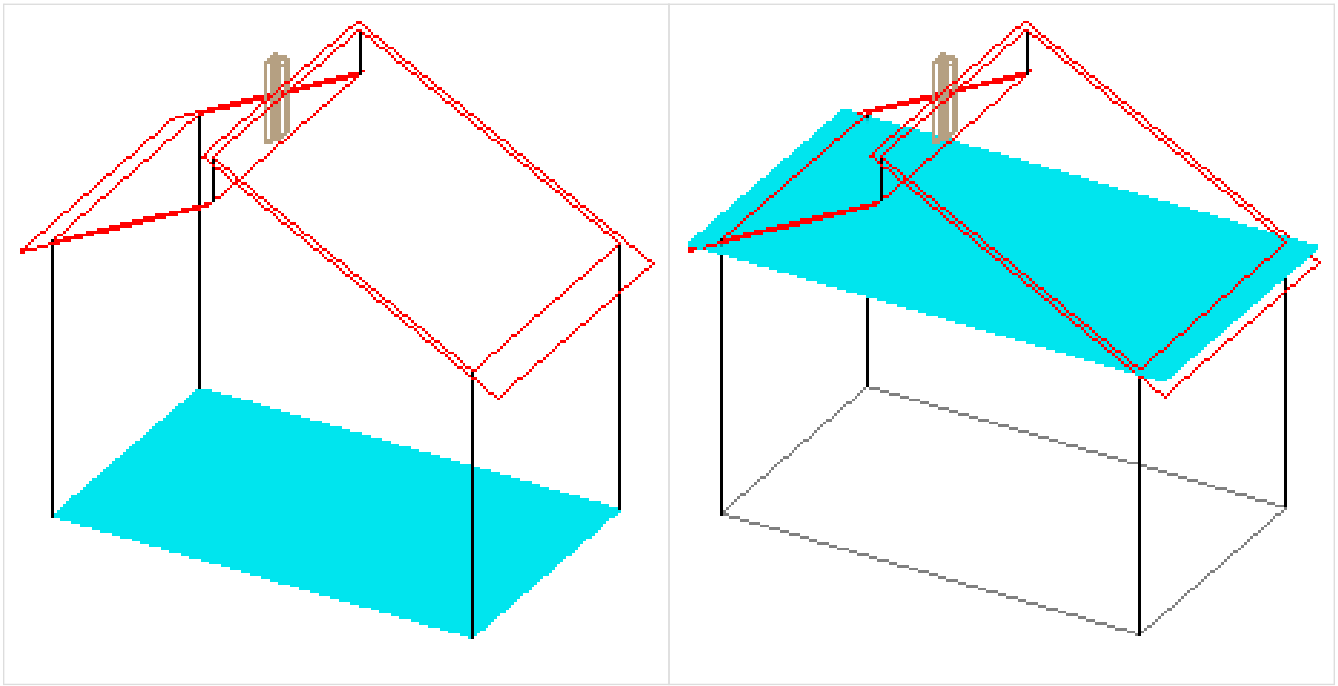


*Figure 63. Modeling a construction (building) in LOD0 (source: Karlsruhe Institute of Technology (KIT), courtesy of Franz-Josef Kaiser).*

Note that at LOD0 a `construction object` also has a `Space` geometry but that geometry is limited to no more than one GM_Point.

#### LOD1

In LOD1, a `construction object` consists of the generalized geometric shape of the object (GM_Solid) as represented by its' boundary sufaces. The boundary surfaces are AbstractThematicSurfaces identified by the `boundary` attribute. Optionally, a GM_MultiCurve representing the TerrainIntersectionCurve can also be specified.

#### LOD2

This geometric representation is refined in LOD2 by additional GM_MultiSurface, GM_MultiCurve, and GM_Solid geometries, used for modelling architectural details like roof overhangs, columns, or antennas.

In LOD2 and higher LODs the outer facade of a `Construction Object` can also be differentiated semantically by the classes AbstractConstructiveElement and AbstractInstallation. An AbstractConstructiveElement is a structural part of the `Construction Objects` exterior shell. At LOD2 only the `boundary` surface is displayed. An AbstractInstallation is a non-structural part of the `Construction Objects` exterior shell. As with AbstractConstructiveElement, only the `boundary` surface is displayed at LOD2.

The `boundary` surfaces are abstractThematicSurfaces identified using the `boundary` attribute of AbstractConstructiveElement and AbstractInstallation.

At LOD2 these boundary surfaces are represented as subclasses of the AbstractConstructionSurface class. Thes subclasses represent surfaces with a special function. These subclasses are:

- roof (RoofSurface),
- ground plate (GroundSurface),
- external walls (WallSurface),
- interior walls (InteriorWallSurface)
- interior floors (FloorSurface)
- exterior floors (OuterFloorSurface),
- interior ceiling (CeilingSurface),
- exterior ceiling (OuterCeilingSurface),
- Closure Surface (ClosureSurface).

| NOTE | LOD2 and LOD3 needs some work. It would be helpful if we could identify which Thematic Surface subclasses apply at each LOD. |
|---|---|

### LOD3

In LOD3, the openings in BuildingConstructiveElement objects (doors and windows) can be represented as thematic objects. In LOD4, the highest level of resolution, also the interior of a building, composed of several rooms, is represented in the building model by the class Room. This enlargement allows a virtual accessibility of buildings, e.g. for visitor information in a museum (“Location Based Services“), the examination of accommodation standards or the presentation of daylight illumination of a building. The aggregation of rooms according to arbitrary, user defined criteria (e.g. for defining the rooms corresponding to a certain storey) is achieved by employing the general grouping concept provided by CityGML (cf. chapter 10.3.6). Interior installations of a building, i.e. objects within a building which (in contrast to furniture) cannot be moved, are represented by the class IntBuildingInstallation. If an installation is attached to a specific room (e.g. radiators or lamps), they are associated with the Room class, otherwise (e.g. in case of rafters or pipes) with _AbstractBuilding. A Room may have the attributes class, function and usage whose value can be defined in code lists (chapter 10.3.8 and annex C.1). The class attribute allows a classification of rooms with respect to the stated function, e.g. commercial or private rooms, and occurs only once. The function attribute is intended to express the main purpose of the room, e.g. living room, kitchen. The attribute usage can be used if the way the object is actually used differs from the function. Both attributes can occur multiple times.

The visible surface of a room is represented geometrically as a Solid or MultiSurface. Semantically, the surface can be structured into specialised _BoundarySurfaces, representing floor (FloorSurface), ceiling (CeilingSur-face), and interior walls (InteriorWallSurface). Room furniture, like tables and chairs, can be represented in the CityGML building model with the class BuildingFurniture. A BuildingFurniture may have the attributes class, function and usage. Annexes G.1 to G.6 provide example CityGML documents containing a single building model which is subsequently refined from a coarse LOD0 representation up to a semantically rich and geomet-ric-topologically sound LOD4 model including the building interior.

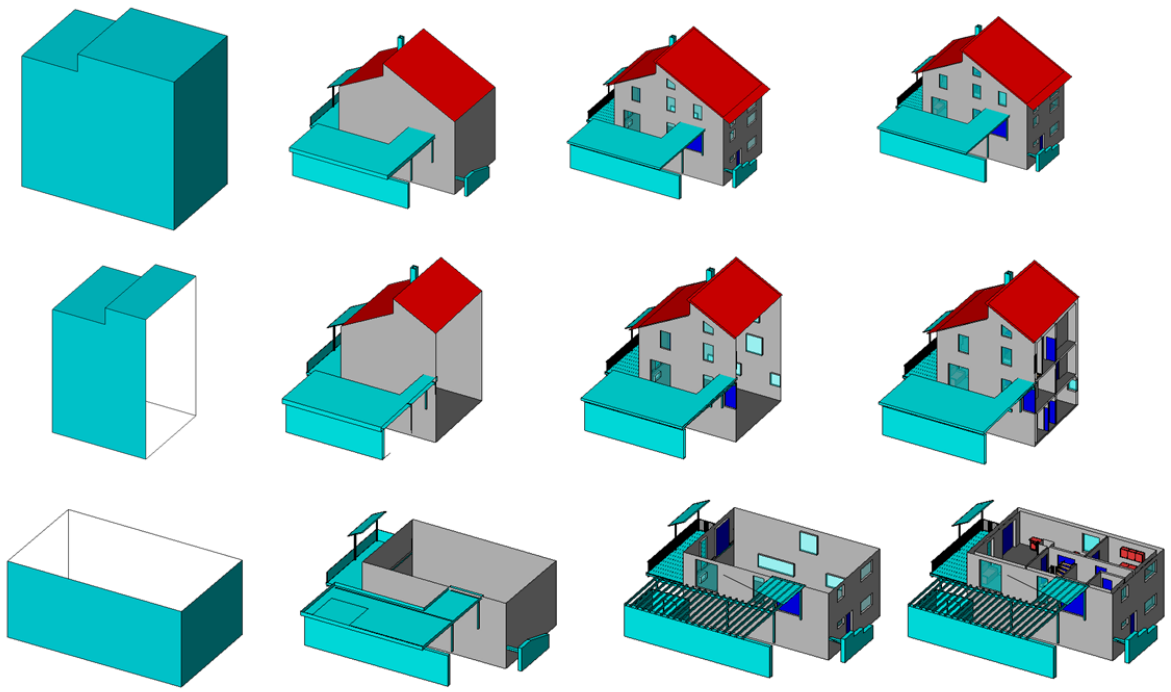


*Figure 64. Building model in LOD1 – LOD4 (source: Karlsruhe Institute of Technology (KIT), courtesy of Franz-Josef Kaiser).<o:p></o:p>*

### 8.7.7. Spaces and Surfaces

The semantic structure of a `Construction Model` is shown in Figure 33. The «FeatureType» classes in this model fall into two categories; `Spaces` and `Surfaces`. `Spaces` are descended from the AbstractOccupiedSpace class and inherit their geometry from this class. `Surfaces` are decended from the AbstractThematicSurface class and inherit their geometry from that class.

#### Occupied Spaces (class AbstractOccupiedSpace)

##### Abstract Construction (class AbstractConstruction)

The AbstractConstruction class is used for free-standing, self-supporting constructions. It represents the construction as a whole. A construction composed of structural segments differing in, for example, the number of storeys or the roof type has to be separated into one Construction having one or more additional Construction Elements (see Figure 43). The geometry and non-spatial properties of the central part of the construction should be represented in the aggregating Construction feature.

The abstract class AbstractConstruction contains properties for `Construction Object` attributes. These properties describe:

- `conditionOfConstruction`:
- `dateOfConstruction:
- `dateOfDemolition`:
- `constructionEvent`:
- `elevation`:
- `height`:
- `occupancy`:
- `boundary`: An association with the AbstractThematicSurface which defines the boundary of this `Construction Object`.

AbstractConstruction is a subclass of AbstractOccupiedSpace and inherits the geometry properties from that class.

##### Other Construction (class OtherConstruction)

***Under construction***

##### Abstract Construction Element (class AbstractConstructionElement)

The class AbstractConstructionElement is derived from AbstractOccupiedSpace. Subclasses of this class are used to identify the constiuent parts of a `Construc tion Object` (see Figure 43). A `construction element` object should be uniquely related to exactly one `construction` or `construction element` object.

AbstractConstructionElement includes properties which describe:

- `isStructualElement`:

- `filling`:
- `boundary`: An association with the AbstractThematicSurface which defines the boundary of this `Construction Element Object`.

AbstractConstructionElement is a subclass of AbstractOccupiedSpace and inherits the geometry properties from that class.

*Figure 65. Examples of buildings consisting of one and two building parts (source: City of Coburg)*

#### Abstract Furniture (class AbstractFurniture)

`Construction Objects` may have `Furnitures` and `Installations`. An AbstractFurniture is a movable part of a `Construction`, such as a chair or table. An AbstractFurniture object should be uniquely related to exactly one `Construction Object`. Its geometry may be represented by an explicit geometry or an ImplicitGeometry object. Following the concept of ImplicitGeometry the geometry of a prototype bridge furniture is stored only once in a local coordinate system and referenced by other `furniture` features.

#### Abstract Installation (class AbstractInstallation)

An AbstractInstallation is an object inside a `Construction Object` with a specialised function or semantic meaning. In contrast to `Furniture`, AbstractInstallation are permanently attached to the structure and cannot be moved. Examples for AbstractInstallation are stairways, railings and heaters. An AbstractInstallation object should be uniquely related to exactly one `Construction Object`. Its geometry may be represented by an explicit geometry or an ImplicitGeometry object. Following the concept of ImplicitGeometry the geometry of a prototype bridge furniture is stored only once in a local coordinate system and referenced by other `furniture` features.

An AbstractInstallation optionally has attributes `class`, `function` and `usage`. The attribute `class`, which can only occur once, represents a general classification of the `Installation Object`. With the attributes `function` and `usage`, nominal and real functions of a `Installation Object` can be described. For all three attributes the list of feasible values can be specified in a code list.

AbstractInstallation is a subclass of AbstractOccupiedSpace and inherits the geometry properties from that class.

#### Abstract Filling Element (class AbsrtactFillingElement)

The class AbsrtactFillingElement is the abstract base class for semantically describing openings like doors or windows in outer or inner boundary surfaces like walls and roofs. Openings only exist in models of LOD3. Each _Opening is associated with a GM_MultiSurface geometry. Alternatively, the geometry may be given as ImplicitGeometry object. Following the concept of ImplicitGeometry the geometry of a prototype opening is stored only once in a local coordinate system and referenced by other opening features.

======= Window (class Window)

The class Window is used for modelling windows in the exterior shell of a `Construction Object`, or between adjacent `Construction Elements`. The formal difference between the classes Window and Door is that – in normal cases – Windows are not specifically intended for the transit of people or vehicles.

======= Door (class Door)

The class Door is used for modelling doors in the exterior shell of a `Construction Object`, or between adjacent `Construction Elements`. Doors can be used by people to enter or leave a `Construction Object` or `Construction Element`. In contrast to a ClosureSurface a door may be closed, blocking the transit of people. A Door may be assigned zero or more addresses. The corresponding Address-PropertyType is defined within the CityGML core module.

### Thematic Surfaces (class AbstractThematicSurface)

***Under construction***

#### Filling Surfaces (class AbstractFillingSurface)

***Under construction***

======= Door Surface (class DoorSurface)

***Under construction***

======= Window Surface (class WindowSurface)

***Under construction***

#### Construction Surfaces (class AbstractConstructionSurface)

***Under construction***

======= Roof (class RoofSurface),

***Under construction***

======= Ground Plate (class GroundSurface)

The ground plate of a `Construction` or `Constructive Element` is modelled by the class GroundSurface. The polygon defining the ground plate is congruent with the construction's footprint. However, the

surface normal of the ground plate is pointing downwards.

======= External Walls (class WallSurface),

All parts of the `Construction` facade belonging to the outer shell can be modelled by the class WallSurface.

======= Interior Walls (class InteriorWallSurface)

***Under construction***

======= Interior Floors (class FloorSurface)

***Under construction***

======= Exterior Floors (class OuterFloorSurface),

A mostly horizontal surface belonging to the outer shell of the `Construction Object` and with the orientation pointing upwards can be modeled as an OuterFloorSurface.

======= Interior Ceiling (class CeilingSurface),

***Under construction***

======= Exterior Ceiling (class OuterCeilingSurface),

A mostly horizontal surface belonging to the outer shell of the `Construction Object` and having the orientation pointing downwards can be modeled as an OuterCeilingSurface.

**Closure Surface (class ClosureSurface).**

An opening in a `Construction Object` not filled by a door or window can be sealed by a virtual surface called a ClosureSurface. Hence, `Constructions` with open sides can be virtually closed in order to be able to compute their volume.

## 8.7.8. UML Model

The UML diagram of the Construction module is depicted in Figure 33.

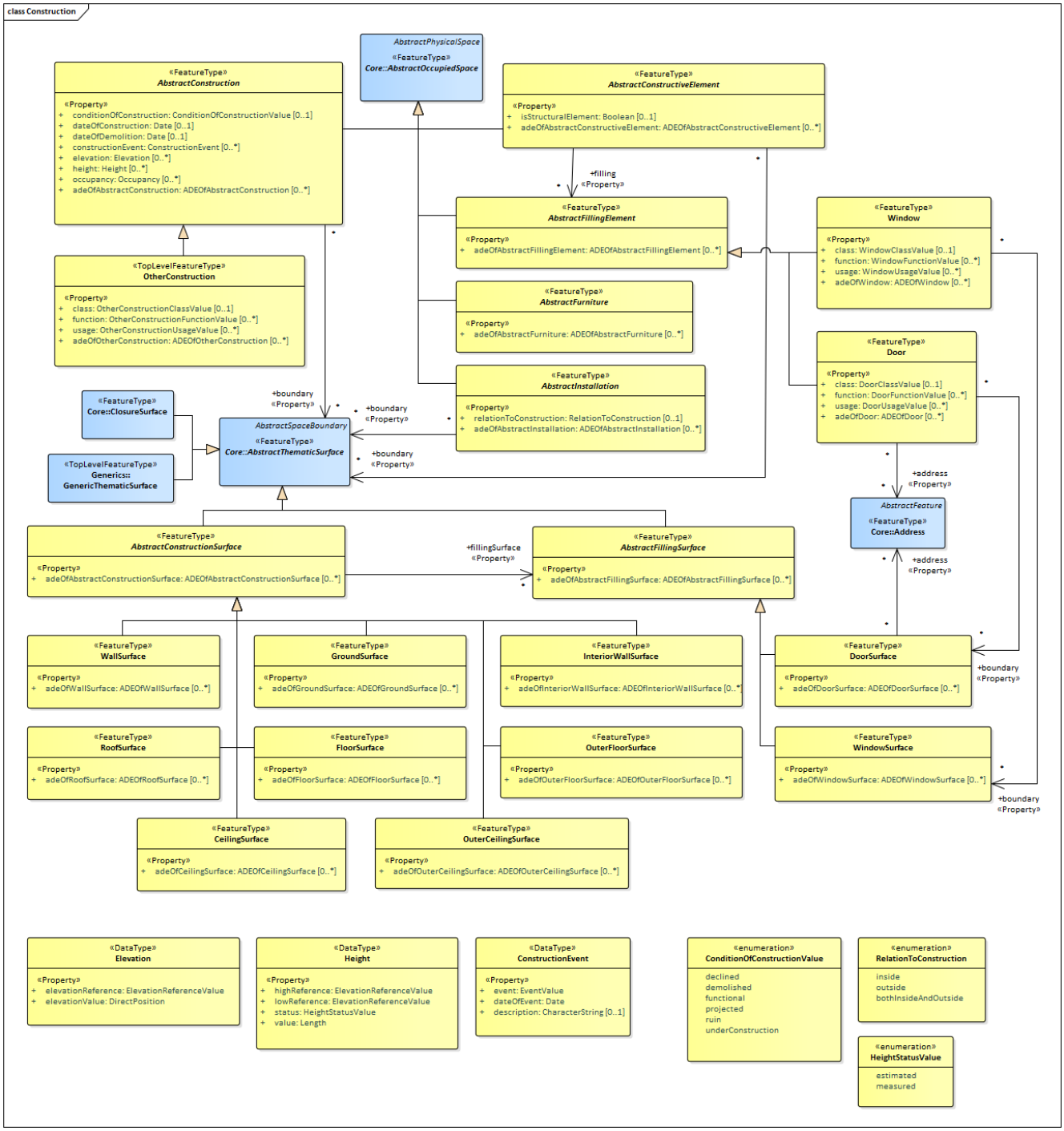



*Figure 66. UML diagram of the Construction Model.*

The ADE data types provided for the Construction module are illustrated in Figure 34.

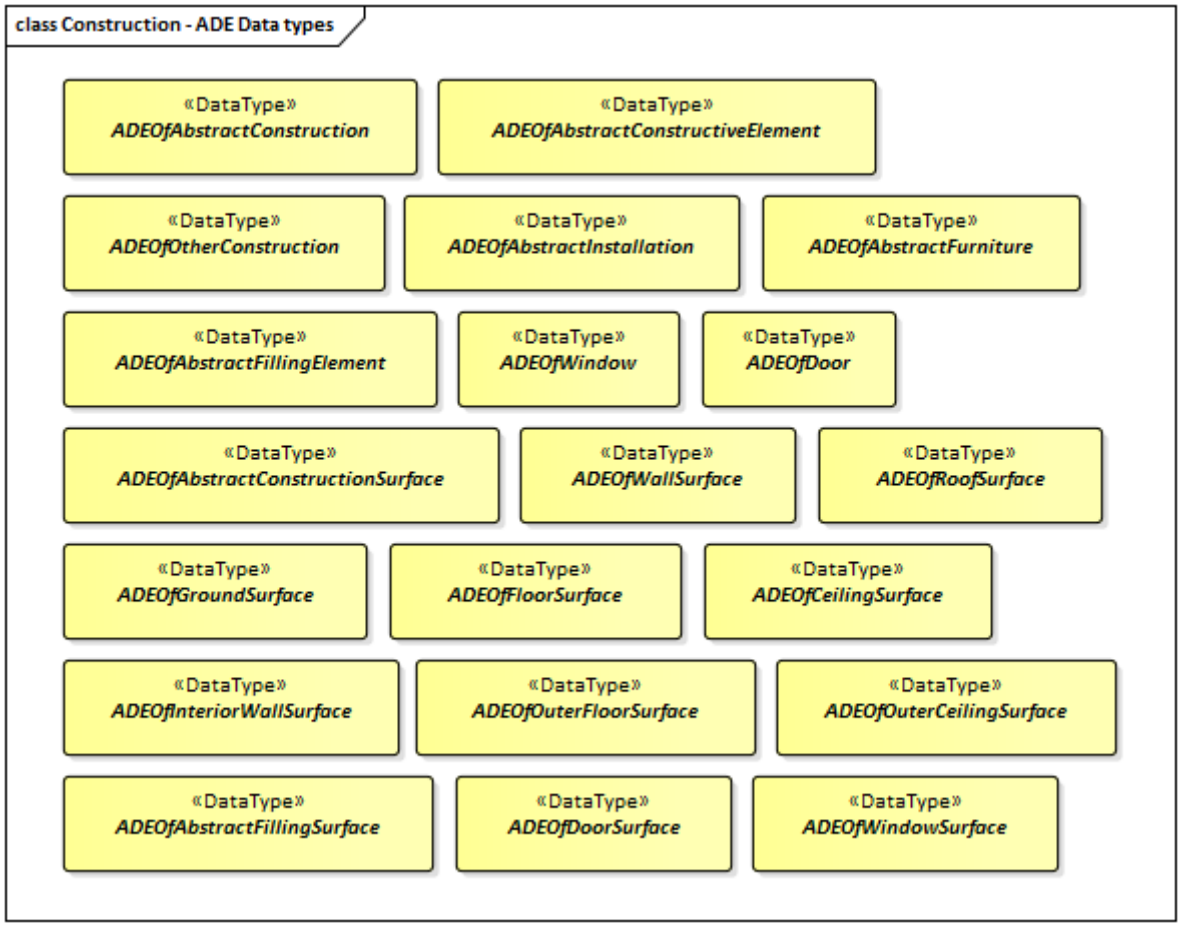



*Figure 67. ADE classes of the CityGML Construction module.*

The Code Lists provided for the Construction module are illustrated in Figure 35.

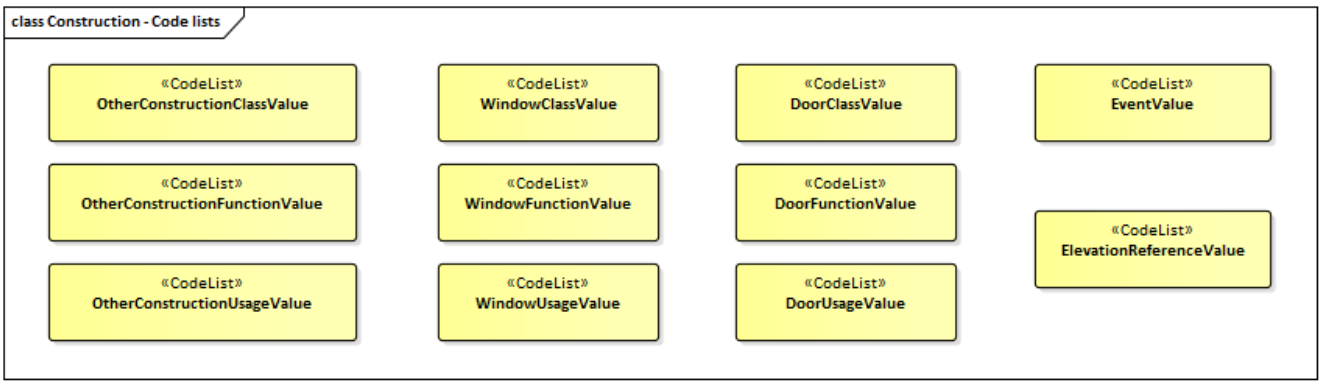



*Figure 68. Codelists from the CityGML Construction module.*

### 8.7.9. Examples

# 8.8. Dynamizer

| Contributors |
| --- |
| C. Heazel - first draft |

### 8.8.1. Synopsis

The Dynamizer module provides the concepts that enable representation of time-varying data for city object properties. In particular, they allow sensor readings to be integrated into 3D city models.

Dynamizers inject timeseries data for an individual attribute of the city object. In order to represent dynamic (time-dependent) variations of its value, the timeseries data overrides the static value of the attribute.

### 8.8.2. Key Concepts

**Dynamizer**: An object that injects timeseries data for an individual attribute of the city object in which it is included.

**Timeseries**: The period of time designated by a first and last time stamp. A time series does not have to be continuous. In that case, each continuous period within the time series is itself a time series. A non-continuous time series is essentially a collection of time series.

### 8.8.3. Discussion

The Dynamizer module provides the concepts that enable representation of time-varying data for city object properties as well as for integrating sensors with 3D city models. Dynamizers are objects that inject timeseries data for an individual attribute of the city object in which the Dynamizer is included. In order to represent dynamic (time-dependent) variations of its value, the timeseries data overrides the static value of the referenced city object attribute.

The dynamic values may be given by retrieving observation results directly from external sensor/IoT services using a sensor connection (e.g. OGC SensorThings API, Sensor Observation Service, or other sensor data platforms including MQTT). Alternatively, the dynamic values may be provided as atomic timeseries that represent time-varying data of a specific data type for a single contiguous time interval. The data can be provided in:

- external tabulated files, such as CSV or Excel sheets,
- external files that format timeseries data according to the OGC TimeseriesML Standard or the OGC Observations & Measurements standards,
- or inline as embedded time-value-pairs.

Furthermore, timeseries data can also be aggregated to form composite timeseries with non-overlapping time intervals.

By using the Dynamizer module, fast changes over a short or longer time period with respect to cities and city models can be represented. This includes:

- variations of spatial properties such as change of a feature's geometry, both in respect to shape and to location (e.g. moving objects),
- variations of thematic attributes such as changes of physical quantities like energy demands, temperatures, solar irradiation, traffic density, pollution concentration, or overpressure on building walls,

- and variations with respect to sensor or real-time data resulting from simulations or measurements.

### 8.8.4. UML Model

The UML diagram of the Dynamizer module is depicted in Figure 69.

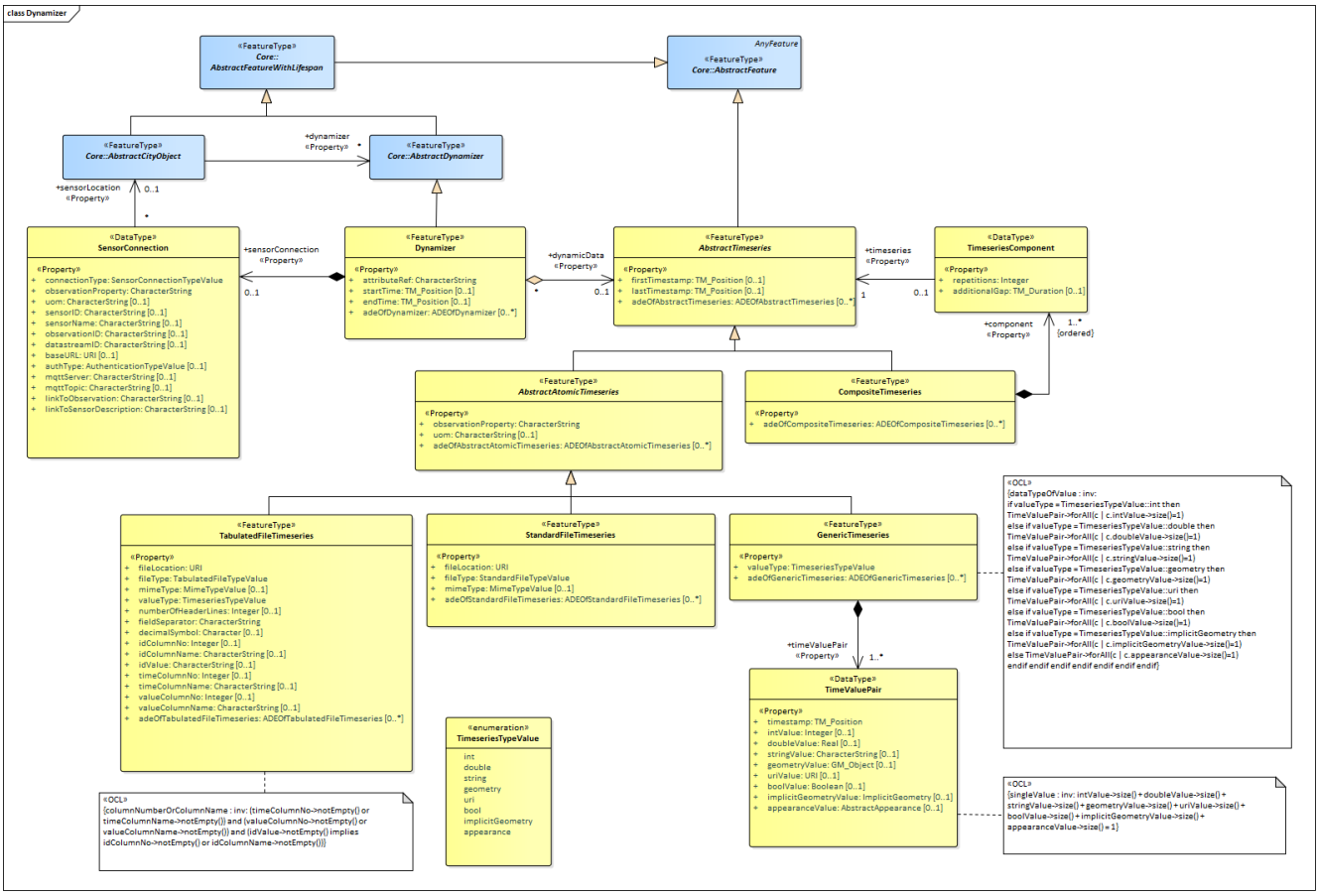



*Figure 69. UML diagram of the Dynamizer Model.*

The ADE data types provided for the Dynamizer module are illustrated in Figure 70.

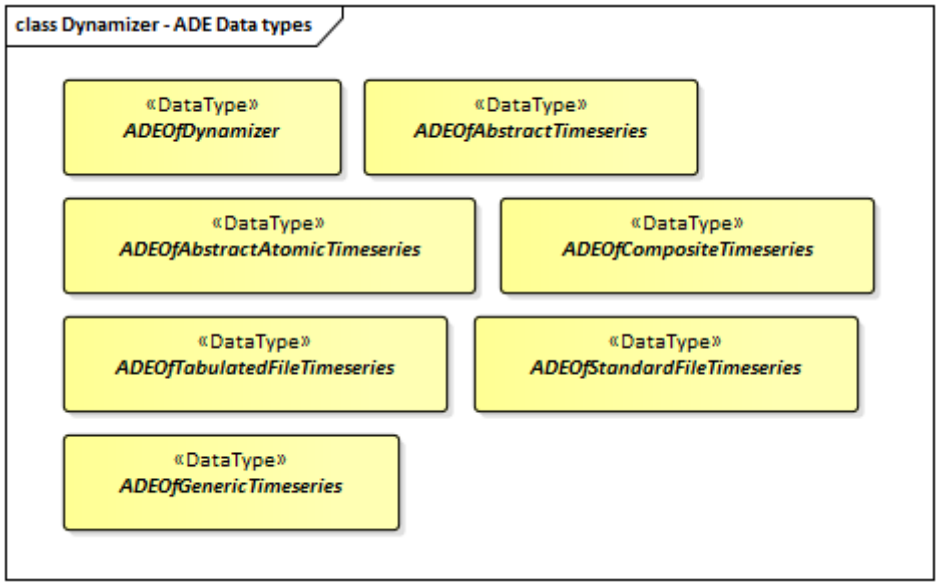



*Figure 70. ADE classes of the CityGML Dynamizer module.*

The Code Lists provided for the Dynamizer module are illustrated in Figure 71.

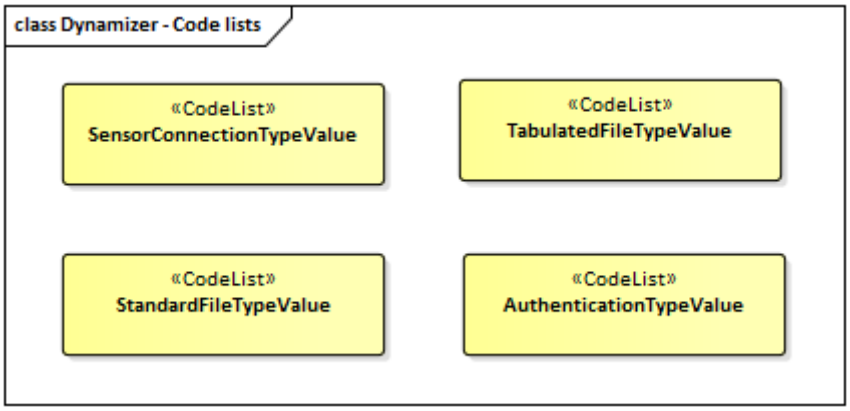



*Figure 71. Codelists from the CityGML Dynamizer module.*

### 8.8.5. Examples

# 8.9. Generics

| Contributors |
| --- |
| C. Heazel - first draft |
| T. Kutzner - update content from v2 to v3 |

### 8.9.1. Synopsis

The *Generics* module supports application-specific extensions to the CityGML conceptual model. These extensions may be used to model and exchange additional attributes and features not covered by the predefined thematic classes of CityGML. Generic extensions shall only be used if appropriate thematic classes or attributes are not provided by any other CityGML module.

### 8.9.2. Key Concepts

**GenericLogicalSpace**: A space that is not represented by any explicitly modelled AbstractLogicalSpace subclass within CityGML. (See Discussion for further details.)
A type of AbstractLogicalSpace.

**GenericOccupiedSpace**: A space that is not represented by any explicitly modelled AbstractOccupiedSpace subclass within CityGML. (See Discussion for further details.)
A type of AbstractOccupiedSpace.

**GenericUnoccupiedSpace**: A space that is not represented by any explicitly modelled AbstractUnoccupiedSpace subclass within CityGML. (See Discussion for further details.)
A type of AbstractUnoccupiedSpace.

**GenericThematicSurface**: A surface that is not represented by any explicitly modelled AbstractThematicSurface subclass within CityGML. (See Discussion for further details.)
A type of AbstractThematicSurface.

**GenericAttributeSet**: A named collection of generic attributes. (See Discussion for further details.)
A type of AbstractGenericAttribute.

**StringAttribute**, **IntAttribute**, **DoubleAttribute**, **DateAttribute**, **UriAttribute**, **MeasureAttribute**, **CodeAttribute**: Data types used to define generic attributes of the types *String*, *Int*, *Double*, *Date*, *URI*, *Measure*, or *Code*, respectively. (See Discussion for further details.)
A type of AbstractGenericAttribute.

### 8.9.3. Discussion

The *Generics* module provides the representation of generic city objects. These are city objects that are not covered by any explicitly modelled thematic class within the CityGML conceptual model. The Generics module also provides the representation of generic attributes, i.e. attributes that are not explicitly represented in the CityGML conceptual model.

In order to avoid problems concerning semantic interoperability, generic city objects and generic attributes shall only be used if appropriate thematic classes and attributes are not provided by any other CityGML module.

In accordance with the CityGML Space concept defined in the Core module, generic city objects can be represented as generic logical spaces, generic occupied spaces, generic unoccupied spaces, and generic thematic surfaces. In this way, spaces and surfaces can be defined that are not represented in the CityGML Conceptual Model by any explicitly modelled subclass of the classes AbstractLogicalSpace, AbstractOccupiedSpace, AbstractUnoccupiedSpace or AbstractThematicSurface. Generic city objects are represented in the UML model by the top-level feature types *GenericLogicalSpace*, *GenericOccupiedSpace*, *GenericUnoccupiedSpace* and *GenericThematicSurface*.

Generic attributes are defined as name-value pairs and are always associated with a city object. Generic attributes can be of type *String*, *Integer*, *Double*, *Date*, *URI*, *Measure*, and *Code*. In addition, generic attributes can be grouped under a common name as generic attribute sets. In the Core module, the class AbstractCityObject is related to the data type *AbstractGenericAttribute* via the role name *genericAttribute*. This means that each thematic subclass of AbstractCityObject inherits this relation and, thus, the possibility that an arbitrary number of generic attributes can be assigned to each thematic subclass in order to represent additional properties of features not represented by the properties that are explicitly defined in the UML model. This also means that the *Generics* module has a deliberate impact on all CityGML modules that define thematic subclasses of the class AbstractCityObject.

#### GenericLogicalSpace, GenericOccupiedSpace, GenericUnoccupiedSpace, and GenericThematicSurface

These classes are used to define generic city objects by representing them as generic spaces and surfaces in accordance with the CityGML 3 Space concept. All classes represent top level feature types and they are derived from corresponding abstract classes in the Core module:

- *GenericLogicalSpace* is to be used for generic city objects that represent *logical* spaces. The class is derived from the class AbstractLogicalSpace.
- *GenericOccupiedSpace* is to be used for generic city objects that represent *physically occupied*

spaces. The class is derived from the class AbstractOccupiedSpace.

- *GenericUnoccupiedSpace* is to be used for generic city objects that represent *physically unoccupied* spaces. This class is derived from AbstractUnoccupiedSpace.
- *GenericThematicSurface* is to be used for representing surfaces that bound a space. This class is derived from the class AbstractThematicSurface.

Generic spaces and surfaces provide the properties *class*, *function* and *usage* whose values are defined in external code lists:

- The *class* property indicates the specific type of the space or surface, i.e. it allows for object classification within a thematic area (e.g. water pipes in the field of infrastructure or city zones in the field of city planning). This property is optional and can only be provided once.
- The *function* property specifies the intended purpose(s) of the space or surface (e.g. water supply or residential zones). This property is optional and can be provided multiple times.
- The *usage* property specifies the actual usage of the space or surface in cases where the way the object is actually used differs from the intended purpose of the object. This property is optional and can occur multiple times.

The geometry of a generic space or surface can be any of the geometries defined in the Core module for the classes AbstractSpace, AbstractPhysicalSpace, and AbstractThematicSurface. The generic spaces and surfaces inherit the geometries defined for these abstract classes, since they are subclasses thereof. This means:

- All generic spaces can be spatially represented as single point in LOD0, as multi-surface in LOD0/2/3, as solid in LOD1/2/3, and as multi-curve in LOD2/3. Absolute coordinates according to the reference system of the city model must be given for these geometries.
- Generic occupied and unoccupied spaces can, in addition, be complemented with a *TerrainIntersectionCurve* in LOD1/2/3, in order to specify the exact intersection of the DTM with the 3D geometry of these spaces. This is important for 3D visualisation but also for certain applications like driving simulators. For instance, when a city wall (e.g. the Great Wall of China) is to be represented as a *GenericOccupiedSpace*, a smooth transition between the DTM and the city wall needs to be ensured. Above that, both spaces can also be represented by point clouds.
- Generic occupied spaces can, in addition, be spatially represented by an ImplicitGeometry. In this case, a reference point (anchor point) of the object and optionally a transformation matrix must be given. In order to compute the actual location of the object, the transformation of the local coordinates into the reference system of the city model must be processed and the anchor point coordinates must be added. The shape of an ImplicitGeometry can be given as an external resource with a proprietary format, e.g. a VRML or DXF file from a local file system or an external web service. Alternatively the shape can be specified as a 3D geometry with local cartesian coordinates using the property *relativeGeometry*.

#### StringAttribute, IntAttribute, DoubleAttribute, DateAttribute, UriAttribute, MeasureAttribute, and CodeAttribute

These data types are used to define generic attributes of the types *String*, *Integer*, *Double*, *Date*, *URI*, *Measure*, or *Code*, respectively. The data types are derived from the abstract data type *AbstractGenericAttribute* that is defined in the Core module.

Each generic attribute has two mandatory properties:

1. *name* for specifying the name of the generic attribute. The name can be freely chosen.
2. *value* for specifying the value of the generic attribute. The type of the value may be *String*, *Integer*, *Double* (floating point number), *Date*, *URI*, *Measure*, or *Code*, depending on the selected subclass.

A *MeasureAttribute* facilitates the representation of measured values. Its value is of the structured type *Measure* which can additionally provide the units of measure via an optional attribute *uom* (units of measure).

**GenericAttributeSet**

Generic attributes can be grouped under a common name using a *GenericAttributeSet*. *GenericAttributeSet* is a data type that is derived from the data type *AbstractGenericAttribute* defined in the Core module; thus, generic attribute sets represent generic attributes themselves. This means that generic attribute sets may also be contained in other generic attribute sets, facilitating a recursive nesting of arbitrary depth.

The value of a generic attribute set is implicitly the set of contained generic attributes to which *GenericAttributeSet* relates via the role name *genericAttribute*. The *name* property specifies the name for the entire generic attribute set. The optional *codeSpace* property (of type *URI*) can be used to associate the generic attribute set with an authority, e.g. the organisation or community that defines and maintains the generic attribute set and its contained attributes. In this way, generic attribute sets can be clearly distinguished from each other even if they share the same name.

### 8.9.4. UML Model

The UML diagram of the *Generics* module is depicted in Figure 72.

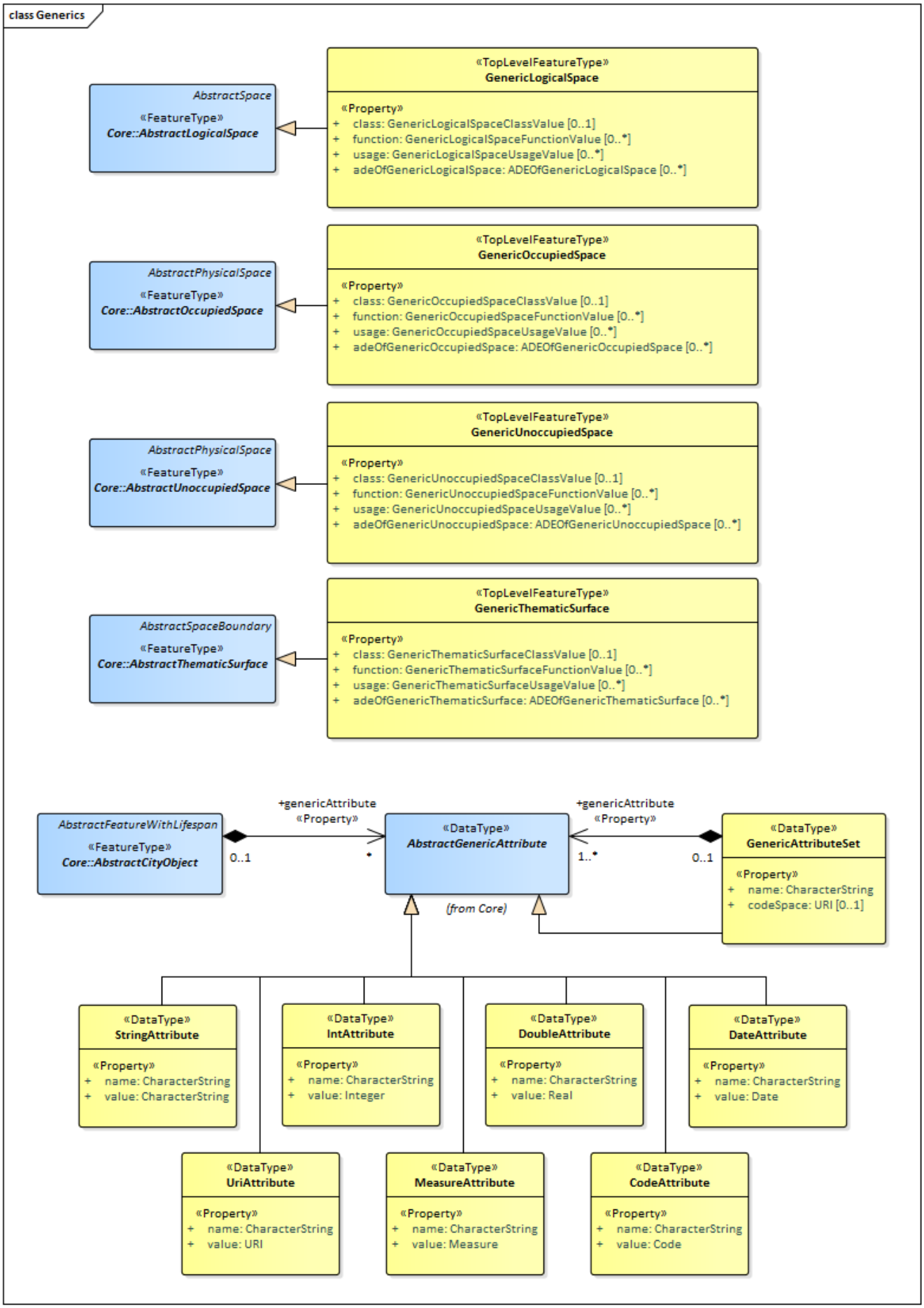



*Figure 72. UML diagram of the Generics Model.*

The ADE data types provided for the *Generics* module are illustrated in Figure 73.

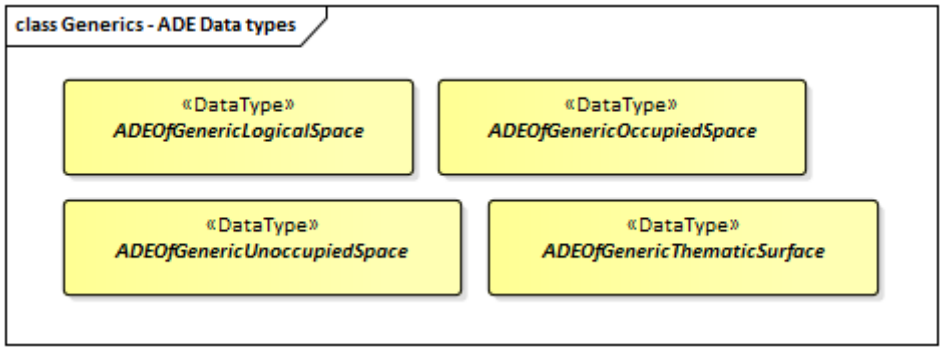



*Figure 73. ADE classes of the CityGML Generics module.*

The Code Lists provided for the *Generics* module are illustrated in Figure 74.

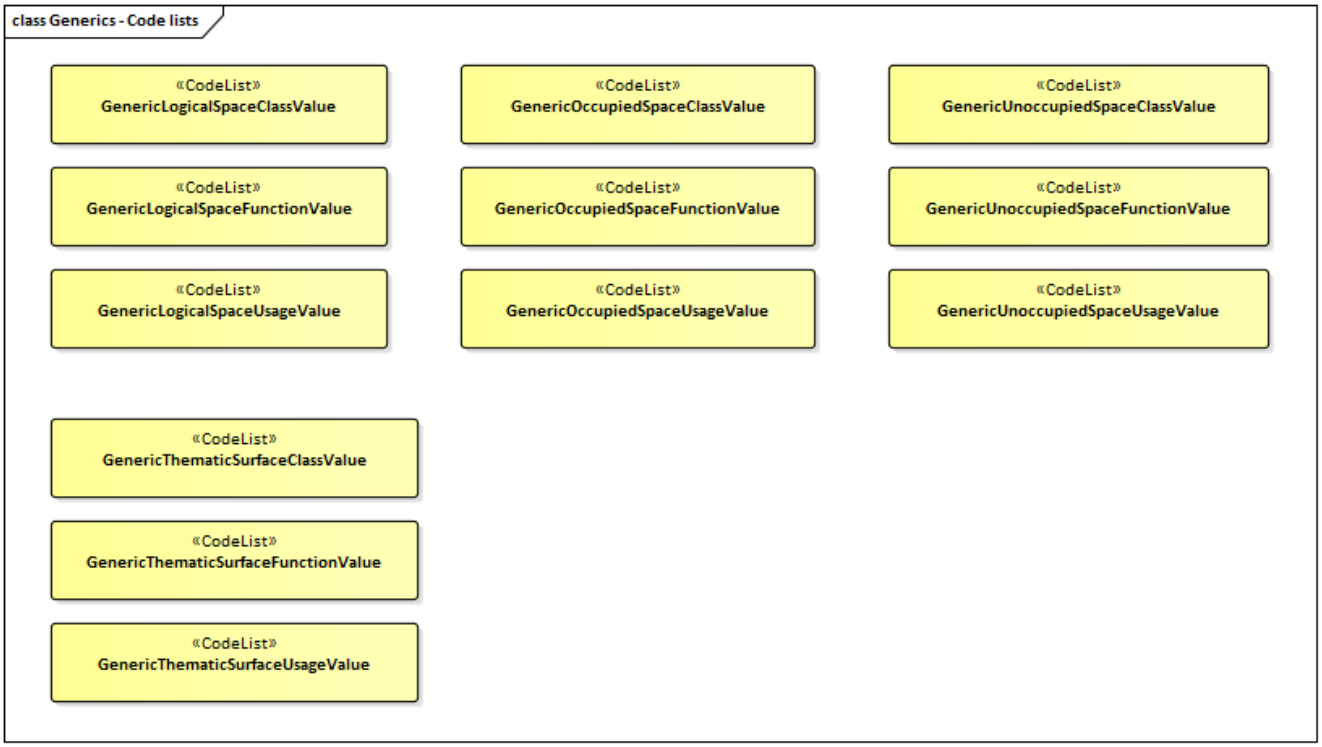



*Figure 74. Codelists from the CityGML Generics module.*

### 8.9.5. Examples

# 8.10. Land Use

| Contributors |
| --- |
| C. Heazel - first draft |

### 8.10.1. Synopsis

The LandUse module defines objects that can be used to describe areas of the earth's surface dedicated to a specific land use or having a specific land cover with or without vegetation, such as sand, rock, mud flats, forest, grasslands, or wetlands (i.e. the physical appearance).

### 8.10.2. Key Concepts

**LandUse**: An area of the earth's surface dedicated to a specific land use or having a specific land cover with or without vegetation, such as sand, rock, mud flats, forest, grasslands, or wetlands.
A type of AbstractThematicSurface.

### 8.10.3. Discussion

LandUse objects can be used to describe areas of the earth's surface dedicated to a specific land use, but also to describe areas of the earth's surface having a specific land cover with or without vegetation, such as sand, rock, mud flats, forest, grasslands, etc (i.e. the physical appearance).

Land use and land cover are different concepts. The first describes human activities on the earth's surface, the second describes its physical and biological cover. However, the two concepts are interlinked and often mixed in practice. Land use objects in CityGML support both concepts: They can be employed to represent parcels, spatial planning objects, recreational objects, and objects describing the physical characteristics of an area in 3D.

Land use objects are represented in the UML model by the top-level feature type LandUse, which is also the only class of the LandUse module.

Every LandUse object may have the attributes `class`, `function`, and `usage`. The `class` attribute is used to represent the classification of land use objects, like settlement area, industrial area, farmland etc., and can occur only once. The possible values can be specified in a code list (cf. [ug_codelist_section]). The attribute `function` defines the purpose of the object or their nature, like e.g. cornfield or heath, while the attribute `usage` can be used, if the way the object is actually used differs from the function. Both the `function` and `usage` attributes can occur multiple times.

The LandUse object is defined for all LOD 0-3 and may have different geometries in any LOD. The surface geometry of a LandUse object is required to have 3D coordinate values. It must be a GM_MultiSurface, which might be assigned appearance properties like textures or colors (see [ug_appearance_section].

LandUse objects can be employed to establish a coherent geometric/semantical tesselation of the earth's surface. In this case topological relations between neighbouring LandUse objects should be made explicit by defining the space boundary only once and by referencing it in the corresponding City Object.

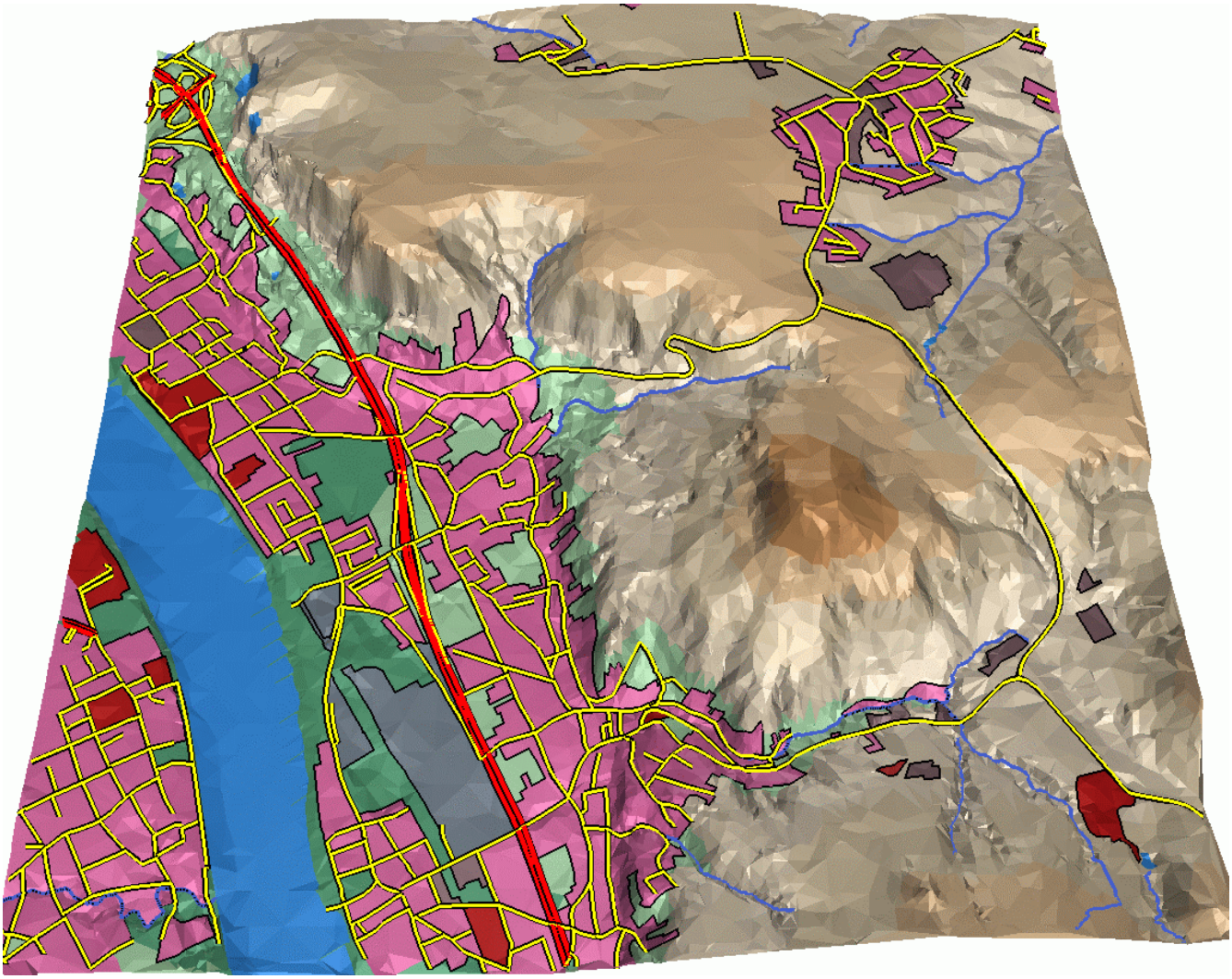


*Figure 75. LOD0 regional model consisting of land use objects in CityGML (source: IGG Uni Bonn).*

### 8.10.4. UML Model

The UML diagram of the LandUse module is depicted in Figure 76.

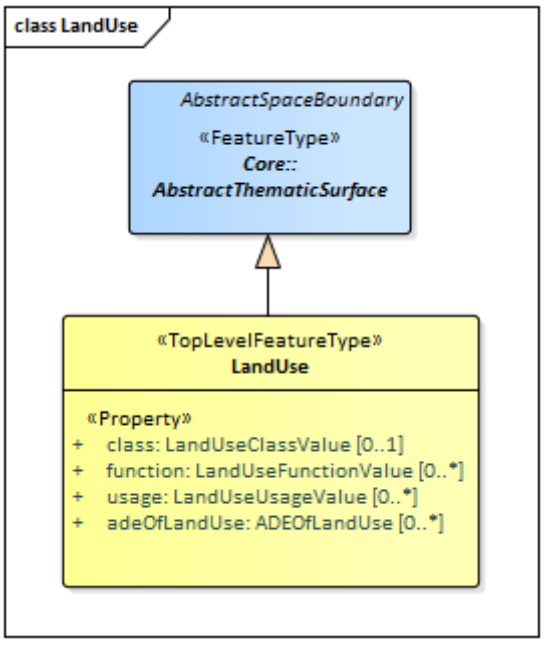



*Figure 76. UML diagram of the Land Use Model.*

The ADE data types provided for the Land Use module are illustrated in Figure 77.

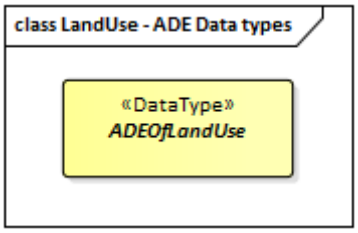



*Figure 77. ADE classes of the CityGML Land Use module.*

The Code Lists provided for the Land Use module are illustrated in Figure 78.

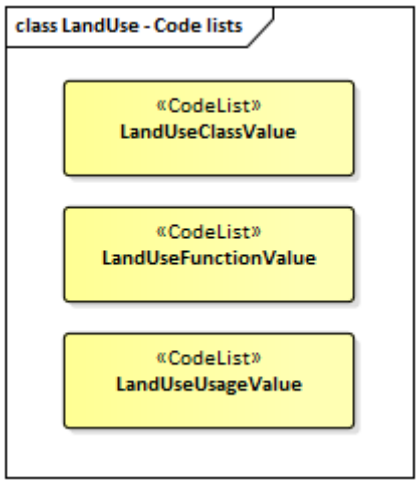



*Figure 78. Codelists from the CityGML Land Use module.*

### 8.10.5. Examples

## 8.11. Point Cloud

| Contributors |
| --- |
| C. Heazel - first draft |

### 8.11.1. Synopsis

The PointCloud module specifies how to encode the geometry of physical spaces and of thematic surfaces as 3D point clouds. In this way, the building hull, a room within a building or a single wall surface can be spatially represented by a point cloud only.

### 8.11.2. Key Concepts

**PointCloud**: A PointCloud is an unordered collection of points that is a sampling of the geometry of a space or space boundary.
A type of AbstractPointCloud.

### 8.11.3. Discussion

The PointCloud module specifies how to encode the geometry of physical spaces and of thematic surfaces as 3D point clouds. In this way, the building hull, a room within a building or a single wall surface can be spatially represented by a point cloud only. The same applies to all other thematic feature types including transportation objects, vegetation, city furniture, etc. Point clouds can either be provided inline within a CityGML file or as reference to external point cloud files of common file types such as LAS or LAZ.

Point clouds are represented in the UML model by the feature type *PointCloud*, which is also the only class of the PointCloud module.

### 8.11.4. UML Model

The UML diagram of the PointCloud module is depicted in Figure 79.

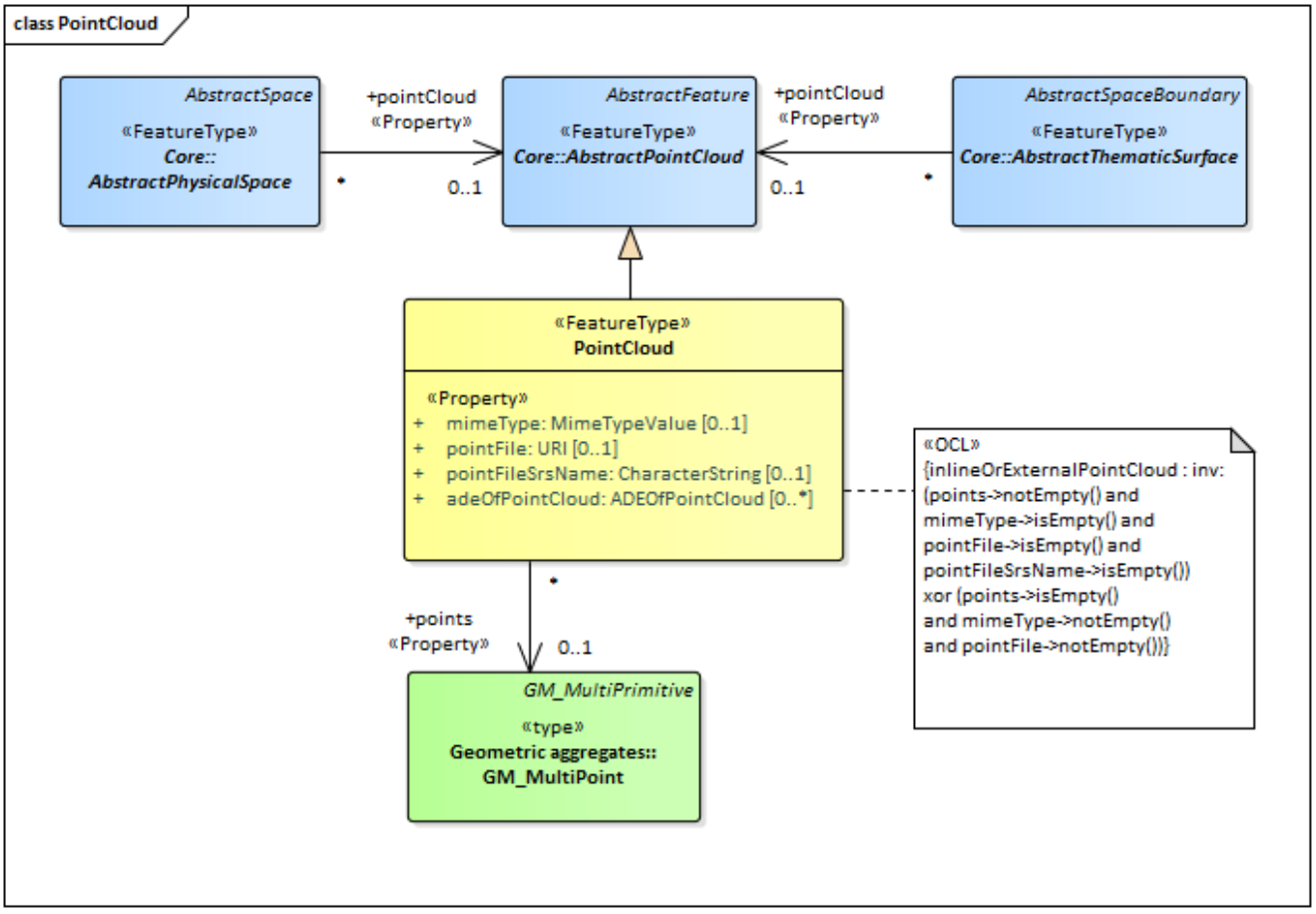



*Figure 79. UML diagram of the Point Cloud Model.*

The ADE data types provided for the Point Cloud module are illustrated in Figure 80.

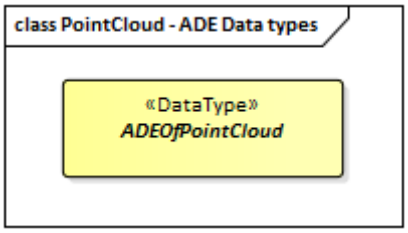



*Figure 80. ADE classes of the CityGML Point Cloud module.*

### 8.11.5. Examples

# 8.12. Relief

| Contributors |
| --- |
| C. Heazel - first draft |

### 8.12.1. Synopsis

The Relief module supports representation of the terrain. CityGML supports terrain representations at different levels of detail, reflecting different accuracies or resolutions. Terrain may be specified as a regular raster or grid, as a TIN, by break lines, and/or by mass points.

### 8.12.2. Key Concepts

**ReliefFeature**: A collection of terrain components representing the Earth's surface, also known as the Digital Terrain Model.
A type of AbstractSpaceBoundary.

**AbstractReliefComponent**: An element of the terrain surface - either a TIN, a raster or grid, mass points or break lines.
A type of AbstractSpaceBoundary.

**TINRelief**: A terrain component as a triangulated irregular network.
A type of AbstractReliefComponent.

**MassPointRelief**: A terrain component as a collection of 3D points.
A type of AbstractReliefComponent.

**BreaklineRelief**: A terrain component with 3D lines. These lines denote break lines or ridge/valley lines.
A type of AbstractReliefComponent.

**RasterRelief**: A terrain component as a regular raster or grid.
A type of AbstractReliefComponent.

### 8.12.3. Discussion

An essential part of a city model is the terrain. The Digital Terrain Model (DTM) of CityGML is provided by the thematic extension module Relief. In CityGML, the terrain is represented by the class ReliefFeature in LOD 0-3. A ReliefFeature consists of one or more entities of the class AbstractReliefComponent. Its validity may be restricted to a certain `extent` defined by an optional GM_Surface. As ReliefFeature and AbstractReliefComponent are derivatives of AbstractCityObject, the corresponding attributes and relations are inherited.

The class ReliefFeature is associated, through AbstractReliefComponent, with different concepts of terrain representations which can coexist. The terrain may be specified:

- as a regular raster or grid (RasterRelief),

- as a TIN (Triangulated Irregular Network - TINReflief),
- by break lines (BreaklineRelief), or
- by mass points (MasspointRelief).

The four types are implemented by the corresponding ISO geometries:

- grids by CV_DiscreteGridPointCoverage,
- break lines by GM_MultiCurve,
- mass points by GM_MultiPoint and
- TINs either by GM_TriangulatedSurface or by GM_Tin.

In case of GM_TriangulatedSurfaces, the triangles are given explicitly while in case of GM_Tin only 3D points are represented, where the triangulation can be reconstructed by standard methods (Delaunay triangulation, cf. Okabe et al. 1992). Break lines are represented by 3D curves. Mass points are simply a set of 3D points.

In a CityGML dataset the four terrain types may be combined in different ways, yielding a high flexibility. First, each type may be represented in different levels of detail, reflecting different accuracies or resolutions. Second, a part of the terrain can be described by the combination of multiple types, for example by a raster and break lines, or by a TIN and break lines. In this case, the break lines must share the geometry with the triangles. Third, neighboring regions may be represented by different types of terrain models. To facilitate this combination, each terrain object is provided with a spatial attribute denoting its extent of validity (Figure 36). In most cases, the extent of validity of a regular raster dataset corresponds to its bounding box. This validity extent is represented by a 2D footprint polygon, which may have holes. This concept enables, for example, the modelling of a terrain by a coarse grid, where some distinguished regions are represented by a detailed, high-accuracy TIN. The boundaries between both types are given by the extent attributes of the corresponding terrain objects.

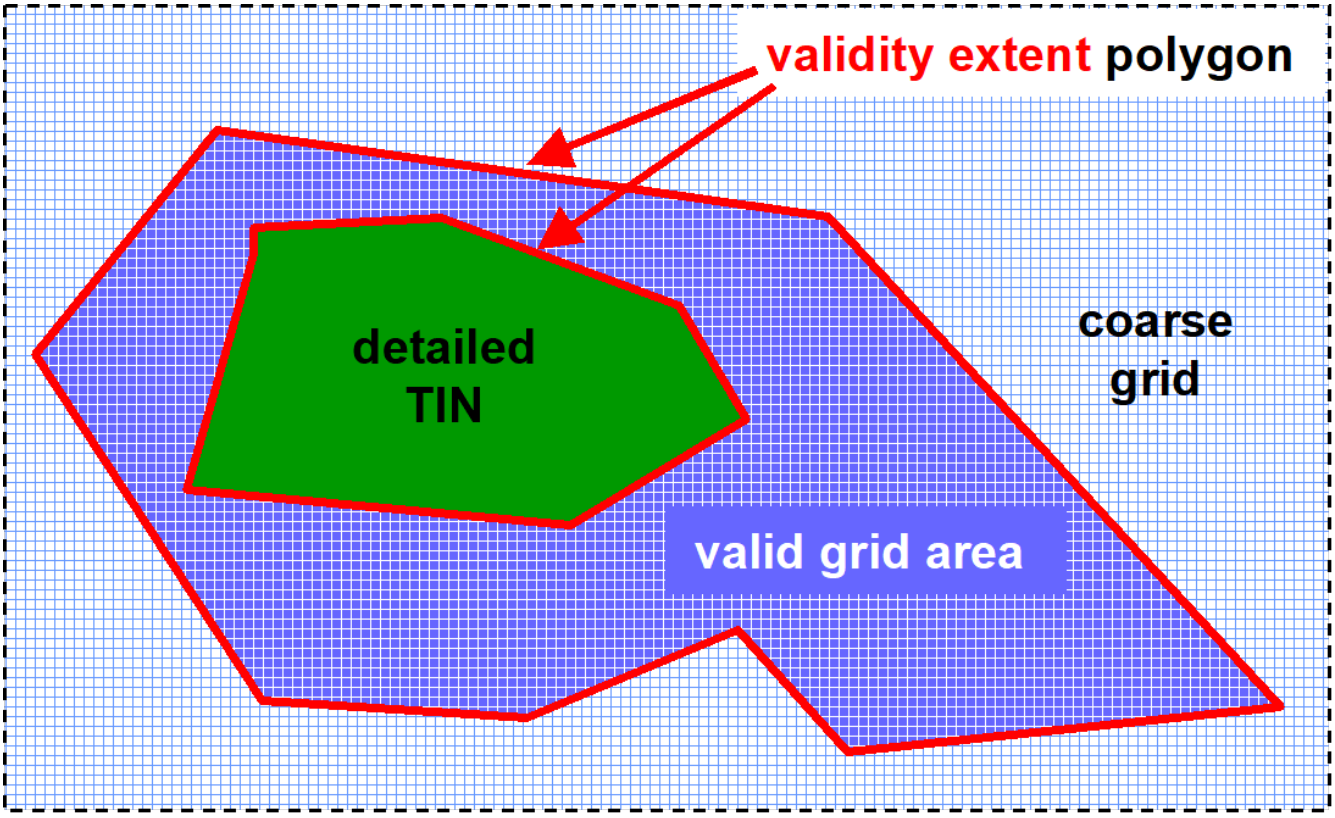



*Figure 81. Nested DTMs in CityGML using validity extent polygons (graphic: IGG Uni Bonn).*

Accuracy and resolution of the DTM are not necessarily dependent on features of other CityGML extenstion modules such as building models. Hence, there is the possibility to integrate building models with higher LOD to a DTM with lower accuracy or resolution.

This approach interacts with the concept of TerrainIntersectionCurve TIC. The TIC can be used like break lines to adjust the DTM to different features such as buildings, bridges, or city furnitures, and hence to ensure a consistent representation of the DTM. If necessary, a retriangulation may have to be processed. A TIC can also be derived by the individual intersection of the DTM and the corresponding feature.

ReliefFeature and its ReliefComponents both have an lod attribute denoting the corresponding level of detail. In most cases, the LOD of a ReliefFeature matches the LOD of its ReliefComponents. However, it is also allowed to specify a ReliefFeature with a high LOD which consists of ReliefComponents where some of them can have a LOD lower than that of the aggregating ReliefFeature. The idea is that, for example, for a LOD3 scene it might be sufficient to use a regular grid in LOD2 with certain higher precision areas defined by ReliefComponents in LOD3. The LOD2 grid and the LOD3 components can easily be integrated using the concept of the validity extent polygon. Therefore, although some of the ReliefComponents would have been classified to a lower LOD, the whole ReliefFeature would be appropriate to use with other LOD3 models which is indicated by setting its lod value to 3.

### 8.12.4. UML Model

The UML diagram of the Relief module is depicted in Figure 82.

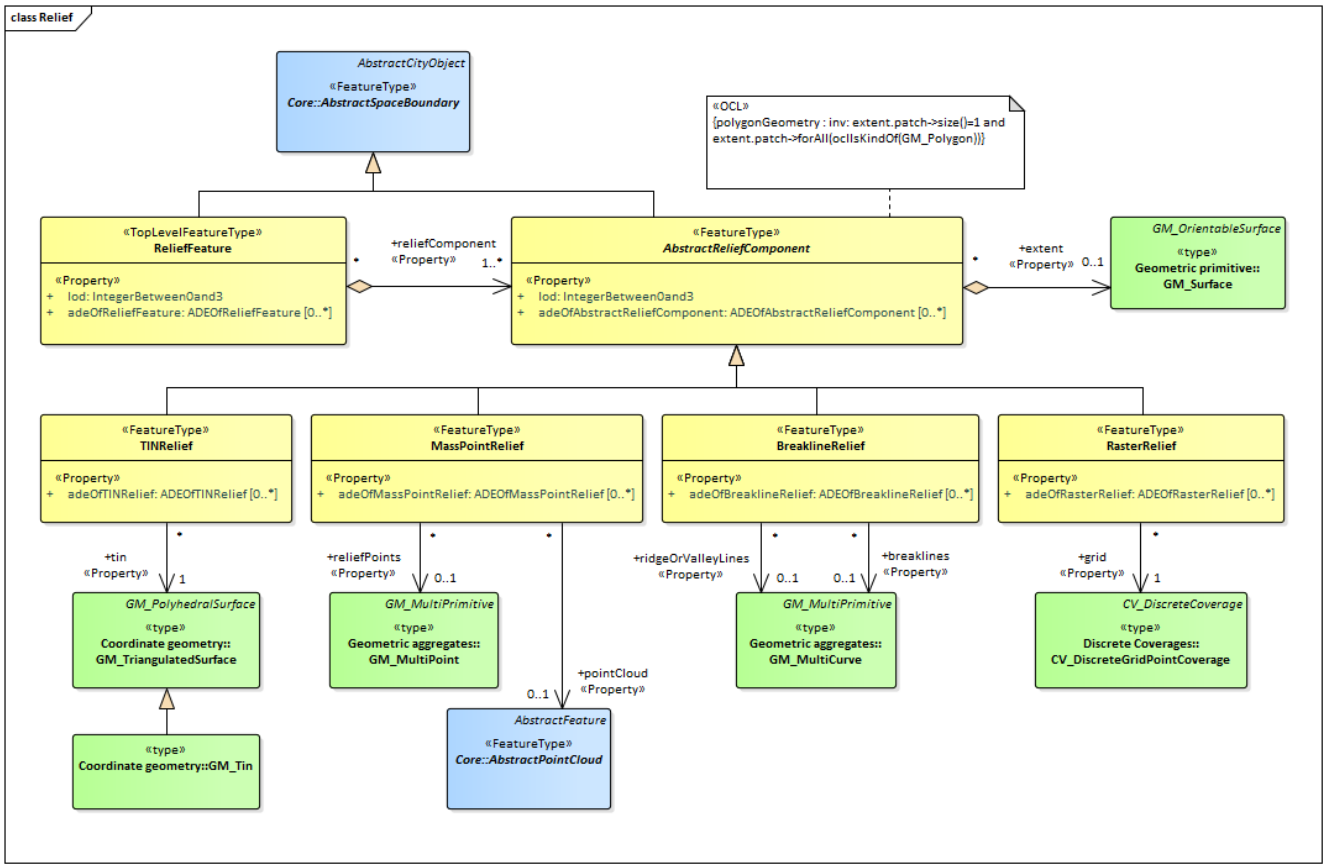



*Figure 82. UML diagram of Relief module.*

The ADE data types provided for the Relief module are illustrated in Figure 83.

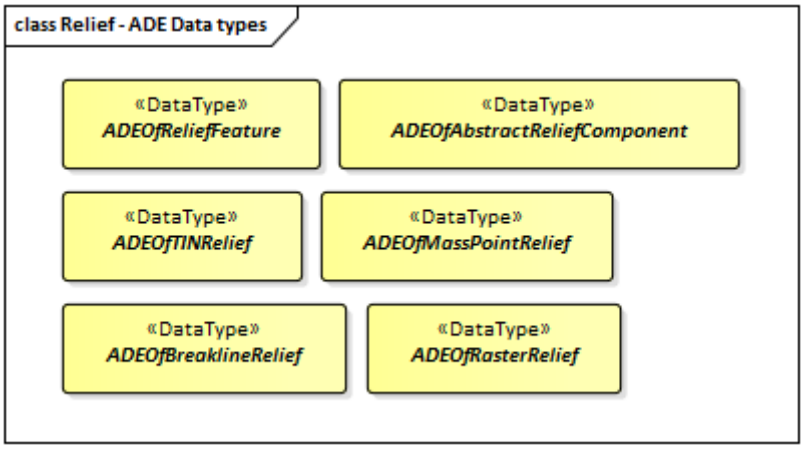



*Figure 83. ADE classes of the CityGML Relief module.*

### 8.12.5. Examples

# 8.13. Transportation

| Contributors |
| --- |
| C. Heazel - first draft |

### 8.13.1. Synopsis

The Transportation module defines central elements of the traffic infrastructure. This includes the transportation classes `road`, `square`, and `track` for the movement of vehicles, bicycles, and pedestrians, the transportation class `railway` for the movement of wheeled vehicles on rails, as well as the transportation class `waterway` for the movement of vessels upon or within water bodies.

### 8.13.2. Key Concepts

**Railway**: Railway represents routes that are utilized by rail vehicles like trams or trains.
A type of AbstractTransportationSpace.

**Road**: Road is intended to be used to represent transportation features that are mainly used by vehicles like cars, for example streets, motorways, and country roads. A type of AbstractTransportationSpace.

**Square**: A Square is an open area commonly found in cities (e.g. a plaza, market square). A type of AbstractTransportationSpace.

**Track**: A Track is a small path mainly used by pedestrians. A type of AbstractTransportationSpace.

**Waterway**: A Waterway is a transportation space used for the movement of vessels upon or within a water body. A type of AbstractTransportationSpace.

**AbstractTransportationSpace**: AbstractTransportationSpace defines the properties and associated concepts which are common to Tracks, Roads, Railways, Squares, and Waterways. The associated concepts include: A type of AbstractUnoccupiedSpace.

- **TrafficSpace**: A TrafficSpace is a space in which traffic (transportation) takes place. This is more than a geometry. It includes properties which may of of interest to anyone who wishes to traverse this space. This includes properties such as overhead clearance, surface type, and direction of travel. A type of AbstractUnoccupiedSpace.
- **Markings**: A Marking is a visible pattern on a transportation area relevant to the structuring or restriction of traffic. Examples are road markings and markings related to railway or waterway traffic. A type of AbstractThematicSurface.
- **Hole**: A Hole is an opening in the surface of a Road, Track or Square such as road damages, manholes or drains. Holes can span multiple transportation objects. A type of AbstractUnoccupiedSpace.
- **AuxillaryTrafficSpace**: a space within the transportation space not intended for traffic purposes. A type of AbstractUnoccupiedSpace.

### 8.13.3. Discussion

The transportation model of CityGML is a multi-functional, multi-scale model focusing on thematic and functional as well as on geometrical/topological aspects. Transportation features are represented as a linear network in LOD0. Starting from LOD1, all transportation features are geometrically described by 3D surfaces. The areal modelling of transportation features allows for

the application of geometric route planning algorithms. This can be useful to determine restrictions and manoeuvres required along a transportation route. This information can also be employed for trajectory planning of mobile robots in the real world or the automatic placement of avatars (virtual people) or vehicle models in 3D visualisations and training simulators.

The main class is AbstractTransportationSpace, which represents, for example, a road, a track, a railway, a waterway, or a square. Figure 37 illustrates the different transportation classes.

**NOTE** Need to add a waterway to figure 57.

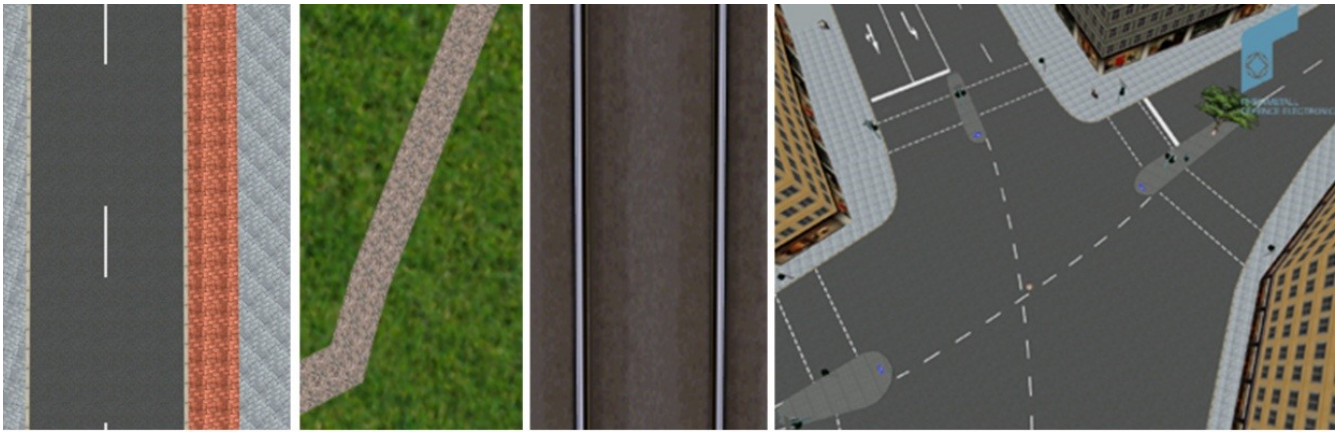


*Figure 84. Transportation classes (from left to right: examples of road, track, rail, waterway, and square) (source: Rheinmetall Defence Electronics)*

Two major concepts associated with all transportation classes are Traffic Space and Auxillary Traffic Space. Traffic Space describes the space that traffic can traverse. Auxillary Traffic Spaces are associated spaces where traffic does not traverse.

[figure-58] depicts an example for a LOD2 Transportation Space configuration within a virtual 3D city model. The Road consists of several Traffic Spaces for the sidewalks, road lanes, parking lots, and of Auxiliary Traffic Spaces below the raised flower beds.

[[figure-58],Figure 1] .Example LOD2 Traffic Spaces in CityGML: a road, which is the aggregation of TrafficAreas and AuxiliaryTrafficAreas (source: City of Solingen, IGG Uni Bonn) image::figures/Figure_58.png[align="center"]

The road itself is represented as a Road object with associated TrafficSpaces and AuxiliaryTrafficSpaces. The TrafficSpaces are those elements which are important in terms of traffic usage such as car driving lanes, pedestrian zones, and cycle lanes. The AuxiliaryTrafficSpaces describe further elements of the road like kerbstones, middle lanes, and green areas.

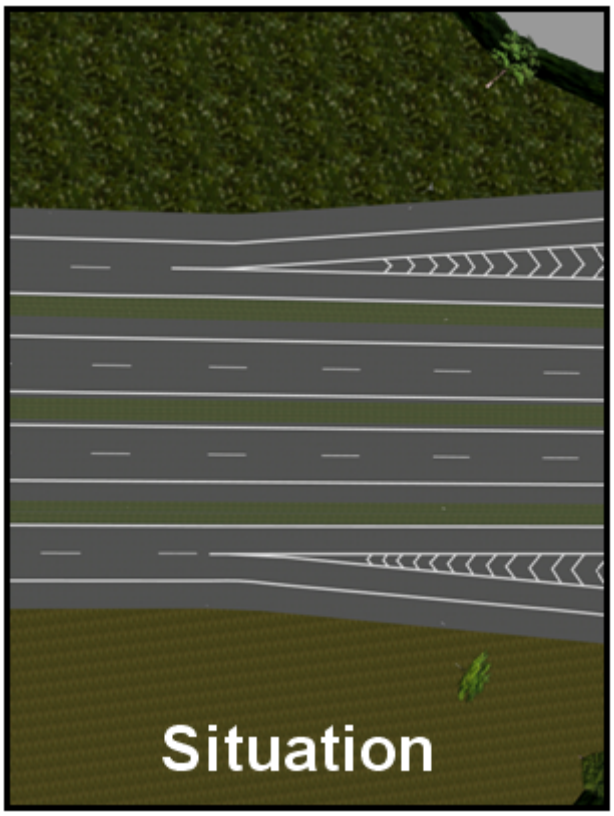
## 8.13.4. Level of Detail

The geometrical representation of the Transportation objects varies through the different levels of detail.

In the coarsest LOD0 the transportation objects are modelled by line objects establishing a linear network. On this abstract level, path finding algorithms or similar analyses can be executed. It also can be used to generate schematic drawings and visualisations of the transport network. Since this

abstract definition of transportation network does not contain explicit descriptions of the transportation objects, it may be task of the viewer application to generate the graphical visualisation, for example by using a library with style-definitions (width, color resp. texture) for each transportation object.

Starting from LOD1 a Transportation object provides an explicit surface geometry. This geometry reflects the actual shape of the object, not just its centerline. In LOD2 to LOD3, the object is further subdivided thematically into Traffic Spaces, which are used by transportation, such as cars, trains, public transport, airplanes, bicycles or pedestrians and into Auxiliary Traffic Spaces, which are of minor importance for transportation purposes. The different representations of a Transportation objects for each LOD are illustrated in Figure 38.



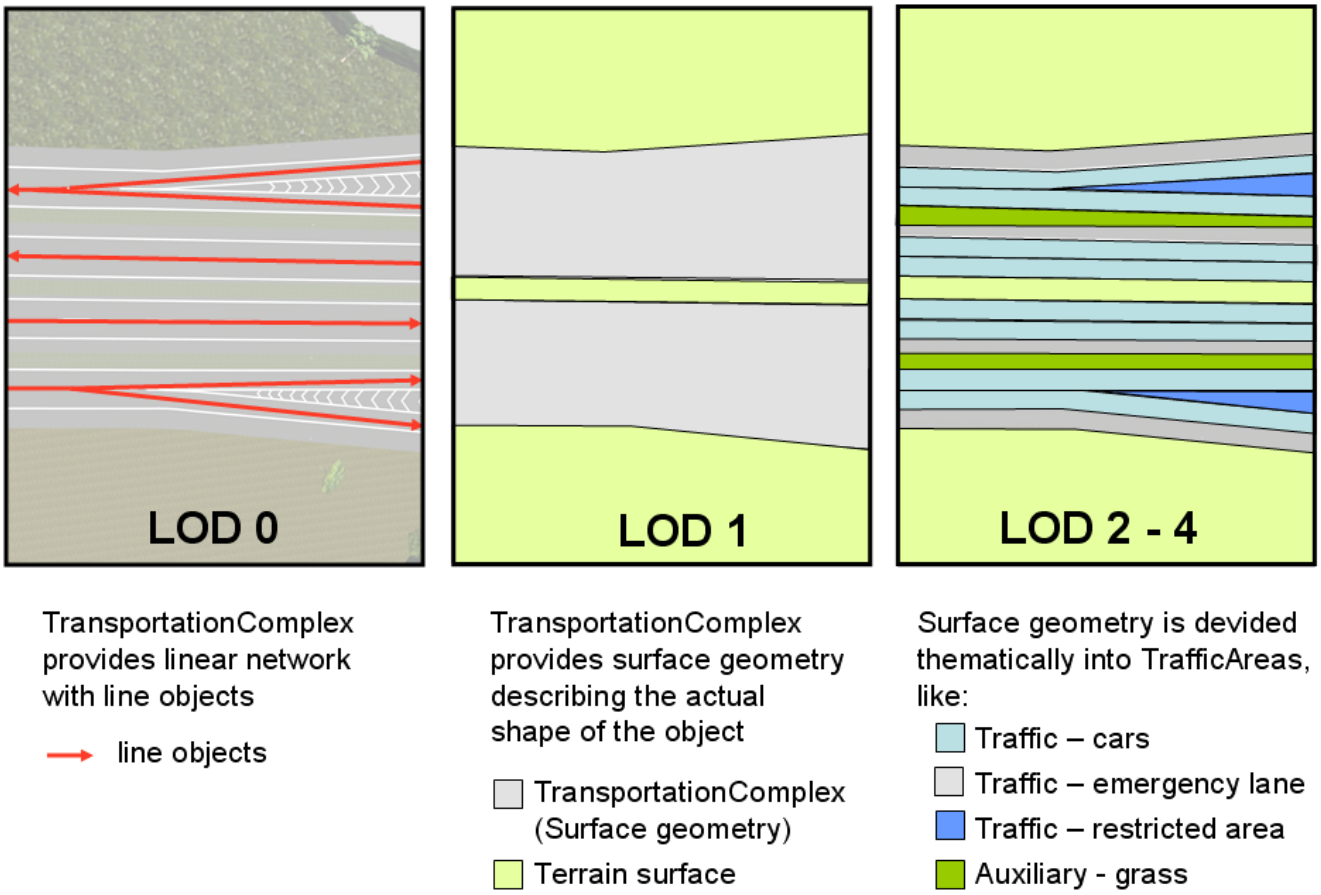



*Figure 85. Transportation Concepts in LOD0, 1, and 2-3 (example shows part of a motorway) (source: Rheinmetall Defence Electronics).*

In LOD0 areal transportation objects, like squares, should be modeled in the same way as in GDF. GDF is the ISO standard for transportation networks and is used in most car navigation systems. In GDF a square is typically represented as a ring surrounding the place and to which the incident roads connect. CityGML does not cover further functional aspects of transportation network models (e.g. speed limits) as it is intended to complement and not replace existing standards like GDF. However, if specific functional aspects have to be associated with CityGML transportation objects, generic attributes provided by CityGML's Generics module can be used.

Moreover, further objects of interest can be added from other information systems by the use of ExternalReferences. For example, GDF datasets, which provide additional information for car navigation, can be used for simulation and visualisation of traffic flows. The values of the object attributes can be augmented or replaced by the use of CodeLists or ADEs. These extensions may be country or user-specific (especially for country-specific road signs and signals).

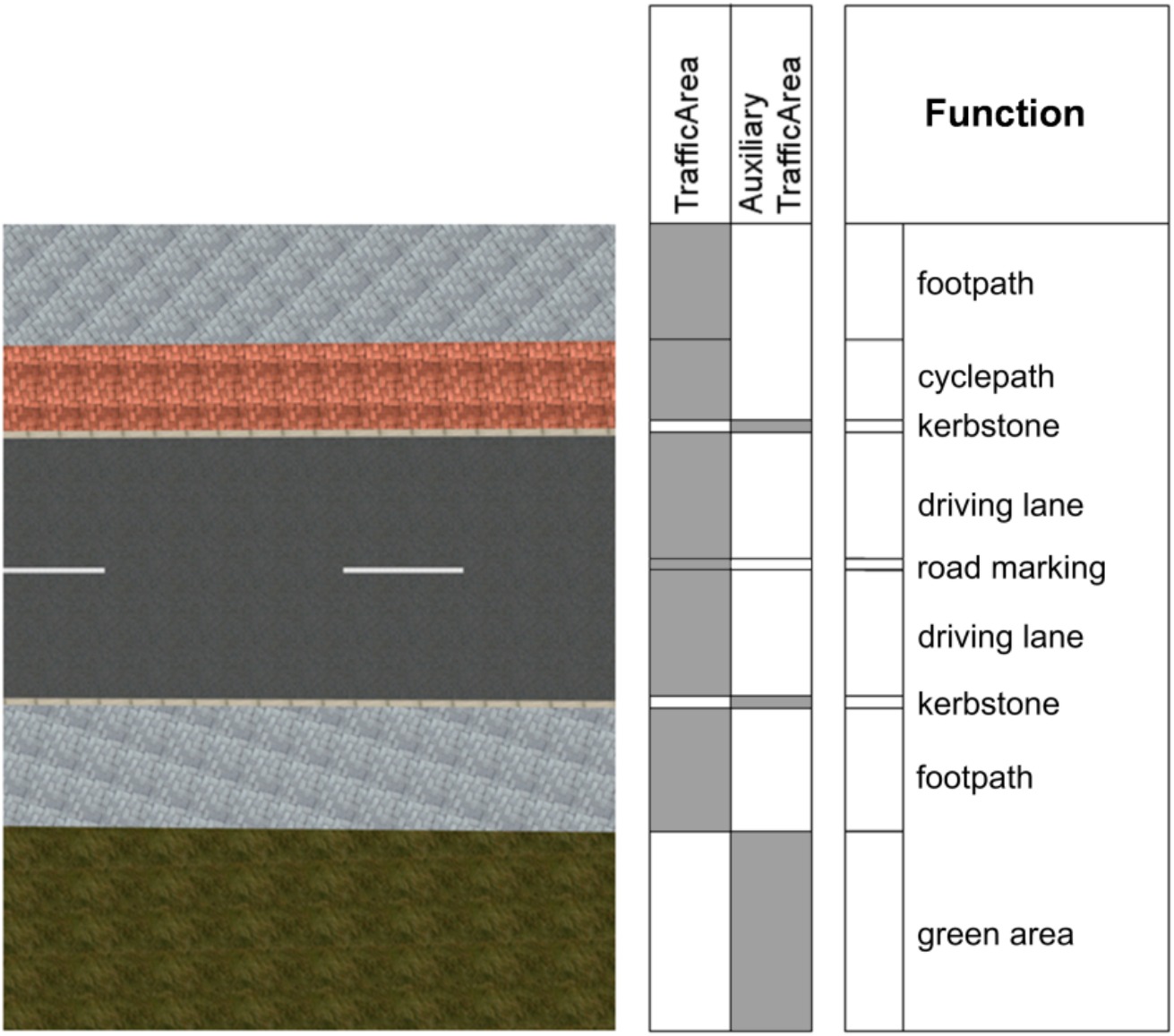



*Figure 86. TransportationComplex in LOD 2-3: representation of a road with a complex cross-section profile (example shows urban road) (source: Rheinmetall Defence Electronics).*

### 8.13.5. UML Model

The UML diagram of the Transportation module is depicted in Figure 87.

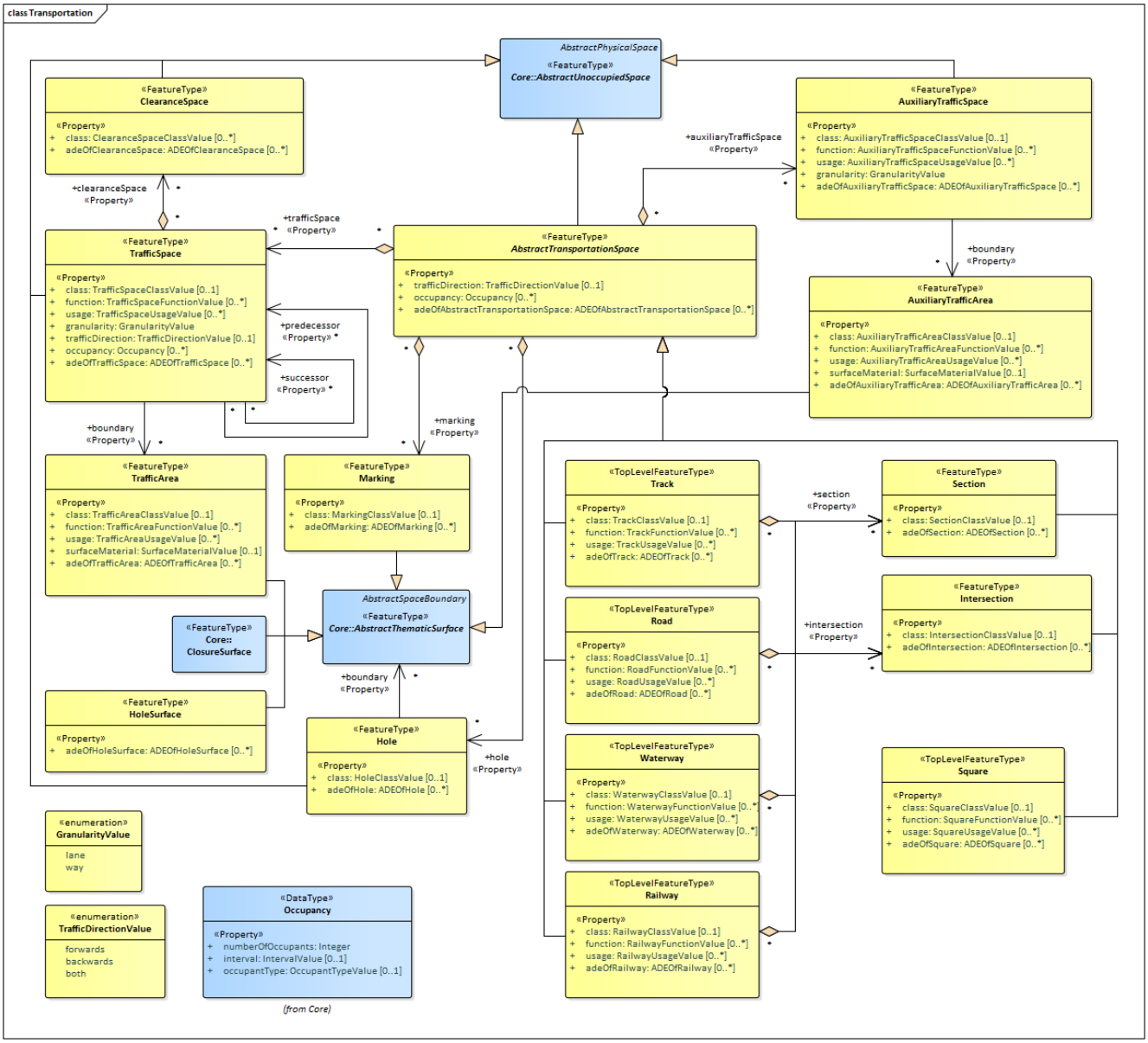



*Figure 87. UML diagram of the Transportation Model.*

The ADE data types provided for the Transportation module are illustrated in Figure 88.

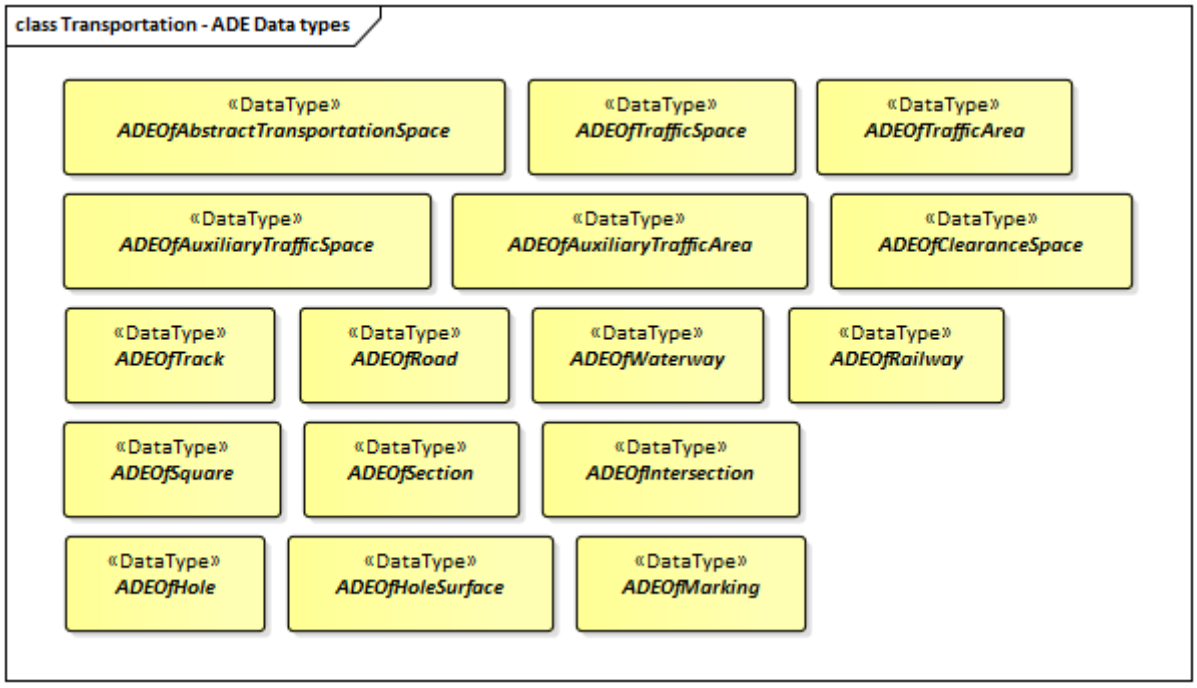



*Figure 88. ADE classes of the CityGML Transportation module.*

The Code Lists provided for the Transportation module are illustrated in Figure 89.

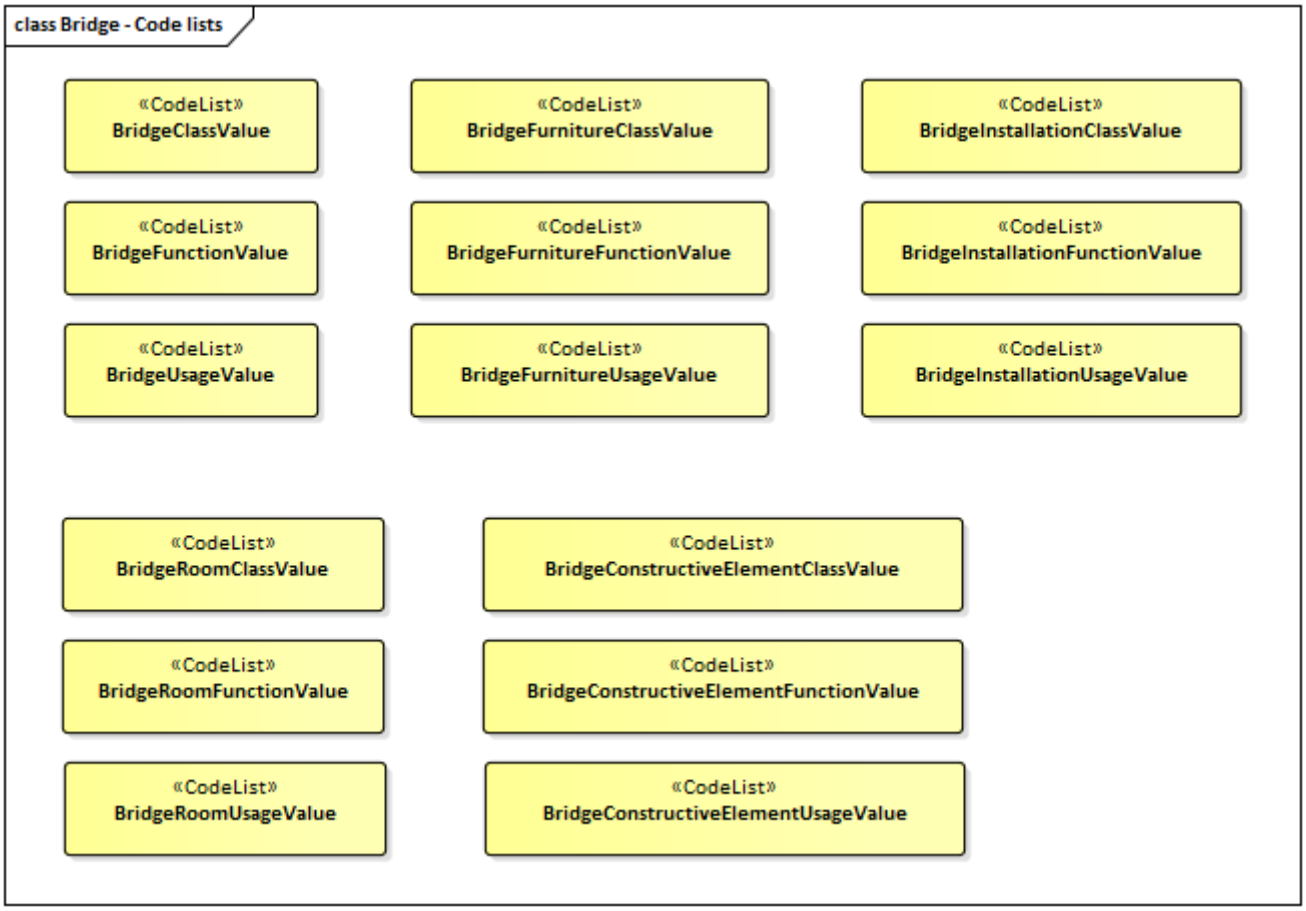



*Figure 89. Codelists from the CityGML Transportation module.*

## 8.13.6. Examples

*Table 7. Examples of TrafficArea*

| Example | Country Road | Motorway Entry |
|---|---|---|
| TransportationComplex – Function | road | road |
| TrafficArea – Usage | car, truck, bus, taxi, motorcycle | car, truck, bus, taxi, motorcycle |
| TrafficArea – Function | driving lane | motorway_entry |
| TrafficArea – SurfaceMaterial | asphalt | concrete |

The following example shows a complex urban crossing. The picture on the left is a screenshot of an editor application for a training simulator, which allows the definition of road networks consisting of transportation objects, external references, buildings and vegetation objects. On the right, the 3D representation of the defined crossing is shown including all referenced static and dynamic models.

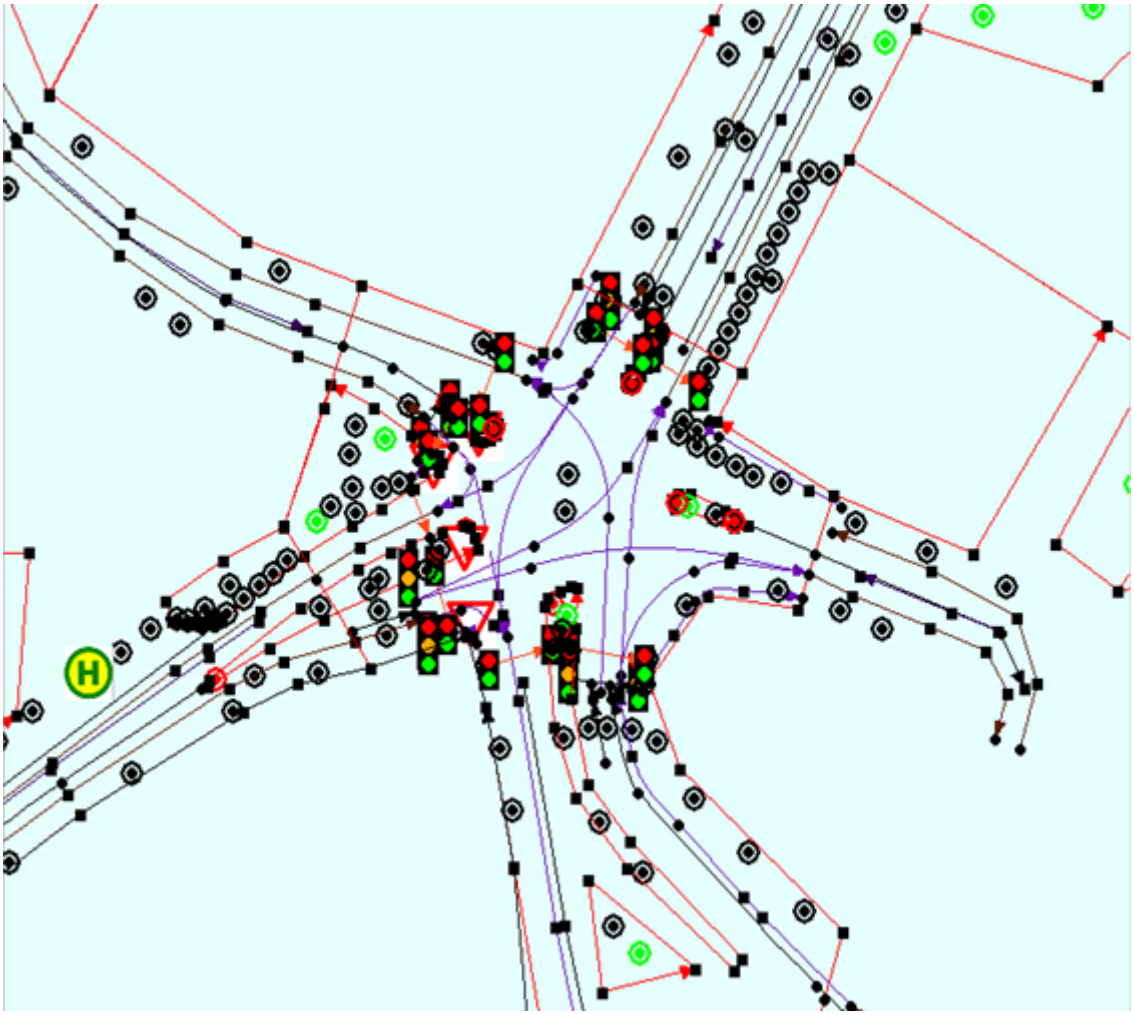


*Figure 90. Complex urban intersection (left: linear transportation network with surface descriptions and external references, right: generated scene) (source: Rheinmetall Defence Electronics).*

# 8.14. Tunnel Model

| Contributors |
| --- |
| C. Heazel - first draft |

| NOTE | This section uses tables to organize mutiple images into rows and columns. The null.png image serves to associate the caption with a figure rather than with the table. This is a kludge, but seems to work. Is there a better way? |
| --- | --- |

## 8.14.1. Synopsis

The Tunnel module supports representation of thematic and spatial aspects of tunnels, tunnel parts, tunnel installations, and interior tunnel structures.

## 8.14.2. Key Concepts

**AbstractTunnel**: Discussion AbstractTunnel is an abstract superclass representing the common attributes and associations of the classes Tunnel and TunnelPart.

A type of AbstractConstruction.

**Tunnel**: (Discussion) A Tunnel represents a horizontal or sloping enclosed passage way of a certain length, mainly underground or underwater.
A type of AbstractTunnel.

**TunnelPart**: (Discussion) A TunnelPart is a physical or functional subdivision of a Tunnel. It would be considered a Tunnel, if it were not part of a collection of other TunnelParts.
A type of AbstractTunnel.

**HollowSpace**: (Discussion) A HollowSpace is a space within a Tunnel or TunnelPart intended for certain functions (e.g. transport or passage ways, service rooms, emergency shelters). A HollowSpace is bounded physically and/or virtually (e.g. by ClosureSurfaces or GenericSurfaces).
A type of AbstractUnoccupiedSpace.

**TunnelInstallation**: (Discussion) A TunnelInstallation is a permanent part of a Tunnel (inside and/or outside) which does not have the significance of a TunnelPart. In contrast to TunnelConstructiveElement, a TunnelInstallation is not essential from a structural point of view. Examples are stairs, antennas or railings.
A type of AbstractInstallation.

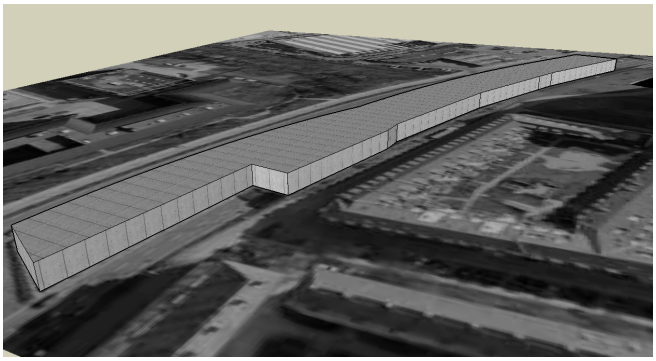
**TunnelFurniture**: (Discussion) A TunnelFurniture is an equipment for occupant use, usually not fixed to the tunnel.
A type of AbstractFurniture.

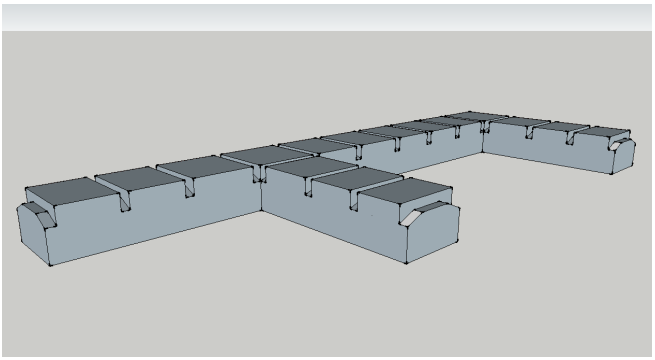
**TunnelConstructiveElement**: (Discussion) A TunnelConstructiveElement is an element of a Tunnel which is essential from a structural point of view. Examples are walls, slabs, beams.
A type of AbstractConstructiveElement.

### 8.14.3. Discussion

The tunnel model is closely related to the building model. It supports the representation of thematic and spatial aspects of tunnels and tunnel parts in four levels of detail, LOD0 to LOD3. The tunnel model of CityGML is defined by the thematic extension module Tunnel. Figure 40 provides examples of tunnel models for each LOD.

| NOTE | This figure is from CityGML 2.0. It needs to be updated for CityGML 3.0 |
|---|---|





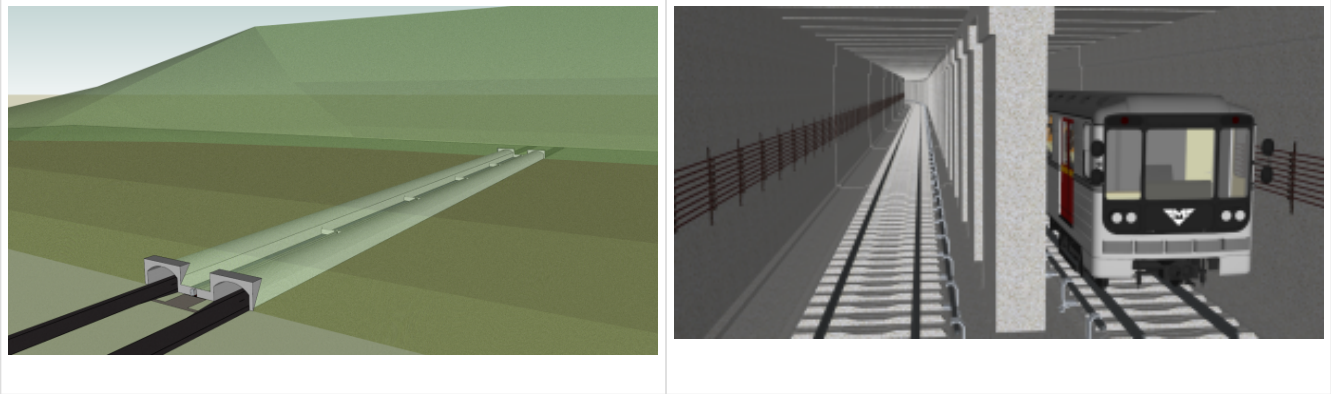


*Figure 91. Examples for tunnel models in LOD1 (upper left), LOD2 (upper right), LOD3 (lower left) and LOD4 (lower right) (source: Google 3D warehouse).*

### Tunnel Class

The Tunnel class is one of the two subclasses of `AbstractTunnel`. If a tunnel only consists of one (homogeneous) part, this class shall be used. A tunnel composed of structural segments, for example tunnel entrance and subway, has to be separated into one tunnel having one or more additional `TunnelParts` (see Figure 41). The geometry and non-spatial properties of the central part of the tunnel should be represented in the aggregating Tunnel feature.

### Tunnel Part Class

If sections of a tunnel differ in geometry and / or attributes, the tunnel can be separated into parts (see Figure 41). Like `Tunnel`, the class TunnelPart is derived from `AbstractTunnel` and inherites all attributes of `AbstractTunnel`. A TunnelPart object should be uniquely related to exactly one tunnel or tunnel part object.

*Figure 92. Example of a tunnel modeled with two tunnel parts (source: Helmut Stracke).*

### Abstract Tunnel Class

The abstract class `AbstractTunnel` contains properties for tunnel attributes, purely geometric representations, and geometric/semantic representations of the tunnel or tunnel part in different levels of detail. The attributes describe:

1. The classification of the tunnel or tunnel part (class),
2. the different functions (function), and
3. the usage (usage).

Spanning the different levels of detail, the tunnel model differs in the complexity and granularity of the geometric representation and the thematic structuring of the model into components with a special semantic meaning. This is illustrated in Figure 42, showing the same tunnel in four different LODs. Some properties of the class `AbstractTunnel` are also associated with certain LODs.

**NOTE** This figure is from CityGML 2.0 and needs to be updated to the CityGML 3.0 LODs.

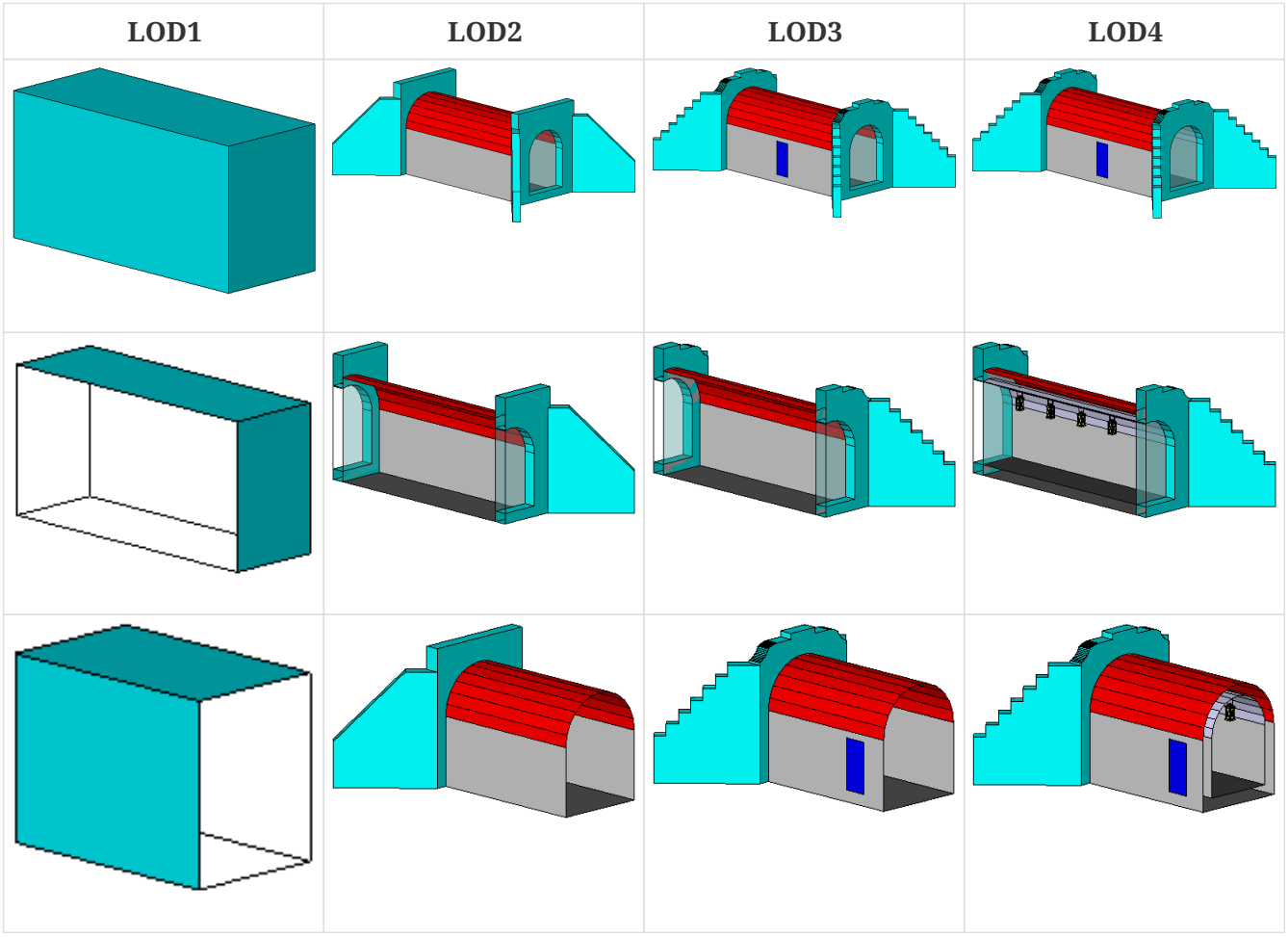



*Figure 93. Tunnel model in LOD1 – LOD4 (source: Karlsruhe Institute of Technology (KIT)).*

Table 8 shows the correspondence of the different geometric and semantic themes of the tunnel model to LODs. In each LOD, the volume of a tunnel can be expressed by a GM_Solid geometry and/or a MultiSurface geometry. The definition of a 3D Terrain Intersection Curve (TIC), used to integrate tunnels from different sources with the Digital Terrain Model, is also possible in all LODs. The TIC can – but does not have to – build closed rings around the tunnel or tunnel parts.

| NOTE | Update the LOD to CityGML 3.0 approach. |
|---|---|

*Table 8. Semantic themes of the class _AbstractTunnel*

| Geometric / semantic theme | Property type | LOD1 | LOD2 | LOD3 | LOD4 |
|---|---|---|---|---|---|
| Building footprint and roof edge | MultiSurfaceType | • | | | |
| | Volume part of the tunnel shell | SolidType | • | • | • |
| • | Surface part of the tunnel shell | MultiSurfaceType | • | • | • |
| • | Terrain intersection curve | MultiCurveType | • | • | • |
| • | Curve part of the tunnel shell | MultiCurveType | | • | • |
| • | Tunnel parts | TunnelPartType | • | • | • |
| • | Boundary surfaces | AbstractBoundarySurfaceType | | • | • |
| • | Tunnel installations | TunnelInstallationType | | • | • |
| • | Openings | AbstractOpeningType | | | • |
| • | Hollow spaces | HollowSpaceType | | | |

### Hollow Space Class

A HollowSpace is a semantic object for modelling the free space inside a tunnel and should be uniquely related to exactly one tunnel or tunnel part object. It should be closed (if necessary by using ClosureSurface) and the geometry normally will be described by a GM_Solid (lod3Solid property). However, if the topological correctness of the boundary cannot be guaranteed, the geometry can alternatively be given as a GM_MultiSurface (lod3MultiSurface property). The surface normals of the outer shell of GM_Solid must point outwards. This is important if appearences should be assigned to HollowSpace surfaces. In this case, textures and colors must be placed on the backside of the corresponding surfaces in order to be visible from the inside of the hollow space.

In addition to the geometrical representation, different parts of the visible surface of a hollow space can be modelled by specialised boundary surfaces (FloorSurface, CeilingSurface, InteriorWallSurface, and ClosureSurface, cf. Section 8.7).

#### Tunnel Installation Class

A TunnelInstallation is permanent part of a Tunnel (inside or outside) which does not have the significance of a TunnelPart. In contrast to TunnelConstrucitveElement, a TunnelInstallation is not aessential from a structural point of view. Examples are stairs, antennas or railings. A TunnelInstallation optionally has attributes `class`, `function` and `usage`. The attribute `class` - which can only occur once - represents a general classification of the installation. With the attributes `function` and `usage`, nominal and real functions of a tunnel installation can be described. For all three attributes the list of feasible values can specified in a code list.

For the geometrical representation of a TunnelInstallation, an arbitrary geometry object from the subset of ISO classes shown in Table 3 can be used. Alternatively, the geometry may be given as ImplicitGeometry object (see [prototypic-objects-section]. Following the concept of ImplicitGeometry the geometry of a prototype tunnel installation is stored only once in a local coordinate system and referenced by other tunnel installation features. The visible surfaces of a tunnel installation can be semantically classified using the concept of boundary surfaces. A TunnelInstallation object should be uniquely related to exactly one tunnel or tunnel part object.

#### Tunnel Furniture Class

Hollow spaces may have TunnelFurniture. A TunnelFurniture is a movable part of a hollow space. A TunnelFurniture object should be uniquely related to exactly one hollow space. Its geometry may be represented by an explicit geometry or an ImplicitGeometry object. Following the concept of ImplicitGeometry, the geometry of a prototype tunnel furniture is stored only once in a local coordinate system and referenced by other tunnel furniture features.

#### Tunnel Constructive Element Class

A TunnelConstructiveElement is an element of a Tunnel which is essential from a structural point of view. Examples are walls, slabs, and beams. A TunnelConstructiveElement optionally has attributes `class`, `function` and `usage`. The attribute `class` - which can only occur once - represents a general classification of the element. With the attributes `function` and `usage`, nominal and real functions of a tunnel constructuve element can be described. For all three attributes the list of feasible values can specified in a code list.

### 8.14.4. Level of Detail

| NOTE | The term Boundary Surface is never defined. |
|---|---|

The geometric representation and semantic structure of an AbstractTunnel is shown in Figure 43. The model is successively refined from LOD0 to LOD3. Therefore, not all components of a tunnel model are represented equally in each LOD and not all aggregation levels are allowed in each LOD. In CityGML, all object classes are associated to the LODs with respect to the proposed minimum acquisition criteria for each LOD. An object can be represented simultaneously in different LODs by providing distinct geometries for the corresponding LODs.

In LOD1, a tunnel model consists of a geometric representation of the tunnel volume. Optionally, a GM_MultiCurve representing the TerrainIntersectionCurve can be specified. The geometric representation is refined in LOD2 by additional GM_MultiSurface and GM_MultiCurve geometries.

In LOD2 and higher LODs the outer structure of a tunnel can also be differentiated semantically by the class TunnelInstallation. The TunnelInstallation class is used for non-structural tunnel elements, like outer stairs, which strongly affecting the outer appearance of a tunnel. A TunnelInstallation may have the attributes class, function and usage.

The visible surface of a hollow space is represented geometrically as a GM_MultiCurve or GM_MultiSurface. Semantically, the surface can be structured into specialised `BoundarySurfaces`, representing floor (FloorSurface), ceiling (Ceil-ingSurface), and interior walls (InteriorWallSurface). Hollow space furniture, like movable equipment in control areas, can be represented in the CityGML tunnel model with the class TunnelFurniture. A TunnelFurniture may have the attributes class, function and usage.

### 8.14.5. UML Model

The UML diagram of the Tunnel module is depicted in Figure 43. The Tunnel module inherits concepts from the Construction module (cf. Section 8.7). The Construction module defines objects that are common to all types of construction, such as the different surface types and the openings.

The UML diagram of the tunnel model is shown in Figure 43. The pivotal class of the model is `AbstractTunnel`, which is a subclass of the thematic class `AbstractConstruction` (and transitively of the root class `CityObject`). `AbstractTunnel` is specialized either to a `Tunnel` or to a `TunnelPart`. Since an `AbstractTunnel` consists of `TunnelParts`, which again are `AbstractTunnels`, an aggregation hierarchy of arbitrary depth may be realized. As subclass of the root class `CityObject`, an `AbstractTunnel` inherits all properties from `CityObject` such as the feature properties (`name` etc.) and the CityGML specific properties like ExternalReferences (cf. Section 8.1). Further properties not explicitly covered by `AbstractTunnel` may be modelled as generic attributes provided by the CityGML Generics module (cf. Section 8.9) or using the CityGML Application Domain Extension mechanism (cf. [ug-model-ade-section]).

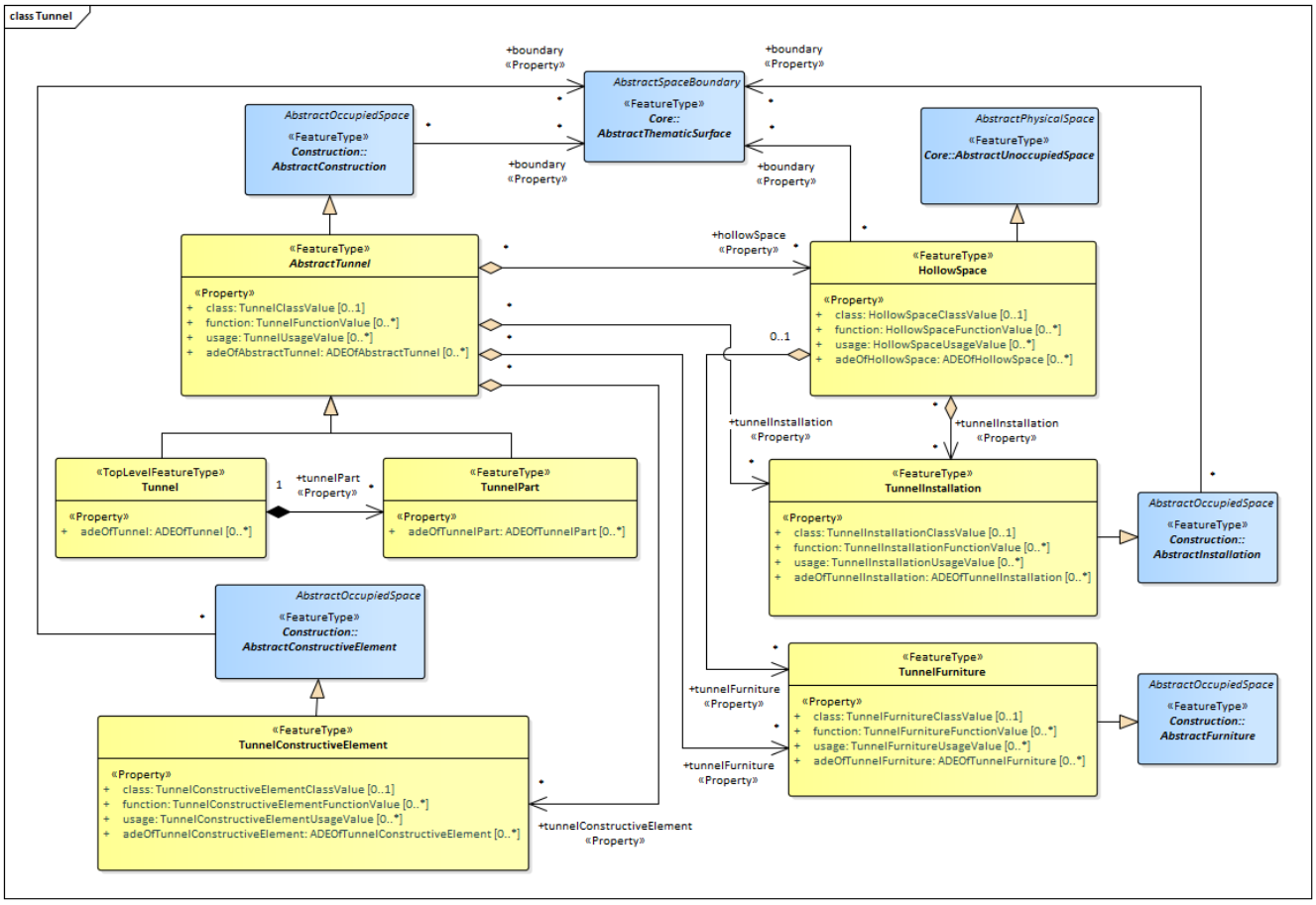



*Figure 94. UML diagram of the Tunnel Model.*

The ADE data types provided for the Tunnel module are illustrated in Figure 44.

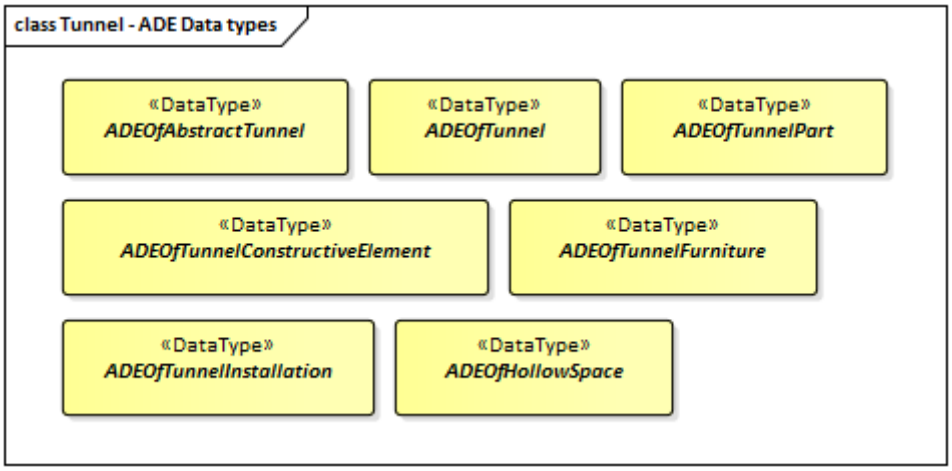



*Figure 95. ADE classes of the CityGML Tunnel module.*

The Code Lists provided for the Tunnel module are illustrated in Figure 45.

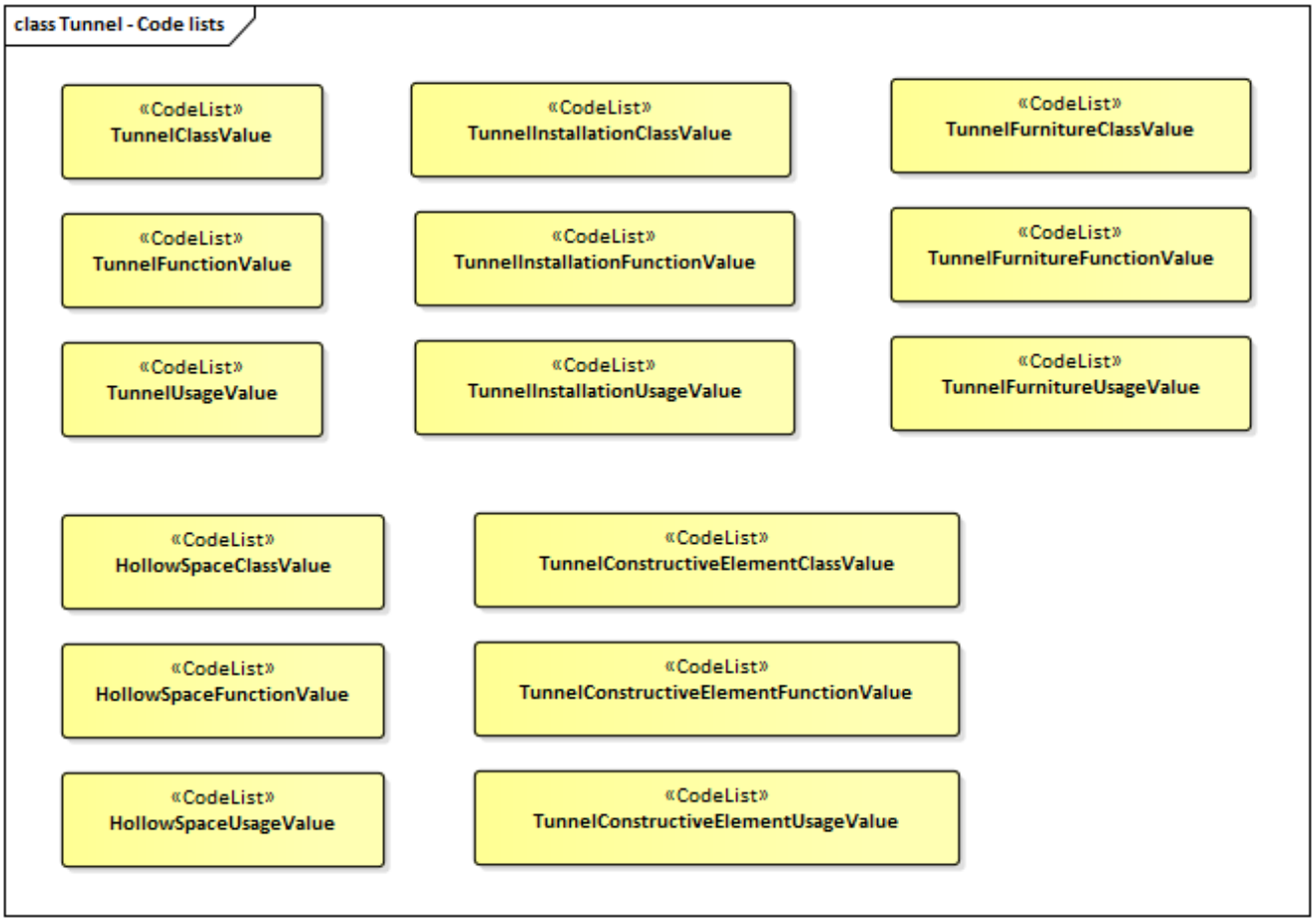



*Figure 96. Codelists from the CityGML Tunnel module.*

### 8.14.6. Examples

The example in Figure 46 shows a pedestrian underpass in the city centre of Karlsruhe, Germany. On the left side of Figure 46, a photo illustrates the real world situation. Both entrances of the underpass are marked in the photo by dashed rectangles. On the right side of the figure, the CityGML tunnel model is shown. The terrain surrounding the tunnel has been virtually cut out of model in order to visualize the entire tunnel with its subsurface body. The same underpass is illustrated in Figure 47 from a different perspective. The camera is positioned in front of the left entrance (black dashed rectangle in Figure 46) and pointing in the direction of the right entrance (white dashed rectangle in Figure 46). On the right side of Figure 47, the tunnel model is shown from the same perspective. Again holes are cut in the terrain surface in order to make the subsurface part of the tunnel visible. An LOD1 representation of the nearby buildings is shown in the background of the model.

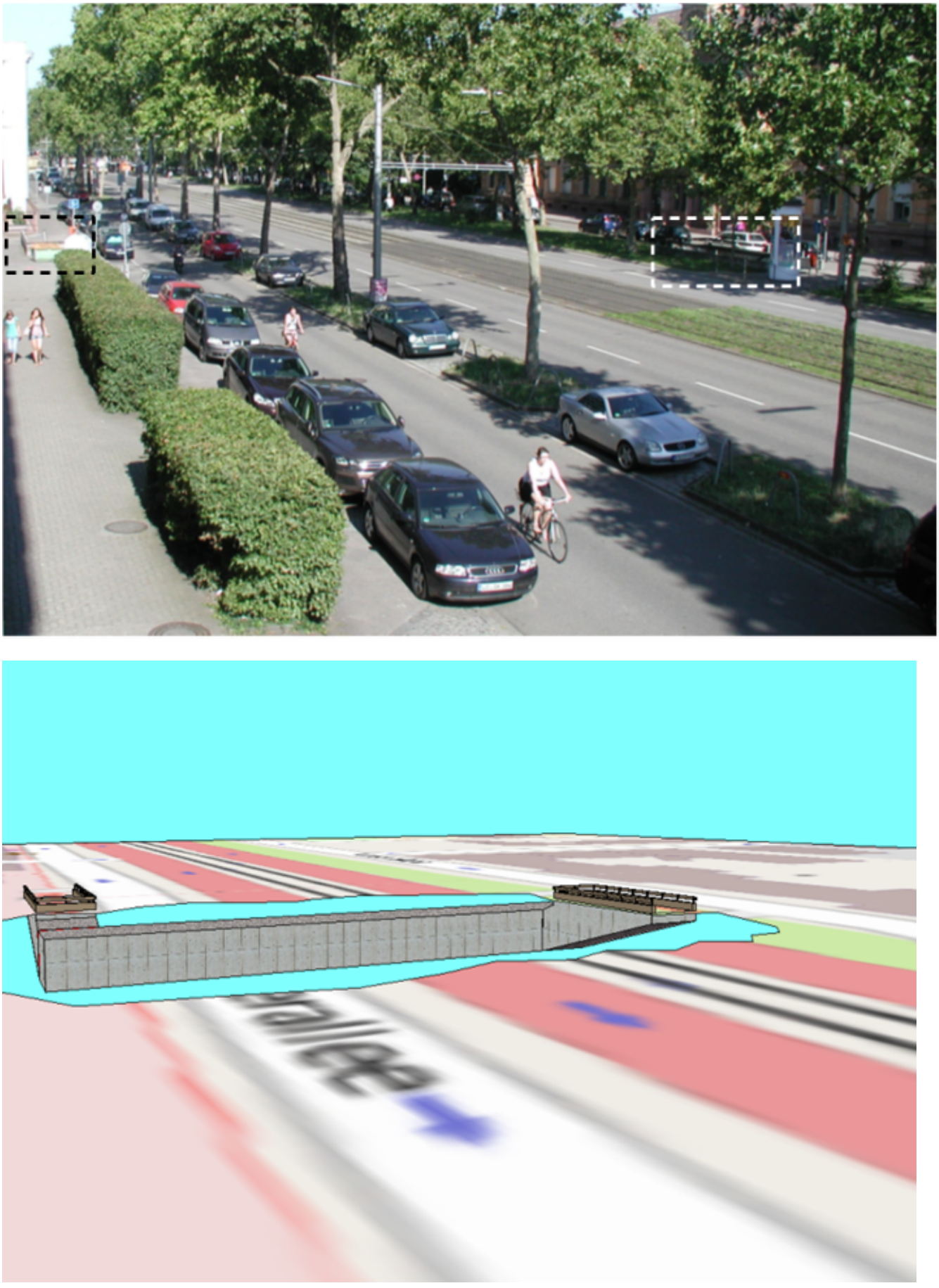


*Figure 97. Example of a tunnel modeled in LOD3 (real situation on the left side; CityGML model on the right side) (source: Karlsruhe Institute of Technology (KIT), courtesy of City of Karlsruhe).*

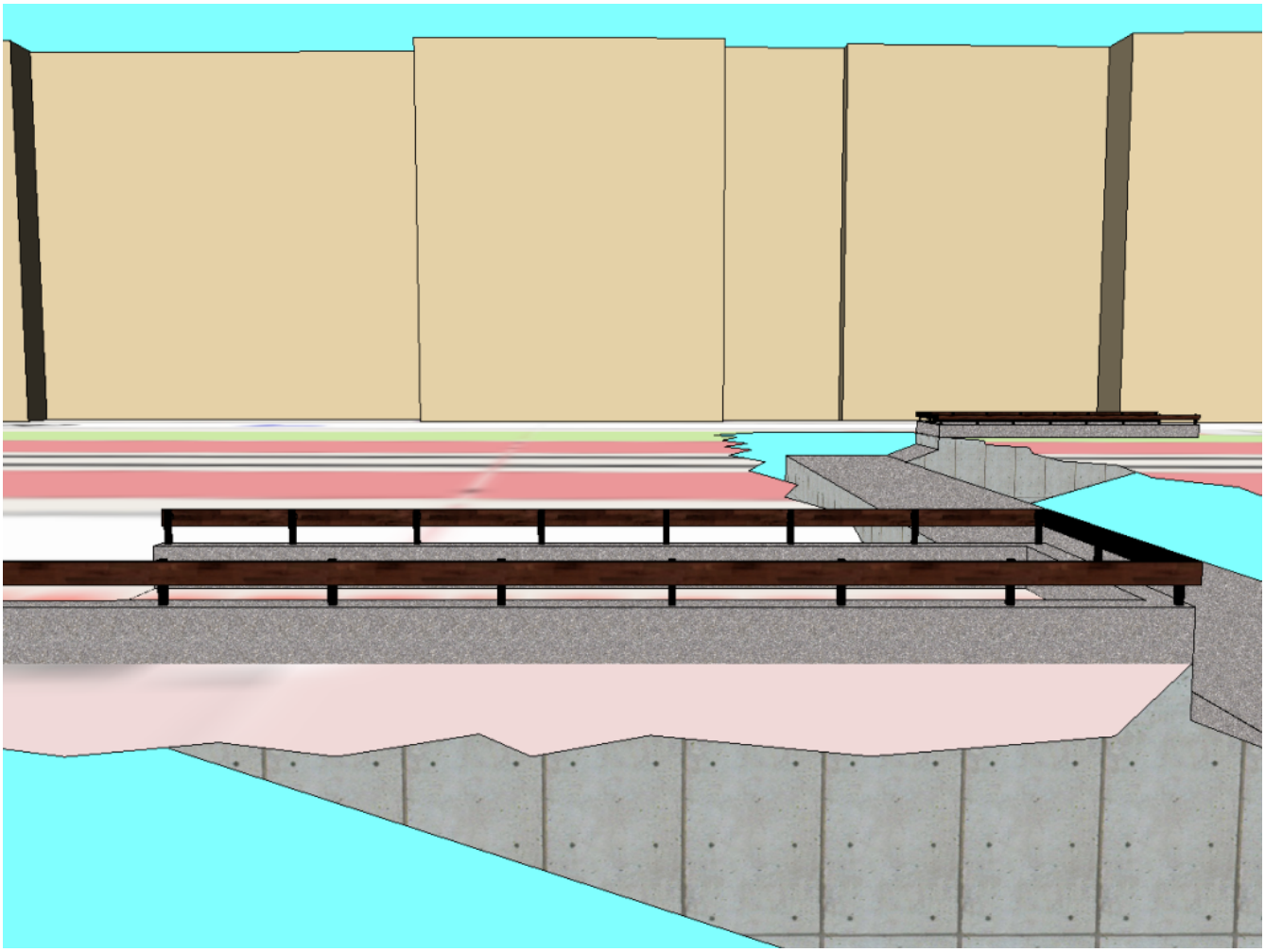


*Figure 98. The same LOD3 tunnel shown from a different perspective. The camera is positioned in front of the left entrance and pointing in the direction of the right entrance. (real situation on the left side; CityGML model on the right side). The model on the right also includes an LOD1 representation of the nearby buildings in the background (painted in light brown) (source: Karlsruhe Institute of Technology (KIT), courtesy of City of Karlsruhe).*

The model is subdivided into one Tunnel (the actual underpass) and two TunnelParts (both entrances). The tunnel and tunnel parts are bounded by GroundSurface, WallSurface, RoofSurface. ClosureSurface objects are used to virtually seal the tunnel entrances. For safety reasons each of the two entrances has railings which are modeled as TunnelInstallation. Due to the high geometrical accuracy and the semantic richness, the model is classified as LOD3.

# 8.15. Vegetation

| Contributors |
| --- |
| C. Heazel - first draft |

## 8.15.1. Synopsis

The Vegetation module defines the concepts to represent vegetation within city models. Vegetation can be represented either as solitary vegetation objects, such as trees, bushes and ferns, or as vegetation areas that are covered by plants of a given species or a typical mixture of plant species, such as forests, steppes and wet meadows.

### 8.15.2. Key Concepts

**SolitaryVegetationObject**: A SolitaryVegetationObject represents individual vegetation objects, e.g. trees or bushes.
A type of AbstractOccupiedSpace.

**Plant Cover**: A PlantCover represents a space covered by vegetation.
A type of AbstractOccupiedSpace.

### 8.15.3. Discussion

Vegetation features are important components of a 3D city model, since they support the recognition of the surrounding environment. By the analysis and visualisation of vegetation objects, statements on their distribution, structure and diversification can be made. Habitats can be analysed and impacts on the fauna can be derived. The vegetation model may be used as a basis for simulations of, for example forest fire, urban aeration or micro climate. The model could be used, for example to examine forest damage, to detect obstacles (e.g. con-cerning air traffic) or to perform analysis tasks in the field of environmental protection.

The vegetation model of CityGML distinguishes between solitary vegetation objects like trees and vegetation areas, which represent biotopes like forests or other plant communities (Figure 99). Single vegetation objects are modelled by the class `SolitaryVegetationObject`, whereas for areas filled with a specific vegetation the class `PlantCover` is used. The geometry representation of a PlantCover feature may be a MultiSurface or a MultiSolid, depending on the vertical extent of the vegetation. For example regarding forests, a MultiSolid representation might be more appropriate.

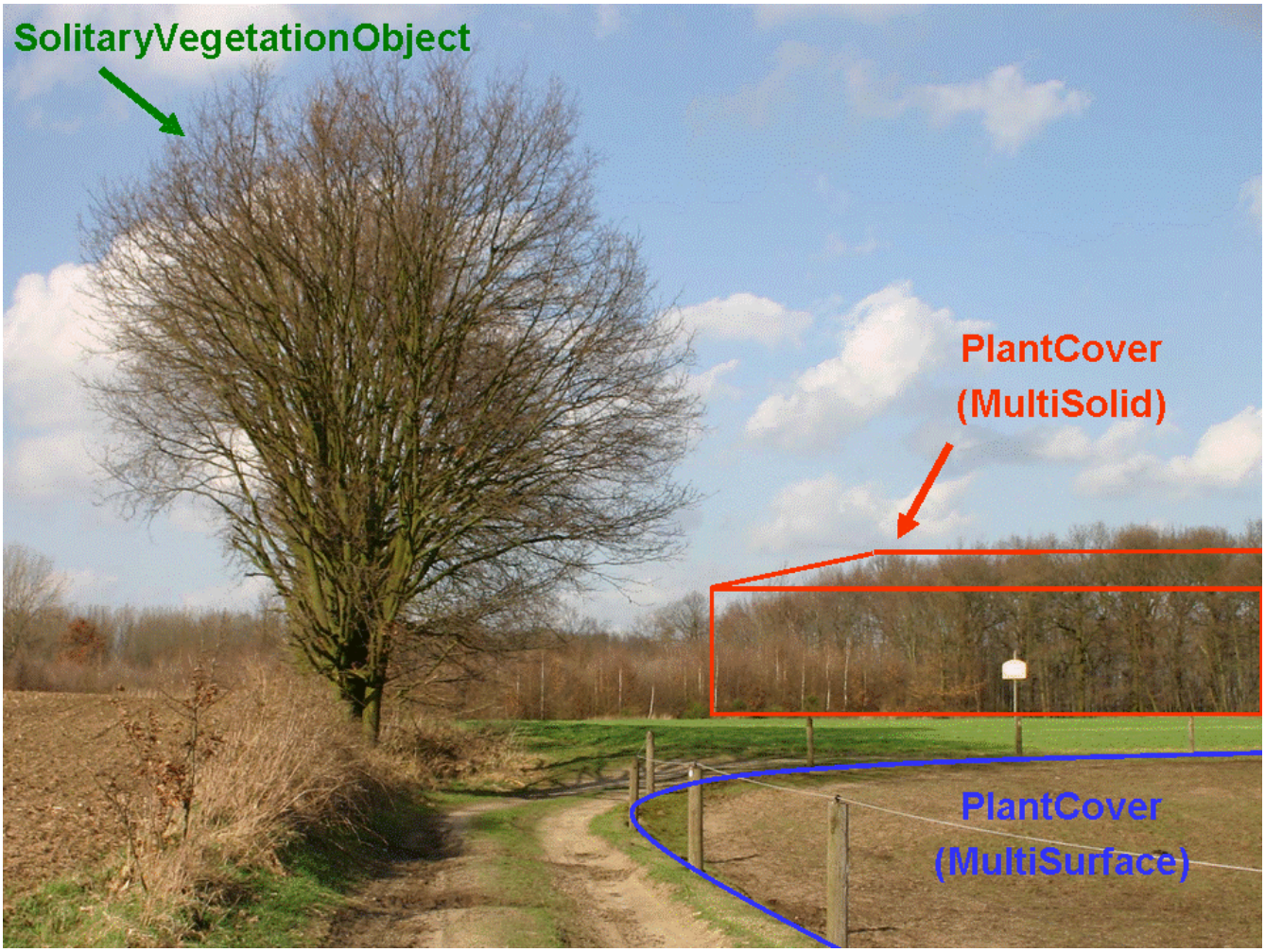
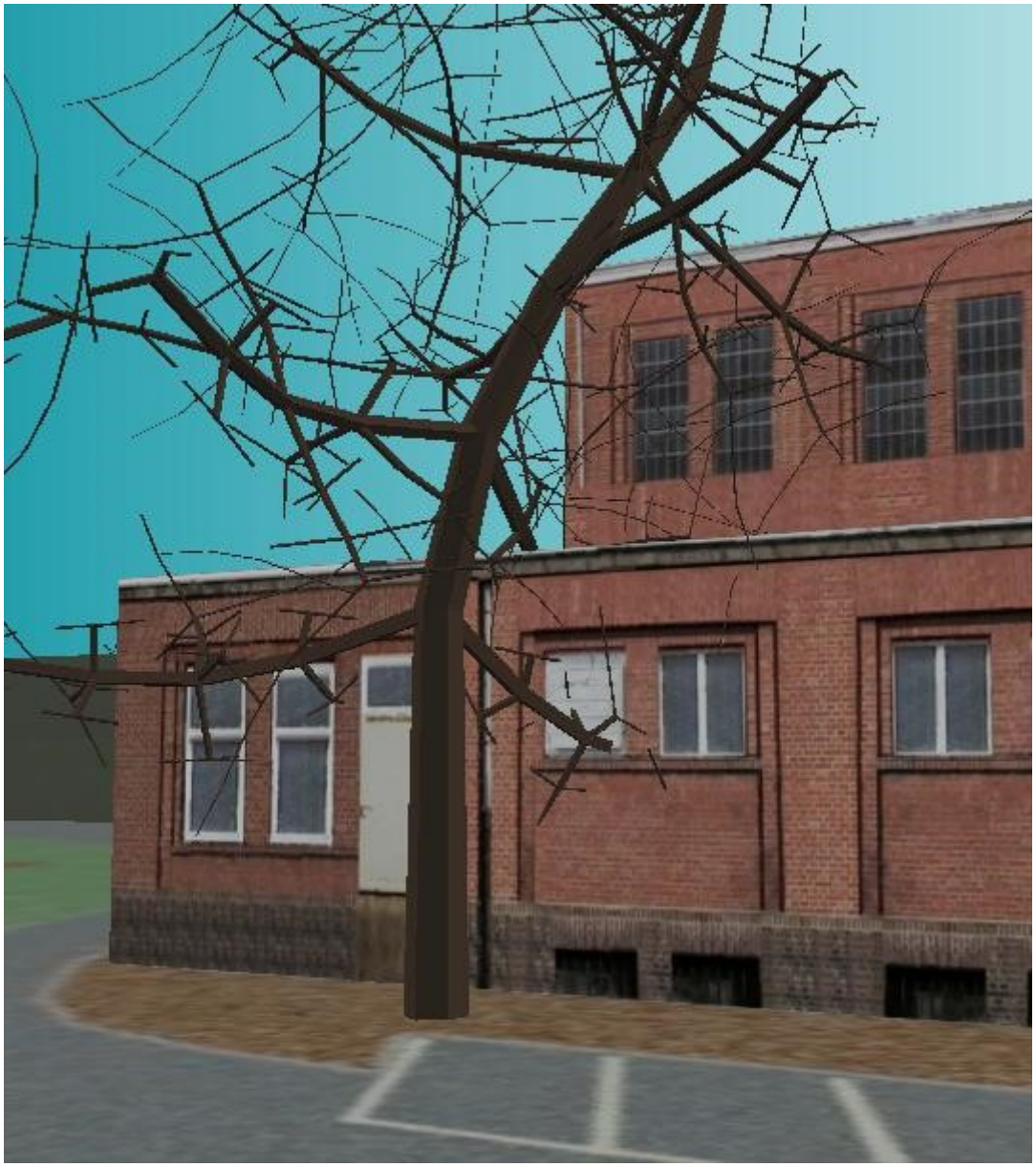



*Figure 99. Example for vegetation objects of the classes SolitaryVegetationObject and PlantCover (graphic: District of Recklinghausen).*

### 8.15.4. Level of Detail

The geometry of a SolitaryVegetationObject may be defined in LOD 1-3 explicitly by a GML geometry having absolute coordinates, or prototypically by an ImplicitGeometry. Solitary vegetation objects probably are one of the most important features where implicit geometries are appropriate, since the shape of most types of vegetation objects, such as trees of the same species, can be treated as identical in most cases. Furthermore, season dependent appearances may by mapped using ImplicitGeometry. For visualisation purposes, only the content of the library object defining the object's shape and appearance has to be swapped (see [figure-65]).



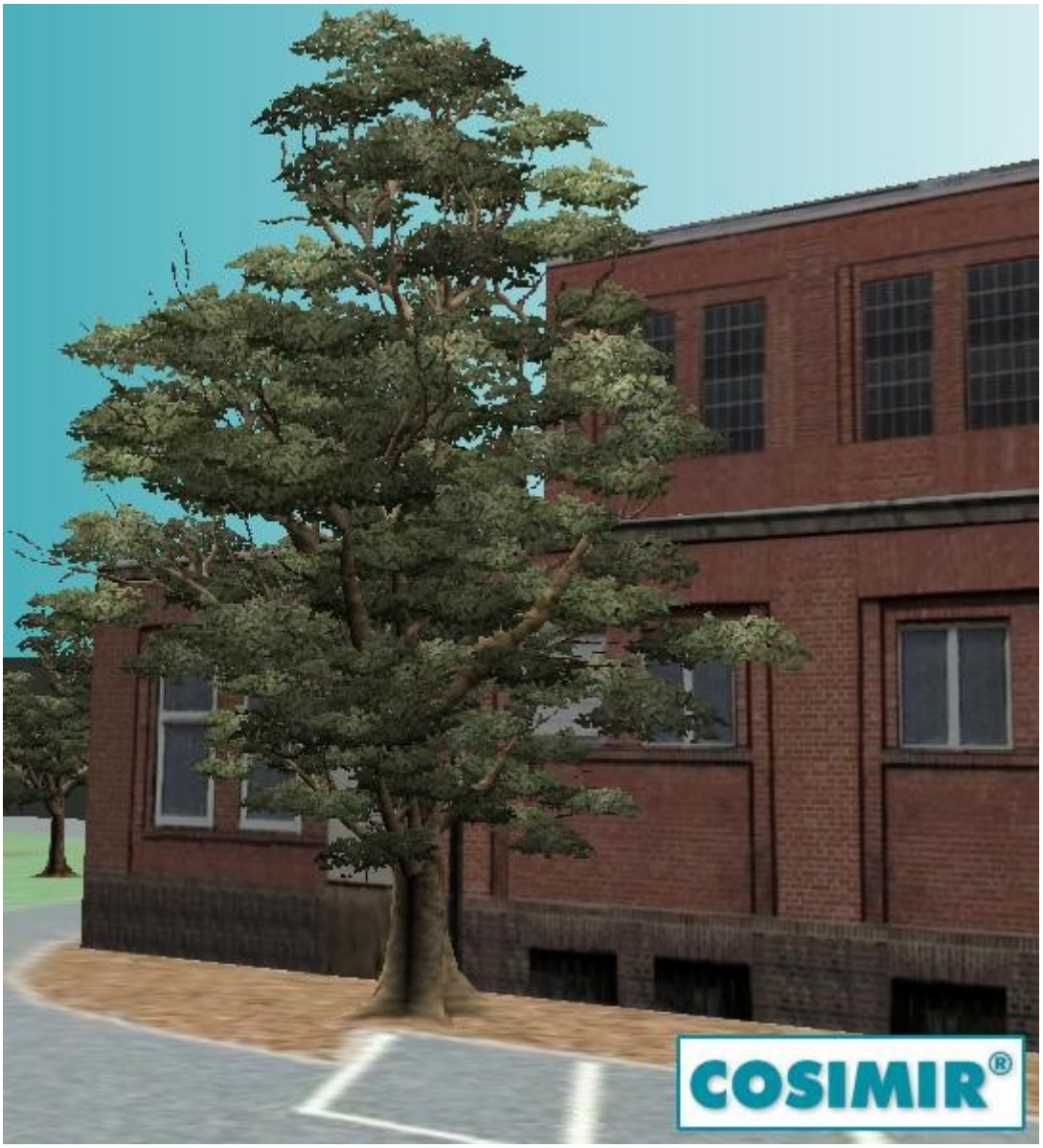



*Figure 100. Visualisation of a vegetation object in different seasons (source: District of Recklinghausen).*

A SolitaryVegetationObject or a PlantCover may have a different geometry in each LOD. Whereas a SolitaryVegetationObject is associated with the `Geometry` class representing an arbitrary geometry, a PlantCover is restricted to be either a `MultiSolid` or a `MultiSurface`. An example of a PlantCover modeled as MultiSolid is a 'solid forest model' (see Figure 101).

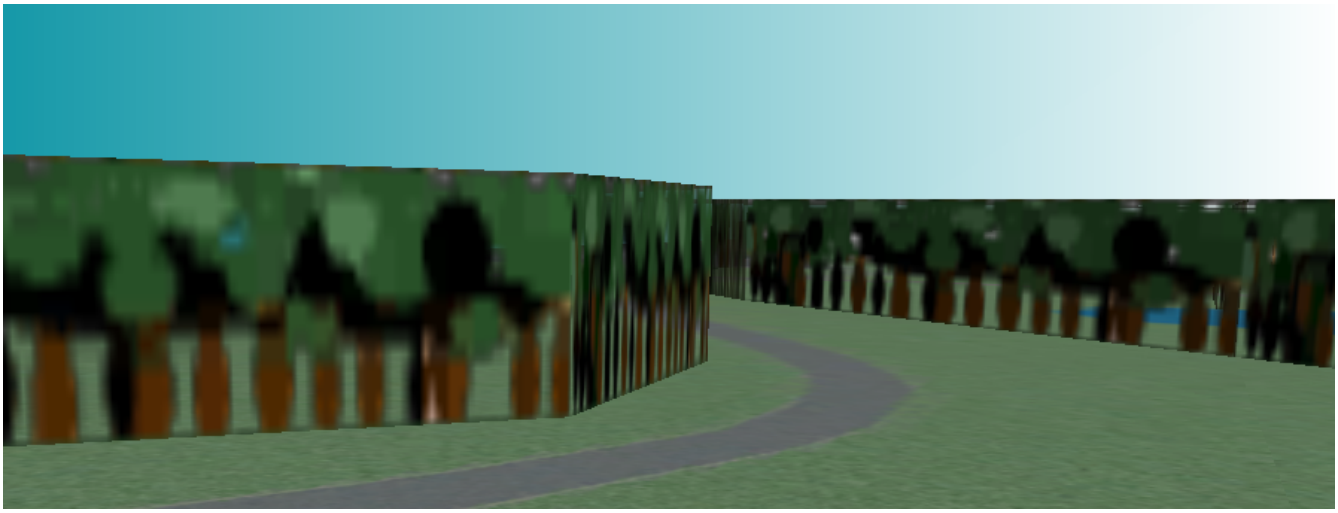


*Figure 101. Example for the visualisation/modelling of a solid forest (source: District of Recklinghausen).*

### 8.15.5. UML Model

The UML diagram of the Vegetation module is depicted in Figure 102.

A SolitaryVegetationObject may have the attributes `class`, `function`, `usage`, `species`, `height`, `trunkDiameter`, `crownDiameter`, `rootBallDiameter`, and `maxRootBallDepth`. The attribute `class` contains the classification of the object or plant habit, e.g. tree, bush, grass, and can occur only once. The attribute `species` defines the species' name, for example "Abies alba", and can occur at most once. The optional attributes `function` and `usage` denotes the intended respectively real purpose of the object, for example botanical monument, and can occur multiple times. The possible attribute values for `class`, `species`, `function`, and `usage` can be provided in a code list. The attribute `height` contains the relative height of the object. The attributes `crownDiameter` and `trunkDiameter` represent the plant crown and trunk diameter respectively. The trunk diameter is often used in regulations of municipal cadastre (e.g. tree management rules). The attributes `rootBallDiameter` and `maxRootBallDepth` represent the diameter of the root ball and its' maximum depth respectively.

A PlantCover feature may have the attributes `class`, `function`, `usage`, `averageHeight`, `minHeight`, and `maxHeight`. The plant community of a PlantCover is represented by the attribute `class`. The values of this attribute can be specified in a code list whose values should not only describe one plant type or species, but denote a typical mixture of plant types in a plant community. This information can be used in particular to generate realistic 3D visualisations, where the PlantCover region is automatically, perhaps randomly, filled with a corresponding mixture of 3D plant objects. The attributes `function` and `usage` indicate the intended respectively real purpose of the object, for example national forest, and can occur multiple times. The attributes `averageHeight`, `minHeight`, and `maxHeight` denotes the minimum, maximum, and average relative vegetation heights.

Since SolitaryVegetationObject and PlantCover are subclasses of AbstractOccupiedSpace, they are Features. As such they inherit the attribute `name` from the AbstractFeature class and an ExternalReference to a corresponding object. That external object may be in an external information system which may contains botanical information from public environmental agencies.

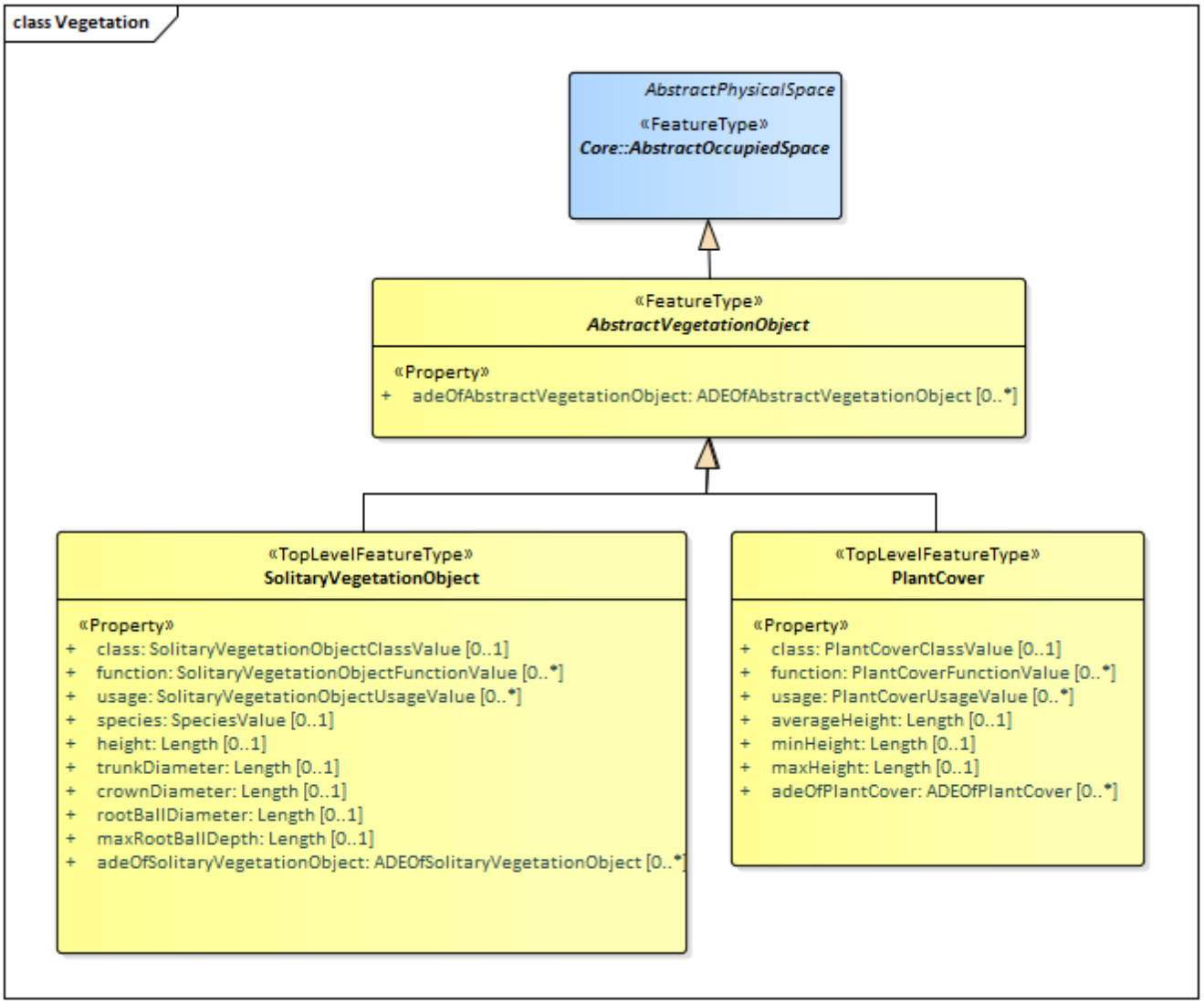



*Figure 102. UML diagram of the Vegetation Model.*

The ADE data types provided for the Vegetation module are illustrated in Figure 103.

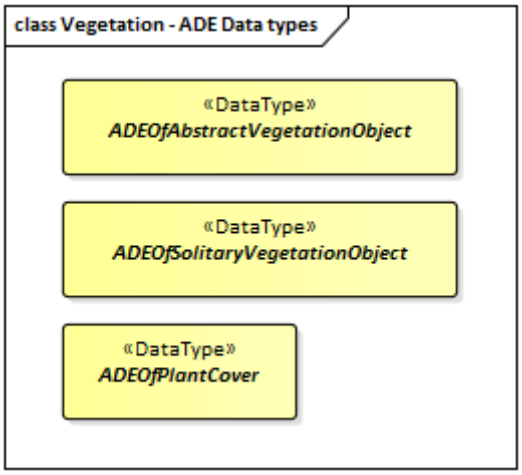



*Figure 103. ADE classes of the CityGML Vegetation module.*

The Code Lists provided for the Vegetation module are illustrated in Figure 104.

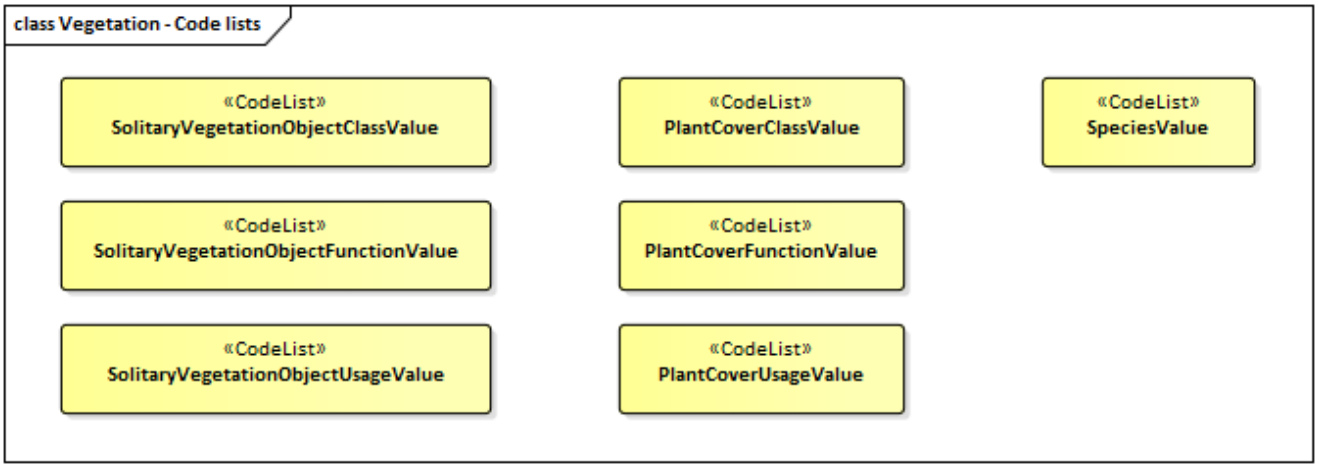



*Figure 104. Codelists from the CityGML Vegetation module.*

### 8.15.6. Examples

The following two excerpts of a CityGML dataset contain a solitary tree (SolitaryVegetationObject) and a plant community (PlantCover). The solitary tree has the attributes: class = 1070 (deciduous tree), species = 1040 (Fagus/beech), height = 8 m, trunkDiameter = 0.7 m, crownDiameter = 8.0 m. The plant community has the attributes: class =1180 (isoeto-nanojuncetea), averageHeight = 0.5 m.

| **NOTE** | include examples, GML or other? |
|---|---|

## 8.16. Versioning

| Contributors |
|---|
| C. Heazel - first draft |

### 8.16.1. Synopsis

The Versioning module defines the concepts that enable encoding representing multiple versions of a city model.

### 8.16.2. Key Concepts

**Version**: Version represents a defined state of a city model consisting of the dedicated versions of all city object instances that belong to the respective city model version. Versions can have names, a description and can be labeled with an arbitrary number of user defined tags.
A type of AbstractVersion.

**VersionTransition**: VersionTransition describes the change of the state of a city model from one version to another. Version transitions can have names, a description and can be further qualified by a type and a reason.
A type of AbstractVersionTransition.

## 8.16.3. Discussion

The Versioning module defines the concepts that enable encoding representing multiple versions of a city model. A specific version represents a defined state of a city model consisting of the dedicated versions of all city object instances that belong to the respective city model version. Each version can be complemented by version transitions that describe the change of the state of a city model from one version to another and that give the reason for the change and the modifications applied. In addition, the Versioning module introduces bitemporal timestamps for all objects. This allows for providing all objects with information on 1) the time period a specific version of an object is an integral part of the 3D city model and 2) the lifespan a specific version of an object exists in the real world.

By using the Versioning module, slow changes over a long time period with respect to cities and city models can be represented. This includes the creation and termination of objects (e.g. construction or demolition of sites, planting of trees, construction of new roads), structural changes of objects (e.g. extension of buildings), and changes in the status of an object (e.g. change of building owner, change of the traffic direction of a road to a one-way street). In this way, the history or evolution of cities and city models can be modelled, parallel or alternative versions of cities and city models can be managed, and changes of geometries and thematic properties of individual city objects over time can be tracked.

## 8.16.4. UML Model

The UML diagram of the Versioning module is depicted in Figure 105.

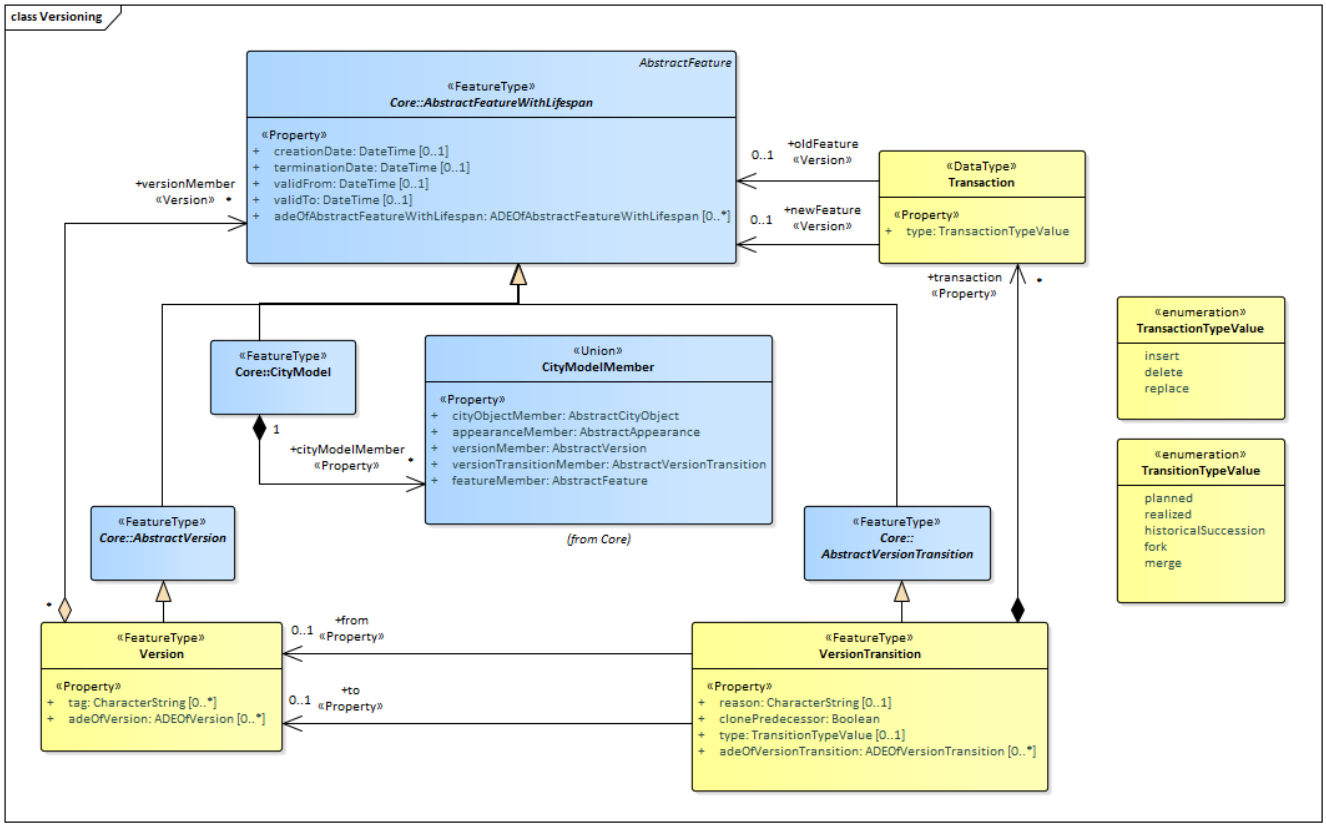



*Figure 105. UML diagram of the Versioning Model.*

The ADE data types provided for the Versioning module are illustrated in Figure 106.

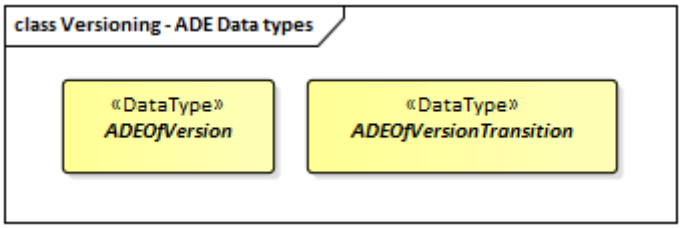



*Figure 106. ADE classes of the CityGML Versioning module.*

### 8.16.5. Examples

## 8.17. Waterbodies

| Contributors |
| --- |
| C. Heazel - first draft |

### 8.17.1. Synopsis

The WaterBody module provides the representation of significant and permanent or semi-permanent accumulations of surface water, usually covering a part of the Earth. Examples of such water bodies that can be modelled with CityGML are rivers, canals, lakes, and basins.

### 8.17.2. Key Concepts

**WaterBody**: A WaterBody represents significant and permanent or semi-permanent accumulations of surface water, usually covering a part of the Earth.

**WaterSurface**: A WaterSurface represents the upper exterior interface between a water body and the atmosphere. Water Surface is a boundary property of a Water Body.

**WaterGroundSurface**: A WaterGroundSurface represents the exterior boundary surface of the submerged bottom of a water body. Water Ground Surface is a boundary property of a Water Body.

### 8.17.3. Discussion

Waters have always played an important role in urbanisation processes and cities were built preferably at rivers and places where landfall seemed to be easy. Obviously, water is essential for human alimentation and sanitation. Water bodies present the most economical way of transportation and are barriers at the same time, that avoid instant access to other locations. Bridging waterways caused the first efforts of construction and resulted in high-tech bridges of today. The landscapes of many cities are dominated by water, which directly relates to 3D city models. Furthermore, water bodies are important for urban life as subject of recreation and possible hazards as e.g. floods.

The distinct character of water bodies compared with the permanence of buildings, roadways, and terrain is considered in this thematic model. Water bodies are dynamic surfaces. Tides occur regularly, but irregular events predominate with respect to natural forces, for example flood events. The visible water surface changes in height and its covered area with the necessity to model its semantics and geometry distinct from adjacent objects like terrain or buildings.

This first modelling approach of water bodies fulfils the requirements of 3D city models. It does not inherit any hydrological or other dynamic aspects. In these terms it does not claim to be complete. However, the semantic and geometric description given here allows further enhancements of dynamics and conceptually different descriptions. The water bodies model of CityGML is embraced by the extension module WaterBody.

The water bodies model represents the thematic aspects and three-dimensional geometry of rivers, canals, lakes, and basins. In the LOD 1-3 water bodies are bounded by distinct thematic surfaces (see Figure 107). These surfaces are the obligatory WaterSurface, defined as the boundary between water and air, and the optional WaterGroundSurface, defined as the boundary between water and underground (e.g. DTM or floor of a 3D basin object).

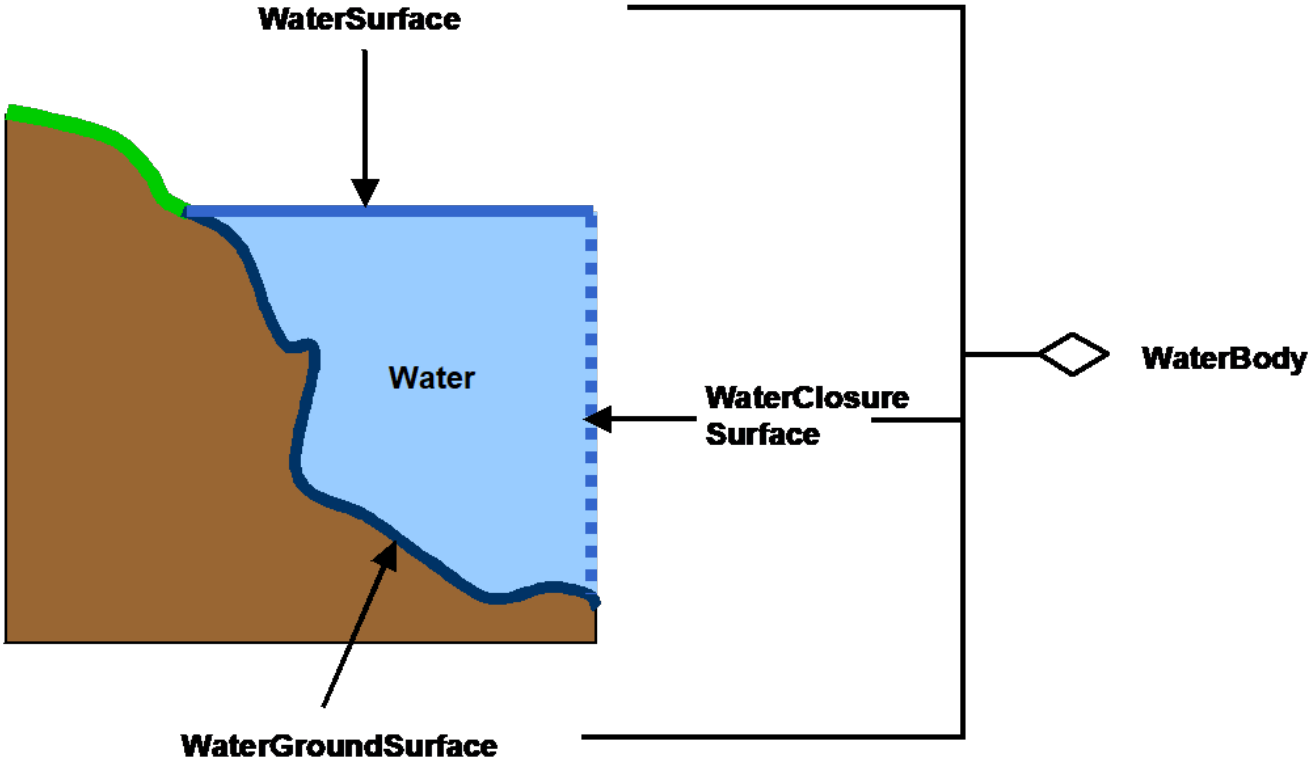



*Figure 107. Illustration of a water body defined in CityGML (graphic: IGG Uni Bonn).*

1. Boundary class “Air to Water”. The WaterSurface is mandatory to the model and usually is registered using photogrammetric analysis or mapping exploration. The representation may vary due to tidal flats or changing water levels, which can be reflected by including different static water surfaces having different waterLevels, as for example highest flooding event, mean sea level, or minimum water level. This offers the opportunity to describe significant water surfaces due to levels that are important for certain representations e.g. in tidal zones.
2. Boundary class “Water to Ground”. The WaterGroundSurface may be known by sonar exploration or other depth measurements. The WaterSurface objects together with the WaterGroundSurface objects enclose the WaterBody as a volume.

## 8.17.4. Level of Detail

Both LOD0 and LOD1 represent a low level of illustration and high grade of generalisation. Here the rivers are modelled as GM_MultiCurve geometry and brooks are omitted. Seas, oceans and lakes with significant extent are represented as a GM_MultiSurface ([figure-56]). Every WaterBody may

be assigned a combination of geometries of different types. Linear water bodies are represented as a network of 3D curves. Each curve is composed of straight line segments, where the line orientation denotes the flow direction (water flows from the first point of a curve to the last). Areal objects like lakes or seas are represented by 3D surface geometries of the water surface.

Starting from LOD1 water bodies may also be modelled as water filled volumes represented by GM_Solids. If a water body is represented by a GM_Solids in LOD2 or higher, the surface geometries of the corresponding thematic WaterGroundSurface, and WaterSurface objects must coincide with the exterior shell of the GM_Solid. This can be ensured, if for each LOD X the respective lodXSolid representation (where X is between 2 and 3) does not redundantly define the geometry, but instead references the corresponding polygons of the lodXSurface elements (where X is between 2 and 3) of WaterGroundSurface and WaterSurface.

LOD2 to LOD3 demand a higher grade of detail and therefore any WaterBody can be outlined by <<thematic-surface-concept,thematic surfaces> or a solid composed of the surrounding <<thematic-surface-concept,thematic surfaces>.

Every object of the class WaterSurface and WaterGroundSurface must have at least one associated surface geometry. This means, that every WaterSurface and WaterGroundSurface feature within a CityGML instance document must contain at least one of the following properties: lod1MultiSurface, lod2MultiSurface, lod3MultiSurface.

The water body model implicitly includes the concept of TerrainIntersectionCurves (TIC), e.g. to specify the exact intersection of the DTM with the 3D geometry of a WaterBody or to adjust a WaterBody or WaterSurface to the surrounding DTM. The rings defining the WaterSurface polygons implicitly delineate the intersection of the water body with the terrain or basin.

### 8.17.5. UML Model

The UML diagram of the water body model is depicted in Figure 108. Each WaterBody object may have the attributes `class`, `function` and `usage` whose possible values can be enumerated in code lists. The attribute `class` defines the classification of the object, e.g. lake, river, or fountain and can occur only once. The attribute `function` contains the purpose of the object like, for example national waterway or public swimming, while the attribute `usage` defines the actual usages, e.g. whether the water body is navigable. The latter two attributes can occur multiple times.

Since WaterBody is a subclass of AbstractOccupiedSpace and hence a feature, it inherits the attribute `name` from the AbstractFeature class. The WaterBody can be differentiated semantically by the class AbstractWaterBoundarySurface. A AbstractWaterBoundarySurface is a part of the water body's exterior shell with a special function like WaterSurface or WaterGroundSurface. As with any CityObject, WaterBody objects as well as WaterSurface and WaterGroundSurface may be assigned ExternalReferences and may be augmented by generic attributes using CityGML's Generics module (see [ug_model_generics_section]).

The optional attribute `waterLevel` of a WaterSurface can be used to describe the water level, for which the given 3D surface geometry was acquired. This is especially important when the water body is influenced by the tide. The allowed values can be defined in a corresponding code list.

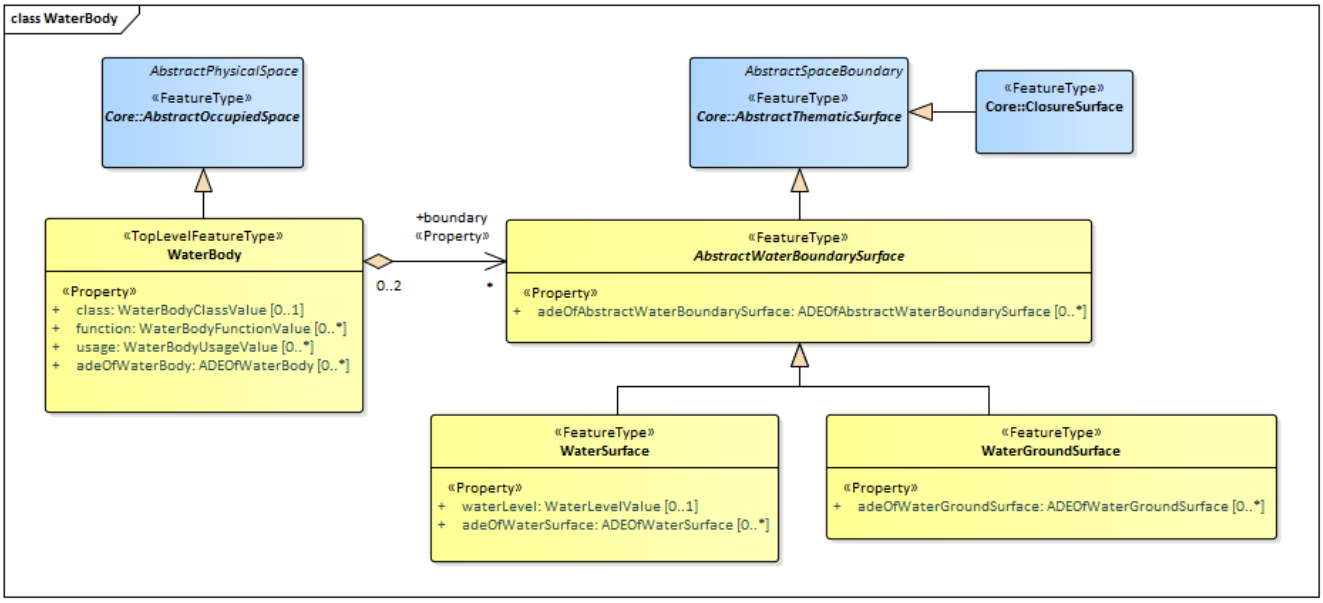



*Figure 108. UML diagram of the Water Body Model.*

The ADE data types provided for the Water Body module are illustrated in Figure 109.

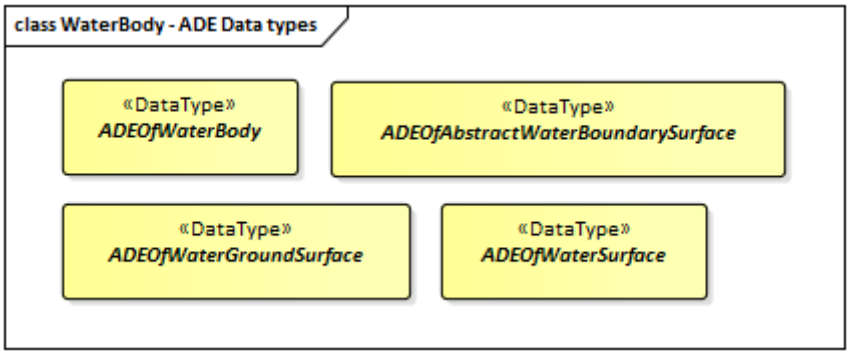



*Figure 109. ADE classes of the CityGML Water Body module.*

The Code Lists provided for the Water Body module are illustrated in Figure 110.

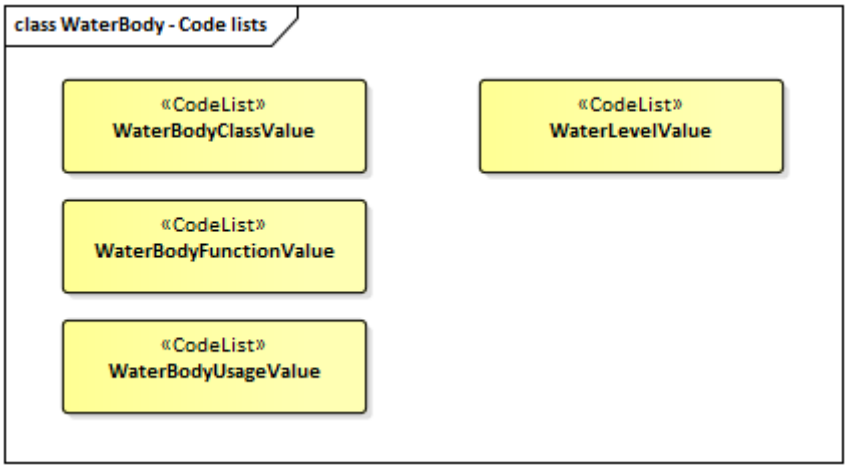



*Figure 110. Codelists from the CityGML Water Body module.*

## 8.17.6. Examples

# 8.18. Extensions

| Contributors |
| --- |
| TBD |

CityGML has been designed as an application independent information model and exchange format for 3D city and landscape models. However, specific applications typically have additional information needs to be modeled and exchanged. In general, there are two different approaches to combine city model data and application data:

1. Embed the CityGML objects into a (larger) application framework and establish the connection between application data and CityGML data within the application framework. For example, CityGML data fragments may be embedded into the application's XML data files or stored as attributes of application objects according to the application's data model.
2. Incorporate application specific information into the CityGML instance documents. This approach is especially feasible if the application specific information follows essentially the same structure as defined by the CityGML schema. This is the case, if the application data could be represented by additional attributes of CityGML objects and only some new feature types would have to be defined.

In the following, we will focus on the second option, as only this approach lies within the scope of the CityGML 3.0 Standard. Generic attributes and objects have already been introduced as a first possibility to support the exchange of application specific data (see [ug_generics_section]). Whereas they allow to extend CityGML without changing its schema definitions, this flexibility has some disadvantages:

- Generic attributes and objects may occur arbitrarily in the CityGML instance documents, but there is no formal specification of the names, datatypes, and multiplicities. Thus, there is no guarantee for an application that a specific instance of a generic attribute is included a minimum or maximum number of times per CityGML feature. Unlike the predefined CityGML objects, the concrete layout and occurrence of generic objects and attributes cannot be validated by a parser. This may reduce semantic interoperability.
- Naming conflicts of generic attributes or objects can occur, if the CityGML instance documents should be augmented by specific information from different applications simultaneously.
- There is only a limited number of predefined data types that can be used for generic attributes. Also the structure of generic objects might not be appropriate to represent more complex objects.

If application specific information are well-structured, it is desirable to represent them in a systematic way, i. e. by the definition of an extra formal schema based on the CityGML schema definitions. Such an schema is called a CityGML Application Domain Extension (ADE). It allows to validate instance documents both against the extended CityGML and the ADE schema and therefore helps to maintain semantic and syntactic interoperability between different systems working in the same application field. In order to prevent naming conflicts, every ADE has to be defined within its own namespace which must differ from the namespaces associated with the CityGML modules. An ADE schema may extend one or more CityGML module schemas. The relevant CityGML module schemas have to be imported by the ADE schema.

The ADE concept defines a special way of extending existing CityGML feature types which allows to use different ADEs within the same instance document simultaneously. For example, the specification of ADEs can be useful in the following application fields: cultural heritage (extension of abstract class _CityObject e.g. by time period information and monument protection status); representation of subsurface objects (tunnel, underpass) or city lighting (light sources like street lamps and house lights); real estate management (economic parameters of the CityGML features; inclusion of attributes defined for real estate assets as defined by OSCRE); utility networks (as topographic features); additional building properties as defined by the U.S. national building information model standard (NBIMS).

### 8.18.1. Technical principle of ADEs

| | |
|---|---|
| **NOTE** | This section is no longer current but does look like it would be useful. Can we update it? |

Each ADE is specified by its own XML schema file. The target namespace is provided by the information community who specifies the CityGML ADE. This is typically not the OGC or the SIG 3D. The namespace should be in the control of this information community and must be given as a previously unused and globally unique URI. This URI will be used in CityGML ADE instance documents to distinguish extensions from CityGML base elements. As the URI refers to the information community it also denotes the originator of the employed ADE.

The ADE's XML schema file must be available (or accessible on the Internet) to everybody creating and parsing CityGML instance documents including these ADE specific augmentations.

An ADE XML schema can define various extensions to CityGML. However, all extensions shall belong to one of the two following categories:

1. New feature types are defined within the ADE namespace and are based on CityGML abstract or con-crete classes. In general, this mechanism follows the same principles as the definition of application schemas for GML. This means, that new feature types have to be derived from existing (here: CityGML) feature types. For example, new feature types could be defined by classes derived from the abstract classes like _CityObject or _AbstractBuilding or the concrete class CityFurniture. The new fea-ture types then automatically inherit all properties (i.e. attributes) and associations (i.e. relations) from the respective CityGML superclasses.
2. Existing CityGML feature types are extended by application specific properties (in the ADE namespace). These properties may have simple or complex data types. Also geometries or embedded features (feature properties) are possible. The latter can also be used to model relations to other features.

   In this case, extension of the CityGML feature type is not being realised by the inheritance mechanism of XML schema. Instead, every CityGML feature type provides a "hook" in its XML schema definition, that allows to attach additional properties to it by ADEs. This "hook" is implemented as a GML proper-ty of the form "_GenericApplicationPropertyOf<Featuretypename>" where <Featuretypename> is equal to the name of the feature type definition in which it is included. The datatype for these kinds of properties is always "xsd:anyType" from the XSD namespace. The minimum occurrence of the "_GenericApplicationPropertyOf<Featuretypename>" is 0 and the maximum occurrence

unbounded. This means, that the CityGML schema allows that every CityGML feature may have an arbitrary num-ber of additional properties with arbitrary XML content with the name "_GenericApplication-PropertyOf<Featuretypename>". For example, the last property in the definition of the CityGML fea-ture type LandUse is the element _GenericApplicationPropertyOfLandUse (cf. chapter 10.10.1).

Such properties are called "hooks" to attach application specific properties, because they are used as the head of a substitution group by ADEs. Whenever an ADE wants to add an extra property to an existing CityGML feature type, it should declare the respective element with the appropriate dataype within the ADE namespace. In the element declaration this element shall be explicitly assigned to the substitution group defined by the corresponding "_GenericApplicationPropertyOf<Featuretypename>" in the cor-responding CityGML module namespace. An example is given in the following subsection.

By following this concept, it is possible to specify different ADEs for different information communi-ties. Every ADE may add their specific properties to the same CityGML feature type as they all can be-long to the same substitution group. This allows to have CityGML instance documents where CityGML features contain additional information from different ADEs simultaneously.

Please note that usage of ADEs introduces an extra level of complexity as data files may contain mixed infor-mation (features, properties) from different namespaces, not only from the GML and CityGML module namespaces. However, extended CityGML instance documents are quite easy to handle by applications that are not "schema-aware", i.e. applications that do not parse and interpret GML application schemas in a generic way. These applications can simply skip anything from a CityGML instance document that is not from a CityGML module or GML namespace. Thus, a building is still represented by the <bldg:Building> element with the standard CityGML properties, but with possibly some extra properties from different namespaces. Also features from a different namespace than those declared by CityGML modules or GML could be skipped (e.g. by a viewer application).

## 8.18.2. Example ADE

In this section, the ADE mechanism is illustrated by a short example, which deals with the application of virtual 3D city models to generate noise pollution maps. In our example, two extensions of CityGML are required for this task: buildings have to be extended to represent a "noise reflection correction" value and the number of inhabitants. As a new feature type noise barriers have to be defined which also have a "noise reflection correc-tion" value.

The XSD schema which has to be defined to implement this model declares a new namespace for the noise extension (http://www.citygml.org/ade/noise_de/2.0). Additionally, the namespaces of the extended CityGML modules are declared (for corresponding prefixes see chapter 4.3 and chapter 7), and the respective schema definition files are imported. The XML schema adds the elements buildingReflectionCorrection and buildingHabitants, both being members of the substitution group bldg:_GenericApplicationPropertyOf-AbstractBuilding which is defined by the CityGML Building module. Thus, both elements may be used as child elements of CityGML building features. Noise barriers are represented as NoiseCityFurnitureSegment elements. The corresponding type NoiseCityFurnitureSegmentType is defined as subtype of the CityGML abstract type

core:AbstractCityObjectType provided by the CityGML Core module, applying the usual subtyping mechanism of XML and XSD. A further element noiseCityFurnitureSegmentProperty is added as a member of the substitu-tion group frn:_GenericApplicationPropertyOfCityFurniture. By this means, noise barriers may be modelled as child elements of CityGML city furniture objects.

The XSD file for this example CityGML Noise ADE is given by the following excerpt (the complete CityGML Noise ADE is given in Annex H):

**NOTE** insert example here (GML?)

An example for a feature collection in a corresponding instance document is depicted below. Two CityGML buildings contain application specific properties distinguished from CityGML properties by the namespace prefix noise. The other properties, function and geometry, are defined by corresponding CityGML modules. In addition to the buildings, a noise barrier as child of a city furniture element is included in the feature collection. Please note, that the order of the child elements in the sequence is not arbitrary: the child elements defined by an ADE subschema have to occur after the child elements defined by CityGML modules. There is, however, no specific order of the ADE properties.

**NOTE** insert example here (GML?)

# Chapter 9. CityGML Data Dictionary

The CityGML UML model is the normative definition of the CityGML Conceptual Model. The Data Dictionary tables in this section were software generated from the UML model. As such, this section provides a normative representation of the CityGML Conceptual Model.

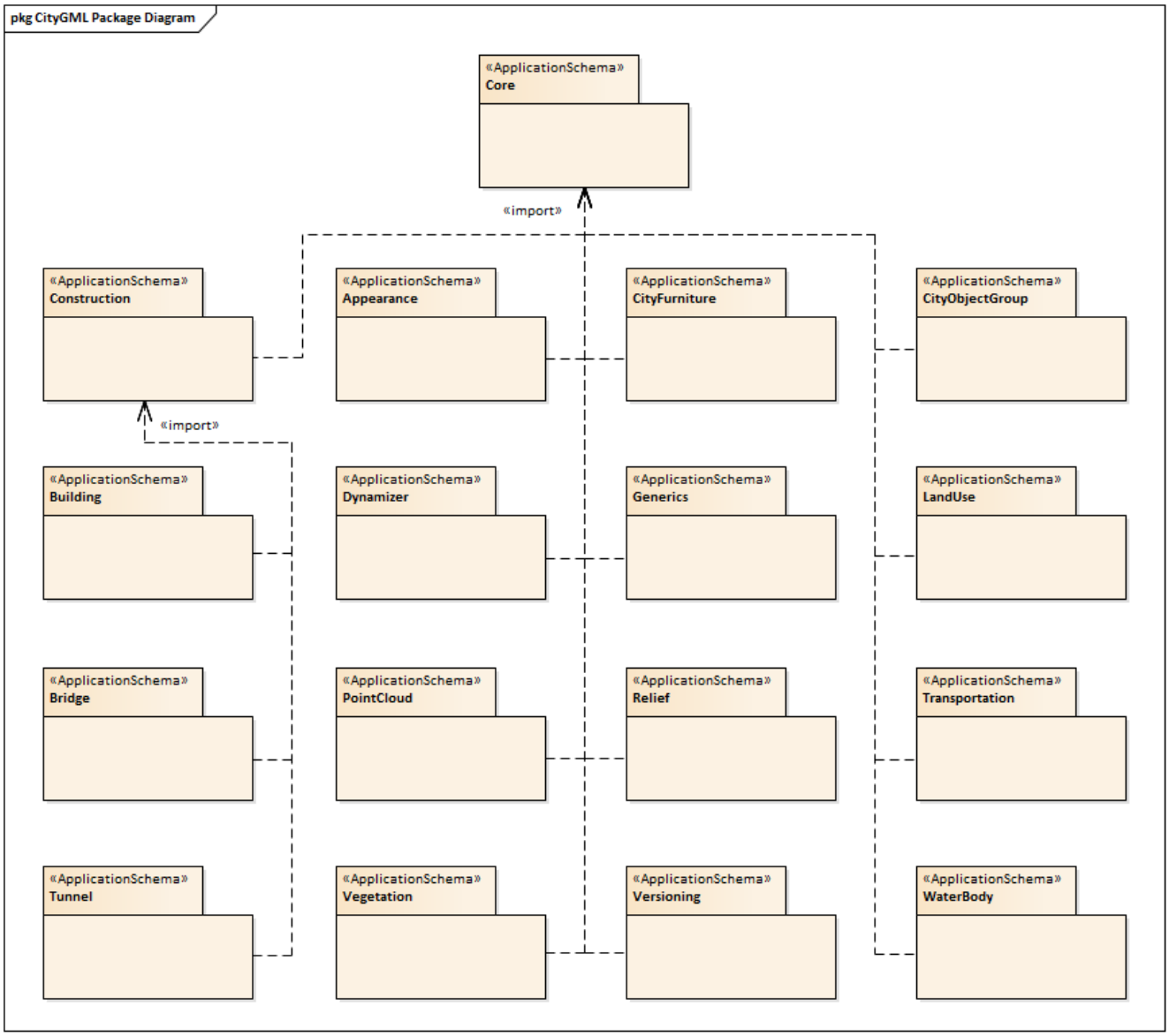



*Figure 111. CityGML UML Packages*

## 9.1. ISO Classes

The following classes are defined in ISO standards and used by the CityGML Conceptual Model.

### 9.1.1. Class AnyFeature (ISO 19109:2015)

| **AnyFeature** |
| --- |

| | |
|---|---|
| Definition: | AnyFeature is an abstract class that is the generalization of all feature types. AnyFeature is an instance of the «metaclass» FeatureType [cf. ISO 19109]. |
| Subclass Of: | none |
| StereoType: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| | FeatureType [1..1] | |

Note: Unless otherwise specified, all attribute and role names have the stereotype «Property»

### 9.1.2. Class CV_DiscreteGridPointCoverage (ISO 19123:2005)

**CV_DiscreteGridPointCoverage**

| | |
|---|---|
| Definition: | A coverage that returns the same feature attribute values for every direct position within any single spatial object, temporal object or spatiotemporal object in its domain. |
| Subclass Of: | CV_DiscreteCoverage |
| StereoType: | «type» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| element | CV_GridPointValuePair [1..*] | |
| valueAssignment | CV_GridValuesMatrix [1..1] | |

Note: Unless otherwise specified, all attribute and role names have the stereotype «Property»

### 9.1.3. Class DirectPosition (ISO 19107: 2003)

**DirectPosition**

| | |
|---|---|
| Definition: | DirectPosition object data types (Figure 14) hold the coordinates for a position within some coordinate reference system. The coordinate reference system is described in ISO 19111. Since DirectPositions, as data types, will often be included in larger objects (such as GM_Objects) that have references to ISO19111::SC_CRS, the DirectPosition::cordinateReferenceSystem may be left NULL if this particular DirectPosition is included in a larger object with such a reference to a SC_CRS. In this case, the DirectPosition::cordinateReferenceSystem is implicitly assumed to take on the value of the containing object's SC_CRS. |
| Subclass Of: | None |
| StereoType: | None |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| CRS | CRS [0..1] | |
| CRS | SC_CRS [0..1] | |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| coordinate | Sequence<Number> [1..1] | |
| dimension | Integer [1..1] | |

Note: Unless otherwise specified, all attribute and role names have the stereotype «Property»

### 9.1.4. Class GM_Object (ISO 19107: 2003)

**GM_Object**

| | |
|---|---|
| Definition: | GM_Object is the root class of the geometric object taxonomy and supports interfaces common to all geographically referenced geometric objects. GM_Object instances are sets of direct positions in a particular coordinate reference system. A GM_Object can be regarded as an infinite set of points that satisfies the set operation interfaces for a set of direct positions, TransfiniteSet<DirectPosition>. Since an infinite collection class cannot be implemented directly, a Boolean test for inclusion shall be provided by the GM_Object interface. This international standard concentrates on vector geometry classes, but future work may use GM_Object as a root class without modification. NOTE As a type, GM_Object does not have a well-defined default state or value representation as a data type. Instantiated subclasses of GM_Object will. |
| Subclass Of: | none |
| StereoType: | «type» |
| Constraint: | dimension() > boundary().dimension (Invariant): |
| Constraint: | boundary().notEmpty() implies boundary().dimension() = dimension() -1 (Invariant): |
| Constraint: | boundary().isEmpty() = isCycle() (Invariant): |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| | Geometry [1..1] | |
| | TransfiniteSet<DirectPosition> [1..1] | |
| | CV_DomainObject [1..1] | |
| CRS | CRS [0..1] | |
| CRS | SC_CRS [0..1] | |

Note: Unless otherwise specified, all attribute and role names have the stereotype «Property»

### 9.1.5. Class GM_MultiCurve (ISO 19107: 2003)

**GM_MultiCurve**

| | |
|---|---|
| Definition: | An aggregate class containing only instances of GM_OrientableCurve. The association role “element” shall be the set of GM_OrientableCurves contained in this GM_MultiCurve. |
| Subclass Of: | GM_MultiPrimitive |
| StereoType: | «type» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| length | Length [1..1] | |

Note: Unless otherwise specified, all attribute and role names have the stereotype «Property»

### 9.1.6. Class GM_MultiPoint (ISO 19107:2003)

**GM_MultiPoint**

| | |
|---|---|
| Definition: | GM_MultiPoint is an aggregate class containing only points. The association role “element” shall be the set of GM_Points contained in this GM_MultiPoint. |
| Subclass Of: | GM_MultiPrimitive |
| StereoType: | «type» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| position | Set<DirectPosition> [1..1] | |

Note: Unless otherwise specified, all attribute and role names have the stereotype «Property»

### 9.1.7. Class GM_MultiSurface (ISO 19107:2003)

**GM_MultiSurface**

| | |
|---|---|
| Definition: | An aggregate class containing only instances of GM_OrientableSurface. The association role “element” shall be the set of GM_OrientableSurfaces contained in this GM_MultiSurface. |
| Subclass Of: | GM_MultiPrimitive |
| StereoType: | «type» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| area | Area [1..1] | |
| perimeter | Length [1..1] | |

Note: Unless otherwise specified, all attribute and role names have the stereotype «Property»

### 9.1.8. Class GM_Point (ISO 19107:2003)

**GM_Point**

| | |
|---|---|
| Definition: | GM_Point is the basic data type for a geometric object consisting of one and only one point. |
| Subclass Of: | GM_Primitive |
| StereoType: | «type» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| | Point [1..1] | |
| composite | GM_CompositePoint [0..*] | |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| position | DirectPosition [1..1] | The attribute "position" shall be the DirectPosition of this GM_Point. GM_Point::position [1] : DirectPosition NOTE In most cases, the state of a GM_Point is fully determined by its position attribute. The only exception to this is if the GM_Point has been subclassed to provide additional non-geometric information such as symbology. |

Note: Unless otherwise specified, all attribute and role names have the stereotype «Property»

### 9.1.9. Class GM_Solid (ISO 19107:2003)

**GM_Solid**

| | |
|---|---|
| Definition: | GM_Solid, a subclass of GM_Primitive, is the basis for 3-dimensional geometry. The extent of a solid is defined by the boundary surfaces. |
| Subclass Of: | GM_Primitive |
| StereoType: | «type» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| composite | GM_CompositeSolid [0..*]<br>Solid [1..1] | |

Note: Unless otherwise specified, all attribute and role names have the stereotype «Property»

### 9.1.10. Class GM_Surface (ISO 19107:2003)

**GM_Surface**

| | |
|---|---|
| Definition: | GM_Surface is a subclass of GM_Primitive and is the basis for 2-dimensional geometry. Unorientable surfaces such as the Möbius band are not allowed. The orientation of a surface chooses an "up" direction through the choice of the upward normal, which, if the surface is not a cycle, is the side of the surface from which the exterior boundary appears counterclockwise. Reversal of the surface orientation reverses the curve orientation of each boundary component, and interchanges the conceptual "up" and "down" direction of the surface. If the surface is the boundary of a solid, the "up" direction is usually outward. For closed surfaces, which have no boundary, the up direction is that of the surface patches, which must be consistent with one another. Its included GM_SurfacePatches describe the interior structure of a GM_Surface. NOTE Other than the restriction on orientability, no other "validity" condition is required for GM_Surface. |
| Subclass Of: | GM_OrientableSurface |
| StereoType: | «type» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| | GM_GenericSurface [1..1]<br>Building [0..*] | |

Note: Unless otherwise specified, all attribute and role names have the stereotype «Property»

### 9.1.11. Class GM_Tin (ISO 19107:2003)

**GM_Tin**

| | |
|---|---|
| Definition: | A GM_Tin is a GM_TriangulatedSurface that uses the Delaunay algorithm or a similar algorithm complemented with consideration for breaklines, stoplines and maximum length of triangle sides (Figure 22). These networks satisfy the Delaunay criterion away from the modifications: For each triangle in the network, the circle passing through its vertexes does not contain, in its interior, the vertex of any other triangle. |
| Subclass Of: | GM_TriangulatedSurface |
| StereoType: | «type» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| breakLines | Set<GM_LineString> [1..1] | |
| controlPoint | GM_Position [3..*] | |
| maxLength | Distance [1..1] | |
| stopLines | Set<GM_LineString> [1..1] | |

Note: Unless otherwise specified, all attribute and role names have the stereotype «Property»

### 9.1.12. Class GM_TriangulatedSurface (ISO 19107:2003)

**GM_TriangulatedSurface**

| | |
|---|---|
| Definition: | A GM_TriangulatedSurface is a GM_PolyhedralSurface that is composed only of triangles (GM_Triangle). There is no restriction on how the triangulation is derived. |
| Subclass Of: | GM_PolyhedralSurface |
| StereoType: | «type» |

Note: Unless otherwise specified, all attribute and role names have the stereotype «Property»

### 9.1.13. Class SC_CRS (ISO 19111:2019)

**SC_CRS**

| | |
|---|---|
| Definition: | Coordinate reference system which is usually single but may be compound. |
| Subclass Of: | IO_IdentifiedObjectBase, RS_ReferenceSystem |
| StereoType: | «type» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| coordOperationTo | CC_CoordinateOperation [0..*] | Not-navigable association from a Coordinate Operation that uses ths CRS as its targetCRS. |
| grid | CV_ReferenceableGrid [0..*] | |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| scope | CharacterString [1..*] | Description of usage, or limitations of usage, for which this CRS is valid. If unknown, enter "not known". |

Note: Unless otherwise specified, all attribute and role names have the stereotype «Property»

### 9.1.14. Class TM_Position (ISO 19108:2006)

**TM_Position**

| | |
|---|---|
| Definition: | TM_Position is a union class that consists of one of the data types listed as its attributes. Date, Time, and DateTime are basic data types defined in ISO/TS 19103. |
| Subclass Of: | None |
| StereoType: | «Union» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| anyOther | TM_TemporalPosition [1..1] | |
| date8601 | Date [1..1] | |
| time8601 | Time [1..1] | |
| dateTime8601 | DateTime [1..1] | |

Note: Unless otherwise specified, all attribute and role names have the stereotype «Property»

# 9.2. Core

| | |
|---|---|
| Description: | The Core module defines the basic components of the CityGML data model. The Core module defines abstract base classes that define the core properties of more specialized thematic classes defined in other modules. The Core module also defines concrete classes that are common to other modules, for example basic data types. |
| Parent Package: | CityGML |
| Stereotype: | «ApplicationSchema» |

## 9.2.1. Classes

**AbstractAppearance**

| | |
|---|---|
| Definition: | AbstractAppearance is the abstract superclass to represent any kind of appearance objects. |
| Subclass of: | AbstractFeatureWithLifespan |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfAbstractAppearance | ADEOfAbstractAppearance [0..*] | Augments AbstractAppearance with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractCityObject**

| | |
|---|---|
| Definition: | AbstractCityObject is the abstract superclass of all thematic classes within the CityGML Conceptual Model. |
| Subclass of: | AbstractFeatureWithLifespan |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| generalizesTo | AbstractCityObject [*] | Relates generalized representations of the same real-world object in different Levels of Detail to the city object. The direction of this relation is from the city object to the corresponding generalized city objects. |
| genericAttribute | AbstractGenericAttribute [*] | Relates generic attributes to the city object. |
| dynamizer | AbstractDynamizer [*] | Relates Dynamizer objects to the city object. These allow timeseries data to override static attribute values of the city object. |
| appearance | AbstractAppearance [*] | Relates appearances to the city object. |
| externalReference | ExternalReference [*] | References external objects in other information systems that have a relation to the city object. |
| relatedTo | AbstractCityObject [*] | Relates other city objects to the city object. It also describes how the city objects are related to each other. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| relativeToTerrain | RelativeToTerrain [0..1] | Describes the vertical position of the city object relative to the surrounding terrain. |
| relativeToWater | RelativeToWater [0..1] | Describes the vertical position of the city object relative to the surrounding water surface. |
| adeOfAbstractCityObject | ADEOfAbstractCityObject [0..*] | Augments AbstractCityObject with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractDynamizer**

| | |
|---|---|
| Definition: | AbstractDynamizer is the abstract superclass to represent Dynamizer objects. |
| Subclass of: | AbstractFeatureWithLifespan |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfAbstractDynamizer | ADEOfAbstractDynamizer [0..*] | Augments AbstractDynamizer with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractFeature**

| | |
|---|---|
| Definition: | AbstractFeature is the abstract superclass of all feature types within the CityGML Conceptual Model. |
| Subclass of: | AnyFeature |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| featureID | ID [1..1] | Specifies the unique identifier of the feature that is valid in the instance document within which it occurs. |
| identifier | ScopedName [0..1] | Specifies the unique identifier of the feature that is valid globally. |
| name | GenericName [0..*] | Specifies the name of the feature. |
| description | CharacterString [0..1] | Provides further information on the feature. |
| adeOfAbstractFeature | ADEOfAbstractFeature [0..*] | Augments AbstractFeature with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractFeatureWithLifespan**

| | |
|---|---|
| Definition: | AbstractFeatureWithLifespan is the base class for all CityGML features. This class allows the optional specification of the real-world and database times for the existence of each feature. |
| Subclass of: | AbstractFeature |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| creationDate | DateTime [0..1] | Indicates the date at which a CityGML feature was added to the CityModel. |
| terminationDate | DateTime [0..1] | Indicates the date at which a CityGML feature was removed from the CityModel. |
| validFrom | DateTime [0..1] | Indicates the date at which a CityGML feature started to exist in the real world. |
| validTo | DateTime [0..1] | Indicates the date at which a CityGML feature ended to exist in the real world. |
| adeOfAbstractFeatureWithLifespan | ADEOfAbstractFeatureWithLifespan [0..*] | Augments AbstractFeatureWithLifespan with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractLogicalSpace**

| | |
|---|---|
| Definition: | AbstractLogicalSpace is the abstract superclass for all types of logical spaces. Logical space refers to spaces that are not bounded by physical surfaces but are defined according to thematic considerations. |
| Subclass of: | AbstractSpace |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfAbstractLogicalSpace | ADEOfAbstractLogicalSpace [0..*] | Augments AbstractLogicalSpace with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractOccupiedSpace**

| | |
|---|---|
| Definition: | AbstractOccupiedSpace is the abstract superclass for all types of physically occupied spaces. Occupied space refers to spaces that are partially or entirely filled with matter. |
| Subclass of: | AbstractPhysicalSpace |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| lod3ImplicitRepresentation | ImplicitGeometry [0..1] | Relates to an implicit geometry that represents the occupied space in Level of Detail 3. |
| lod1ImplicitRepresentation | ImplicitGeometry [0..1] | Relates to an implicit geometry that represents the occupied space in Level of Detail 1. |
| lod2ImplicitRepresentation | ImplicitGeometry [0..1] | Relates to an implicit geometry that represents the occupied space in Level of Detail 2. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfAbstractOccupiedSpace | ADEOfAbstractOccupiedSpace [0..*] | Augments AbstractOccupiedSpace with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractPhysicalSpace**

| | |
|---|---|
| Definition: | AbstractPhysicalSpace is the abstract superclass for all types of physical spaces. Physical space refers to spaces that are fully or partially bounded by physical objects. |
| Subclass of: | AbstractSpace |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| lod3TerrainIntersectionCurve | GM_MultiCurve [0..1] | Relates to a 3D MultiCurve geometry that represents the terrain intersection curve of the physical space in Level of Detail 3. |
| lod2TerrainIntersectionCurve | GM_MultiCurve [0..1] | Relates to a 3D MultiCurve geometry that represents the terrain intersection curve of the physical space in Level of Detail 2. |
| pointCloud | AbstractPointCloud [0..1] | Relates to a 3D PointCloud that represents the physical space. |
| lod1TerrainIntersectionCurve | GM_MultiCurve [0..1] | Relates to a 3D MultiCurve geometry that represents the terrain intersection curve of the physical space in Level of Detail 1. |

| Attribute | Value type and multiplicity | Definition |
| --- | --- | --- |
| adeOfAbstractPhysicalSpace | ADEOfAbstractPhysicalSpace [0..*] | Augments AbstractPhysicalSpace with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractPointCloud**

| | |
| --- | --- |
| Definition: | AbstractPointCloud is the abstract superclass to represent PointCloud objects. |
| Subclass of: | AbstractFeature |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
| --- | --- | --- |
| adeOfAbstractPointCloud | ADEOfAbstractPointCloud [0..*] | Augments AbstractPointCloud with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractSpace**

| | |
| --- | --- |
| Definition: | AbstractSpace is the abstract superclass for all types of spaces. A space is an entity of volumetric extent in the real world. |
| Subclass of: | AbstractCityObject |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| lod2MultiCurve | GM_MultiCurve [0..1] | Relates to a 3D MultiCurve geometry that represents the space in Level of Detail 2. |
| lod0MultiCurve | GM_MultiCurve [0..1] | Relates to a 3D MultiCurve geometry that represents the space in Level of Detail 0. |
| lod0MultiSurface | GM_MultiSurface [0..1] | Relates to a 3D MultiSurface geometry that represents the space in Level of Detail 0. |
| lod2MultiSurface | GM_MultiSurface [0..1] | Relates to a 3D MultiSurface geometry that represents the space in Level of Detail 2. |
| lod3MultiSurface | GM_MultiSurface [0..1] | Relates to a 3D MultiSurface geometry that represents the space in Level of Detail 3. |
| lod0Point | GM_Point [0..1] | Relates to a 3D Point geometry that represents the space in Level of Detail 0. |
| lod3Solid | GM_Solid [0..1] | Relates to a 3D Solid geometry that represents the space in Level of Detail 3. |
| lod3MultiCurve | GM_MultiCurve [0..1] | Relates to a 3D MultiCurve geometry that represents the space in Level of Detail 3. |
| lod2Solid | GM_Solid [0..1] | Relates to a 3D Solid geometry that represents the space in Level of Detail 2. |
| boundary | AbstractSpaceBoundary [*] | Relates to surfaces that bound the space. |
| lod1Solid | GM_Solid [0..1] | Relates to a 3D Solid geometry that represents the space in Level of Detail 1. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| spaceType | SpaceType [0..1] | Specifies the degree of openness of a space. |
| volume | QualifiedVolume [0..*] | Specifies qualified volumes related to the space. |
| area | QualifiedArea [0..*] | Specifies qualified areas related to the space. |
| adeOfAbstractSpace | ADEOfAbstractSpace [0..*] | Augments AbstractSpace with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractSpaceBoundary**

| | |
|---|---|
| Definition: | AbstractSpaceBoundary is the abstract superclass for all types of space boundaries. A space boundary is an entity with areal extent in the real world. Space boundaries are objects that bound a Space. They also realize the contact between adjacent spaces. |
| Subclass of: | AbstractCityObject |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfAbstractSpaceBoundary | ADEOfAbstractSpaceBoundary [0..*] | Augments AbstractSpaceBoundary with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractThematicSurface**

| | |
|---|---|
| Definition: | AbstractThematicSurface is the abstract superclass for all types of thematic surfaces. |
| Subclass of: | AbstractSpaceBoundary |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| lod1MultiSurface | GM_MultiSurface [0..1] | Relates to a 3D MultiSurface geometry that represents the thematic surface in Level of Detail 1. |
| pointCloud | AbstractPointCloud [0..1] | Relates to a 3D PointCloud that represents the thematic surface. |
| lod0MultiCurve | GM_MultiCurve [0..1] | Relates to a 3D MultiCurve geometry that represents the thematic surface in Level of Detail 0. |
| lod3MultiSurface | GM_MultiSurface [0..1] | Relates to a 3D MultiSurface geometry that represents the thematic surface in Level of Detail 3. |
| lod0MultiSurface | GM_MultiSurface [0..1] | Relates to a 3D MultiSurface geometry that represents the thematic surface in Level of Detail 0. |
| lod2MultiSurface | GM_MultiSurface [0..1] | Relates to a 3D MultiSurface geometry that represents the thematic surface in Level of Detail 2. |

| Attribute | Value type and multiplicity | Definition |
| --- | --- | --- |
| area | QualifiedArea [0..*] | Specifies qualified areas related to the thematic surface. |
| adeOfAbstractThematicSurface | ADEOfAbstractThematicSurface [0..*] | Augments AbstractThematicSurface with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractUnoccupiedSpace**

| | |
| --- | --- |
| Definition: | AbstractUnoccupiedSpace is the abstract superclass for all types of physically unoccupied spaces. Unoccupied space refers to spaces that are entirely or mostly free of matter. |
| Subclass of: | AbstractPhysicalSpace |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
| --- | --- | --- |
| adeOfAbstractUnoccupiedSpace | ADEOfAbstractUnoccupiedSpace [0..*] | Augments AbstractUnoccupiedSpace with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractVersion**

| | |
| --- | --- |
| Definition: | AbstractVersion is the abstract superclass to represent Version objects. |
| Subclass of: | AbstractFeatureWithLifespan |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
| --- | --- | --- |
| adeOfAbstractVersion | ADEOfAbstractVersion [0..*] | Augments AbstractVersion with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractVersionTransition**

| | |
|---|---|
| Definition: | AbstractVersionTransition is the abstract superclass to represent VersionTransition objects. |
| Subclass of: | AbstractFeatureWithLifespan |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfAbstractVersionTransition | ADEOfAbstractVersionTransition [0..*] | Augments AbstractVersionTransition with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Address**

| | |
|---|---|
| Definition: | Address represents an address of a city object. |
| Subclass of: | AbstractFeature |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| multiPoint | GM_MultiPoint [0..1] | Relates to the MultiPoint geometry of the Address. The geometry relates the address spatially to a city object. |
| xalAddress | XALAddress [1..1] | Relates an OASIS address object to the Address. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfAddress | ADEOfAddress [0..*] | Augments the Address with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**CityModel**

| | |
|---|---|
| Definition: | CityModel is the container for all objects belonging to a city model. |
| Subclass of: | AbstractFeatureWithLifespan |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| cityModelMember | CityModelMember [*] | Relates to all objects that are part of the CityModel. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| engineeringCRS | EngineeringCRS [0..1] | Specifies the local engineering coordinate reference system of the CityModel that can be provided inline the CityModel instead of referencing a well-known CRS definition. The definition of an engineering CRS requires an anchor point which relates the origin of the local coordinate system to a point on the earth's surface in order to facilitate the transformation of coordinates from the local engineering CRS. |
| adeOfCityModel | ADEOfCityModel [0..*] | Augments the CityModel with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**CityObjectRelation**

| | |
|---|---|
| Definition: | CityObjectRelation represents a specific relation from the city object in which the relation is included to another city object. |
| Subclass of: | None |
| Stereotype: | «ObjectType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| genericAttribute | AbstractGenericAttribute [*] | Relates generic attributes to the CityObjectRelation. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| relationType | RelationTypeValue [1..1] | Indicates the specific type of the CityObjectRelation. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**ClosureSurface**

| | |
|---|---|
| Definition: | ClosureSurface is a special type of thematic surface used to close holes in volumetric objects. Closure surfaces are virtual (non-physical) surfaces. |
| Subclass of: | AbstractThematicSurface |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfClosureSurface | ADEOfClosureSurface [0..*] | Augments the ClosureSurface with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**ImplicitGeometry**

| | |
|---|---|
| Definition: | ImplicitGeometry is a geometry representation where the shape is stored only once as a prototypical geometry. Examples are a tree or other vegetation object, a traffic light or a traffic sign. This prototypic geometry object can be re-used or referenced many times, wherever the corresponding feature occurs in the 3D city model. |
| Subclass of: | None |
| Stereotype: | «ObjectType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| relativeGeometry | GM_Object [0..1] | Relates to a prototypical geometry in a local coordinate system stored inline with the city model. |
| referencePoint | GM_Point [1..1] | Relates to a 3D Point geometry that represents the base point of the object in the world coordinate system. |
| appearance | AbstractAppearance [*] | Relates appearances to the ImplicitGeometry. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| objectID | ID [1..1] | Specifies the unique identifier of the ImplicitGeometry. |
| transformationMatrix | TransformationMatrix4x4 [1..1] | Specifies the mathematical transformation (translation, rotation, and scaling) between the prototypical geometry and the actual spatial position of the object. |
| mimeType | MimeTypeValue [0..1] | Specifies the MIME type of the external file that stores the prototypical geometry. |
| libraryObject | URI [0..1] | Specifies the URI that points to the prototypical geometry stored in an external file. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.2.2. Data Types

**AbstractGenericAttribute**

| | |
|---|---|
| Definition: | AbstractGenericAttribute is the abstract superclass for all types of generic attributes. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractAppearance**

| | |
|---|---|
| Definition: | ADEOfAbstractAppearance acts as a hook to define properties within an ADE that are to be added to AbstractAppearance. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractCityObject**

| | |
|---|---|
| Definition: | ADEOfAbstractCityObject acts as a hook to define properties within an ADE that are to be added to AbstractCityObject. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractDynamizer**

| | |
|---|---|
| Definition: | ADEOfAbstractDynamizer acts as a hook to define properties within an ADE that are to be added to AbstractDynamizer. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractFeature**

| | |
|---|---|
| Definition: | ADEOfAbstractFeature acts as a hook to define properties within an ADE that are to be added to AbstractFeature. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractFeatureWithLifespan**

| | |
|---|---|
| Definition: | ADEOfAbstractFeatureWithLifespan acts as a hook to define properties within an ADE that are to be added to AbstractFeatureWithLifespan. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractLogicalSpace**

| | |
|---|---|
| Definition: | ADEOfAbstractLogicalSpace acts as a hook to define properties within an ADE that are to be added to AbstractLogicalSpace. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractOccupiedSpace**

| | |
|---|---|
| Definition: | ADEOfAbstractOccupiedSpace acts as a hook to define properties within an ADE that are to be added to AbstractOccupiedSpace. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractPhysicalSpace**

| | |
|---|---|
| Definition: | ADEOfAbstractPhysicalSpace acts as a hook to define properties within an ADE that are to be added to AbstractPhysicalSpace. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractPointCloud**

| | |
|---|---|
| Definition: | ADEOfAbstractPointCloud acts as a hook to define properties within an ADE that are to be added to AbstractPointCloud. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractSpace**

| | |
|---|---|
| Definition: | ADEOfAbstractSpace acts as a hook to define properties within an ADE that are to be added to AbstractSpace. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractSpaceBoundary**

| | |
|---|---|
| Definition: | ADEOfAbstractSpaceBoundary acts as a hook to define properties within an ADE that are to be added to AbstractSpaceBoundary. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractThematicSurface**

| | |
|---|---|
| Definition: | ADEOfAbstractThematicSurface acts as a hook to define properties within an ADE that are to be added to AbstractThematicSurface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractUnoccupiedSpace**

| | |
|---|---|
| Definition: | ADEOfAbstractUnoccupiedSpace acts as a hook to define properties within an ADE that are to be added to AbstractUnoccupiedSpace. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractVersion**

| | |
|---|---|
| Definition: | ADEOfAbstractVersion acts as a hook to define properties within an ADE that are to be added to AbstractVersion. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractVersionTransition**

| | |
|---|---|
| Definition: | ADEOfAbstractVersionTransition acts as a hook to define properties within an ADE that are to be added to AbstractVersionTransition. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAddress**

| | |
|---|---|
| Definition: | ADEOfAddress acts as a hook to define properties within an ADE that are to be added to an Address. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfCityModel**

| | |
|---|---|
| Definition: | ADEOfCityModel acts as a hook to define properties within an ADE that are to be added to a CityModel. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfClosureSurface**

| | |
|---|---|
| Definition: | ADEOfClosureSurface acts as a hook to define properties within an ADE that are to be added to a ClosureSurface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ExternalReference**

| | |
|---|---|
| Definition: | ExternalReference is a reference to a corresponding object in another information system, for example in the German cadastre (ALKIS), the German topographic information system (ATKIS), or the OS UK MasterMap®. |
| Subclass of: | None |
| Stereotype: | «DataType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| targetResource | URI [1..1] | Specifies the URI that points to the object in the external information system. |
| informationSystem | URI [0..1] | Specifies the URI that points to the external information system. |
| relationType | URI [0..1] | Specifies a URI that additionally qualifies the ExternalReference. The URI can point to a definition from an external ontology (e.g. the sameAs relation from OWL) and allows for mapping the ExternalReference to RDF triples. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Occupancy**

| | |
|---|---|
| Definition: | Occupancy is an application-dependent indication of what is contained by a feature. |
| Subclass of: | None |
| Stereotype: | «DataType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| numberOfOccupants | Integer [1..1] | Indicates the number of occupants contained by a feature. |
| interval | IntervalValue [0..1] | Indicates the time period the occupants are contained by a feature. |
| occupantType | OccupantTypeValue [0..1] | Indicates the specific type of the occupants that are contained by a feature. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**QualifiedArea**

| | |
|---|---|
| Definition: | QualifiedArea is an application-dependent measure of the area of a space or of a thematic surface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| area | Area [1..1] | Specifies the value of the QualifiedArea. |
| typeOfArea | QualifiedAreaTypeValue [1..1] | Indicates the specific type of the QualifiedArea. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**QualifiedVolume**

| | |
|---|---|
| Definition: | QualifiedVolume is an application-dependent measure of the volume of a space. |
| Subclass of: | None |
| Stereotype: | «DataType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| volume | Volume [1..1] | Specifies the value of the QualifiedVolume. |
| typeOfVolume | QualifiedVolumeTypeValue [1..1] | Indicates the specific type of the QualifiedVolume. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**XALAddress**

| | |
|---|---|
| Definition: | XALAddress represents address details according to the OASIS xAL standard. |
| Subclass of: | None |
| Stereotype: | «DataType» |

### 9.2.3. Basic Types

**Code**

| | |
|---|---|
| Definition: | Code is a basic type for a String-based term, keyword, or name that can additionally have a code space. |
| Subclass of: | None |
| Stereotype: | «BasicType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| codeSpace | URI [0..1] | Associates the Code with an authority that controls the term, keyword, or name. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**DoubleBetween0and1**

| | |
|---|---|
| Definition: | DoubleBetween0and1 is a basic type for values, which are greater or equal than 0 and less or equal than 1. The type is used for color encoding, for example. |
| Subclass of: | None |
| Stereotype: | «BasicType» |
| Constraint: | valueBetween0and1 (OCL): inv: DoubleBetween0and1.allInstances() → forAll(p \| p > = 0 and p < = 1) |

**DoubleBetween0and1List**

| | |
|---|---|
| Definition: | DoubleBetween0and1List is a basic type that represents a list of double values greater or equal than 0 and less or equal than 1. The type is used for color encoding, for example. |
| Subclass of: | None |
| Stereotype: | «BasicType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| list | DoubleBetween0and1 [1..1] | Specifies the list of double values. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**DoubleList**

| | |
|---|---|
| Definition: | DoubleList is an ordered sequence of double values. |
| Subclass of: | None |
| Stereotype: | «BasicType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| list | Real [1..1] | Specifies the list of double values. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**DoubleOrNilReasonList**

| | |
|---|---|
| Definition: | DoubleOrNilReasonList is a basic type that represents a list of double values and/or nil reasons. |
| Subclass of: | None |
| Stereotype: | «BasicType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| list | DoubleOrNilReason [1..1] | Specifies the list of double values and/or nil reasons. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**ID**

| | |
|---|---|
| Definition: | ID is a basic type that represents a unique identifier. |
| Subclass of: | None |
| Stereotype: | «BasicType» |

**IntegerBetween0and3**

| | |
|---|---|
| Definition: | IntegerBetween0and3 is a basic type for integer values, which are greater or equal than 0 and less or equal than 3. The type is used for encoding the LOD number. |
| Subclass of: | None |
| Stereotype: | «BasicType» |
| Constraint: | valueBetween0and3 (OCL): inv: IntegerBetween0and3.allInstances() → forAll(p \| p > = 0 and p < = 3) |

**MeasureOrNilReasonList**

| | |
|---|---|
| Definition: | MeasureOrNilReasonList is a basic type that represents a list of double values and/or nil reasons together with a unit of measurement. |
| Subclass of: | DoubleOrNilReasonList |
| Stereotype: | «BasicType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| uom | UnitOfMeasure [1..1] | Specifies the unit of measurement of the double values. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**TransformationMatrix2x2**

| | |
|---|---|
| Definition: | TransformationMatrix2x2 is a 2 by 2 matrix represented as a list of four double values in row major order. |
| Subclass of: | DoubleList |
| Stereotype: | «BasicType» |
| Constraint: | lengthOfList (OCL): inv: list→size() = 4 |

**TransformationMatrix3x4**

| | |
|---|---|
| Definition: | TransformationMatrix3x4 is a 3 by 4 matrix represented as a list of twelve double values in row major order. |
| Subclass of: | DoubleList |
| Stereotype: | «BasicType» |
| Constraint: | lengthOfList (OCL): inv: list→size() = 12 |

**TransformationMatrix4x4**

| | |
|---|---|
| Definition: | TransformationMatrix4x4 is a 4 by 4 matrix represented as a list of sixteen double values in row major order. |
| Subclass of: | DoubleList |
| Stereotype: | «BasicType» |
| Constraint: | lengthOfList (OCL): inv: list → size() = 16 |

### 9.2.4. Unions

**CityModelMember**

| | |
|---|---|
| Definition: | CityModelMember is a union type that enumerates the different types of objects that can occur as members of a city model. |
| Stereotype: | «Union» |

| Member name | Type | Definition |
|---|---|---|
| cityObjectMember | AbstractCityObject [1..1] | Specifies the city objects that are part of the CityModel. |
| appearanceMember | AbstractAppearance [1..1] | Specifies the appearances of the CityModel. |
| versionMember | AbstractVersion [1..1] | Specifies the different versions of the CityModel. |
| versionTransitionMember | AbstractVersionTransition [1..1] | Specifies the transitions between the different versions of the CityModel. |
| featureMember | AbstractFeature [1..1] | Specifies the feature objects that are part of the CityModel. It allows to include objects that are not derived from a class defined in the CityGML conceptual model, but from the ISO 19109 class AnyFeature. |

**DoubleOrNilReason**

| | |
|---|---|
| Definition: | DoubleOrNilReason is a union type that allows for choosing between a double value and a nil reason. |
| Stereotype: | «Union» |

| Member name | Type | Definition |
|---|---|---|
| value | Real [1..1] | Specifies the double value. |
| nilReason | NilReason [1..1] | Specifies the nil reason. |

**NilReason**

| | |
|---|---|
| Definition: | NilReason is a union type that allows for choosing between two different types of nil reason. |
| Stereotype: | «Union» |

| Member name | Type | Definition |
|---|---|---|
| nilReasonEnumeration | NilReasonEnumeration [1..1] | Indicates a nil reason that is provided in a code list. |
| URI | URI [1..1] | Specifies a URI that points to a resource that describes the nil reason. |

### 9.2.5. Code Lists

**IntervalValue**

| | |
|---|---|
| Definition: | IntervalValue is a code list used to specify a time period. |
| Stereotype: | «CodeList» |

**MimeTypeValue**

| | |
|---|---|
| Definition: | MimeTypeValue is a code list used to specify the MIME type of a referenced resource. |
| Stereotype: | «CodeList» |

**NilReasonEnumeration**

| | |
|---|---|
| Definition: | NilReasonEnumeration is a code list that enumerates the different nil reasons. |
| Stereotype: | «CodeList» |

**OccupantTypeValue**

| | |
|---|---|
| Definition: | OccupantTypeValue is a code list used to classify occupants. |
| Stereotype: | «CodeList» |

**OtherRelationTypeValue**

| | |
|---|---|
| Definition: | OtherRelationTypeValue is a code list used to classify other types of city object relations. |
| Stereotype: | «CodeList» |

**QualifiedAreaTypeValue**

| | |
|---|---|
| Definition: | QualifiedAreaTypeValue is a code list used to specify area types. |
| Stereotype: | «CodeList» |

**QualifiedVolumeTypeValue**

| | |
|---|---|
| Definition: | QualifiedVolumeTypeValue is a code list used to specify volume types. |
| Stereotype: | «CodeList» |

**RelationTypeValue**

| | |
|---|---|
| Definition: | RelationTypeValue is a code list used to classify city object relations. |
| Stereotype: | «CodeList» |

**TemporalRelationTypeValue**

| | |
|---|---|
| Definition: | TemporalRelationTypeValue is a code list used to classify temporal city object relations. |
| Stereotype: | «CodeList» |

| **TopologicalRelationTypeValue** | |
|---|---|
| Definition: | TopologicalRelationTypeValue is a code list used to classify topological city object relations. |
| Stereotype: | «CodeList» |

### 9.2.6. Enumerations

| **RelativeToTerrain** | |
|---|---|
| Definition: | RelativeToTerrain enumerates the spatial relations of a city object relative to terrain in a qualitative way. |
| StereoType: | <<Enumeration>> |

| Literal value | Definition |
|---|---|
| entirelyAboveTerrain | Indicates that the city object is located entirely above the terrain. |
| substantiallyAboveTerrain | Indicates that the city object is for the most part located above the terrain. |
| substantiallyAboveAndBelowTerrain | Indicates that the city object is located half above the terrain and half below the terrain. |
| substantiallyBelowTerrain | Indicates that the city object is for the most part located below the terrain. |
| entirelyBelowTerrain | Indicates that the city object is located entirely below the terrain. |

| **RelativeToWater** | |
|---|---|
| Definition: | RelativeToWater enumerates the spatial relations of a city object relative to the water surface in a qualitative way. |
| StereoType: | <<Enumeration>> |

| Literal value | Definition |
|---|---|
| entirelyAboveWaterSurface | Indicates that the city object is located entirely above the water surface. |
| substantiallyAboveWaterSurface | Indicates that the city object is for the most part located above the water surface. |
| substantiallyAboveAndBelowWaterSurface | Indicates that the city object is located half above the water surface and half below the water surface. |
| substantiallyBelowWaterSurface | Indicates that the city object is for the most part located below the water surface. |
| entirelyBelowWaterSurface | Indicates that the city object is located entirely below the water surface. |
| temporarilyAboveAndBelowWaterSurface | Indicates that the city object is temporarily located above or below the water level, because the height of the water surface is varying. |

**SpaceType**

| | |
|---|---|
| Definition: | SpaceType is an enumeration that characterises a space according to its closure properties. |
| StereoType: | <<Enumeration>> |

| Literal value | Definition |
|---|---|
| closed | Indicates that the space has boundaries at the bottom, at the top, and on all sides. |
| open | Indicates that the space has at maximum a boundary at the bottom. |
| semiOpen | Indicates that the space has a boundary at the bottom and on at least one side. |

# 9.3. Appearance

| | |
|---|---|
| Description: | The Appearance module supports the modelling of the observable surface properties of CityGML features in the form of textures and material. |
| Parent Package: | CityGML |
| Stereotype: | «ApplicationSchema» |

## 9.3.1. Classes

**AbstractSurfaceData**

| | |
|---|---|
| Definition: | AbstractSurfaceData is the abstract superclass for different kinds of textures and material. |
| Subclass of: | AbstractFeature |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| isFront | Boolean [0..1] | Indicates whether the texture or material is assigned to the front side or the back side of the surface geometry object. |
| adeOfAbstractSurfaceData | ADEOfAbstractSurfaceData [0..*] | Augments AbstractSurfaceData with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractTexture**

| | |
|---|---|
| Definition: | AbstractTexture is the abstract superclass to represent the common attributes of the classes ParameterizedTexture and GeoreferencedTexture. |
| Subclass of: | AbstractSurfaceData |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| imageURI | URI [1..1] | Specifies the URI that points to the external image data file. |
| mimeType | MimeTypeValue [0..1] | Specifies the MIME type of the external point cloud file. |
| textureType | TextureType [0..1] | Indicates the specific type of the texture. |
| wrapMode | WrapMode [0..1] | Specifies the behaviour of the texture when the texture is smaller than the surface to which it is applied. |
| borderColor | ColorPlusOpacity [0..1] | Specifies the color of that part of the surface that is not covered by the texture. |
| adeOfAbstractTexture | ADEOfAbstractTexture [0..*] | Augments AbstractTexture with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Appearance**

| | |
|---|---|
| Definition: | An Appearance is a collection of surface data, i.e. observable properties for surface geometry objects in the form of textures and material. |
| Subclass of: | AbstractAppearance |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| surfaceData | AbstractSurfaceData [*] | Relates to the surface data that are part of the Appearance. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| theme | CharacterString [0..1] | Specifies the topic of the Appearance. Each Appearance contains surface data for one theme only. Examples of themes are infrared radiation, noise pollution, or earthquake-induced structural stress. |
| adeOfAppearance | ADEOfAppearance [0..*] | Augments the Appearance with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**GeoreferencedTexture**

| | |
|---|---|
| Definition: | A GeoreferencedTexture is a texture that uses a planimetric projection. It contains an implicit parameterization that is either stored within the image file, an accompanying world file or specified using the orientation and referencePoint elements. |
| Subclass of: | AbstractTexture |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| referencePoint | GM_Point [0..1] | Relates to the 2D Point geometry that represents the center of the upper left image pixel in world space. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| preferWorldFile | Boolean [0..1] | Indicates whether the georeference from the image file or the accompanying world file should be preferred. |
| orientation | TransformationMatrix2x2 [0..1] | Specifies the rotation and scaling of the image in form of a 2x2 matrix. |
| target | URI [0..*] | Specifies the URI that points to the surface geometry objects to which the texture is applied. |
| adeOfGeoreferencedTexture | ADEOfGeoreferencedTexture [0..*] | Augments the GeoreferencedTexture with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**ParameterizedTexture**

| | |
|---|---|
| Definition: | A ParameterizedTexture is a texture that uses texture coordinates or a transformation matrix for parameterization. |
| Subclass of: | AbstractTexture |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| textureParameterization | AbstractTextureParameterization [*] | Relates to the texture coordinates or transformation matrices used for parameterization. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfParameterizedTexture | ADEOfParameterizedTexture [0..*] | Augments the ParameterizedTexture with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**TextureAssociation**

| | |
|---|---|
| Definition: | TextureAssociation denotes the relation of a texture to a surface geometry object. |
| Subclass of: | None |
| Stereotype: | «ObjectType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| target | URI [1..1] | Specifies the URI that points to the surface geometry object to which the texture is applied. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**X3DMaterial**

| | |
|---|---|
| Definition: | X3DMaterial defines properties for surface geometry objects based on the material definitions from the X3D and COLLADA standards. |
| Subclass of: | AbstractSurfaceData |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| ambientIntensity | DoubleBetween0and1 [0..1] | Specifies the minimum percentage of diffuseColor that is visible regardless of light sources. |
| diffuseColor | Color [0..1] | Specifies the color of the light diffusely reflected by the surface geometry object. |
| emissiveColor | Color [0..1] | Specifies the color of the light emitted by the surface geometry object. |
| specularColor | Color [0..1] | Specifies the color of the light directly reflected by the surface geometry object. |
| shininess | DoubleBetween0and1 [0..1] | Specifies the sharpness of the specular highlight. |
| transparency | DoubleBetween0and1 [0..1] | Specifies the degree of transparency of the surface geometry object. |
| isSmooth | Boolean [0..1] | Specifies which interpolation method is used for the shading of the surface geometry object. If the attribute is set to true, vertex normals should be used for shading (Gouraud shading). Otherwise, normals should be constant for a surface patch (flat shading). |
| target | URI [0..*] | Specifies the URI that points to the surface geometry objects to which the material is applied. |
| adeOfX3DMaterial | ADEOfX3DMaterial [0..*] | Augments the X3DMaterial with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.3.2. Data Types

**AbstractTextureParameterization**

| | |
|---|---|
| Definition: | AbstractTextureParameterization is the abstract superclass for different kinds of texture parameterizations. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractSurfaceData**

| | |
|---|---|
| Definition: | ADEOfAbstractSurfaceData acts as a hook to define properties within an ADE that are to be added to AbstractSurfaceData. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractTexture**

| | |
|---|---|
| Definition: | ADEOfAbstractTexture acts as a hook to define properties within an ADE that are to be added to AbstractTexture. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAppearance**

| | |
|---|---|
| Definition: | ADEOfAppearance acts as a hook to define properties within an ADE that are to be added to an Appearance. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfGeoreferencedTexture**

| | |
|---|---|
| Definition: | ADEOfGeoreferencedTexture acts as a hook to define properties within an ADE that are to be added to a GeoreferencedTexture. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfParameterizedTexture**

| | |
|---|---|
| Definition: | ADEOfParameterizedTexture acts as a hook to define properties within an ADE that are to be added to a ParameterizedTexture. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfX3DMaterial**

| | |
|---|---|
| Definition: | ADEOfX3DMaterial acts as a hook to define properties within an ADE that are to be added to an X3DMaterial. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**TexCoordGen**

| | |
|---|---|
| Definition: | TexCoordGen defines texture parameterization using a transformation matrix. |
| Subclass of: | AbstractTextureParameterization |
| Stereotype: | «DataType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| crs | SC_CRS [0..1] | Relates to the coordinate reference system of the transformation matrix. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| worldToTexture | TransformationMatrix3x4 [1..1] | Specifies the 3x4 transformation matrix that defines the transformation between world coordinates and texture coordinates. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**TexCoordList**

| | |
|---|---|
| Definition: | TexCoordList defines texture parameterization using texture coordinates. |
| Subclass of: | AbstractTextureParameterization |
| Stereotype: | «DataType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| textureCoordinates | DoubleList [1..*] | Specifies the coordinates of texture used for parameterization. The texture coordinates are provided separately for each LinearRing of the surface geometry object. |
| ring | URI [1..*] | Specifies the URIs that point to the LinearRings that are parameterized using the given texture coordinates. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.3.3. Basic Types

**Color**

| | |
|---|---|
| Definition: | Color is a list of three double values between 0 and 1 defining an RGB color value. |
| Subclass of: | DoubleBetween0and1List |
| Stereotype: | «BasicType» |
| Constraint: | lengthOfList (OCL): inv: list→size() = 3 |

**ColorPlusOpacity**

| | |
|---|---|
| Definition: | Color is a list of four double values between 0 and 1 defining an RGBA color value. Opacity value of 0 means transparent. |
| Subclass of: | DoubleBetween0and1List |
| Stereotype: | «BasicType» |
| Constraint: | lengthOfList (OCL): inv: list→size() = 3 or list→size() = 4 |

### 9.3.4. Unions

none

### 9.3.5. Code Lists

none

### 9.3.6. Enumerations

**TextureType**

| | |
|---|---|
| Definition: | TextureType enumerates the different texture types. |
| StereoType: | <<Enumeration>> |

| Literal value | Definition |
|---|---|
| specific | Indicates that the texture is specific to a single surface. |
| typical | Indicates that the texture is characteristic of a surface and can be used repeatedly. |
| unknown | Indicates that the texture type is not known. |

**WrapMode**

| | |
|---|---|
| Definition: | WrapMode enumerates the different fill modes for textures. |
| StereoType: | <<Enumeration>> |

| Literal value | Definition |
|---|---|
| none | Indicates that the texture is applied to the surface "as is". The part of the surface that is not covered by the texture is shown fully transparent. [cf. COLLADA] |
| wrap | Indicates that the texture is repeated until the surface is fully covered. [cf. COLLADA] |
| mirror | Indicates that the texture is repeated and mirrored. [cf. COLLADA] |
| clamp | Indicates that the texture is stretched to the edges of the surface. [cf. COLLADA] |
| border | Indicates that the texture is applied to the surface "as is". The part of the surface that is not covered by the texture is filled with the RGBA color that is specified in the attribute borderColor. [cf. COLLADA] |

## 9.4. CityFurniture

| | |
|---|---|
| Description: | The CityFurniture module supports representation of city furniture objects. City furniture objects are immovable objects like lanterns, traffic signs, advertising columns, benches, or bus stops that can be found in traffic areas, residential areas, on squares, or in built-up areas. |
| Parent Package: | CityGML |
| Stereotype: | «ApplicationSchema» |

### 9.4.1. Classes

**CityFurniture**

| | |
|---|---|
| Definition: | CityFurniture is an object or piece of equipment installed in the outdoor environment for various purposes. Examples include street signs, traffic signals, street lamps, benches, fountains. |
| Subclass of: | AbstractOccupiedSpace |
| Stereotype: | «TopLevelFeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | CityFurnitureClassValue [0..1] | Indicates the specific type of the CityFurniture. |
| function | CityFurnitureFunctionValue [0..*] | Specifies the intended purposes of the CityFurniture. |
| usage | CityFurnitureUsageValue [0..*] | Specifies the actual uses of the CityFurniture. |
| adeOfCityFurniture | ADEOfCityFurniture [0..*] | Augments the CityFurniture with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.4.2. Data Types

**ADEOfCityFurniture**

| | |
|---|---|
| Definition: | ADEOfCityFurniture acts as a hook to define properties within an ADE that are to be added to a CityFurniture. |
| Subclass of: | None |
| Stereotype: | «DataType» |

### 9.4.3. Basic Types

none

### 9.4.4. Unions

none

### 9.4.5. Code Lists

| **CityFurnitureClassValue** | |
|---|---|
| Definition: | CityFurnitureClassValue is a code list used to further classify a CityFurniture. |
| Stereotype: | «CodeList» |

| **CityFurnitureFunctionValue** | |
|---|---|
| Definition: | CityFurnitureFunctionValue is a code list that enumerates the different purposes of a CityFurniture. |
| Stereotype: | «CodeList» |

| **CityFurnitureUsageValue** | |
|---|---|
| Definition: | CityFurnitureUsageValue is a code list that enumerates the different uses of a CityFurniture. |
| Stereotype: | «CodeList» |

### 9.4.6. Enumerations

none

## 9.5. CityObjectGroup

| Description: | The CityObjectGroup module supports grouping of city objects. Arbitrary city objects may be aggregated in groups according to user-defined criteria. A group may be further classified by application-specific attributes. |
|---|---|
| Parent Package: | CityGML |
| Stereotype: | «ApplicationSchema» |

### 9.5.1. Classes

| **CityObjectGroup** |
|---|

| | |
|---|---|
| Definition: | A CityObjectGroup represents an application-specific aggregation of city objects according to some user-defined criteria. Examples for groups are the buildings in a specific region, the result of a query, or objects put together for visualization purposes. Each member of a group may be qualified by a role name, reflecting the role each city object plays in the context of the group. |
| Subclass of: | AbstractLogicalSpace |
| Stereotype: | «TopLevelFeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| parent | AbstractCityObject [0..1] | Relates to a city object to which the CityObjectGroup belongs. |
| groupMember | AbstractCityObject [*] | Relates to the city objects that are part of the CityObjectGroup. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | CityObjectGroupClassValue [0..1] | Indicates the specific type of the CityObjectGroup. |
| function | CityObjectGroupFunctionValue [0..*] | Specifies the intended purposes of the CityObjectGroup. |
| usage | CityObjectGroupUsageValue [0..*] | Specifies the actual usages of the CityObjectGroup. |
| adeOfCityObjectGroup | ADEOfCityObjectGroup [0..*] | Augments the CityObjectGroup with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Role**

| | |
|---|---|
| Definition: | Role qualifies the function of a city object within the CityObjectGroup. |
| Subclass of: | None |
| Stereotype: | «ObjectType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| role | CharacterString [0..1] | Describes the role the city object plays within the CityObjectGroup. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.5.2. Data Types

**ADEOfCityObjectGroup**

| | |
|---|---|
| Definition: | ADEOfCityObjectGroup acts as a hook to define properties within an ADE that are to be added to a CityObjectGroup. |
| Subclass of: | None |
| Stereotype: | «DataType» |

### 9.5.3. Basic Types

none

### 9.5.4. Unions

none

### 9.5.5. Code Lists

**CityObjectGroupClassValue**

| | |
|---|---|
| Definition: | CityObjectGroupClassValue is a code list used to further classify a CityObjectGroup. |
| Stereotype: | «CodeList» |

**CityObjectGroupFunctionValue**

| | |
|---|---|
| Definition: | CityObjectGroupFunctionValue is a code list that enumerates the different purposes of a CityObjectGroup. |
| Stereotype: | «CodeList» |

**CityObjectGroupUsageValue**

| | |
|---|---|
| Definition: | CityObjectGroupUsageValue is a code list that enumerates the different uses of a CityObjectGroup. |
| Stereotype: | «CodeList» |

### 9.5.6. Enumerations

none

## 9.6. Dynamizer

| | |
|---|---|
| Description: | The Dynamizer module supports the injection of timeseries data for individual attributes of CityGML features. Timeseries data can either be retrieved from external Sensor APIs (e.g. OGC SensorThings API, OGC Sensor Observation Services, MQTT, proprietary platforms), external standardized timeseries files (e.g. OGC TimeseriesML or OGC Observations & Measurements), external tabulated files (e.g CSV) or can be represented inline as basic time-value pairs. |
| Parent Package: | CityGML |
| Stereotype: | «ApplicationSchema» |

### 9.6.1. Classes

**AbstractAtomicTimeseries**

| | |
|---|---|
| Definition: | AbstractAtomicTimeseries represents the attributes and relationships that are common to all kinds of atomic timeseries (GenericTimeseries, TabulatedFileTimeseries, StandardFileTimeseries). An atomic timeseries represents time-varying data of a specific data type for a single contiguous time interval. |
| Subclass of: | AbstractTimeseries |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| observationProperty | CharacterString [1..1] | Specifies the phenomenon for which the atomic timeseries provides observation values. |
| uom | CharacterString [0..1] | Specifies the unit of measurement of the observation values. |
| adeOfAbstractAtomicTimeseries | ADEOfAbstractAtomicTimeseries [0..*] | Augments AbstractAtomicTimeseries with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractTimeseries**

| | |
|---|---|
| Definition: | AbstractTimeseries is the abstract superclass representing any type of timeseries data. |
| Subclass of: | AbstractFeature |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| firstTimestamp | TM_Position [0..1] | Specifies the beginning of the timeseries. |
| lastTimestamp | TM_Position [0..1] | Specifies the end of the timeseries. |
| adeOfAbstractTimeseries | ADEOfAbstractTimeseries [0..*] | Augments AbstractTimeseries with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**CompositeTimeseries**

| | |
|---|---|
| Definition: | A CompositeTimeseries is a (possibly recursive) aggregation of atomic and composite timeseries. The components of a composite timeseries must have non-overlapping time intervals. |
| Subclass of: | AbstractTimeseries |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| component | TimeseriesComponent [1..*] | Relates to the atomic and composite timeseries that are part of the CompositeTimeseries. The referenced timeseries are sequentially ordered. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfCompositeTimeseries | ADEOfCompositeTimeseries [0..*] | Augments the CompositeTimeseries with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Dynamizer**

| | |
|---|---|
| Definition: | A Dynamizer is an object that injects timeseries data for an individual attribute of the city object in which it is included. The timeseries data overrides the static value of the referenced city object attribute in order to represent dynamic (time-dependent) variations of its value. |
| Subclass of: | AbstractDynamizer |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| dynamicData | AbstractTimeseries [0..1] | Relates to the timeseries data that is given either inline within a CityGML dataset or by a link to an external file containing timeseries data. |
| sensorConnection | SensorConnection [0..1] | Relates to the sensor API that delivers timeseries data. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| attributeRef | CharacterString [1..1] | Specifies the attribute of a CityGML feature whose value is overridden or replaced by the (dynamic) values specified by the Dynamizer. |
| startTime | TM_Position [0..1] | Specifies the beginning of the time span for which the Dynamizer provides dynamic values. |
| endTime | TM_Position [0..1] | Specifies the end of the time span for which the Dynamizer provides dynamic values. |
| adeOfDynamizer | ADEOfDynamizer [0..*] | Augments the Dynamizer with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**GenericTimeseries**

| | |
|---|---|
| Definition: | A GenericTimeseries represents time-varying data in the form of embedded time-value-pairs of a specific data type for a single contiguous time interval. |
| Subclass of: | AbstractAtomicTimeseries |
| Stereotype: | «FeatureType» |
| Constraint: | dataTypeOfValue (OCL): inv:<br>if valueType = TimeseriesTypeValue::integer then<br>TimeValuePair→forAll(c \| c.intValue→size()=1)<br>else if valueType = TimeseriesTypeValue::double then<br>TimeValuePair→forAll(c \| c.doubleValue→size()=1)<br>else if valueType = TimeseriesTypeValue::string then<br>TimeValuePair→forAll(c \| c.stringValue→size()=1)<br>else if valueType = TimeseriesTypeValue::geometry then<br>TimeValuePair→forAll(c \| c.geometryValue→size()=1)<br>else if valueType = TimeseriesTypeValue::uri then<br>TimeValuePair→forAll(c \| c.uriValue→size()=1)<br>else if valueType = TimeseriesTypeValue::bool then<br>TimeValuePair→forAll(c \| c.boolValue→size()=1)<br>else if valueType = TimeseriesTypeValue::implicitGeometry then<br>TimeValuePair→forAll(c \| c.implicitGeometryValue→size()=1)<br>else TimeValuePair→forAll(c \| c.appearanceValue→size()=1) |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| timeValuePair | TimeValuePair [1..*] | Relates to the time-value-pairs that are part of the GenericTimeseries. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| valueType | TimeseriesTypeValue [1..1] | Indicates the specific type of all time-value-pairs that are part of the GenericTimeseries. |
| adeOfGenericTimeseries | ADEOfGenericTimeseries [0..*] | Augments the GenericTimeseries with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**StandardFileTimeseries**

| | |
|---|---|
| Definition: | A StandardFileTimeseries represents time-varying data for a single contiguous time interval. The data is provided in an external file referenced in the StandardFileTimeseries. The data within the external file is encoded according to a dedicated format for the representation of timeseries data such as using the OGC TimeseriesML or OGC Observations & Measurements Standard. The data type of the data has to be specified within the external file. |
| Subclass of: | AbstractAtomicTimeseries |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| fileLocation | URI [1..1] | Specifies the URI that points to the external timeseries file. |
| fileType | StandardFileTypeValue [1..1] | Specifies the format used to represent the timeseries data. |
| mimeType | MimeTypeValue [0..1] | Specifies the MIME type of the external timeseries file. |
| adeOfStandardFileTimeseries | ADEOfStandardFileTimeseries [0..*] | Augments the StandardFileTimeseries with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**TabulatedFileTimeseries**

| | |
|---|---|
| Definition: | A TabulatedFileTimeseries represents time-varying data of a specific data type for a single contiguous time interval. The data is provided in an external file referenced in the TabulatedFileTimeseries. The file contains table structured data using an appropriate file format such as comma-separated values (CSV), Microsoft Excel (XLSX) or Google Spreadsheet. The timestamps and the values are given in specific columns of the table. Each row represents a single time-value-pair. A subset of rows can be selected using the idColumn and idValue attributes. |
| Subclass of: | AbstractAtomicTimeseries |
| Stereotype: | «FeatureType» |
| Constraint: | columnNumberOrColumnName (OCL): inv: (timeColumnNo→notEmpty() or timeColumnName→notEmpty()) and (valueColumnNo→notEmpty() or valueColumnName→notEmpty()) and (idValue→notEmpty() implies idColumnNo→notEmpty() or + idColumnName→notEmpty()) |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| fileLocation | URI [1..1] | Specifies the URI that points to the external timeseries file. |
| fileType | TabulatedFileTypeValue [1..1] | Specifies the format used to represent the timeseries data. |
| mimeType | MimeTypeValue [0..1] | Specifies the MIME type of the external timeseries file. |
| valueType | TimeseriesTypeValue [1..1] | Indicates the specific type of the timeseries data. |
| numberOfHeaderLines | Integer [0..1] | Indicates the number of lines at the beginning of the tabulated file that represent headers. |
| fieldSeparator | CharacterString [1..1] | Indicates which symbol is used to separate the individual values in the tabulated file. |
| decimalSymbol | Character [0..1] | Indicates which symbol is used to separate the integer part from the fractional part of a decimal number. |
| idColumnNo | Integer [0..1] | Specifies the number of the column that stores the identifier of the time-value-pair. |
| idColumnName | CharacterString [0..1] | Specifies the name of the column that stores the identifier of the time-value-pair. |
| idValue | CharacterString [0..1] | Specifies the value of the identifier for which the time-value-pairs are to be selected. |
| timeColumnNo | Integer [0..1] | Specifies the number of the column that stores the timestamp of the time-value-pair. |
| timeColumnName | CharacterString [0..1] | Specifies the name of the column that stores the timestamp of the time-value-pair. |
| valueColumnNo | Integer [0..1] | Specifies the number of the column that stores the value of the time-value-pair. |
| valueColumnName | CharacterString [0..1] | Specifies the name of the column that stores the value of the time-value-pair. |
| adeOfTabulatedFileTimeseries | ADEOfTabulatedFileTimeseries [0..*] | Augments the TabulatedFileTimeseries with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.6.2. Data Types

**ADEOfAbstractAtomicTimeseries**

| | |
|---|---|
| Definition: | ADEOfAbstractAtomicTimeseries acts as a hook to define properties within an ADE that are to be added to AbstractAtomicTimeseries. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractTimeseries**

| | |
|---|---|
| Definition: | ADEOfAbstractTimeseries acts as a hook to define properties within an ADE that are to be added to AbstractTimeseries. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfCompositeTimeseries**

| | |
|---|---|
| Definition: | ADEOfCompositeTimeseries acts as a hook to define properties within an ADE that are to be added to a CompositeTimeseries. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfDynamizer**

| | |
|---|---|
| Definition: | ADEOfDynamizer acts as a hook to define properties within an ADE that are to be added to a Dynamizer. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfGenericTimeseries**

| | |
|---|---|
| Definition: | ADEOfGenericTimeseries acts as a hook to define properties within an ADE that are to be added to a GenericTimeseries. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfStandardFileTimeseries**

| | |
|---|---|
| Definition: | ADEOfStandardFileTimeseries acts as a hook to define properties within an ADE that are to be added to a StandardFileTimeseries. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfTabulatedFileTimeseries**

| | |
|---|---|
| Definition: | ADEOfTabulatedFileTimeseries acts as a hook to define properties within an ADE that are to be added to a TabulatedFileTimeseries. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**SensorConnection**

| | |
|---|---|
| Definition: | A SensorConnection provides all details that are required to retrieve a specific datastream from an external sensor web service. This data type comprises the service type (e.g. OGC SensorThings API, OGC Sensor Observation Services, MQTT, proprietary platforms), the URL of the sensor service, the identifier for the sensor or thing, and its observed property as well as information about the required authentication method. |
| Subclass of: | None |
| Stereotype: | «DataType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| sensorLocation | AbstractCityObject [0..1] | Relates the sensor to the city object where it is located. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| connectionType | SensorConnectionTypeValue [1..1] | Indicates the type of Sensor API to which the SensorConnection refers. |
| observationProperty | CharacterString [1..1] | Specifies the phenomenon for which the SensorConnection provides observations. |
| uom | CharacterString [0..1] | Specifies the unit of measurement of the observations. |
| sensorID | CharacterString [0..1] | Specifies the unique identifier of the sensor from which the SensorConnection retrieves observations. |
| sensorName | CharacterString [0..1] | Specifies the name of the sensor from which the SensorConnection retrieves observations. |
| observationID | CharacterString [0..1] | Specifies the unique identifier of the observation that is retrieved by the SensorConnection. |
| datastreamID | CharacterString [0..1] | Specifies the datastream that is retrieved by the SensorConnection. |
| baseURL | URI [0..1] | Specifies the base URL of the Sensor API request. |
| authType | AuthenticationTypeValue [0..1] | Specifies the type of authentication required to be able to access the Sensor API. |
| mqttServer | CharacterString [0..1] | Specifies the name of the MQTT Server. This attribute is relevant when the MQTT Protocol is used to connect to a Sensor API. |
| mqttTopic | CharacterString [0..1] | Names the specific datastream that is retrieved by the SensorConnection. This attribute is relevant when the MQTT Protocol is used to connect to a Sensor API. |
| linkToObservation | CharacterString [0..1] | Specifies the complete URL to the observation request. |
| linkToSensorDescription | CharacterString [0..1] | Specifies the complete URL to the sensor description request. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**TimeseriesComponent**

| | |
|---|---|
| Definition: | TimeseriesComponent represents an element of a CompositeTimeseries. |
| Subclass of: | None |
| Stereotype: | «DataType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| timeseries | AbstractTimeseries [1..1] | Relates a timeseries to the TimeseriesComponent. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| repetitions | Integer [1..1] | Specifies how often the timeseries that is referenced by the TimeseriesComponent should be iterated. |
| additionalGap | TM_Duration [0..1] | Specifies how much extra time is added after all repetitions as an additional gap. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**TimeValuePair**

| | |
|---|---|
| Definition: | A TimeValuePair represents a value that is valid for a given timepoint. For each TimeValuePair, only one of the value properties can be used mutually exclusive. Which value property has to be provided depends on the selected value type in the GenericTimeSeries feature, in which the TimeValuePair is included. |
| Subclass of: | None |
| Stereotype: | «DataType» |
| Constraint: | singleValue (OCL): inv:<br>intValue→size() + doubleValue→size() + stringValue→size() + geometryValue→size() + uriValue→size() + boolValue→size() + implicitGeometryValue→size() + appearanceValue→size() = 1 |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| timestamp | TM_Position [1..1] | Specifies the timepoint at which the value of the TimeValuePair is valid. |
| intValue | Integer [0..1] | Specifies the "Integer" value of the TimeValuePair. |
| doubleValue | Real [0..1] | Specifies the "Double" value of the TimeValuePair. |
| stringValue | CharacterString [0..1] | Specifies the "String" value of the TimeValuePair. |
| geometryValue | GM_Object [0..1] | Specifies the geometry value of the TimeValuePair. |
| uriValue | URI [0..1] | Specifies the "URI" value of the TimeValuePair. |
| boolValue | Boolean [0..1] | Specifies the "Boolean" value of the TimeValuePair. |
| implicitGeometryValue | ImplicitGeometry [0..1] | Specifies the "ImplicitGeometry" value of the TimeValuePair. |
| appearanceValue | AbstractAppearance [0..1] | Specifies the "Appearance" value of the TimeValuePair. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.6.3. Basic Types

none

### 9.6.4. Unions

none

### 9.6.5. Code Lists

**AuthenticationTypeValue**

| | |
|---|---|
| Definition: | AuthenticationTypeValue is a code list used to specify the authentication method to be used to access the referenced sensor service. Each value provides enough information such that a software application could determine the required access credentials. |
| Stereotype: | «CodeList» |

**SensorConnectionTypeValue**

| | |
|---|---|
| Definition: | SensorConnectionTypeValue is a code list used to specify the type of the referenced sensor service. Each value provides enough information such that a software application would be able to identify the API type and version. |
| Stereotype: | «CodeList» |

**StandardFileTypeValue**

| | |
|---|---|
| Definition: | StandardFileTypeValue is a code list used to specify the type of the referenced external timeseries data file. Each value provides information about the standard and version. |
| Stereotype: | «CodeList» |

**TabulatedFileTypeValue**

| | |
|---|---|
| Definition: | TabulatedFileTypeValue is a code list used to specify the data format of the referenced external tabulated data file. |
| Stereotype: | «CodeList» |

### 9.6.6. Enumerations

**TimeseriesTypeValue**

| | |
|---|---|
| Definition: | TimeseriesTypeValue enumerates the possible value types for GenericTimeseries and TimeValuePair. |
| StereoType: | <<Enumeration>> |

| Literal value | Definition |
|---|---|
| int | Indicates that the values of the GenericTimeseries are of type "Integer". |
| double | Indicates that the values of the GenericTimeseries are of type "Double". |
| string | Indicates that the values of the GenericTimeseries are of type "String". |
| geometry | Indicates that the values of the GenericTimeseries are geometries. |
| uri | Indicates that the values of the GenericTimeseries are of type "URI". |
| bool | Indicates that the values of the GenericTimeseries are of type "Boolean". |
| implicitGeometry | Indicates that the values of the GenericTimeseries are of type "ImplicitGeometry". |
| appearance | Indicates that the values of the GenericTimeseries are of type "Appearance". |

# 9.7. Generics

| | |
|---|---|
| Description: | The Generics module supports application-specific extensions to the CityGML data model. These extensions may be used to model and exchange additional attributes and features not covered by the predefined thematic classes of CityGML. Generic extensions shall only be used if appropriate thematic classes or attributes are not provided by any other CityGML module. |
| Parent Package: | CityGML |
| Stereotype: | «ApplicationSchema» |

## 9.7.1. Classes

**GenericLogicalSpace**

| | |
|---|---|
| Definition: | A GenericLogicalSpace is a space that is not represented by any explicitly modelled AbstractLogicalSpace subclass within CityGML. |
| Subclass of: | AbstractLogicalSpace |
| Stereotype: | «TopLevelFeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | GenericLogicalSpaceClassValue [0..1] | Indicates the specific type of the GenericLogicalSpace. |
| function | GenericLogicalSpaceFunctionValue [0..*] | Specifies the intended purposes of the GenericLogicalSpace. |
| usage | GenericLogicalSpaceUsageValue [0..*] | Specifies the actual uses of the GenericLogicalSpace. |
| adeOfGenericLogicalSpace | ADEOfGenericLogicalSpace [0..*] | Augments the GenericLogicalSpace with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**GenericOccupiedSpace**

| | |
|---|---|
| Definition: | A GenericOccupiedSpace is a space that is not represented by any explicitly modelled AbstractOccupiedSpace subclass within CityGML. |
| Subclass of: | AbstractOccupiedSpace |
| Stereotype: | «TopLevelFeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | GenericOccupiedSpaceClassValue [0..1] | Indicates the specific type of the GenericOccupiedSpace. |
| function | GenericOccupiedSpaceFunctionValue [0..*] | Specifies the intended purposes of the GenericOccupiedSpace. |
| usage | GenericOccupiedSpaceUsageValue [0..*] | Specifies the actual uses of the GenericOccupiedSpace. |
| adeOfGenericOccupiedSpace | ADEOfGenericOccupiedSpace [0..*] | Augments the GenericOccupiedSpace with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**GenericThematicSurface**

| | |
|---|---|
| Definition: | A GenericThematicSurface is a surface that is not represented by any explicitly modelled AbstractThematicSurface subclass within CityGML. |
| Subclass of: | AbstractThematicSurface |
| Stereotype: | «TopLevelFeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | GenericThematicSurfaceClassValue [0..1] | Indicates the specific type of the GenericThematicSurface. |
| function | GenericThematicSurfaceFunctionValue [0..*] | Specifies the intended purposes of the GenericThematicSurface. |
| usage | GenericThematicSurfaceUsageValue [0..*] | Specifies the actual uses of the GenericThematicSurface. |
| adeOfGenericThematicSurface | ADEOfGenericThematicSurface [0..*] | Augments the GenericThematicSurface with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**GenericUnoccupiedSpace**

| | |
|---|---|
| Definition: | A GenericUnoccupiedSpace is a space that is not represented by any explicitly modelled AbstractUnoccupiedSpace subclass within CityGML. |
| Subclass of: | AbstractUnoccupiedSpace |
| Stereotype: | «TopLevelFeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | GenericUnoccupiedSpaceClassValue [0..1] | Indicates the specific type of the GenericUnoccupiedSpace. |
| function | GenericUnoccupiedSpaceFunctionValue [0..*] | Specifies the intended purposes of the GenericUnoccupiedSpace. |
| usage | GenericUnoccupiedSpaceUsageValue [0..*] | Specifies the actual uses of the GenericUnoccupiedSpace. |
| adeOfGenericUnoccupiedSpace | ADEOfGenericUnoccupiedSpace [0..*] | Augments the GenericUnoccupiedSpace with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.7.2. Data Types

**ADEOfGenericLogicalSpace**

| | |
|---|---|
| Definition: | ADEOfGenericLogicalSpace acts as a hook to define properties within an ADE that are to be added to a GenericLogicalSpace. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfGenericOccupiedSpace**

| | |
|---|---|
| Definition: | ADEOfGenericOccupiedSpace acts as a hook to define properties within an ADE that are to be added to a GenericOccupiedSpace. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfGenericThematicSurface**

| | |
|---|---|
| Definition: | ADEOfGenericThematicSurface acts as a hook to define properties within an ADE that are to be added to a GenericThematicSurface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfGenericUnoccupiedSpace**

| | |
|---|---|
| Definition: | ADEOfGenericUnoccupiedSpace acts as a hook to define properties within an ADE that are to be added to a GenericUnoccupiedSpace. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**CodeAttribute**

| | |
|---|---|
| Definition: | CodeAttribute is a data type used to define generic attributes of type "Code". |
| Subclass of: | AbstractGenericAttribute |
| Stereotype: | «DataType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| name | CharacterString [1..1] | Specifies the name of the CodeAttribute. |
| value | Code [1..1] | Specifies the "Code" value. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**DateAttribute**

| | |
|---|---|
| Definition: | DateAttribute is a data type used to define generic attributes of type "Date". |
| Subclass of: | AbstractGenericAttribute |
| Stereotype: | «DataType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| name | CharacterString [1..1] | Specifies the name of the DateAttribute. |
| value | Date [1..1] | Specifies the "Date" value. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**DoubleAttribute**

| | |
|---|---|
| Definition: | DoubleAttribute is a data type used to define generic attributes of type "Double". |
| Subclass of: | AbstractGenericAttribute |
| Stereotype: | «DataType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| name | CharacterString [1..1] | Specifies the name of the DoubleAttribute. |
| value | Real [1..1] | Specifies the "Double" value. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**GenericAttributeSet**

| | |
|---|---|
| Definition: | A GenericAttributeSet is a named collection of generic attributes. |
| Subclass of: | AbstractGenericAttribute |
| Stereotype: | «DataType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| genericAttribute | AbstractGenericAttribute [1..*] | Relates to the generic attributes that are part of the GenericAttributeSet. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| name | CharacterString [1..1] | Specifies the name of the GenericAttributeSet. |
| codeSpace | URI [0..1] | Associates the GenericAttributeSet with an authority that maintains the collection of generic attributes. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**IntAttribute**

| | |
|---|---|
| Definition: | IntAttribute is a data type used to define generic attributes of type "Integer". |
| Subclass of: | AbstractGenericAttribute |
| Stereotype: | «DataType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| name | CharacterString [1..1] | Specifies the name of the IntAttribute. |
| value | Integer [1..1] | Specifies the "Integer" value. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**MeasureAttribute**

| | |
|---|---|
| Definition: | MeasureAttribute is a data type used to define generic attributes of type "Measure". |
| Subclass of: | AbstractGenericAttribute |
| Stereotype: | «DataType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| name | CharacterString [1..1] | Specifies the name of the MeasureAttribute. |
| value | Measure [1..1] | Specifies the value of the MeasureAttribute. The value is of type "Measure", which can additionally provide the units of measure. [cf. ISO 19103] |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**StringAttribute**

| | |
|---|---|
| Definition: | StringAttribute is a data type used to define generic attributes of type "String". |
| Subclass of: | AbstractGenericAttribute |
| Stereotype: | «DataType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| name | CharacterString [1..1] | Specifies the name of the StringAttribute. |
| value | CharacterString [1..1] | Specifies the "String" value. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**UriAttribute**

| | |
|---|---|
| Definition: | UriAttribute is a data type used to define generic attributes of type "URI". |
| Subclass of: | AbstractGenericAttribute |
| Stereotype: | «DataType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| name | CharacterString [1..1] | Specifies the name of the UriAttribute. |
| value | URI [1..1] | Specifies the "URI" value. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.7.3. Basic Types

none

### 9.7.4. Unions

none

### 9.7.5. Code Lists

**GenericLogicalSpaceClassValue**

| | |
|---|---|
| Definition: | GenericLogicalSpaceClassValue is a code list used to further classify a GenericLogicalSpace. |
| Stereotype: | «CodeList» |

**GenericLogicalSpaceFunctionValue**

| | |
|---|---|
| Definition: | GenericLogicalSpaceFunctionValue is a code list that enumerates the different purposes of a GenericLogicalSpace. |
| Stereotype: | «CodeList» |

**GenericLogicalSpaceUsageValue**

| | |
|---|---|
| Definition: | GenericLogicalSpaceUsageValue is a code list that enumerates the different uses of a GenericLogicalSpace. |
| Stereotype: | «CodeList» |

**GenericOccupiedSpaceClassValue**

| | |
|---|---|
| Definition: | GenericOccupiedSpaceClassValue is a code list used to further classify a GenericOccupiedSpace. |
| Stereotype: | «CodeList» |

**GenericOccupiedSpaceFunctionValue**

| | |
|---|---|
| Definition: | GenericOccupiedSpaceFunctionValue is a code list that enumerates the different purposes of a GenericOccupiedSpace. |
| Stereotype: | «CodeList» |

**GenericOccupiedSpaceUsageValue**

| | |
|---|---|
| Definition: | GenericOccupiedSpaceUsageValue is a code list that enumerates the different uses of a GenericOccupiedSpace. |
| Stereotype: | «CodeList» |

**GenericThematicSurfaceClassValue**

| | |
|---|---|
| Definition: | GenericThematicSurfaceClassValue is a code list used to further classify a GenericThematicSurface. |
| Stereotype: | «CodeList» |

**GenericThematicSurfaceFunctionValue**

| | |
|---|---|
| Definition: | GenericThematicSurfaceFunctionValue is a code list that enumerates the different purposes of a GenericThematicSurface. |
| Stereotype: | «CodeList» |

**GenericThematicSurfaceUsageValue**

| | |
|---|---|
| Definition: | GenericThematicSurfaceUsageValue is a code list that enumerates the different uses of a GenericThematicSurface. |
| Stereotype: | «CodeList» |

**GenericUnoccupiedSpaceClassValue**

| | |
|---|---|
| Definition: | GenericUnoccupiedSpaceClassValue is a code list used to further classify a GenericUnoccupiedSpace. |
| Stereotype: | «CodeList» |

**GenericUnoccupiedSpaceFunctionValue**

| | |
|---|---|
| Definition: | GenericUnoccupiedSpaceFunctionValue is a code list that enumerates the different purposes of a GenericUnoccupiedSpace. |
| Stereotype: | «CodeList» |

**GenericUnoccupiedSpaceUsageValue**

| | |
|---|---|
| Definition: | GenericUnoccupiedSpaceUsageValue is a code list that enumerates the different uses of a GenericUnoccupiedSpace. |
| Stereotype: | «CodeList» |

### 9.7.6. Enumerations

none

## 9.8. LandUse

| | |
|---|---|
| Description: | The LandUse module supports representation of areas of the earth's surface dedicated to a specific land use. |
| Parent Package: | CityGML |
| Stereotype: | «ApplicationSchema» |

### 9.8.1. Classes

**LandUse**

| | |
|---|---|
| Definition: | A LandUse object is an area of the earth's surface dedicated to a specific land use or having a specific land cover with or without vegetation, such as sand, rock, mud flats, forest, grasslands, or wetlands. |
| Subclass of: | AbstractThematicSurface |
| Stereotype: | «TopLevelFeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | LandUseClassValue [0..1] | Indicates the specific type of the LandUse. |
| function | LandUseFunctionValue [0..*] | Specifies the intended purposes of the LandUse. |
| usage | LandUseUsageValue [0..*] | Specifies the actual uses of the LandUse. |
| adeOfLandUse | ADEOfLandUse [0..*] | Augments the LandUse with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.8.2. Data Types

| ADEOfLandUse | |
|---|---|
| Definition: | ADEOfLandUse acts as a hook to define properties within an ADE that are to be added to a LandUse. |
| Subclass of: | None |
| Stereotype: | «DataType» |

### 9.8.3. Basic Types

none

### 9.8.4. Unions

none

### 9.8.5. Code Lists

| LandUseClassValue | |
|---|---|
| Definition: | LandUseClassValue is a code list used to further classify a LandUse. |
| Stereotype: | «CodeList» |

| LandUseFunctionValue | |
|---|---|
| Definition: | LandUseFunctionValue is a code list that enumerates the different purposes of a LandUse. |
| Stereotype: | «CodeList» |

| LandUseUsageValue | |
|---|---|
| Definition: | LandUseUsageValue is a code list that enumerates the different uses of a LandUse. |
| Stereotype: | «CodeList» |

### 9.8.6. Enumerations

none

# 9.9. PointCloud

| | |
|---|---|
| Description: | The PointCloud module supports representation of CityGML features by a collection of points. |
| Parent Package: | CityGML |
| Stereotype: | «ApplicationSchema» |

## 9.9.1. Classes

**PointCloud**

| | |
|---|---|
| Definition: | A PointCloud is an unordered collection of points that is a sampling of the geometry of a space or space boundary. |
| Subclass of: | AbstractPointCloud |
| Stereotype: | «FeatureType» |
| Constraint: | inlineOrExternalPointCloud (OCL): inv: (points→notEmpty() and mimeType→isEmpty() and pointFile→isEmpty() and pointFileSrsName→isEmpty()) xor (points→isEmpty() and mimeType→notEmpty() and pointFile→notEmpty()) |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| points | GM_MultiPoint [0..1] | Relates to the 3D MultiPoint geometry of the PointCloud stored inline with the city model. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| mimeType | MimeTypeValue [0..1] | Specifies the MIME type of the external point cloud file. |
| pointFile | URI [0..1] | Specifies the URI that points to the external point cloud file. |
| pointFileSrsName | CharacterString [0..1] | Indicates the coordinate reference system used by the external point cloud file. |
| adeOfPointCloud | ADEOfPointCloud [0..*] | Augments the PointCloud with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

## 9.9.2. Data Types

**ADEOfPointCloud**

| | |
|---|---|
| Definition: | ADEOfPointCloud acts as a hook to define properties within an ADE that are to be added to a PointCloud. |
| Subclass of: | None |
| Stereotype: | «DataType» |

### 9.9.3. Basic Types

none

### 9.9.4. Unions

none

### 9.9.5. Code Lists

none

### 9.9.6. Enumerations

none

## 9.10. Relief

| | |
|---|---|
| Description: | The Relief module supports representation of the terrain. CityGML supports terrain representations at different levels of detail, reflecting different accuracies or resolutions. Terrain may be specified as a regular raster or grid, as a TIN, by break lines, and/or by mass points. |
| Parent Package: | CityGML |
| Stereotype: | «ApplicationSchema» |

### 9.10.1. Classes

**AbstractReliefComponent**

| | |
|---|---|
| Definition: | An AbstractReliefComponent represents an element of the terrain surface - either a TIN, a raster or grid, mass points or break lines. |
| Subclass of: | AbstractSpaceBoundary |
| Stereotype: | «FeatureType» |
| Constraint: | polygonGeometry (OCL): inv: extent.patch→size()=1 and extent.patch→forAll(oclIsKindOf(GM_Polygon)) |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| extent | GM_Surface [0..1] | Indicates the geometrical extent of the terrain component. The geometrical extent is provided as a 2D Surface geometry. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| lod | IntegerBetween0and3 [1..1] | Indicates the Level of Detail of the terrain component. |
| adeOfAbstractReliefComponent | ADEOfAbstractReliefComponent [0..*] | Augments AbstractReliefComponent with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**BreaklineRelief**

| | |
|---|---|
| Definition: | A BreaklineRelief represents a terrain component with 3D lines. These lines denote break lines or ridge/valley lines. |
| Subclass of: | AbstractReliefComponent |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| breaklines | GM_MultiCurve [0..1] | Relates to the 3D MultiCurve geometry of the MassPointRelief. This association role is used to represent break lines. |
| ridgeOrValleyLines | GM_MultiCurve [0..1] | Relates to the 3D MultiCurve geometry of the MassPointRelief. This association role is used to represent ridge or valley lines. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfBreaklineRelief | ADEOfBreaklineRelief [0..*] | Augments the BreaklineRelief with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**MassPointRelief**

| | |
|---|---|
| Definition: | A MassPointRelief represents a terrain component as a collection of 3D points. |
| Subclass of: | AbstractReliefComponent |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| pointCloud | AbstractPointCloud [0..1] | Relates to the 3D PointCloud of the MassPointRelief. |
| reliefPoints | GM_MultiPoint [0..1] | Relates to the 3D MultiPoint geometry of the MassPointRelief. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfMassPointRelief | ADEOfMassPointRelief [0..*] | Augments the MassPointRelief with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**RasterRelief**

| | |
|---|---|
| Definition: | A RasterRelief represents a terrain component as a regular raster or grid. |
| Subclass of: | AbstractReliefComponent |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| grid | CV_DiscreteGridPointCoverage [1] | Relates to the DiscreteGridPointCoverage of the RasterRelief. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfRasterRelief | ADEOfRasterRelief [0..*] | Augments the RasterRelief with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**ReliefFeature**

| | |
|---|---|
| Definition: | A ReliefFeature is a collection of terrain components representing the Earth's surface, also known as the Digital Terrain Model. |
| Subclass of: | AbstractSpaceBoundary |
| Stereotype: | «TopLevelFeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| reliefComponent | AbstractReliefComponent [1..*] | Relates to the terrain components that are part of the ReliefFeature. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| lod | IntegerBetween0and3 [1..1] | Indicates the Level of Detail of the ReliefFeature. |
| adeOfReliefFeature | ADEOfReliefFeature [0..*] | Augments the ReliefFeature with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**TINRelief**

| | |
|---|---|
| Definition: | A TINRelief represents a terrain component as a triangulated irregular network. |
| Subclass of: | AbstractReliefComponent |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| tin | GM_TriangulatedSurface [1] | Relates to the triangulated surface of the TINRelief. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfTINRelief | ADEOfTINRelief [0..*] | Augments the TINRelief with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.10.2. Data Types

**ADEOfAbstractReliefComponent**

| | |
|---|---|
| Definition: | ADEOfAbstractReliefComponent acts as a hook to define properties within an ADE that are to be added to AbstractReliefComponent. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfBreaklineRelief**

| | |
|---|---|
| Definition: | ADEOfBreaklineRelief acts as a hook to define properties within an ADE that are to be added to a BreaklineRelief. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfMassPointRelief**

| | |
|---|---|
| Definition: | ADEOfMassPointRelief acts as a hook to define properties within an ADE that are to be added to a MassPointRelief. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfRasterRelief**

| | |
|---|---|
| Definition: | ADEOfRasterRelief acts as a hook to define properties within an ADE that are to be added to a RasterRelief. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfReliefFeature**

| | |
|---|---|
| Definition: | ADEOfReliefFeature acts as a hook to define properties within an ADE that are to be added to a ReliefFeature. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfTINRelief**

| | |
|---|---|
| Definition: | ADEOfTINRelief acts as a hook to define properties within an ADE that are to be added to a TINRelief. |
| Subclass of: | None |
| Stereotype: | «DataType» |

### 9.10.3. Basic Types

none

### 9.10.4. Unions

none

### 9.10.5. Code Lists

none

### 9.10.6. Enumerations

none

## 9.11. Transportation

| | |
|---|---|
| Description: | The Transportation module supports representation of the transportation infrastructure. Transportation features include roads, tracks, waterways, railways, and squares. Transportation features may be represented as a network and/or as a collection of spaces or surface elements embedded in a three-dimensional space. |
| Parent Package: | CityGML |
| Stereotype: | «ApplicationSchema» |

### 9.11.1. Classes

**AbstractTransportationSpace**

| | |
|---|---|
| Definition: | AbstractTransportationSpace is the abstract superclass of transportation objects such as Roads, Tracks, Railways, Waterways or Squares. |
| Subclass of: | AbstractUnoccupiedSpace |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| marking | Marking [*] | Relates to the markings that are part of the transportation space. |
| trafficSpace | TrafficSpace [*] | Relates to the traffic spaces that are part of the transportation space. |
| auxiliaryTrafficSpace | AuxiliaryTrafficSpace [*] | Relates to the auxiliary traffic spaces that are part of the transportation space. |
| hole | Hole [*] | Relates to the holes that are part of the transportation space. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| trafficDirection | TrafficDirectionValue [0..1] | Indicates the direction of traffic flow relative to the corresponding linear geometry representation. |
| occupancy | Occupancy [0..*] | Provides information on the residency of persons, vehicles, or other moving features in the transportation space. |
| adeOfAbstractTransportationSpace | ADEOfAbstractTransportationSpace [0..*] | Augments AbstractTransportationSpace with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AuxiliaryTrafficArea**

| | |
|---|---|
| Definition: | An AuxiliaryTrafficArea is the ground surface of an AuxiliaryTrafficSpace. |
| Subclass of: | AbstractThematicSurface |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | AuxiliaryTrafficAreaClassValue [0..1] | Indicates the specific type of the AuxiliaryTrafficArea. |
| function | AuxiliaryTrafficAreaFunctionValue [0..*] | Specifies the intended purposes of the AuxiliaryTrafficArea. |
| usage | AuxiliaryTrafficAreaUsageValue [0..*] | Specifies the actual uses of the AuxiliaryTrafficArea. |
| surfaceMaterial | SurfaceMaterialValue [0..1] | Specifies the type of pavement of the AuxiliaryTrafficArea. |
| adeOfAuxiliaryTrafficArea | ADEOfAuxiliaryTrafficArea [0..*] | Augments the AuxiliaryTrafficArea with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AuxiliaryTrafficSpace**

| | |
|---|---|
| Definition: | An AuxiliaryTrafficSpace is a space within the transportation space not intended for traffic purposes. |
| Subclass of: | AbstractUnoccupiedSpace |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| boundary | AuxiliaryTrafficArea [*] | Relates to the auxiliary traffic areas that bound the AuxiliaryTrafficSpace. This relation is inherited from the Core module. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | AuxiliaryTrafficSpaceClassValue [0..1] | Indicates the specific type of the AuxiliaryTrafficSpace. |
| function | AuxiliaryTrafficSpaceFunctionValue [0..*] | Specifies the intended purposes of the AuxiliaryTrafficSpace. |
| usage | AuxiliaryTrafficSpaceUsageValue [0..*] | Specifies the actual uses of the AuxiliaryTrafficSpace. |
| granularity | GranularityValue [1..1] | Defines whether auxiliary traffic spaces are represented by individual ways or by individual lanes, depending on the desired level of spatial and semantic decomposition. |
| adeOfAuxiliaryTrafficSpace | ADEOfAuxiliaryTrafficSpace [0..*] | Augments the AuxiliaryTrafficSpace with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**ClearanceSpace**

| | |
|---|---|
| Definition: | A ClearanceSpace represents the actual free space above a TrafficArea within which a mobile object can move without contacting an obstruction. |
| Subclass of: | AbstractUnoccupiedSpace |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | ClearanceSpaceClassValue [0..*] | Indicates the specific type of the ClearanceSpace. |
| adeOfClearanceSpace | ADEOfClearanceSpace [0..*] | Augments the ClearanceSpace with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Hole**

| | |
|---|---|
| Definition: | A Hole is an opening in the surface of a Road, Track or Square such as road damages, manholes or drains. Holes can span multiple transportation objects. |
| Subclass of: | AbstractUnoccupiedSpace |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| boundary | AbstractThematicSurface [*] | Relates to the surfaces that bound the Hole. This relation is inherited from the Core module. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | HoleClassValue [0..1] | Indicates the specific type of the Hole. |
| adeOfHole | ADEOfHole [0..*] | Augments the Hole with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**HoleSurface**

| | |
|---|---|
| Definition: | A HoleSurface is a representation of the ground surface of a hole. |
| Subclass of: | AbstractThematicSurface |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfHoleSurface | ADEOfHoleSurface [0..*] | Augments the HoleSurface with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Intersection**

| | |
|---|---|
| Definition: | An Intersection is a transportation space that is a shared segment of multiple Road, Track, Railway, or Waterway objects (e.g. a crossing of two roads or a level crossing of a road and a railway). |
| Subclass of: | AbstractTransportationSpace |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | IntersectionClassValue [0..1] | Indicates the specific type of the Intersection. |
| adeOfIntersection | ADEOfIntersection [0..*] | Augments the Intersection with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Marking**

| | |
|---|---|
| Definition: | A Marking is a visible pattern on a transportation area relevant to the structuring or restriction of traffic. Examples are road markings and markings related to railway or waterway traffic. |
| Subclass of: | AbstractThematicSurface |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | MarkingClassValue [0..1] | Indicates the specific type of the Marking. |
| adeOfMarking | ADEOfMarking [0..*] | Augments the Marking with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Railway**

| | |
|---|---|
| Definition: | A Railway is a transportation space used by wheeled vehicles on rails. |
| Subclass of: | AbstractTransportationSpace |
| Stereotype: | «TopLevelFeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| intersection | Intersection [*] | Relates to the intersections that are part of the Railway. |
| section | Section [*] | Relates to the sections that are part of the Railway. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | RailwayClassValue [0..1] | Indicates the specific type of the Railway. |
| function | RailwayFunctionValue [0..*] | Specifies the intended purposes of the Railway. |
| usage | RailwayUsageValue [0..*] | Specifies the actual uses of the Railway. |
| adeOfRailway | ADEOfRailway [0..*] | Augments the Railway with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Road**

| | |
|---|---|
| Definition: | A Road is a transportation space used by vehicles, bicycles and/or pedestrians. |
| Subclass of: | AbstractTransportationSpace |
| Stereotype: | «TopLevelFeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| intersection | Intersection [*] | Relates to the intersections that are part of the Road. |
| section | Section [*] | Relates to the sections that are part of the Road. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | RoadClassValue [0..1] | Indicates the specific type of the Road. |
| function | RoadFunctionValue [0..*] | Specifies the intended purposes of the Road. |
| usage | RoadUsageValue [0..*] | Specifies the actual uses of the Road. |
| adeOfRoad | ADEOfRoad [0..*] | Augments the Road with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Section**

| | |
|---|---|
| Definition: | A Section is a transportation space that is a segment of a Road, Railway, Track, or Waterway. |
| Subclass of: | AbstractTransportationSpace |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | SectionClassValue [0..1] | Indicates the specific type of the Section. |
| adeOfSection | ADEOfSection [0..*] | Augments the Section with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Square**

| | |
|---|---|
| Definition: | A Square is a transportation space for unrestricted movement for vehicles, bicycles and/or pedestrians. This includes plazas as well as large sealed surfaces such as parking lots. |
| Subclass of: | AbstractTransportationSpace |
| Stereotype: | «TopLevelFeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | SquareClassValue [0..1] | Indicates the specific type of the Square. |
| function | SquareFunctionValue [0..*] | Specifies the intended purposes of the Square. |
| usage | SquareUsageValue [0..*] | Specifies the actual uses of the Square. |
| adeOfSquare | ADEOfSquare [0..*] | Augments the Square with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Track**

| | |
|---|---|
| Definition: | A Track is a small path mainly used by pedestrians. Tracks can be segmented into Sections and Intersections. |
| Subclass of: | AbstractTransportationSpace |
| Stereotype: | «TopLevelFeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| section | Section [*] | Relates to the sections that are part of the Track. |
| intersection | Intersection [*] | Relates to the intersections that are part of the Track. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | TrackClassValue [0..1] | Indicates the specific type of the Track. |
| function | TrackFunctionValue [0..*] | Specifies the intended purposes of the Track. |
| usage | TrackUsageValue [0..*] | Specifies the actual uses of the Track. |
| adeOfTrack | ADEOfTrack [0..*] | Augments the Track with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**TrafficArea**

| | |
|---|---|
| Definition: | A TrafficArea is the ground surface of a TrafficSpace. Traffic areas are the surfaces upon which traffic actually takes place. |
| Subclass of: | AbstractThematicSurface |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | TrafficAreaClassValue [0..1] | Indicates the specific type of the TrafficArea. |
| function | TrafficAreaFunctionValue [0..*] | Specifies the intended purposes of the TrafficArea. |
| usage | TrafficAreaUsageValue [0..*] | Specifies the actual uses of the TrafficArea. |
| surfaceMaterial | SurfaceMaterialValue [0..1] | Specifies the type of pavement of the TrafficArea. |
| adeOfTrafficArea | ADEOfTrafficArea [0..*] | Augments the TrafficArea with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**TrafficSpace**

| | |
|---|---|
| Definition: | A TrafficSpace is a space in which traffic takes place. Traffic includes the movement of entities such as trains, vehicles, pedestrians, ships, or other transportation types. |
| Subclass of: | AbstractUnoccupiedSpace |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| successor | TrafficSpace [*] | Indicates the successor(s) of the TrafficSpace. |
| clearanceSpace | ClearanceSpace [*] | Relates to the clearance spaces that are part of the TrafficSpace. |
| predecessor | TrafficSpace [*] | Indicates the predecessor(s) of the TrafficSpace. |
| boundary | TrafficArea [*] | Relates to the traffic areas that bound the TrafficSpace. This relation is inherited from the Core module. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | TrafficSpaceClassValue [0..1] | Indicates the specific type of the TrafficSpace. |
| function | TrafficSpaceFunctionValue [0..*] | Specifies the intended purposes of the TrafficSpace. |
| usage | TrafficSpaceUsageValue [0..*] | Specifies the actual uses of the TrafficSpace. |
| granularity | GranularityValue [1..1] | Defines whether traffic spaces are represented by individual ways or by individual lanes, depending on the desired level of spatial and semantic decomposition. |
| trafficDirection | TrafficDirectionValue [0..1] | Indicates the direction of traffic flow relative to the corresponding linear geometry representation. |
| occupancy | Occupancy [0..*] | Provides information on the residency of persons, vehicles, or other moving features in the TrafficSpace. |
| adeOfTrafficSpace | ADEOfTrafficSpace [0..*] | Augments the TrafficSpace with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Waterway**

| | |
|---|---|
| Definition: | A Waterway is a transportation space used for the movement of vessels upon or within a water body. |
| Subclass of: | AbstractTransportationSpace |
| Stereotype: | «TopLevelFeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| intersection | Intersection [*] | Relates to the intersections that are part of the Waterway. |
| section | Section [*] | Relates to the sections that are part of the Waterway. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | WaterwayClassValue [0..1] | Indicates the specific type of the Waterway. |
| function | WaterwayFunctionValue [0..*] | Specifies the intended purposes of the Waterway. |
| usage | WaterwayUsageValue [0..*] | Specifies the actual uses of the Waterway. |
| adeOfWaterway | ADEOfWaterway [0..*] | Augments the Waterway with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.11.2. Data Types

**ADEOfAbstractTransportationSpace**

| | |
|---|---|
| Definition: | ADEOfAbstractTransportationSpace acts as a hook to define properties within an ADE that are to be added to AbstractTransportationSpace. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAuxiliaryTrafficArea**

| | |
|---|---|
| Definition: | ADEOfAuxiliaryTrafficArea acts as a hook to define properties within an ADE that are to be added to an AuxiliaryTrafficArea. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAuxiliaryTrafficSpace**

| | |
|---|---|
| Definition: | ADEOfAuxiliaryTrafficSpace acts as a hook to define properties within an ADE that are to be added to an AuxiliaryTrafficSpace. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfClearanceSpace**

| | |
|---|---|
| Definition: | ADEOfClearanceSpace acts as a hook to define properties within an ADE that are to be added to a ClearanceSpace. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfHole**

| | |
|---|---|
| Definition: | ADEOfHole acts as a hook to define properties within an ADE that are to be added to a Hole. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfHoleSurface**

| | |
|---|---|
| Definition: | ADEOfHoleSurface acts as a hook to define properties within an ADE that are to be added to a HoleSurface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfIntersection**

| | |
|---|---|
| Definition: | ADEOfIntersection acts as a hook to define properties within an ADE that are to be added to an Intersection. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfMarking**

| | |
|---|---|
| Definition: | ADEOfMarking acts as a hook to define properties within an ADE that are to be added to a Marking. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfRailway**

| | |
|---|---|
| Definition: | ADEOfRailway acts as a hook to define properties within an ADE that are to be added to a Railway. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfRoad**

| | |
|---|---|
| Definition: | ADEOfRoad acts as a hook to define properties within an ADE that are to be added to a Road. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfSection**

| | |
|---|---|
| Definition: | ADEOfSection acts as a hook to define properties within an ADE that are to be added to a Section. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfSquare**

| | |
|---|---|
| Definition: | ADEOfSquare acts as a hook to define properties within an ADE that are to be added to a Square. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfTrack**

| | |
|---|---|
| Definition: | ADEOfTrack acts as a hook to define properties within an ADE that are to be added to a Track. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfTrafficArea**

| | |
|---|---|
| Definition: | ADEOfTrafficArea acts as a hook to define properties within an ADE that are to be added to a TrafficArea. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfTrafficSpace**

| | |
|---|---|
| Definition: | ADEOfTrafficSpace acts as a hook to define properties within an ADE that are to be added to a TrafficSpace. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfWaterway**

| | |
|---|---|
| Definition: | ADEOfWaterway acts as a hook to define properties within an ADE that are to be added to a Waterway. |
| Subclass of: | None |
| Stereotype: | «DataType» |

### 9.11.3. Basic Types

none

### 9.11.4. Unions

none

### 9.11.5. Code Lists

**AuxiliaryTrafficAreaClassValue**

| | |
|---|---|
| Definition: | AuxiliaryTrafficAreaClassValue is a code list used to further classify an AuxiliaryTrafficArea. |
| Stereotype: | «CodeList» |

**AuxiliaryTrafficAreaFunctionValue**

| | |
|---|---|
| Definition: | AuxiliaryTrafficAreaFunctionValue is a code list that enumerates the different purposes of an AuxiliaryTrafficArea. |
| Stereotype: | «CodeList» |

**AuxiliaryTrafficAreaUsageValue**

| | |
|---|---|
| Definition: | AuxiliaryTrafficAreaUsageValue is a code list that enumerates the different uses of an AuxiliaryTrafficArea. |
| Stereotype: | «CodeList» |

**AuxiliaryTrafficSpaceClassValue**

| | |
|---|---|
| Definition: | AuxiliaryTrafficSpaceClassValue is a code list used to further classify an AuxiliaryTrafficSpace. |
| Stereotype: | «CodeList» |

**AuxiliaryTrafficSpaceFunctionValue**

| | |
|---|---|
| Definition: | AuxiliaryTrafficSpaceFunctionValue is a code list that enumerates the different purposes of an AuxiliaryTrafficSpace. |
| Stereotype: | «CodeList» |

**AuxiliaryTrafficSpaceUsageValue**

| | |
|---|---|
| Definition: | AuxiliaryTrafficSpaceUsageValue is a code list that enumerates the different uses of an AuxiliaryTrafficSpace. |
| Stereotype: | «CodeList» |

**ClearanceSpaceClassValue**

| | |
|---|---|
| Definition: | ClearanceSpaceClassValue is a code list used to further classify a ClearanceSpace. |
| Stereotype: | «CodeList» |

**HoleClassValue**

| | |
|---|---|
| Definition: | HoleClassValue is a code list used to further classify a Hole. |
| Stereotype: | «CodeList» |

**IntersectionClassValue**

| | |
|---|---|
| Definition: | IntersectionClassValue is a code list used to further classify an Intersection. |
| Stereotype: | «CodeList» |

**MarkingClassValue**

| | |
|---|---|
| Definition: | MarkingClassValue is a code list used to further classify a Marking. |
| Stereotype: | «CodeList» |

**RailwayClassValue**

| | |
|---|---|
| Definition: | RailwayClassValue is a code list used to further classify a Railway. |
| Stereotype: | «CodeList» |

**RailwayFunctionValue**

| | |
|---|---|
| Definition: | RailwayFunctionValue is a code list that enumerates the different purposes of a Railway. |
| Stereotype: | «CodeList» |

**RailwayUsageValue**

| | |
|---|---|
| Definition: | RailwayUsageValue is a code list that enumerates the different uses of a Railway. |
| Stereotype: | «CodeList» |

**RoadClassValue**

| | |
|---|---|
| Definition: | RoadClassValue is a code list used to further classify a Road. |
| Stereotype: | «CodeList» |

**RoadFunctionValue**

| | |
|---|---|
| Definition: | RoadFunctionValue is a code list that enumerates the different purposes of a Road. |
| Stereotype: | «CodeList» |

**RoadUsageValue**

| | |
|---|---|
| Definition: | RoadUsageValue is a code list that enumerates the different uses of a Road. |
| Stereotype: | «CodeList» |

**SectionClassValue**

| | |
|---|---|
| Definition: | SectionClassValue is a code list used to further classify a Section. |
| Stereotype: | «CodeList» |

**SquareClassValue**

| | |
|---|---|
| Definition: | SquareClassValue is a code list used to further classify a Square. |
| Stereotype: | «CodeList» |

**SquareFunctionValue**

| | |
|---|---|
| Definition: | SquareFunctionValue is a code list that enumerates the different purposes of a Square. |
| Stereotype: | «CodeList» |

**SquareUsageValue**

| | |
|---|---|
| Definition: | SquareUsageValue is a code list that enumerates the different uses of a Square. |
| Stereotype: | «CodeList» |

**SurfaceMaterialValue**

| | |
|---|---|
| Definition: | SurfaceMaterialValue is a code list that enumerates the different surface materials. |
| Stereotype: | «CodeList» |

**TrackClassValue**

| | |
|---|---|
| Definition: | TrackClassValue is a code list used to further classify a Track. |
| Stereotype: | «CodeList» |

**TrackFunctionValue**

| | |
|---|---|
| Definition: | TrackFunctionValue is a code list that enumerates the different purposes of a Track. |
| Stereotype: | «CodeList» |

**TrackUsageValue**

| | |
|---|---|
| Definition: | TrackUsageValue is a code list that enumerates the different uses of a Track. |
| Stereotype: | «CodeList» |

**TrafficAreaClassValue**

| | |
|---|---|
| Definition: | TrafficAreaClassValue is a code list used to further classify a TrafficArea. |
| Stereotype: | «CodeList» |

**TrafficAreaFunctionValue**

| | |
|---|---|
| Definition: | TrafficAreaFunctionValue is a code list that enumerates the different purposes of a TrafficArea. |
| Stereotype: | «CodeList» |

**TrafficAreaUsageValue**

| | |
|---|---|
| Definition: | TrafficAreaUsageValue is a code list that enumerates the different uses of a TrafficArea. |
| Stereotype: | «CodeList» |

**TrafficSpaceClassValue**

| | |
|---|---|
| Definition: | TrafficSpaceClassValue is a code list used to further classify a TrafficSpace. |
| Stereotype: | «CodeList» |

**TrafficSpaceFunctionValue**

| | |
|---|---|
| Definition: | TrafficSpaceFunctionValue is a code list that enumerates the different purposes of a TrafficSpace. |
| Stereotype: | «CodeList» |

**TrafficSpaceUsageValue**

| | |
|---|---|
| Definition: | TrafficSpaceUsageValue is a code list that enumerates the different uses of a TrafficSpace. |
| Stereotype: | «CodeList» |

**WaterwayClassValue**

| | |
|---|---|
| Definition: | WaterwayClassValue is a code list used to further classify a Waterway. |
| Stereotype: | «CodeList» |

**WaterwayFunctionValue**

| | |
|---|---|
| Definition: | WaterwayFunctionValue is a code list that enumerates the different purposes of a Waterway. |
| Stereotype: | «CodeList» |

**WaterwayUsageValue**

| | |
|---|---|
| Definition: | WaterwayUsageValue is a code list that enumerates the different uses of a Waterway. |
| Stereotype: | «CodeList» |

### 9.11.6. Enumerations

**GranularityValue**

| | |
|---|---|
| Definition: | GranularityValue enumerates the different levels of granularity in which transportation objects are represented. |
| StereoType: | <<Enumeration>> |

| Literal value | Definition |
|---|---|
| lane | Indicates that the individual lanes of the transportation object are represented. |
| way | Indicates that the individual (carriage)ways of the transportation object are represented. |

**TrafficDirectionValue**

| | |
|---|---|
| Definition: | TrafficDirectionValue enumerates the allowed directions of travel of a mobile object. |
| StereoType: | <<Enumeration>> |

| Literal value | Definition |
|---|---|
| forwards | Indicates that traffic flows in the direction of the corresponding linear geometry. |
| backwards | Indicates that traffic flows in the opposite direction of the corresponding linear geometry. |
| both | Indicates that traffic flows in both directions. |

## 9.12. Vegetation

| | |
|---|---|
| Description: | The Vegetation module supports representation of vegetation objects with vegetation-specific thematic classes. CityGML's vegetation model distinguishes between solitary vegetation objects like trees, and vegetation areas which represent biotopes like forests or other plant communities. |
| Parent Package: | CityGML |

Stereotype: «ApplicationSchema»

### 9.12.1. Classes

**AbstractVegetationObject**

Definition: AbstractVegetationObject is the abstract superclass for all kinds of vegetation objects.

Subclass of: AbstractOccupiedSpace

Stereotype: «FeatureType»

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfAbstractVegetationObject | ADEOfAbstractVegetationObject [0..*] | Augments AbstractVegetationObject with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**PlantCover**

Definition: A PlantCover represents a space covered by vegetation.

Subclass of: AbstractVegetationObject

Stereotype: «TopLevelFeatureType»

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | PlantCoverClassValue [0..1] | Indicates the specific type of the PlantCover. |
| function | PlantCoverFunctionValue [0..*] | Specifies the intended purposes of the PlantCover. |
| usage | PlantCoverUsageValue [0..*] | Specifies the actual uses of the PlantCover. |
| averageHeight | Length [0..1] | Specifies the average height of the PlantCover. |
| minHeight | Length [0..1] | Specifies the minimum height of the PlantCover. |
| maxHeight | Length [0..1] | Specifies the maximum height of the PlantCover. |
| adeOfPlantCover | ADEOfPlantCover [0..*] | Augments the PlantCover with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**SolitaryVegetationObject**

| | |
|---|---|
| Definition: | A SolitaryVegetationObject represents individual vegetation objects, e.g. trees or bushes. |
| Subclass of: | AbstractVegetationObject |
| Stereotype: | «TopLevelFeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | SolitaryVegetationObjectClassValue [0..1] | Indicates the specific type of the SolitaryVegetationObject. |
| function | SolitaryVegetationObjectFunctionValue [0..*] | Specifies the intended purposes of the SolitaryVegetationObject. |
| usage | SolitaryVegetationObjectUsageValue [0..*] | Specifies the actual uses of the SolitaryVegetationObject. |
| species | SpeciesValue [0..1] | Indicates the botanical name of the SolitaryVegetationObject. |
| height | Length [0..1] | Distance between the highest point of the vegetation object and the lowest point of the terrain at the bottom of the object. |
| trunkDiameter | Length [0..1] | Specifies the diameter of the SolitaryCityObject's trunk. |
| crownDiameter | Length [0..1] | Specifies the diameter of the SolitaryCityObject's crown. |
| rootBallDiameter | Length [0..1] | Specifies the diameter of the SolitaryCityObject's root ball. |
| maxRootBallDepth | Length [0..1] | Specifies the vertical distance between the lowest point of the SolitaryVegetationObject's root ball and the terrain surface. |
| adeOfSolitaryVegetationObject | ADEOfSolitaryVegetationObject [0..*] | Augments the SolitaryVegetationObject with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.12.2. Data Types

**ADEOfAbstractVegetationObject**

| | |
|---|---|
| Definition: | ADEOfAbstractVegetationObject acts as a hook to define properties within an ADE that are to be added to AbstractVegetationObject. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfPlantCover**

| | |
|---|---|
| Definition: | ADEOfPlantCover acts as a hook to define properties within an ADE that are to be added to a PlantCover. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfSolitaryVegetationObject**

| | |
|---|---|
| Definition: | ADEOfSolitaryVegetationObject acts as a hook to define properties within an ADE that are to be added to a SolitaryVegetationObject. |
| Subclass of: | None |
| Stereotype: | «DataType» |

### 9.12.3. Basic Types

none

### 9.12.4. Unions

none

### 9.12.5. Code Lists

**PlantCoverClassValue**

| | |
|---|---|
| Definition: | PlantCoverClassValue is a code list used to further classify a PlantCover. |
| Stereotype: | «CodeList» |

**PlantCoverFunctionValue**

| | |
|---|---|
| Definition: | PlantCoverFunctionValue is a code list that enumerates the different purposes of a PlantCover. |
| Stereotype: | «CodeList» |

**PlantCoverUsageValue**

| | |
|---|---|
| Definition: | PlantCoverUsageValue is a code list that enumerates the different uses of a PlantCover. |
| Stereotype: | «CodeList» |

**SolitaryVegetationObjectClassValue**

| | |
|---|---|
| Definition: | SolitaryVegetationObjectClassValue is a code list used to further classify a SolitaryVegetationObject. |
| Stereotype: | «CodeList» |

**SolitaryVegetationObjectFunctionValue**

| | |
|---|---|
| Definition: | SolitaryVegetationObjectFunctionValue is a code list that enumerates the different purposes of a SolitaryVegetationObject. |
| Stereotype: | «CodeList» |

**SolitaryVegetationObjectUsageValue**

| | |
|---|---|
| Definition: | SolitaryVegetationObjectUsageValue is a code list that enumerates the different uses of a SolitaryVegetationObject. |
| Stereotype: | «CodeList» |

**SpeciesValue**

| | |
|---|---|
| Definition: | A SpeciesValue is a code list that enumerates the species of a SolitaryVegetationObject. |
| Stereotype: | «CodeList» |

### 9.12.6. Enumerations

none

# 9.13. Versioning

| | |
|---|---|
| Description: | The Versioning module supports representation of multiple versions of CityGML features within a single CityGML model. In addition, also the version transitions and transactions that lead to the different versions can be represented. |
| Parent Package: | CityGML |
| Stereotype: | «ApplicationSchema» |

## 9.13.1. Classes

**Version**

| | |
|---|---|
| Definition: | Version represents a defined state of a city model consisting of the dedicated versions of all city object instances that belong to the respective city model version. Versions can have names, a description and can be labeled with an arbitrary number of user defined tags. |
| Subclass of: | AbstractVersion |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| versionMember «Version» | AbstractFeatureWithLifespan [*] | Relates to all city objects that are part of the city model version. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| tag | CharacterString [0..*] | Allows for adding keywords to the city model version. |
| adeOfVersion | ADEOfVersion [0..*] | Augments the Version with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**VersionTransition**

| | |
|---|---|
| Definition: | VersionTransition describes the change of the state of a city model from one version to another. Version transitions can have names, a description and can be further qualified by a type and a reason. |
| Subclass of: | AbstractVersionTransition |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| from | Version [0..1] | Relates to the predecessor version of the VersionTransition. |
| transaction | Transaction [*] | Relates to all transactions that have been applied as part of the VersionTransition. |
| to | Version [0..1] | Relates to the successor version of the VersionTransition. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| reason | CharacterString [0..1] | Specifies why the VersionTransition has been carried out. |
| clonePredecessor | Boolean [1..1] | Indicates whether the set of city object instances belonging to the successor version of the city model is either explicitly enumerated within the successor version object (attribute clonePredecessor=false), or has to be derived from the modifications of the city model provided as a list of transactions on the city object versions contained in the predecessor version (attribute clonePredecessor=true). |
| type | TransitionTypeValue [0..1] | Indicates the specific type of the VersionTransition. |
| adeOfVersionTransition | ADEOfVersionTransition [0..*] | Augments the VersionTransition with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.13.2. Data Types

**ADEOfVersion**

| | |
|---|---|
| Definition: | ADEOfVersion acts as a hook to define properties within an ADE that are to be added to a Version. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfVersionTransition**

| | |
|---|---|
| Definition: | ADEOfVersionTransition acts as a hook to define properties within an ADE that are to be added to a VersionTransition. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**Transaction**

| | |
|---|---|
| Definition: | Transaction represents a modification of the city model by the creation, termination, or replacement of a specific city object. While the creation of a city object also marks its first object version, the termination marks the end of existence of a real world object and, hence, also terminates the final version of a city object. The replacement of a city object means that a specific version of it is replaced by a new version. |
| Subclass of: | None |
| Stereotype: | «DataType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| newFeature «Version» | AbstractFeatureWithLifespan [0..1] | Relates to the version of the city object subsequent to the Transaction. |
| oldFeature «Version» | AbstractFeatureWithLifespan [0..1] | Relates to the version of the city object prior to the Transaction. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| type | TransactionTypeValue [1..1] | Indicates the specific type of the Transaction. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.13.3. Basic Types

none

### 9.13.4. Unions

none

### 9.13.5. Code Lists

none

### 9.13.6. Enumerations

**TransactionTypeValue**

| | |
|---|---|
| Definition: | TransactionTypeValue enumerates the three possible types of transactions: insert, delete, or replace. |
| StereoType: | <<Enumeration>> |

| Literal value | Definition |
|---|---|
| insert | Indicates that the feature referenced from the Transaction via the "newFeature" association has been newly created; the association "oldFeature" is empty in this case. |
| delete | Indicates that the feature referenced from the Transaction via the "oldFeature" association ceases to exist; the association "newFeature" is empty in this case. |
| replace | Indicates that the feature referenced from the Transaction via the "oldFeature" association has been replaced by the feature referenced via the "newFeature" association. |

**TransitionTypeValue**

| | |
|---|---|
| Definition: | TransitionTypeValue enumerates the different kinds of version transitions. “planned” and “fork” should be used in cases when from one city model version multiple successor versions are being created. “realized” and “merge” should be used when different city model versions are converging into a common successor version. |
| StereoType: | <<Enumeration>> |

| Literal value | Definition |
|---|---|
| planned | Indicates that the successor version of the city model represents a planning state for a possible future of the city. |
| realized | Indicates that the predecessor version is the chosen one from a number of possible planning versions. |
| historicalSuccession | Indicates that the successor version reflects updates on the city model over time (historical timeline). It shall only be used for at most one version transition outgoing from a city model version. |
| fork | Indicates other reasons to create alternative city model versions, for example, when different parties are updating parts of the city model or to reflect the results of different simulation runs. |
| merge | Indicates other reasons to converge multiple versions back into a common city model version. |

# 9.14. WaterBody

| | |
|---|---|
| Description: | The WaterBody module supports representation of the thematic aspects and 3D geometry of rivers, canals, lakes, and basins. It does, however, not inherit any hydrological or other dynamic aspects of fluid flow. |
| Parent Package: | CityGML |
| Stereotype: | «ApplicationSchema» |

## 9.14.1. Classes

**AbstractWaterBoundarySurface**

| | |
|---|---|
| Definition: | AbstractWaterBoundarySurface is the abstract superclass for all kinds of thematic surfaces bounding a water body. |
| Subclass of: | AbstractThematicSurface |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfAbstractWaterBoundarySurface | ADEOfAbstractWaterBoundarySurface [0..*] | Augments AbstractWaterBoundarySurface with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**WaterBody**

| | |
|---|---|
| Definition: | A WaterBody represents significant and permanent or semi-permanent accumulations of surface water, usually covering a part of the Earth. |
| Subclass of: | AbstractOccupiedSpace |
| Stereotype: | «TopLevelFeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| boundary | AbstractWaterBoundarySurface [*] | |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | WaterBodyClassValue [0..1] | Indicates the specific type of the WaterBody. |
| function | WaterBodyFunctionValue [0..*] | Specifies the intended purposes of the WaterBody. |
| usage | WaterBodyUsageValue [0..*] | Specifies the actual uses of the WaterBody. |
| adeOfWaterBody | ADEOfWaterBody [0..*] | Augments the WaterBody with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**WaterGroundSurface**

| | |
|---|---|
| Definition: | A WaterGroundSurface represents the exterior boundary surface of the submerged bottom of a water body. |
| Subclass of: | AbstractWaterBoundarySurface |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfWaterGroundSurface | ADEOfWaterGroundSurface [0..*] | Augments the WaterGroundSurface with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**WaterSurface**

| | |
|---|---|
| Definition: | A WaterSurface represents the upper exterior interface between a water body and the atmosphere. |
| Subclass of: | AbstractWaterBoundarySurface |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| waterLevel | WaterLevelValue [0..1] | Specifies the level of the WaterSurface. |
| adeOfWaterSurface | ADEOfWaterSurface [0..*] | Augments the WaterSurface with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.14.2. Data Types

**ADEOfAbstractWaterBoundarySurface**

| | |
|---|---|
| Definition: | ADEOfAbstractWaterBoundarySurface acts as a hook to define properties within an ADE that are to be added to AbstractWaterBoundarySurface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfWaterBody**

| | |
|---|---|
| Definition: | ADEOfWaterBody acts as a hook to define properties within an ADE that are to be added to a WaterBody. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfWaterGroundSurface**

| | |
|---|---|
| Definition: | ADEOfWaterGroundSurface acts as a hook to define properties within an ADE that are to be added to a WaterGroundSurface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfWaterSurface**

| | |
|---|---|
| Definition: | ADEOfWaterSurface acts as a hook to define properties within an ADE that are to be added to a WaterSurface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

### 9.14.3. Basic Types

none

### 9.14.4. Unions

none

### 9.14.5. Code Lists

**WaterBodyClassValue**

| | |
|---|---|
| Definition: | WaterBodyClassValue is a code list used to further classify a WaterBody. |
| Stereotype: | «CodeList» |

**WaterBodyFunctionValue**

| | |
|---|---|
| Definition: | WaterBodyFunctionValue is a code list that enumerates the different purposes of a WaterBody. |
| Stereotype: | «CodeList» |

**WaterBodyUsageValue**

| | |
|---|---|
| Definition: | WaterBodyUsageValue is a code list that enumerates the different uses of a WaterBody. |
| Stereotype: | «CodeList» |

**WaterLevelValue**

| | |
|---|---|
| Definition: | WaterLevelValue is a code list that enumerates the different levels of a water surface. |
| Stereotype: | «CodeList» |

### 9.14.6. Enumerations

none

## 9.15. Construction

| | |
|---|---|
| Description: | The Construction module supports representation of key elements of different types of constructions. These key elements include construction surfaces (e.g floor and ceiling), windows and doors, constructive elements (e.g. beams and slabs), installations, and furniture. |
| Parent Package: | CityGML |
| Stereotype: | «ApplicationSchema» |

### 9.15.1. Classes

**AbstractConstruction**

| | |
|---|---|
| Definition: | AbstractConstruction is the abstract superclass for objects that are manufactured by humans from construction materials, are connected to earth, and are intended to be permanent. A connection with the ground also exists when the construction rests by its own weight on the ground or is moveable limited on stationary rails or if the construction is intended to be used mainly stationary. |
| Subclass of: | AbstractOccupiedSpace |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| boundary | AbstractThematicSurface [*] | Relates to the surfaces that bound the construction. This relation is inherited from the Core module. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| conditionOfConstruction | ConditionOfConstructionValue [0..1] | Indicates the life-cycle status of the construction. [cf. INSPIRE] |
| dateOfConstruction | Date [0..1] | Indicates the date at which the construction was completed. |
| dateOfDemolition | Date [0..1] | Indicates the date at which the construction was demolished. |
| constructionEvent | ConstructionEvent [0..*] | Describes specific events in the life-time of the construction. |
| elevation | Elevation [0..*] | Specifies qualified elevations of the construction in relation to a well-defined surface which is commonly taken as origin (e.g. geoid or water level). [cf. INSPIRE] |
| height | Height [0..*] | Specifies qualified heights of the construction above ground or below ground. [cf. INSPIRE] |
| occupancy | Occupancy [0..*] | Provides qualified information on the residency of persons, animals, or other moveable objects in the construction. |
| adeOfAbstractConstruction | ADEOfAbstractConstruction [0..*] | Augments AbstractConstruction with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractConstructionSurface**

| | |
|---|---|
| Definition: | AbstractConstructionSurface is the abstract superclass for different kinds of surfaces that bound a construction. |
| Subclass of: | AbstractThematicSurface |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| fillingSurface | AbstractFillingSurface [*] | Relates to the surfaces that seal the openings of the construction surface. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfAbstractConstructionSurface | ADEOfAbstractConstructionSurface [0..*] | Augments AbstractConstructionSurface with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractConstructiveElement**

| | |
|---|---|
| Definition: | AbstractConstructiveElement is the abstract superclass for the representation of volumetric elements of a construction. Examples are walls, beams, slabs. |
| Subclass of: | AbstractOccupiedSpace |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| boundary | AbstractThematicSurface [*] | Relates to the surfaces that bound the constructive element. This relation is inherited from the Core module. |
| filling | AbstractFillingElement [*] | Relates to the elements that fill the opening of the constructive element. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| isStructuralElement | Boolean [0..1] | Indicates whether the constructive element is essential from a structural point of view. A structural element cannot be omitted without collapsing of the construction. Examples are pylons and anchorages of bridges. |
| adeOfAbstractConstructiveElement | ADEOfAbstractConstructiveElement [0..*] | Augments AbstractConstructiveElement with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractFillingElement**

| | |
|---|---|
| Definition: | AbstractFillingElement is the abstract superclass for different kinds of elements that fill the openings of a construction. |
| Subclass of: | AbstractOccupiedSpace |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
| --- | --- | --- |
| adeOfAbstractFillingElement | ADEOfAbstractFillingElement [0..*] | Augments AbstractFillingElement with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractFillingSurface**

| | |
| --- | --- |
| Definition: | AbstractFillingSurface is the abstract superclass for different kinds of surfaces that seal openings filled by filling elements. |
| Subclass of: | AbstractThematicSurface |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
| --- | --- | --- |
| adeOfAbstractFillingSurface | ADEOfAbstractFillingSurface [0..*] | Augments AbstractFillingSurface with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractFurniture**

| | |
| --- | --- |
| Definition: | AbstractFurniture is the abstract superclass for the representation of furniture objects of a construction. |
| Subclass of: | AbstractOccupiedSpace |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
| --- | --- | --- |
| adeOfAbstractFurniture | ADEOfAbstractFurniture [0..*] | Augments AbstractFurniture with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractInstallation**

| | |
|---|---|
| Definition: | AbstractInstallation is the abstract superclass for the representation of installation objects of a construction. |
| Subclass of: | AbstractOccupiedSpace |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| boundary | AbstractThematicSurface [*] | Relates to the surfaces that bound the installation. This relation is inherited from the Core module. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| relationToConstruction | RelationToConstruction [0..1] | Indicates whether the installation is located inside and/or outside of the construction. |
| adeOfAbstractInstallation | ADEOfAbstractInstallation [0..*] | Augments AbstractInstallation with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**CeilingSurface**

| | |
|---|---|
| Definition: | A CeilingSurface is a surface that represents the interior ceiling of a construction. An example is the ceiling of a room. |
| Subclass of: | AbstractConstructionSurface |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfCeilingSurface | ADEOfCeilingSurface [0..*] | Augments the CeilingSurface with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Door**

| | |
|---|---|
| Definition: | A Door is a construction for closing an opening intended primarily for access or egress or both. [cf. ISO 6707-1] |
| Subclass of: | AbstractFillingElement |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| address | Address [*] | Relates to the addresses that are assigned to the Door. |
| boundary | DoorSurface [*] | Relates to the door surfaces that bound the Door. This relation is inherited from the Core module. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | DoorClassValue [0..1] | Indicates the specific type of the Door. |
| function | DoorFunctionValue [0..*] | Specifies the intended purposes of the Door. |
| usage | DoorUsageValue [0..*] | Specifies the actual uses of the Door. |
| adeOfDoor | ADEOfDoor [0..*] | Augments the Door with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**DoorSurface**

| | |
|---|---|
| Definition: | A DoorSurface is either a boundary surface of a Door feature or a surface that seals an opening filled by a door. |
| Subclass of: | AbstractFillingSurface |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| address | Address [*] | Relates to the addresses that are assigned to the DoorSurface. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfDoorSurface | ADEOfDoorSurface [0..*] | Augments the DoorSurface with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**FloorSurface**

| | |
|---|---|
| Definition: | A FloorSurface is surface that represents the interior floor of a construction. An example is the floor of a room. |
| Subclass of: | AbstractConstructionSurface |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfFloorSurface | ADEOfFloorSurface [0..*] | Augments the FloorSurface with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**GroundSurface**

| | |
|---|---|
| Definition: | A GroundSurface is a surface that represents the ground plate of a construction. The polygon defining the ground plate is congruent with the footprint of the construction. |
| Subclass of: | AbstractConstructionSurface |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfGroundSurface | ADEOfGroundSurface [0..*] | Augments the GroundSurface with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**InteriorWallSurface**

| | |
|---|---|
| Definition: | An InteriorWallSurface is a surface that is visible from inside a construction. An example is the wall of a room. |
| Subclass of: | AbstractConstructionSurface |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfInteriorWallSurface | ADEOfInteriorWallSurface [0..*] | Augments the InteriorWallSurface with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**OtherConstruction**

| | |
|---|---|
| Definition: | An OtherConstruction is a construction that is not covered by any of the other subclasses of AbstractConstruction. |
| Subclass of: | AbstractConstruction |
| Stereotype: | «TopLevelFeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | OtherConstructionClassValue [0..1] | Indicates the specific type of the OtherConstruction. |
| function | OtherConstructionFunctionValue [0..*] | Specifies the intended purposes of the OtherConstruction. |
| usage | OtherConstructionUsageValue [0..*] | Specifies the actual uses of the OtherConstruction. |
| adeOfOtherConstruction | ADEOfOtherConstruction [0..*] | Augments the OtherConstruction with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**OuterCeilingSurface**

| | |
|---|---|
| Definition: | An OuterCeilingSurface is a surface that belongs to the outer building shell with the orientation pointing downwards. An example is the ceiling of a loggia. |
| Subclass of: | AbstractConstructionSurface |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
| --- | --- | --- |
| adeOfOuterCeilingSurface | ADEOfOuterCeilingSurface [0..*] | Augments the OuterCeilingSurface with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**OuterFloorSurface**

| | |
| --- | --- |
| Definition: | An OuterFloorSurface is a surface that belongs to the outer construction shell with the orientation pointing upwards. An example is the floor of a loggia. |
| Subclass of: | AbstractConstructionSurface |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
| --- | --- | --- |
| adeOfOuterFloorSurface | ADEOfOuterFloorSurface [0..*] | Augments the OuterFloorSurface with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**RoofSurface**

| | |
| --- | --- |
| Definition: | A RoofSurface is a surface that delimits major roof parts of a construction. |
| Subclass of: | AbstractConstructionSurface |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
| --- | --- | --- |
| adeOfRoofSurface | ADEOfRoofSurface [0..*] | Augments the RoofSurface with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**WallSurface**

| | |
|---|---|
| Definition: | A WallSurface is a surface that is part of the building facade visible from the outside. |
| Subclass of: | AbstractConstructionSurface |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfWallSurface | ADEOfWallSurface [0..*] | Augments the WallSurface with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Window**

| | |
|---|---|
| Definition: | A Window is a construction for closing an opening in a wall or roof, primarily intended to admit light and/or provide ventilation. [cf. ISO 6707-1] |
| Subclass of: | AbstractFillingElement |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| boundary | WindowSurface [*] | Relates to the window surfaces that bound the Window. This relation is inherited from the Core module. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | WindowClassValue [0..1] | Indicates the specific type of the Window. |
| function | WindowFunctionValue [0..*] | Specifies the intended purposes of the Window. |
| usage | WindowUsageValue [0..*] | Specifies the actual uses of the Window. |
| adeOfWindow | ADEOfWindow [0..*] | Augments the Window with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**WindowSurface**

| | |
|---|---|
| Definition: | A WindowSurface is either a boundary surface of a Window feature or a surface that seals an opening filled by a window. |
| Subclass of: | AbstractFillingSurface |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfWindowSurface | ADEOfWindowSurface [0..*] | Augments the WindowSurface with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

## 9.15.2. Data Types

**ADEOfAbstractConstruction**

| | |
|---|---|
| Definition: | ADEOfAbstractConstruction acts as a hook to define properties within an ADE that are to be added to AbstractConstruction. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractConstructionSurface**

| | |
|---|---|
| Definition: | ADEOfAbstractConstructionSurface acts as a hook to define properties within an ADE that are to be added to AbstractConstructionSurface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractConstructiveElement**

| | |
|---|---|
| Definition: | ADEOfAbstractConstructiveElement acts as a hook to define properties within an ADE that are to be added to AbstractConstructiveElement. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractFillingElement**

| | |
|---|---|
| Definition: | ADEOfAbstractFillingElement acts as a hook to define properties within an ADE that are to be added to AbstractFillingElement. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractFillingSurface**

| | |
|---|---|
| Definition: | ADEOfAbstractFillingSurface acts as a hook to define properties within an ADE that are to be added to AbstractFillingSurface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractFurniture**

| | |
|---|---|
| Definition: | ADEOfAbstractFurniture acts as a hook to define properties within an ADE that are to be added to AbstractFurniture. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractInstallation**

| | |
|---|---|
| Definition: | ADEOfAbstractInstallation acts as a hook to define properties within an ADE that are to be added to AbstractInstallation. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfCeilingSurface**

| | |
|---|---|
| Definition: | ADEOfCeilingSurface acts as a hook to define properties within an ADE that are to be added to a CeilingSurface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfDoor**

| | |
|---|---|
| Definition: | ADEOfDoor acts as a hook to define properties within an ADE that are to be added to a Door. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfDoorSurface**

| | |
|---|---|
| Definition: | ADEOfDoorSurface acts as a hook to define properties within an ADE that are to be added to a DoorSurface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfFloorSurface**

| | |
|---|---|
| Definition: | ADEOfFloorSurface acts as a hook to define properties within an ADE that are to be added to a FloorSurface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfGroundSurface**

| | |
|---|---|
| Definition: | ADEOfGroundSurface acts as a hook to define properties within an ADE that are to be added to a GroundSurface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfInteriorWallSurface**

| | |
|---|---|
| Definition: | ADEOfInteriorWallSurface acts as a hook to define properties within an ADE that are to be added to an InteriorWallSurface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfOtherConstruction**

| | |
|---|---|
| Definition: | ADEOfOtherConstruction acts as a hook to define properties within an ADE that are to be added to an OtherConstruction. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfOuterCeilingSurface**

| | |
|---|---|
| Definition: | ADEOfOuterCeilingSurface acts as a hook to define properties within an ADE that are to be added to an OuterCeilingSurface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfOuterFloorSurface**

| | |
|---|---|
| Definition: | ADEOfOuterFloorSurface acts as a hook to define properties within an ADE that are to be added to an OuterFloorSurface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfRoofSurface**

| | |
|---|---|
| Definition: | ADEOfRoofSurface acts as a hook to define properties within an ADE that are to be added to a RoofSurface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfWallSurface**

| | |
|---|---|
| Definition: | ADEOfWallSurface acts as a hook to define properties within an ADE that are to be added to a WallSurface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfWindow**

| | |
|---|---|
| Definition: | ADEOfWindow acts as a hook to define properties within an ADE that are to be added to a Window. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfWindowSurface**

| | |
|---|---|
| Definition: | ADEOfWindowSurface acts as a hook to define properties within an ADE that are to be added to a WindowSurface. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ConstructionEvent**

| | |
|---|---|
| Definition: | A ConstructionEvent is a data type used to describe a specific event that is associated with a construction. Examples are the issuing of a building permit or the renovation of a building. |
| Subclass of: | None |
| Stereotype: | «DataType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| event | EventValue [1..1] | Indicates the specific event type. |
| dateOfEvent | Date [1..1] | Specifies the date at which the event took or will take place. |
| description | CharacterString [0..1] | Provides additional information on the event. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Elevation**

| | |
|---|---|
| Definition: | Elevation is a data type that includes the elevation value itself and information on how this elevation was measured. [cf. INSPIRE] |
| Subclass of: | None |
| Stereotype: | «DataType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| elevationReference | ElevationReferenceValue [1..1] | Specifies the level from which the elevation was measured. [cf. INSPIRE] |
| elevationValue | DirectPosition [1..1] | Specifies the value of the elevation. [cf. INSPIRE] |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Height**

| | |
|---|---|
| Definition: | Height represents a vertical distance (measured or estimated) between a low reference and a high reference. [cf. INSPIRE] |
| Subclass of: | None |
| Stereotype: | «DataType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| highReference | ElevationReferenceValue [1..1] | Indicates the high point used to calculate the value of the height. [cf. INSPIRE] |
| lowReference | ElevationReferenceValue [1..1] | Indicates the low point used to calculate the value of the height. [cf. INSPIRE] |
| status | HeightStatusValue [1..1] | Indicates the way the height has been captured. [cf. INSPIRE] |
| value | Length [1..1] | Specifies the value of the height above or below ground. [cf. INSPIRE] |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.15.3. Basic Types

none

### 9.15.4. Unions

none

### 9.15.5. Code Lists

**DoorClassValue**

| | |
|---|---|
| Definition: | DoorClassValue is a code list used to further classify a Door. |
| Stereotype: | «CodeList» |

**DoorFunctionValue**

| | |
|---|---|
| Definition: | DoorFunctionValue is a code list that enumerates the different purposes of a Door. |
| Stereotype: | «CodeList» |

**DoorUsageValue**

| | |
|---|---|
| Definition: | DoorUsageValue is a code list that enumerates the different uses of a Door. |
| Stereotype: | «CodeList» |

**ElevationReferenceValue**

| | |
|---|---|
| Definition: | ElevationReferenceValue is a code list that enumerates the different elevation reference levels used to measure construction heights. |
| Stereotype: | «CodeList» |

**EventValue**

| | |
|---|---|
| Definition: | EventValue is a code list that enumerates the different events of a construction. |
| Stereotype: | «CodeList» |

**OtherConstructionClassValue**

| | |
|---|---|
| Definition: | OtherConstructionClassValue is a code list used to further classify an OtherConstruction. |
| Stereotype: | «CodeList» |

**OtherConstructionFunctionValue**

| | |
|---|---|
| Definition: | OtherConstructionFunctionValue is a code list that enumerates the different purposes of an OtherConstruction. |
| Stereotype: | «CodeList» |

**OtherConstructionUsageValue**

| | |
|---|---|
| Definition: | OtherConstructionUsageValue is a code list that enumerates the different uses of an OtherConstruction. |
| Stereotype: | «CodeList» |

**WindowClassValue**

| | |
|---|---|
| Definition: | WindowClassValue is a code list used to further classify a Window. |
| Stereotype: | «CodeList» |

**WindowFunctionValue**

| | |
|---|---|
| Definition: | WindowFunctionValue is a code list that enumerates the different purposes of a Window. |
| Stereotype: | «CodeList» |

**WindowUsageValue**

| | |
|---|---|
| Definition: | WindowUsageValue is a code list that enumerates the different uses of a Window. |
| Stereotype: | «CodeList» |

### 9.15.6. Enumerations

**ConditionOfConstructionValue**

| | |
|---|---|
| Definition: | ConditionOfConstructionValue enumerates different conditions of a construction. [cf. INSPIRE] |
| StereoType: | <<Enumeration>> |

| Literal value | Definition |
|---|---|
| declined | Indicates that the construction cannot be used under normal conditions, though its main elements (walls, roof) are still present. [cf. INSPIRE] |
| demolished | Indicates that the construction has been demolished. There are no more visible remains. [cf. INSPIRE] |
| functional | Indicates that the construction is functional. [cf. INSPIRE] |
| projected | Indicates that the construction is being designed. Construction works have not yet started. [cf. INSPIRE] |
| ruin | Indicates that the construction has been partly demolished and some main elements (roof, walls) have been destroyed. There are some visible remains of the construction. [cf. INSPIRE] |
| underConstruction | Indicates that the construction is under construction and not yet functional. This applies only to the initial construction works of the construction and not to maintenance work. [cf. INSPIRE] |

**HeightStatusValue**

| | |
|---|---|
| Definition: | HeightStatusValue enumerates the different methods used to capture a height. [cf. INSPIRE] |
| StereoType: | <<Enumeration>> |

| Literal value | Definition |
|---|---|
| estimated | Indicates that the height has been estimated and not measured. [cf. INSPIRE] |
| measured | Indicates that the height has been (directly or indirectly) measured. [cf. INSPIRE] |

**RelationToConstruction**

| | |
|---|---|
| Definition: | RelationToConstruction is an enumeration used to describe whether an installation is positioned inside and/or outside of a construction. |
| StereoType: | <<Enumeration>> |

| Literal value | Definition |
|---|---|
| inside | Indicates that the installation is positioned inside of the construction. |
| outside | Indicates that the installation is positioned outside of the construction. |
| bothInsideAndOutside | Indicates that the installation is positioned inside as well as outside of the construction. |

# 9.16. Bridge

| | |
|---|---|
| Description: | The Bridge module supports representation of thematic and spatial aspects of bridges, bridge parts, bridge installations, and interior bridge structures. |
| Parent Package: | CityGML |
| Stereotype: | «ApplicationSchema» |

## 9.16.1. Classes

**AbstractBridge**

| | |
|---|---|
| Definition: | AbstractBridge is an abstract superclass representing the common attributes and associations of the classes Bridge and BridgePart. |
| Subclass of: | AbstractConstruction |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| bridgeConstructiveElement | BridgeConstructiveElement [*] | Relates the constructive elements to the Bridge or BridgePart. |
| bridgeInstallation | BridgeInstallation [*] | Relates the installation objects to the Bridge or BridgePart. |
| bridgeFurniture | BridgeFurniture [*] | Relates the furniture objects to the Bridge or BridgePart. |
| bridgeRoom | BridgeRoom [*] | Relates the rooms to the Bridge or BridgePart. |
| address | Address [*] | Relates the addresses to the Bridge or BridgePart. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | BridgeClassValue [0..1] | Indicates the specific type of the Bridge or BridgePart. |
| function | BridgeFunctionValue [0..*] | Specifies the intended purposes of the Bridge or BridgePart. |
| usage | BridgeUsageValue [0..*] | Specifies the actual uses of the Bridge or BridgePart. |
| isMovable | Boolean [0..1] | Indicates whether the Bridge or BridgePart can be moved to allow for watercraft to pass. |
| adeOfAbstractBridge | ADEOfAbstractBridge [0..*] | Augments AbstractBridge with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Bridge**

| | |
|---|---|
| Definition: | A Bridge represents a structure that affords the passage of pedestrians, animals, vehicles, and service(s) above obstacles or between two points at a height above ground. [cf. ISO 6707-1] |
| Subclass of: | AbstractBridge |
| Stereotype: | «TopLevelFeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| bridgePart | BridgePart [*] | Relates the bridge parts to the Bridge. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfBridge | ADEOfBridge [0..*] | Augments the Bridge with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**BridgeConstructiveElement**

| | |
|---|---|
| Definition: | A BridgeConstructiveElement is an element of a bridge which is essential from a structural point of view. Examples are pylons, anchorages, slabs, beams. |
| Subclass of: | AbstractConstructiveElement |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | BridgeConstructiveElementClassValue [0..1] | Indicates the specific type of the BridgeConstructiveElement. |
| function | BridgeConstructiveElementFunctionValue [0..*] | Specifies the intended purposes of the BridgeConstructiveElement. |
| usage | BridgeConstructiveElementUsageValue [0..*] | Specifies the actual uses of the BridgeConstructiveElement. |
| adeOfBridgeConstructiveElement | ADEOfBridgeConstructiveElement [0..*] | Augments the BridgeConstructiveElement with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**BridgeFurniture**

| | |
|---|---|
| Definition: | A BridgeFurniture is an equipment for occupant use, usually not fixed to the bridge. [cf. ISO 6707-1] |
| Subclass of: | AbstractFurniture |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | BridgeFurnitureClassValue [0..1] | Indicates the specific type of the BridgeFurniture. |
| function | BridgeFurnitureFunctionValue [0..*] | Specifies the intended purposes of the BridgeFurniture. |
| usage | BridgeFurnitureUsageValue [0..*] | Specifies the actual uses of the BridgeFurniture. |
| adeOfBridgeFurniture | ADEOfBridgeFurniture [0..*] | Augments the BridgeFurniture with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**BridgeInstallation**

| | |
|---|---|
| Definition: | A BridgeInstallation is a permanent part of a Bridge (inside and/or outside) which does not have the significance of a BridgePart. In contrast to BridgeConstructiveElements, a BridgeInstallation is not essential from a structural point of view. Examples are stairs, antennas or railways. |
| Subclass of: | AbstractInstallation |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | BridgeInstallationClassValue [0..1] | Indicates the specific type of the BridgeInstallation. |
| function | BridgeInstallationFunctionValue [0..*] | Specifies the intended purposes of the BridgeInstallation. |
| usage | BridgeInstallationUsageValue [0..*] | Specifies the actual uses of the BridgeInstallation. |
| adeOfBridgeInstallation | ADEOfBridgeInstallation [0..*] | Augments the BridgeInstallation with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**BridgePart**

| | |
|---|---|
| Definition: | A BridgePart is a physical or functional subdivision of a Bridge. It would be considered a Bridge, if it were not part of a collection of other BridgeParts. |
| Subclass of: | AbstractBridge |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfBridgePart | ADEOfBridgePart [0..*] | Augments the BridgePart with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**BridgeRoom**

| | |
|---|---|
| Definition: | A BridgeRoom is a space within a Bridge or BridgePart intended for human occupancy (e.g. a place of work or recreation) and/or containment (storage) of animals or things. A BridgeRoom is bounded physically and/or virtually (e.g. by ClosureSurfaces or GenericSurfaces). |
| Subclass of: | AbstractUnoccupiedSpace |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| bridgeInstallation | BridgeInstallation [*] | Relates to the installation objects to the BridgeRoom. |
| boundary | AbstractThematicSurface [*] | Relates to the surfaces that bound the BridgeRoom. This relation is inherited from the Core module. |
| bridgeFurniture | BridgeFurniture [*] | Relates the furniture objects to the BridgeRoom. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | BridgeRoomClassValue [0..1] | Indicates the specific type of the BridgeRoom. |
| function | BridgeRoomFunctionValue [0..*] | Specifies the intended purposes of the BridgeRoom. |
| usage | BridgeRoomUsageValue [0..*] | Specifies the actual uses of the BridgeRoom. |
| adeOfBridgeRoom | ADEOfBridgeRoom [0..*] | Augments the BridgeRoom with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.16.2. Data Types

**ADEOfAbstractBridge**

| | |
|---|---|
| Definition: | ADEOfAbstractBridge acts as a hook to define properties within an ADE that are to be added to AbstractBridge. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfBridge**

| | |
|---|---|
| Definition: | ADEOfBridge acts as a hook to define properties within an ADE that are to be added to a Bridge. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfBridgeConstructiveElement**

| | |
|---|---|
| Definition: | ADEOfBridgeConstructiveElement acts as a hook to define properties within an ADE that are to be added to a BridgeConstructiveElement. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfBridgeFurniture**

| | |
|---|---|
| Definition: | ADEOfBridgeFurniture acts as a hook to define properties within an ADE that are to be added to a BridgeFurniture. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfBridgeInstallation**

| | |
|---|---|
| Definition: | ADEOfBridgeInstallation acts as a hook to define properties within an ADE that are to be added to a BridgeInstallation. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfBridgePart**

| | |
|---|---|
| Definition: | ADEOfBridgePart acts as a hook to define properties within an ADE that are to be added to a BridgePart. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfBridgeRoom**

| | |
|---|---|
| Definition: | ADEOfBridgeRoom acts as a hook to define properties within an ADE that are to be added to a BridgeRoom. |
| Subclass of: | None |
| Stereotype: | «DataType» |

### 9.16.3. Basic Types

none

### 9.16.4. Unions

none

### 9.16.5. Code Lists

**BridgeClassValue**

| | |
|---|---|
| Definition: | BridgeClassValue is a code list used to further classify a Bridge. |
| Stereotype: | «CodeList» |

**BridgeConstructiveElementClassValue**

| | |
|---|---|
| Definition: | BridgeConstructiveElementClassValue is a code list used to further classify a BridgeConstructiveElement. |
| Stereotype: | «CodeList» |

**BridgeConstructiveElementFunctionValue**

| | |
|---|---|
| Definition: | BridgeConstructiveElementFunctionValue is a code list that enumerates the different purposes of a BridgeConstructiveElement. |
| Stereotype: | «CodeList» |

**BridgeConstructiveElementUsageValue**

| | |
|---|---|
| Definition: | BridgeConstructiveElementUsageValue is a code list that enumerates the different uses of a BridgeConstructiveElement. |
| Stereotype: | «CodeList» |

**BridgeFunctionValue**

| | |
|---|---|
| Definition: | BridgeFunctionValue is a code list that enumerates the different purposes of a Bridge. |
| Stereotype: | «CodeList» |

**BridgeFurnitureClassValue**

| | |
|---|---|
| Definition: | BridgeFurnitureClassValue is a code list used to further classify a BridgeFurniture. |
| Stereotype: | «CodeList» |

**BridgeFurnitureFunctionValue**

| | |
|---|---|
| Definition: | BridgeFurnitureFunctionValue is a code list that enumerates the different purposes of a BridgeFurniture. |
| Stereotype: | «CodeList» |

**BridgeFurnitureUsageValue**

| | |
|---|---|
| Definition: | BridgeFurnitureUsageValue is a code list that enumerates the different uses of a BridgeFurniture. |
| Stereotype: | «CodeList» |

**BridgeInstallationClassValue**

| | |
|---|---|
| Definition: | BridgeInstallationClassValue is a code list used to further classify a BridgeInstallation. |
| Stereotype: | «CodeList» |

**BridgeInstallationFunctionValue**

| | |
|---|---|
| Definition: | BridgeInstallationFunctionValue is a code list that enumerates the different purposes of a BridgeInstallation. |
| Stereotype: | «CodeList» |

**BridgeInstallationUsageValue**

| | |
|---|---|
| Definition: | BridgeInstallationUsageValue is a code list that enumerates the different uses of a BridgeInstallation. |
| Stereotype: | «CodeList» |

**BridgeRoomClassValue**

| | |
|---|---|
| Definition: | BridgeRoomClassValue is a code list used to further classify a BridgeRoom. |
| Stereotype: | «CodeList» |

**BridgeRoomFunctionValue**

| | |
|---|---|
| Definition: | BridgeRoomFunctionValue is a code list that enumerates the different purposes of a BridgeRoom. |
| Stereotype: | «CodeList» |

**BridgeRoomUsageValue**

| | |
|---|---|
| Definition: | BridgeRoomUsageValue is a code list that enumerates the different uses of a BridgeRoom. |
| Stereotype: | «CodeList» |

**BridgeUsageValue**

| | |
|---|---|
| Definition: | BridgeUsageValue is a code list that enumerates the different uses of a Bridge. |
| Stereotype: | «CodeList» |

### 9.16.6. Enumerations

none

# 9.17. Building

| | |
|---|---|
| Description: | The Building module supports representation of thematic and spatial aspects of buildings, building parts, building installations, building subdivisions, and interior building structures. |
| Parent Package: | CityGML |
| Stereotype: | «ApplicationSchema» |

## 9.17.1. Classes

**AbstractBuilding**

| | |
|---|---|
| Definition: | AbstractBuilding is an abstract superclass representing the common attributes and associations of the classes Building and BuildingPart. |
| Subclass of: | AbstractConstruction |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| buildingFurniture | BuildingFurniture [*] | Relates the furniture objects to the Building or BuildingPart. |
| buildingRoom | BuildingRoom [*] | Relates the rooms to the Building or BuildingPart. |
| buildingInstallation | BuildingInstallation [*] | Relates the installation objects to the Building or BuildingPart. |
| buildingSubdivision | AbstractBuildingSubdivision [*] | Relates the logical subdivisions to the Building or BuildingPart. |
| buildingConstructiveElement | BuildingConstructiveElement [*] | Relates the constructive elements to the Building or BuildingPart. |
| address | Address [*] | Relates the addresses to the Building or BuildingPart. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | BuildingClassValue [0..1] | Indicates the specific type of the Building or BuildingPart. |
| function | BuildingFunctionValue [0..*] | Specifies the intended purposes of the Building or BuildingPart. |
| usage | BuildingUsageValue [0..*] | Specifies the actual uses of the Building or BuildingPart. |
| roofType | RoofTypeValue [0..1] | Indicates the shape of the roof of the Building or BuildingPart. |
| storeysAboveGround | Integer [0..1] | Indicates the number of storeys positioned above ground level. |
| storeysBelowGround | Integer [0..1] | Indicates the number of storeys positioned below ground level. |
| storeyHeightsAboveGround | MeasureOrNilReasonList [0..1] | Lists the heights of each storey above ground. The first value in the list denotes the height of the storey closest to the ground level, the last value denotes the height furthest away. |
| storeyHeightsBelowGround | MeasureOrNilReasonList [0..1] | Lists the height of each storey below ground. The first value in the list denotes the height of the storey closest to the ground level, the last value denotes the height furthest away. |
| adeOfAbstractBuilding | ADEOfAbstractBuilding [0..*] | Augments AbstractBuilding with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**AbstractBuildingSubdivision**

| | |
|---|---|
| Definition: | AbstractBuildingSubdivision is the abstract superclass for different kinds of logical building subdivisions. |
| Subclass of: | AbstractLogicalSpace |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| buildingRoom | BuildingRoom [*] | Relates the rooms to the building subdivision. |
| buildingFurniture | BuildingFurniture [*] | Relates the furniture objects to the building subdivision. |
| buildingConstructiveElement | BuildingConstructiveElement [*] | Relates the constructive elements to the building subdivision. |
| buildingInstallation | BuildingInstallation [*] | Relates the installation objects to the building subdivision. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | BuildingSubdivisionClassValue [0..1] | Indicates the specific type of the building subdivision. |
| function | BuildingSubdivisionFunctionValue [0..*] | Specifies the intended purposes of the building subdivision. |
| usage | BuildingSubdivisionUsageValue [0..*] | Specifies the actual uses of the building subdivision. |
| elevation | Elevation [0..*] | Specifies qualified elevations of the building subdivision in relation to a well-defined surface which is commonly taken as origin (e.g. geoid or water level). [cf. INSPIRE] |
| sortKey | Real [0..1] | Defines an order among the objects that belong to the building subdivision. An example is the sorting of storeys. |
| adeOfAbstractBuildingSubdivision | ADEOfAbstractBuildingSubdivision [0..*] | Augments AbstractBuildingSubdivision with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Building**

| | |
|---|---|
| Definition: | A Building is a free-standing, self-supporting construction that is roofed, usually walled, and can be entered by humans and is normally designed to stand permanently in one place. It is intended for human occupancy (e.g. a place of work or recreation), habitation and/or shelter of humans, animals or things. |
| Subclass of: | AbstractBuilding |
| Stereotype: | «TopLevelFeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| buildingPart | BuildingPart [*] | Relates the building parts to the Building. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfBuilding | ADEOfBuilding [0..*] | Augments the Building with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**BuildingConstructiveElement**

| | |
|---|---|
| Definition: | A BuildingConstructiveElement is an element of a Building which is essential from a structural point of view. Examples are walls, slabs, staircases, beams. |
| Subclass of: | AbstractConstructiveElement |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | BuildingConstructiveElementClassValue [0..1] | Indicates the specific type of the BuildingConstructiveElement. |
| function | BuildingConstructiveElementFunctionValue [0..*] | Specifies the intended purposes of the BuildingConstructiveElement. |
| usage | BuildingConstructiveElementUsageValue [0..*] | Specifies the actual uses of the BuildingConstructiveElement. |
| adeOfBuildingConstructiveElement | ADEOfBuildingConstructiveElement [0..*] | Augments the BuildingConstructiveElement with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**BuildingFurniture**

| | |
|---|---|
| Definition: | A BuildingFurniture is an equipment for occupant use, usually not fixed to the building. [cf. ISO 6707-1] |
| Subclass of: | AbstractFurniture |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | BuildingFurnitureClassValue [0..1] | Indicates the specific type of the BuildingFurniture. |
| function | BuildingFurnitureFunctionValue [0..*] | Specifies the intended purposes of the BuildingFurniture. |
| usage | BuildingFurnitureUsageValue [0..*] | Specifies the actual uses of the BuildingFurniture. |
| adeOfBuildingFurniture | ADEOfBuildingFurniture [0..*] | Augments the BuildingFurniture with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**BuildingInstallation**

| | |
|---|---|
| Definition: | A BuildingInstallation is a permanent part of a Building (inside and/or outside) which has not the significance of a BuildingPart. Examples are stairs, antennas, balconies or small roofs. |
| Subclass of: | AbstractInstallation |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | BuildingInstallationClassValue [0..1] | Indicates the specific type of the BuildingInstallation. |
| function | BuildingInstallationFunctionValue [0..*] | Specifies the intended purposes of the BuildingInstallation. |
| usage | BuildingInstallationUsageValue [0..*] | Specifies the actual uses of the BuildingInstallation. |
| adeOfBuildingInstallation | ADEOfBuildingInstallation [0..*] | Augments the BuildingInstallation with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**BuildingPart**

| | |
|---|---|
| Definition: | A BuildingPart is a physical or functional subdivision of a Building. It would be considered a Building, if it were not part of a collection of other BuildingParts. |
| Subclass of: | AbstractBuilding |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfBuildingPart | ADEOfBuildingPart [0..*] | Augments the BuildingPart with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**BuildingRoom**

| | |
|---|---|
| Definition: | A BuildingRoom is a space within a Building or BuildingPart intended for human occupancy (e.g. a place of work or recreation) and/or containment of animals or things. A BuildingRoom is bounded physically and/or virtually (e.g. by ClosureSurfaces or GenericSurfaces). |
| Subclass of: | AbstractUnoccupiedSpace |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| buildingInstallation | BuildingInstallation [*] | Relates the installation objects to the BuildingRoom. |
| buildingFurniture | BuildingFurniture [*] | Relates the furniture objects to the BuildingRoom. |
| boundary | AbstractThematicSurface [*] | Relates to the surfaces that bound the BuildingRoom. This relation is inherited from the Core module. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | BuildingRoomClassValue [0..1] | Indicates the specific type of the BuildingRoom. |
| function | BuildingRoomFunctionValue [0..*] | Specifies the intended purposes of the BuildingRoom. |
| usage | BuildingRoomUsageValue [0..*] | Specifies the actual uses of the BuildingRoom. |
| roomHeight | RoomHeight [0..*] | Specifies qualified heights of the BuildingRoom. |
| adeOfBuildingRoom | ADEOfBuildingRoom [0..*] | Augments the BuildingRoom with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**BuildingUnit**

| | |
|---|---|
| Definition: | A BuildingUnit is a logical subdivision of a Building. BuildingUnits are formed according to some homogeneous property like function, ownership, management, or accessibility. They may be separately sold, rented out, inherited, managed, etc. |
| Subclass of: | AbstractBuildingSubdivision |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| storey | Storey [*] | Relates to the storeys on which the BuildingUnit is located. |
| address | Address [*] | Relates to the addresses that are assigned to the BuildingUnit. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfBuildingUnit | ADEOfBuildingUnit [0..*] | Augments the BuildingUnit with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Storey**

| | |
|---|---|
| Definition: | A Storey is typically a horizontal section of a Building. Storeys are not always defined according to the building structure, but can also be defined according to logical considerations. |
| Subclass of: | AbstractBuildingSubdivision |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| boundary | AbstractThematicSurface [*] | Relates to the surfaces that bound the Storey. This relation is inherited from the Core module. |
| buildingUnit | BuildingUnit [*] | Relates to the building units that belong to the Storey. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfStorey | ADEOfStorey [0..*] | Augments the Storey with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.17.2. Data Types

**ADEOfAbstractBuilding**

| | |
|---|---|
| Definition: | ADEOfAbstractBuilding acts as a hook to define properties within an ADE that are to be added to AbstractBuilding. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfAbstractBuildingSubdivision**

| | |
|---|---|
| Definition: | ADEOfAbstractBuildingSubdivision acts as a hook to define properties within an ADE that are to be added to AbstractBuildingSubdivision. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfBuilding**

| | |
|---|---|
| Definition: | ADEOfBuilding acts as a hook to define properties within an ADE that are to be added to a Building. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfBuildingConstructiveElement**

| | |
|---|---|
| Definition: | ADEOfBuildingConstructiveElement acts as a hook to define properties within an ADE that are to be added to a BuildingConstructiveElement. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfBuildingFurniture**

| | |
|---|---|
| Definition: | ADEOfBuildingFurniture acts as a hook to define properties within an ADE that are to be added to a BuildingFurniture. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfBuildingInstallation**

| | |
|---|---|
| Definition: | ADEOfBuildingInstallation acts as a hook to define properties within an ADE that are to be added to a BuildingInstallation. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfBuildingPart**

| | |
|---|---|
| Definition: | ADEOfBuildingPart acts as a hook to define properties within an ADE that are to be added to a BuildingPart. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfBuildingRoom**

| | |
|---|---|
| Definition: | ADEOfBuildingRoom acts as a hook to define properties within an ADE that are to be added to a BuildingRoom. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfBuildingUnit**

| | |
|---|---|
| Definition: | ADEOfBuildingUnit acts as a hook to define properties within an ADE that are to be added to a BuildingUnit. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfStorey**

| | |
|---|---|
| Definition: | ADEOfStorey acts as a hook to define properties within an ADE that are to be added to a Storey. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**RoomHeight**

| | |
|---|---|
| Definition: | The RoomHeight represents a vertical distance (measured or estimated) between a low reference and a high reference. [cf. INSPIRE] |
| Subclass of: | None |
| Stereotype: | «DataType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| highReference | RoomElevationReferenceValue [1..1] | Indicates the high point used to calculate the value of the room height. |
| lowReference | RoomElevationReferenceValue [1..1] | Indicates the low point used to calculate the value of the room height. |
| status | HeightStatusValue [1..1] | Indicates the way the room height has been captured. |
| value | Length [1..1] | Specifies the value of the room height. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.17.3. Basic Types

none

### 9.17.4. Unions

none

### 9.17.5. Code Lists

**BuildingClassValue**

| | |
|---|---|
| Definition: | BuildingClassValue is a code list used to further classify a Building. |
| Stereotype: | «CodeList» |

**BuildingConstructiveElementClassValue**

| | |
|---|---|
| Definition: | BuildingConstructiveElementClassValue is a code list used to further classify a BuildingConstructiveElement. |
| Stereotype: | «CodeList» |

**BuildingConstructiveElementFunctionValue**

| | |
|---|---|
| Definition: | BuildingConstructiveElementFunctionValue is a code list that enumerates the different purposes of a BuildingConstructiveElement. |
| Stereotype: | «CodeList» |

**BuildingConstructiveElementUsageValue**

| | |
|---|---|
| Definition: | BuildingConstructiveElementUsageValue is a code list that enumerates the different uses of a BuildingConstructiveElement. |
| Stereotype: | «CodeList» |

**BuildingFunctionValue**

| | |
|---|---|
| Definition: | BuildingFunctionValue is a code list that enumerates the different purposes of a Building. |
| Stereotype: | «CodeList» |

**BuildingFurnitureClassValue**

| | |
|---|---|
| Definition: | BuildingFurnitureClassValue is a code list used to further classify a BuildingFurniture. |
| Stereotype: | «CodeList» |

**BuildingFurnitureFunctionValue**

| | |
|---|---|
| Definition: | BuildingFurnitureFunctionValue is a code list that enumerates the different purposes of a BuildingFurniture. |
| Stereotype: | «CodeList» |

**BuildingFurnitureUsageValue**

| | |
|---|---|
| Definition: | BuildingFurnitureUsageValue is a code list that enumerates the different uses of a BuildingFurniture. |
| Stereotype: | «CodeList» |

**BuildingInstallationClassValue**

| | |
|---|---|
| Definition: | BuildingInstallationClassValue is a code list used to further classify a BuildingInstallation. |
| Stereotype: | «CodeList» |

**BuildingInstallationFunctionValue**

| | |
|---|---|
| Definition: | BuildingInstallationFunctionValue is a code list that enumerates the different purposes of a BuildingInstallation. |
| Stereotype: | «CodeList» |

**BuildingInstallationUsageValue**

| | |
|---|---|
| Definition: | BuildingInstallationUsageValue is a code list that enumerates the different uses of a BuildingInstallation. |
| Stereotype: | «CodeList» |

**BuildingRoomClassValue**

| | |
|---|---|
| Definition: | BuildingRoomClassValue is a code list used to further classify a BuildingRoom. |
| Stereotype: | «CodeList» |

**BuildingRoomFunctionValue**

| | |
|---|---|
| Definition: | BuildingRoomFunctionValue is a code list that enumerates the different purposes of a BuildingRoom. |
| Stereotype: | «CodeList» |

**BuildingRoomUsageValue**

| | |
|---|---|
| Definition: | BuildingRoomUsageValue is a code list that enumerates the different uses of a BuildingRoom. |
| Stereotype: | «CodeList» |

**BuildingSubdivisionClassValue**

| | |
|---|---|
| Definition: | BuildingSubdivisionClassValue is a code list used to further classify a BuildingSubdivision. |
| Stereotype: | «CodeList» |

**BuildingSubdivisionFunctionValue**

| | |
|---|---|
| Definition: | BuildingSubdivisionFunctionValue is a code list that enumerates the different purposes of a BuildingSubdivision. |
| Stereotype: | «CodeList» |

**BuildingSubdivisionUsageValue**

| | |
|---|---|
| Definition: | BuildingSubdivisionUsageValue is a code list that enumerates the different uses of a BuildingSubdivision. |
| Stereotype: | «CodeList» |

**BuildingUsageValue**

| | |
|---|---|
| Definition: | BuildingUsageValue is a code list that enumerates the different uses of a Building. |
| Stereotype: | «CodeList» |

**RoofTypeValue**

| | |
|---|---|
| Definition: | RoofTypeValue is a code list that enumerates different roof types. |
| Stereotype: | «CodeList» |

**RoomElevationReferenceValue**

| | |
|---|---|
| Definition: | RoomElevationReferenceValue is a code list that enumerates the different elevation reference levels used to measure room heights. |
| Stereotype: | «CodeList» |

### 9.17.6. Enumerations

none

## 9.18. Tunnel

| | |
|---|---|
| Description: | The Tunnel module supports representation of thematic and spatial aspects of tunnels, tunnel parts, tunnel installations, and interior tunnel structures. |
| Parent Package: | CityGML |
| Stereotype: | «ApplicationSchema» |

### 9.18.1. Classes

**AbstractTunnel**

| | |
|---|---|
| Definition: | AbstractTunnel is an abstract superclass representing the common attributes and associations of the classes Tunnel and TunnelPart. |
| Subclass of: | AbstractConstruction |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| hollowSpace | HollowSpace [*] | Relates the hollow spaces to the Tunnel or TunnelPart. |
| tunnelConstructiveElement | TunnelConstructiveElement [*] | Relates the constructive elements to the Tunnel or TunnelPart. |
| tunnelInstallation | TunnelInstallation [*] | Relates the installation objects to the Tunnel or TunnelPart. |
| tunnelFurniture | TunnelFurniture [*] | Relates the furniture objects to the Tunnel or TunnelPart. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | TunnelClassValue [0..1] | Indicates the specific type of the Tunnel or TunnelPart. |
| function | TunnelFunctionValue [0..*] | Specifies the intended purposes of the Tunnel or TunnelPart. |
| usage | TunnelUsageValue [0..*] | Specifies the actual uses of the Tunnel or TunnelPart. |
| adeOfAbstractTunnel | ADEOfAbstractTunnel [0..*] | Augments AbstractTunnel with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**HollowSpace**

| | |
|---|---|
| Definition: | A HollowSpace is a space within a Tunnel or TunnelPart intended for certain functions (e.g. transport or passage ways, service rooms, emergency shelters). A HollowSpace is bounded physically and/or virtually (e.g. by ClosureSurfaces or GenericSurfaces). |
| Subclass of: | AbstractUnoccupiedSpace |
| Stereotype: | «FeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| tunnelInstallation | TunnelInstallation [*] | Relates the installation objects to the HollowSpace. |
| tunnelFurniture | TunnelFurniture [*] | Relates the furniture objects to the HollowSpace. |
| boundary | AbstractThematicSurface [*] | Relates to the surfaces that bound the HollowSpace. This relation is inherited from the Core module. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | HollowSpaceClassValue [0..1] | Indicates the specific type of the HollowSpace. |
| function | HollowSpaceFunctionValue [0..*] | Specifies the intended purposes of the HollowSpace. |
| usage | HollowSpaceUsageValue [0..*] | Specifies the actual uses of the HollowSpace. |
| adeOfHollowSpace | ADEOfHollowSpace [0..*] | Augments the HollowSpace with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**Tunnel**

| | |
|---|---|
| Definition: | A Tunnel represents a horizontal or sloping enclosed passage way of a certain length, mainly underground or underwater. [cf. ISO 6707-1] |
| Subclass of: | AbstractTunnel |
| Stereotype: | «TopLevelFeatureType» |

| Role name | Target class and multiplicity | Definition |
|---|---|---|
| tunnelPart | TunnelPart [*] | Relates the tunnel parts to the Tunnel. |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfTunnel | ADEOfTunnel [0..*] | Augments the Tunnel with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**TunnelConstructiveElement**

| | |
|---|---|
| Definition: | A TunnelConstructiveElement is an element of a Tunnel which is essential from a structural point of view. Examples are walls, slabs, beams. |
| Subclass of: | AbstractConstructiveElement |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | TunnelConstructiveElementClassValue [0..1] | Indicates the specific type of the TunnelConstructiveElement. |
| function | TunnelConstructiveElementFunctionValue [0..*] | Specifies the intended purposes of the TunnelConstructiveElement. |
| usage | TunnelConstructiveElementUsageValue [0..*] | Specifies the actual uses of the TunnelConstructiveElement. |
| adeOfTunnelConstructiveElement | ADEOfTunnelConstructiveElement [0..*] | Augments the TunnelConstructiveElement with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**TunnelFurniture**

| | |
|---|---|
| Definition: | A TunnelFurniture is an equipment for occupant use, usually not fixed to the tunnel. [cf. ISO 6707-1] |
| Subclass of: | AbstractFurniture |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | TunnelFurnitureClassValue [0..1] | Indicates the specific type of the TunnelFurniture. |
| function | TunnelFurnitureFunctionValue [0..*] | Specifies the intended purposes of the TunnelFurniture. |
| usage | TunnelFurnitureUsageValue [0..*] | Specifies the actual uses of the TunnelFurniture. |
| adeOfTunnelFurniture | ADEOfTunnelFurniture [0..*] | Augments the TunnelFurniture with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**TunnelInstallation**

| | |
|---|---|
| Definition: | A TunnelInstallation is a permanent part of a Tunnel (inside and/or outside) which does not have the significance of a TunnelPart. In contrast to TunnelConstructiveElement, a TunnelInstallation is not essential from a structural point of view. Examples are stairs, antennas or railings. |
| Subclass of: | AbstractInstallation |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| class | TunnelInstallationClassValue [0..1] | Indicates the specific type of the TunnelInstallation. |
| function | TunnelInstallationFunctionValue [0..*] | Specifies the intended purposes of the TunnelInstallation. |
| usage | TunnelInstallationUsageValue [0..*] | Specifies the actual uses of the TunnelInstallation. |
| adeOfTunnelInstallation | ADEOfTunnelInstallation [0..*] | Augments the TunnelInstallation with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

**TunnelPart**

| | |
|---|---|
| Definition: | A TunnelPart is a physical or functional subdivision of a Tunnel. It would be considered a Tunnel, if it were not part of a collection of other TunnelParts. |
| Subclass of: | AbstractTunnel |
| Stereotype: | «FeatureType» |

| Attribute | Value type and multiplicity | Definition |
|---|---|---|
| adeOfTunnelPart | ADEOfTunnelPart [0..*] | Augments the TunnelPart with properties defined in an ADE. |

Note: Unless otherwise specified, all attributes and role names have the stereotype «Property».

### 9.18.2. Data Types

**ADEOfAbstractTunnel**

| | |
|---|---|
| Definition: | ADEOfAbstractTunnel acts as a hook to define properties within an ADE that are to be added to AbstractTunnel. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfHollowSpace**

| | |
|---|---|
| Definition: | ADEOfHollowSpace acts as a hook to define properties within an ADE that are to be added to a HollowSpace. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfTunnel**

| | |
|---|---|
| Definition: | ADEOfTunnel acts as a hook to define properties within an ADE that are to be added to a Tunnel. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfTunnelConstructiveElement**

| | |
|---|---|
| Definition: | ADEOfTunnelConstructiveElement acts as a hook to define properties within an ADE that are to be added to a TunnelConstructiveElement. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfTunnelFurniture**

| | |
|---|---|
| Definition: | ADEOfTunnelFurniture acts as a hook to define properties within an ADE that are to be added to a TunnelFurniture. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfTunnelInstallation**

| | |
|---|---|
| Definition: | ADEOfTunnelInstallation acts as a hook to define properties within an ADE that are to be added to a TunnelInstallation. |
| Subclass of: | None |
| Stereotype: | «DataType» |

**ADEOfTunnelPart**

| | |
|---|---|
| Definition: | ADEOfTunnelPart acts as a hook to define properties within an ADE that are to be added to a TunnelPart. |
| Subclass of: | None |
| Stereotype: | «DataType» |

### 9.18.3. Basic Types

none

### 9.18.4. Unions

none

### 9.18.5. Code Lists

**HollowSpaceClassValue**

| | |
|---|---|
| Definition: | HollowSpaceClassValue is a code list used to further classify a HollowSpace. |
| Stereotype: | «CodeList» |

**HollowSpaceFunctionValue**

| | |
|---|---|
| Definition: | HollowSpaceFunctionValue is a code list that enumerates the different purposes of a HollowSpace. |
| Stereotype: | «CodeList» |

**HollowSpaceUsageValue**

| | |
|---|---|
| Definition: | HollowSpaceUsageValue is a code list that enumerates the different uses of a HollowSpace. |
| Stereotype: | «CodeList» |

**TunnelClassValue**

| | |
|---|---|
| Definition: | TunnelClassValue is a code list used to further classify a Tunnel. |
| Stereotype: | «CodeList» |

**TunnelConstructiveElementClassValue**

| | |
|---|---|
| Definition: | TunnelConstructiveElementClassValue is a code list used to further classify a TunnelConstructiveElement. |
| Stereotype: | «CodeList» |

**TunnelConstructiveElementFunctionValue**

| | |
|---|---|
| Definition: | TunnelConstructiveElementFunctionValue is a code list that enumerates the different purposes of a TunnelConstructiveElement. |
| Stereotype: | «CodeList» |

**TunnelConstructiveElementUsageValue**

| | |
|---|---|
| Definition: | TunnelConstructiveElementUsageValue is a code list that enumerates the different uses of a TunnelConstructiveElement. |
| Stereotype: | «CodeList» |

**TunnelFunctionValue**

| | |
|---|---|
| Definition: | TunnelFunctionValue is a code list that enumerates the different purposes of a Tunnel. |
| Stereotype: | «CodeList» |

**TunnelFurnitureClassValue**

| | |
|---|---|
| Definition: | TunnelFurnitureClassValue is a code list used to further classify a TunnelFurniture. |
| Stereotype: | «CodeList» |

**TunnelFurnitureFunctionValue**

| | |
|---|---|
| Definition: | TunnelFurnitureFunctionValue is a code list that enumerates the different purposes of a TunnelFurniture. |
| Stereotype: | «CodeList» |

**TunnelFurnitureUsageValue**

| | |
|---|---|
| Definition: | TunnelFurnitureUsageValue is a code list that enumerates the different uses of a TunnelFurniture. |
| Stereotype: | «CodeList» |

**TunnelInstallationClassValue**

| | |
|---|---|
| Definition: | TunnelInstallationClassValue is a code list used to further classify a TunnelInstallation. |
| Stereotype: | «CodeList» |

**TunnelInstallationFunctionValue**

| | |
|---|---|
| Definition: | TunnelInstallationFunctionValue is a code list that enumerates the different purposes of a TunnelInstallation. |
| Stereotype: | «CodeList» |

| **TunnelInstallationUsageValue** | |
|---|---|
| Definition: | TunnelInstallationUsageValue is a code list that enumerates the different uses of a TunnelInstallation. |
| Stereotype: | «CodeList» |

| **TunnelUsageValue** | |
|---|---|
| Definition: | TunnelUsageValue is a code list that enumerates the different uses of a Tunnel. |
| Stereotype: | «CodeList» |

### 9.18.6. Enumerations

none

# Annex A: Revision History

| Date | Release | Editor | Primary clauses modified | Description |
|---|---|---|---|---|
| 2016-04-28 | 0.1 | G. Editor | all | initial version |

# Annex B: Glossary

**conformance test class**
set of conformance test modules that must be applied to receive a single certificate of conformance
[OGC 08-131r3, definition 4.4]

**feature**
abstraction of real world phenomena
[ISO 19101-1:2014, definition 4.1.11]

**feature attribute**
characteristic of a feature
[ISO 19101-1:2014, definition 4.1.12]

**feature type**
class of features having common characteristics
[ISO 19156:2011, definition 4.7]

**measurement**
set of operations having the object of determining the value of a quantity
[ISO 19101-2:2018, definition 3.21] / [VIM:1993, 2.1]

**model**
abstraction of some aspects of reality
[ISO 19109:2015, definition 4.15]

**observation**
act of measuring or otherwise determining the value of a property
[ISO 19156:2011, definition 4.11]

**observation procedure**
method, algorithm or instrument, or system of these, which may be used in making an observation
[ISO 19156:2011, 4.12]

**observation result**
estimate of the value of a property determined through a known observation procedure
[ISO 19156:2011, 4.14]

**property**
facet or attribute of an object referenced by a name.
[ISO 19143:2010, definition 4.21]

**requirements class**
aggregate of all requirement modules that must all be satisfied to satisfy a conformance test class
[OGC 08-131r3, definition 4.19]

**schema**
formal description of a model
[ISO 19101-1:2014, definition 4.1.34]

**sensor**
type of observation procedure that provides the estimated value of an observed property at its output
[OGC 08-094r1, definition 4.5]

**Standardization Target**
TBD

**timeseries**
sequence of data values which are ordered in time
[OGC 15-043r3]

**universe of discourse**
view of the real or hypothetical world that includes everything of interest
[ISO 19101-1:2014, definition 4.1.38]

**version**
Particular variation of a spatial object
[INSPIRE Glossary]

# B.1. ISO Concepts

The following concepts from the ISO TC211 Harmonized UML model are referenced by the CityGML Conceptual UML model but do not play a major role in its' definition. They are provided here to support a more complete understanding of the model.

**Area**
The measure of the physical extent of any topologically 2-D geometric object. Usually measured in "square" units of length.
[ISO 19103:2015]

**Boolean**
boolean is the mathematical datatype associated with two-valued logic
[ISO 19103:2015]

**CC_CoordinateOperation**
mathematical operation on coordinates that transforms or converts coordinates to another coordinate reference system.
[ISO 19111:2019]

**Character**
symbol from a standard character-set.
[ISO 19103:2015]

**CharacterString**
Characterstring is a family of datatypes which represent strings of symbols from standard character-sets.
[ISO 19103:2015]

**CRS**

Coordinate reference system which is usually single but may be compound.
[ISO 19111:2019]

**CV_DiscreteCoverage**
A subclass of CV_Coverage that returns a single record of values for any direct position within a single geometric object in its spatiotemporal domain.
[ISO 19123:2005]

**CV_DomainObject**
[ISO 19123:2005]

**CV_GridPointValuePair**
[ISO 19123:2005]

**CV_GridValuesMatrix**
The geometry represented by the various offset vectors is in the image plane of the grid.
[ISO 19123:2005]

**CV_ReferenceableGrid**
[ISO 19123:2005]

**Date**
Date gives values for year, month and day. Representation of Date is specified in ISO 8601. Principles for date and time are further discussed in ISO 19108.
[ISO 19103:2015]

**DateTime**
A DateTime is a combination of a date and a time types. Representation of DateTime is specified in ISO 8601. Principles for date and time are further discussed in ISO 19108.
[ISO 19103:2015]

**Distance**
Used as a type for returning distances and possibly lengths.
[ISO 19103:2015]

**Engineering CRS**
A contextually local coordinate reference system which can be divided into two broad categories:

1. earth-fixed systems applied to engineering activities on or near the surface of the earth;
2. CRSs on moving platforms such as road vehicles, vessels, aircraft or spacecraft.
   [ISO 19111:2019]

**Generic Name**
Generic Name is the abstract class for all names in a NameSpace. Each instance of a GenericName is either a LocalName or a ScopedName.
[ISO 19103:2015]

**Geometry**
[ISO 19107:2003]

**GM_CompositePoint**
[ISO 19107:2003]

**GM_CompositeSolid**
set of geometric solids adjoining one another along common boundary geometric surfaces
[ISO 19107:2003]

**GM_GenericSurface**
GM_Surface and GM_SurfacePatch both represent sections of surface geometry, and therefore share a number of operation signatures. These are defined in the interface class GM_GenericSurface.
[ISO 19107:2003]

**GM_LineString**
consists of sequence of line segments, each having a parameterization like the one for GM_LineSegment
[ISO 19107:2003]

**GM_MultiPrimitive**
[ISO 19107:2003]

**GM_OrientableSurface**
a surface and an orientation inherited from GM_OrientablePrimitive. If the orientation is "+", then the GM_OrientableSurface is a GM_Surface. If the orientation is "-", then the GM_OrientableSurface is a reference to a GM_Surface with an upNormal that reverses the direction for this GM_OrientableSurface, the sense of "the top of the surface".
[ISO 19107:2003]

**GM_PolyhedralSurface**
a GM_Surface composed of polygon surfaces (GM_Polygon) connected along their common boundary curves.
[ISO 19107:2003]

**GM_Position**
a union type consisting of either a DirectPosition or of a reference to a GM_Point from which a DirectPosition shall be obtained.
[ISO 19107:2003]

**GM_Primitive**
The abstract root class of the geometric primitives. Its main purpose is to define the basic "boundary" operation that ties the primitives in each dimension together.
[ISO 19107:2003]

**Integer**
An exact integer value, with no fractional part.
[ISO 19103:2015]

**Internet of Things**
The network of physical objects--“things”--that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the Internet.

Wikipedia

**IO_IdentifiedObjectBase**
[ISO 19103:2015]

**Length**
The measure of distance as an integral, i.e. the limit of an infinite sum of distances between points on a curve.
[ISO 19103:2015]

**Measure**
The result from performing the act or process of ascertaining the extent, dimensions, or quantity of some entity.
[ISO 19103:2015]

**Number**
The base type for all number data, giving the basic algebraic operations.
[ISO 19103:2015]

**Point**
GM_Point is the basic data type for a geometric object consisting of one and only one point.
[ISO 19107:2003]

**Real**
The common binary Real finite implementation using base 2.
[ISO 19103:2015]

**RS_ReferenceSystem**
Description of a spatial and temporal reference system used by a dataset.
[ISO 19111:2019]

**Scoped Name**
ScopedName is a composite of a LocalName for locating another NameSpace and a GenericName valid in that NameSpace. ScopedName contains a LocalName as head and a GenericName, which might be a LocalName or a ScopedName, as tail.
[ISO 19103:2015]

**Solid**
GM_Solid, a subclass of GM_Primitive, is the basis for 3-dimensional geometry. The extent of a solid is defined by the boundary surfaces.
[ISO 19107:2003]

**Time**
Time is the designation of an instant on a selected time scale, astronomical or atomic. It is used in the sense of time of day.
[ISO 19103:2015]

**TM_Duration**
[ISO 19108:2006]

**TM_TemporalPosition**
The position of a TM_Instant relative to a TM_ReferenceSystem.
[ISO 19108:2006]

**Unit of Measure**
Any of the systems devised to measure some physical quantity such distance or area or a system devised to measure such things as the passage of time.
[ISO 19103:2015]

**URI**
Uniform Resource Identifier (URI), is a compact string of characters used to identify or name a resource
[ISO 19103:2015]

**Volume**
Volume is the measure of the physical space of any 3-D geometric object.
[ISO 19103:2015]

# B.2. Abbreviated Terms

- 2D Two Dimensional
- 3D Three Dimensional
- AEC Architecture, Engineering, Construction
- ALKIS German National Standard for Cadastral Information
- ATKIS German National Standard for Topographic and Cartographic Information
- BIM Building Information Modeling
- B-Rep Boundary Representation
- bSI buildingSMART International
- CAD Computer Aided Design
- COLLADA Collaborative Design Activity
- CSG Constructive Solid Geometry
- DTM Digital Terrain Model
- DXF Drawing Exchange Format
- EuroSDR European Spatial Data Research Organisation
- ESRI Environmental Systems Research Institute
- FM Facility Management
- GDF Geographic Data Files
- GDI-DE Spatial Data Infrastructure Germany (Geodateninfrastruktur Deutschland)
- GDI NRW Geodata Infrastructure North-Rhine Westphalia
- GML Geography Markup Language

- IAI International Alliance for Interoperability (now buildingSMART International (bSI))
- IETF Internet Engineering Task Force
- IFC Industry Foundation Classes
- IoT Internet of Things
- ISO International Organization for Standardisation
- ISO/TC211 ISO Technical Committee 211
- LOD Levels of Detail
- MQTT
- NBIMS National Building Information Model Standard
- OASIS Organisation for the Advancement of Structured Information Standards
- OGC Open Geospatial Consortium
- OSCRE Open Standards Consortium for Real Estate
- SIG 3D Special Interest Group 3D of the GDI-DE
- TIC Terrain Intersection Curve
- TIN Triangulated Irregular Network
- UML Unified Modeling Language
- URI Uniform Resource Identifier
- VRML Virtual Reality Modeling Language
- W3C World Wide Web Consortium
- W3DS OGC Web 3D Service
- WFS OGC Web Feature Service
- X3D Open Standards XML-enabled 3D file format of the Web 3D Consortium
- XML Extensible Markup Language
- xAL OASIS extensible Address Language

# Annex C: Bibliography


- Open Geospatial Consortium: **The Specification Model — A Standard for Modular specifications**, OGC 08-131
- Agugiaro, G., Benner, J., Cipriano, P., Nouvel, R., 2018: **The Energy Application Domain Extension for CityGML: enhancing interoperability for urban energy simulations**. Open Geospatial Data, Software and Standards, Vol. 3. https://doi.org/10.1186/s40965-018-0042-y
- Becker, T., Nagel, C., Kolbe, T. H., 2011: **Integrated 3D Modeling of Multi-utility Networks and their Interdependencies for Critical Infrastructure Analysis**. In: T. H. Kolbe, G. König, C. Nagel (Eds.): Advances in 3D Geoinformation Sciences. LNG&C, Springer, Berlin. https://doi.org/10.1007/978-3-642-12670-3_1
- Beil, C., Kolbe, T. H., 2017: **CityGML and the streets of New York - A proposal for detailed street space modelling**. In: Proceedings of the 12th International 3D GeoInfo Conference 2017, ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, Vol. IV-4/W5, ISPRS. http://doi.org/10.5194/isprs-annals-IV-4-W5-9-2017
- Biljecki, F., Stoter, J., Ledoux, H., Zlatanova, S., Çöltekin, A., 2015: **Applications of 3D City Models: State of the Art Review**. ISPRS International Journal of Geo-Information, 4(4). https://doi.org/10.3390/ijgi4042842
- Biljecki, F., Kumar, K., Nagel, C., 2018: **CityGML Application Domain Extension (ADE): overview of developments**. Open Geospatial Data, Software and Standards, 3(1). https://doi.org/10.1186/s40965-018-0055-6
- Billen, R., Zaki, C. E., Servières, M., Moreau, G., Hallot, P., 2012: **Developing an ontology of space: Application to 3D city modeling**. In: Leduc, T., Moreau, G., Billen, R. (eds): Usage, usability, and utility of 3D city models — European COST Action TU0801, EDP Sciences, Nantes, Vol. 02007. https://hal.archives-ouvertes.fr/hal-01521445
- Chaturvedi, K., Smyth, C. S., Gesquière, G., Kutzner, T., Kolbe, T. H., 2015: **Managing versions and history within semantic 3D city models for the next generation of CityGML**. In: Selected papers from the 10th International 3DGeoInfo Conference 2015 in Kuala Lumpur, Malaysia, Springer LNG&C, Berlin. https://doi.org/10.1007/978-3-319-25691-7_11
- Chaturvedi, K., Kolbe, T. H., 2016: **Integrating Dynamic Data and Sensors with Semantic 3D City Models in the context of Smart Cities**. In: Proceedings of the 11th International 3D Geoinfo Conference, ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, Vol. IV-2/W1, ISPRS. https://doi.org/10.5194/isprs-annals-IV-2-W1-31-2016
- Chaturvedi, K., Kolbe, T. H., 2017: **Future City Pilot 1 Engineering Report**, Open Geospatial Consortium. OGC Doc. 19-098
- Chaturvedi, K., Kolbe, T. H., 2019: **A Requirement Analysis on Extending Semantic 3D City Models for Supporting Time-dependent Properties**. In: Proceedings of the 4th International Conference on Smart Data and Smart Cities, ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, Vol. IV-4/W9, ISPRS. https://doi.org/10.5194/isprs-annals-IV-4-W9-19-2019
- Elfes, A., 1989: **Using occupancy grids for mobile robot perception and navigation**. Computer 22(6):46–57. https://doi.org/10.1109/2.30720

- Foley, J., van Dam, A., Feiner, S., Hughes, J., 2002: **Computer Graphics: Principles and Practice**. 2nd ed., Addison Wesley
- Gröger, G., Plümer, L., 2012: **CityGML – Interoperable semantic 3D city models**. ISPRS Journal of Photogrammetry and Remote Sensing, Vol. 71, July 2012. https://dx.doi.org/10.1016/j.isprsjprs.2012.04.004
- Gröger, G., Kolbe, T. H., Nagel, C., Häfele, K.-H., 2012: **OGC City Geography Markup Language (CityGML) Encoding Standard, Version 2.0.0**, Open Geospatial Consortium. OGC Doc. 12-019
- Jensen, Christian S. and Dyreson, Curtis E.: **The Consensus Glossary of Temporal Database Concepts**. February 1998 Version. In: Temporal Databases: Research and Practice [online]. Springer Berlin Heidelberg, 1998. p. 367–405. Lecture Notes in Computer Science. Available from: 10.1007/BFb0053710
- Jensen, Christian S. and Snodgrass, Richard T., eds.: **TR-90, Temporal Database Entries for the Springer Encyclopedia of Database Systems**. Technical Report. TimeCenter, 22 May 2008. Available from: http://timecenter.cs.aau.dk/TimeCenterPublications/TR-90.pdf
- Johnson, Tom: **Bitemporal Data**. Elsevier, 2014. ISBN 978-0-12-408067-6. Available from: 10.1016/C2012-0-06609-4
- Kaden, R., Clemen, C., 2017: **Applying Geodetic Coordinate Reference Systems within Building Information Modeling (BIM)**. In: Proceedings of the FIG Working Week 2017, Helsinki, Finland. https://www.fig.net/resources/proceedings/fig_proceedings/fig2017/papers/ts06h/TS06H_kaden_clemen_8967.pdf
- Kolbe, T. H., Gröger, G., 2003: **Towards unified 3D city models**. In: Proceedings of the Joint ISPRS Commission IV Workshop on Challenges in Geospatial Analysis, Integration and Visualization II, Stuttgart, Germany. https://mediatum.ub.tum.de/doc/1145769/
- Kolbe, T. H., 2009: **Representing and Exchanging 3D City Models with CityGML**. In: J. Lee, S. Zlatanova (Eds.), 3D Geo-Information Sciences, Selected Papers of the 3rd International Workshop on 3D Geo-Information in Seoul, Korea. Springer, Berlin. https://doi.org/10.1007/978-3-540-87395-2_2
- Konde, A., Tauscher, H., Biljecki, F., Crawford, J., 2018: **Floor plans in CityGML**. In: Proceedings of the 13th 3D GeoInfo Conference 2018, ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, Vol. IV-4/W6, 25–32, ISPRS. https://doi.org/10.5194/isprs-annals-IV-4-W6-25-2018
- Kutzner, T., Hijazi, I., Kolbe, T. H., 2018: **Semantic Modelling of 3D Multi-utility Networks for Urban Analyses and Simulations – The CityGML Utility Network ADE**. International Journal of 3-D Information Modeling (IJ3DIM) 7(2), 1-34. https://dx.doi.org/10.4018/IJ3DIM.2018040101
- Kutzner, T., Chaturvedi, K. & Kolbe, T. H., 2020: **CityGML 3.0: New Functions Open Up New Applications**. PFG - Journal of Photogrammetry, Remote Sensing and Geoinformation Science, 88, 43–61. https://doi.org/10.1007/s41064-020-00095-z
- Labetski, A., van Gerwen, S., Tamminga, G., Ledoux, H., Stoter, J., 2018: **A proposal for an improved transportation model in CityGML**. In: Proceedings of the 13th 3D GeoInfo Conference 2018, ISPRS Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, Vol. XLII-4/W10, 89–96. https://doi.org/10.5194/isprs-archives-XLII-4-W10-89-2018
- Liu, Ling and Özsu, M. Tamer, eds.: **Encyclopedia of Database Systems**. New York, NY :

Springer New York, 2018. ISBN 978-1-4614-8266-6. Available from: 10.1007/978-1-4614-8265-9

- Löwner, M.-O., Gröger, G., Benner, J., Biljecki, F., Nagel, C., 2016: **Proposal for a new LOD and multi-representation concept for CityGML**. In: Proceedings of the 11th 3D Geoinfo Conference 2016, ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, Vol. IV-2/W1, 3–12. https://doi.org/10.5194/isprs-annals-IV-2-W1-3-2016
- Nouvel, R., Bahu, J. M., Kaden, R., Kaempf, J., Cipriano, P., Lauster, M., Haefele, K.-H., Munoz, E., Tournaire, O, Casper, E., 2015: **Development of the CityGML Application Domain Extension Energy for Urban Energy Simulation**. In: Proceedings of Building Simulation 2015 - 14th Conference of the International Building Performance Simulation Association, IBPSA, 559-564. http://www.ibpsa.org/proceedings/BS2015/p2863.pdf
- Smith, B., Varzi, A. C., 2000: **Fiat and Bona Fide Boundaries**. Philosophy and Phenomenological Research, Vol. 60, No. 2, 401-420. https://doi.org/10.2307/2653492
- Snodgrass, Richard T: **Developing time-oriented database applications in SQL**. San Francisco, California : Morgan Kaufmann Publishers, July 1999. ISBN 1-55860-436-7. Available from: http://www.cs.arizona.edu/rts/tdbbook.pdf[http://www.cs.arizona.edu/rts/tdbbook.pdf]
- Stadler, A., Kolbe, T. H., 2007: **Spatio-semantic Coherence in the Integration of 3D City Models**. In: Proceedings of the 5th International ISPRS Symposium on Spatial Data Quality ISSDQ 2007 in Enschede. http://www.isprs.org/proceedings/XXXVI/2-C43/Session1/paper_Stadler.pdf
- Vretanos, P. A. 2010: **OpenGIS Web Feature Service 2.0 Interface Standard**, Open Geospatial Consortium. OGC Doc. 09-025r1
- OASIS MQTT Technical Committee: **MQTT Version 5.0 Standard**, OASIS, March 7, 2019, Available from OASIS.
- Reed, C., Belayneh T.: **OGC Indexed 3d Scene Layer (I3S) and Scene Layer Package Format Specification**, Open Geospatial Consortium, Available from OGC Doc. 17-014r7
- [[3dtiles_citation, OGC 3D Tiles]]Cozzi, P., Lilley, S., Getz, G. **OGC 3D Tiles Specification 1.0** Open Geospatial Consortium, Available from OGC Doc. 18-053r2
- Burggraf, D.: **OGC KML 2.3**, Open Geospatial Consortium, Available from OGC Doc. 12-007r2
- Bröring, A., Stasch, C., Echterhoff, J.: **OGC® Sensor Observation Service Interface Standard**, Open Geospatial Consortium, Available from OGC Doc. 12-006
- Liang, S., Huang, C., Khalafbeigi, T.: **OGC SensorThings API Part 1: Sensing**, Open Geospatial Consortium, Available from OGC Doc. 15-078r6
- [[3dps_citation, OGC 3D Portrayal Service]]Hagedorn, B., Thum, S., Reitz, T., Coors, V., Gutbell, R.: **OGC® 3D Portrayal Service 1.0**, Open Geospatial Consortium, Available from OGC Doc. 15-001r3.
- Bhatia, S.,Cozzi, P., Knyazev, A., Parisi, T.: **The GL Transmission Format (glTF)**, The Khronos Group, Available from https://www.khronos.org/gltf.